# Precise asymptotics for subgroups of the symmetric group

The critical model, section capacity, and global reduction

**Abstract**

Let $s_n$ be the number of actual subgroups of $S_n$, and let $G_r$ count the subspaces of $\mathbb{F}_2^r$. We prove

$$s_n = n! G_{\lfloor n/2 \rfloor} [y^{\lfloor n/2 \rfloor}] (1 + y/6)^{n \bmod 2} \exp(y/2 + y^2/6 + y^4/384) \big(1 + O(2^{-cn})\big)$$

for a constant $c > 0$, common to both parities. The critical family includes all central lifts and the two odd-degree markers. The upper bound uses an annihilator-aware section capacity theorem, joint-source quotient moments, and complete weighted recurrences that establish boundedness before using it. We prove the necessary orbit-action exhaustion and give explicit finite certificates for its bounded cases. The split $c = 1$ case is a separate application and a check on the retained joint data. A saddle-point formula has relative error $O(n^{-1})$; an elementary expression has four residue-class constants and a first correction of order $n^{-1/4}$.

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# 1 The counting problem and its normalization

We count subgroups as subsets of $S_n$, rather than conjugacy classes or abstract isomorphism classes. All unspecified logarithms in counting estimates are to base two; $\log_e$ denotes the natural logarithm. Roney-Dougal and Tracey [RDT25] prove

$$s_n \le 2^{n^2/16+O(n^{3/2})}. \tag{1.1}$$

Together with the elementary binary construction, this identifies the quadratic exponential scale. The more precise benchmark considered here also records the labelled orbit decomposition.

Put $R = \lfloor n/2 \rfloor$ and $\varepsilon = n - 2R$, and define

$$P(y) = \frac{y}{2} + \frac{y^2}{6} + \frac{y^4}{384}, \qquad c_R = [y^R]e^{P(y)}, \qquad p_n = c_R + \frac{\varepsilon}{6}c_{R-1}, \qquad L_n = n!G_R p_n. \tag{1.2}$$

Here $c_j = 0$ for $j < 0$, $G_R = \sum_k \begin{bmatrix} R \\ k \end{bmatrix}_2$, and $G_0 = 1$. In particular $p_n$ is a coefficient, whereas the normalized number of subgroups in the critical family will be denoted $C_n^{\mathrm{crit}}$. The identity $P'(y)e^{P(y)} = (e^{P(y)})'$ gives

$$Rc_R = \frac{1}{2}c_{R-1} + \frac{1}{3}c_{R-2} + \frac{1}{96}c_{R-4}. \tag{1.3}$$

All coefficients are positive. Useful crude consequences include $c_{R-1}/c_R \le 2R$ and $c_{R-h}/c_R \le (2R)^h$ for $0 \le h \le R$.

**Theorem 1.1** (Critical-family asymptotic)**.** *Consider the subgroups whose even orbits are the four actions in Table 1. In odd degree allow, in addition, either one fixed point or one natural $S_3$ orbit. Require full projection onto each displayed action, and include every subgroup with these orbit projections. Their number is $L_n(1+O(2^{-an}))$ for some $a > 0$. Consequently $s_n \ge L_n(1-O(2^{-an}))$.*

**Theorem 1.2** (Section capacity)**.** *Let $T \le S_s$ be faithful, transitive, and not a 2-group, where $s \ge 24$ is even. Let $A$ be any section of its binary permutation module, with action factoring through a specified quotient $B$ of $T$. For a finite $\mathbb{F}_2B$-module $M$, put*

$$r(M) = \max_X \frac{m_X(\operatorname{Soc} M)}{\dim_{\operatorname{End}_B X} X}, \qquad r(0) = 0,$$

*where $X$ runs over simple modules. There is a subspace $C \le A^B$ such that*

$$2\dim C + 4r(A/C) \le \frac{47s}{48}. \tag{1.4}$$

*The statement concerns the actual quotient $A/C$, including all socle constituents exposed by the quotient.*

The critical model and the capacity estimate are proved in Sections 2 and 5. The complete counting argument gives the following theorem.

**Theorem 1.3** (Precise subgroup asymptotic)**.** *There exist $c, C > 0$ and $n_0$ such that*

$$\left|\frac{s_n}{L_n} - 1\right| \le C2^{-cn} \qquad (n \ge n_0).$$

*The constants are common to both parities. No boundedness assumption on $s_n/L_n$ is needed.*

We first prove complete counting estimates on one complementary source, with the normalizer of each original permutation action retained. Sections 20 and 24 give the exhaustive nonbinary action reduction and the complete binary-error recurrence with ordinary targets. The natural $C_3/A_4$ packets and exact marker collapse complete the disjoint partition in Section 32. The even binary error and the two odd shifted errors are replaced by their strictly forward ordinary rows before induction. One boundedness-first induction on the ordinary count then proves Theorem 1.3; no separate boundedness theorem for the complete binary count is used. The split $c = 1$ calculation in Section 29 is a required application of the joint-source method, with its shared-character and inverse-complement controls retained. Section 33 evaluates $L_n$ and gives the saddle-point and elementary forms of the theorem.

### 1.1 Actual actions and recovered profiles

For a transitive action $U \leq S_w$ write $a(U) = |N_{S_w}(U)|$. If a profile contains $m_U$ copies of each action type, then the number of labelled orbit data is

$$\frac{n!}{\prod_U a(U)^{m_U} m_U!}. \tag{1.5}$$

Indeed, partition the points into the prescribed unordered blocks and choose a conjugate of $U$ on each block. The factors $w!$ cancel. A subgroup full on each block recovers both its actual orbits and their permutation actions, so there is no further overcount.

For later use, let $b = n - w$. Pointing an original orbit of type $U$ has placement factor $n!/(b!a(U))$. If its complete complement is $J \leq S_b$, a fibre bound $z_U(J)$ therefore gives the normalized bound

$$\frac{1}{b!a(U)p_nG_R} \sum_{J \leq S_b} z_U(J). \tag{1.6}$$

In particular a uniform fibre envelope $z_U(b)$ produces the coefficient

$$\Delta(n,b)\frac{z_U(b)}{a(U)}, \qquad \Delta(n,b) = \frac{p_bG_{\lfloor b/2 \rfloor}}{p_nG_R} \tag{1.7}$$

on the complete normalized count $s_b/L_b$. An auxiliary quotient cover does not change $a(U)$.

### 1.2 Algebraic inputs

We use Gaussian coefficients, Goursat's lemma, Burnside's basis theorem, the inflation–restriction sequence, the degree-two Künneth formula, and classification of group extensions by $H^2$. For permutation groups we use the abelian-section bound of Kovács and Praeger [KP89]: an elementary abelian $p$-section of a subgroup of $S_b$ has rank at most $b/p$. Its form for $p$-groups is also recorded as [RDT25, Theorem 2.8]. The module-generator estimates used in Section 5 come from [Tra18, Definition 1.5, Theorem 1.6 and Corollary 3.12]. The finite algebraic constants and the distinction between literal witness verification and exhaustive coverage are specified in Appendix A.

## 2 The complete critical family

The four even actions used in the model are listed below. The indicated kernel is trivial in the first two cases and is the centre in the last two. The resulting quotient is elementary abelian of the indicated rank. The extraspecial action is the faithful transitive action of plus type on eight points, denoted $E_8$. The normalizer and quadratic-form calculations are literal finite calculations;

| Action | Degree | Quotient rank | $a(U)$ |
|---|---|---|---|
| $C_2$ | 2 | 1 | 2 |
| regular $V_4$ | 4 | 2 | 24 |
| natural $D_8$ | 4 | 2 | 8 |
| $E_8 \cong 2_+^{1+4}$ | 8 | 4 | 384 |

Table 1: The critical even actions.

Appendix A describes their witnesses. The theorem about this family does not require the additional assertion that these exhaust equality in a permutation-rank bound.

For a fixed word $D$ in these actions, let $K$ be the product of the specified kernels and put $V = D/K = \mathbb{F}_2^R$. If $t = \dim K$, the squaring map is a vector-valued quadratic map $q : V \longrightarrow K$. Its polarization is the commutator map. Write

$$\phi = \prod_{i \geq 1} (1 - 2^{-i}) > 0.$$

The Gaussian product formula implies

$$2^{k(R-k)} \leq \begin{bmatrix} R \\ k \end{bmatrix}_2 \leq \phi^{-1} 2^{k(R-k)}, \qquad G_R \asymp 2^{\lfloor R^2/4 \rfloor}. \tag{2.1}$$

All constants in this section are absolute.

**Lemma 2.1** (The canonical subgroups)**.** *Uniformly in the critical word, the number of full-projecting subspaces $U \leq V$ is $G_R(1 + O(R2^{-R/2}))$.*

*Proof.* Failure to project onto a quotient coordinate of dimension at most four places $U$ in one of its at most 15 pullback hyperplanes. There are at most $R$ coordinates. Each hyperplane contains $G_{R-1}$ subspaces, and (2.1) gives $G_{R-1}/G_R = O(2^{-R/2})$. The full preimage of a full-projecting $U$ contains $K$. Its projection to each factor therefore contains the specified kernel and surjects onto the quotient, so equals that factor. Different $U$ give different full preimages. □

**Lemma 2.2** (All lifts of a subspace)**.** *Let $U \leq V$ have dimension $k$, and set $Q(U) = \langle q(u) : u \in U \rangle \leq K$. Put $\mathcal{A}(U) = Q(U)^{\perp} \leq K^*$. The number of subgroups of $D$ whose image is exactly $U$ is*

$$\sum_{B \leq \mathcal{A}(U)} 2^{k \dim B} = \sum_{j=0}^{\dim \mathcal{A}(U)} \begin{bmatrix} \dim \mathcal{A}(U) \\ j \end{bmatrix}_2 2^{kj}. \tag{2.2}$$

*If $U$ is full-projecting, all these subgroups project onto the factors.*

*Proof.* For such a subgroup $H$, put $W = H \cap K$. Necessarily $q(U) \subseteq W$. Conversely every element of $\pi^{-1}(U)/W$ then has square one. Such a group is abelian, since $(xy)^{-1} = y^{-1}x^{-1}$ and every element is its own inverse; hence the quotient is elementary abelian. The subgroups with intersection $W$ are the complements to $K/W$ in this elementary abelian extension. They form a torsor under $\mathrm{Hom}(U, K/W)$ and number $2^{k \operatorname{codim}_K W}$. Annihilator duality $B = W^{\perp}$ gives (2.2). Fullness follows again from Burnside's basis theorem on each factor. □

**Lemma 2.3** (Quadratic realization)**.** *Fix a quadratic form $Q$ on a binary vector space. If $q_0$ is a nonsingular plus-type form of dimension $h \in \{2, 4\}$, the surjections $f$ satisfying $q_0 \circ f = Q$ form either the empty set or a torsor for $O^+(h, 2)$. In particular there are at most 72 such maps.*

*Proof.* If a map exists, the radical of the polar form of $Q$ is exactly $\ker f$, since the target polar form is nonsingular and $f$ is onto. Thus every such map has the same kernel and induces an isometry from the same quotient to the target. Any two differ by a unique target isometry. The elementary orders are $|O^+(2, 2)| = 2$ and $|O^+(4, 2)| = 72$. □

**Proposition 2.4** (Uniform exceptional-lift estimate)**.** *Set $d = R/2 - t \geq 0$. For the fixed critical word, the total number of noncanonical full subgroups is $O(G_R 2^{-d})$.*

*Proof.* Fix an $\ell$-space $B \le K^*$, where $\ell \ge 1$. The restrictions to $B$ of the nonabelian coordinate functionals span $B^*$; choose $\ell$ of them as a basis. These are the pivot coordinates. Count ordered maps $\mathbb{F}_2^k \to V$ which are injective and full on each coordinate and which satisfy $B \le \mathcal{A}(U)$. Relations dual to the chosen basis isolate the pivot coordinates. Thus fixing the nonpivot maps fixes every pivot quadratic form jointly. Lemma 2.3 allows at most 72 choices for its surjective coordinate map. Each pivot has target dimension at least two. Thus the ordered-map count is at most $72^{\ell} 2^{(R-2\ell)k}$.

Division by $|\operatorname{GL}(k,2)|$, compared with the unrestricted ordered injection count at least $\phi 2^{Rk}$, gives

$$\#\{U : \dim U = k,\ B \le \mathcal{A}(U),\ U \text{ full}\} \le \phi^{-1} \begin{bmatrix} R \\ k \end{bmatrix}_2 72^{\ell} 2^{-2\ell k}.$$

There are at most $\phi^{-1} 2^{\ell(t-\ell)}$ choices for $B$. Multiplying by its lift weight $2^{k\ell}$ in (2.2), the exceptional mass at dimension $k$ is at most

$$\phi^{-2} \begin{bmatrix} R \\ k \end{bmatrix}_2 \sum_{\ell \ge 1} 72^{\ell} 2^{-\ell(k-t+\ell)}. \tag{2.3}$$

Put $x = k - R/2$. Dividing by $G_R$ introduces a factor at most $C 2^{-x^2}$ by (2.1). The exponent depending on $x$ and $\ell$ becomes

$$-x^2 - \ell x - \ell d - \ell^2 = -(x + \ell/2)^2 - \ell d - 3\ell^2/4.$$

The Gaussian sum over either lattice is uniformly bounded. Also,

$$\sum_{\ell \ge 1} 72^{\ell} 2^{-3\ell^2/4} < \infty.$$

Since $d \ge 0$, the result is $O(2^{-d})$. □

*Proof of Theorem 1.1.* Summing (1.5) over critical profiles of rank $R$ gives $n! c_R$ in even degree, because the two rank-two actions have combined weight $1/24 + 1/8 = 1/6$. The canonical count therefore supplies the main term, with the uniform error of Lemma 2.1.

For multiplicities $(a, v, b, e)$ of the four actions,

$$R = a + 2v + 2b + 4e, \quad t = b + e, \quad d = a/2 + v + e.$$

Inserting the weight $2^{-d}$ of Proposition 2.4 changes the exponential generating polynomial to

$$\widetilde{P}(y) = \frac{y}{2\sqrt{2}} + \frac{7y^2}{48} + \frac{y^4}{768}. \tag{2.4}$$

Write $\widetilde{c}_R = [y^R] e^{\widetilde{P}(y)}$ and use the original positive saddle $\rho = \rho_R$ from Section 33. Positivity and the independent saddle estimate of Theorem 33.2 give

$$\frac{\widetilde{c}_R}{c_R} \ll \sqrt{R} \exp\bigl(\widetilde{P}(\rho) - P(\rho)\bigr) \ll \sqrt{R}\, e^{-R/16} = O(2^{-R/32}). \tag{2.5}$$

Indeed $\widetilde{c}_R \rho^R \le e^{\widetilde{P}(\rho)}$, whereas $c_R \rho^R \gg e^{P(\rho)}/\sqrt{R}$. Also $\widetilde{P}(\rho) \le P(\rho) - \rho^4/768$ and $\rho^4 \ge 48R$ for all sufficiently large $R$, by the saddle equation. The final bound absorbs the square-root factor. No second saddle estimate is needed; the rate is not optimized.

In odd degree the fixed-point option contributes $c_R$. The natural $S_3$ option contributes $c_{R-1}/6$. Any subgroup full on its $S_3$ coordinate and on the binary coordinates contains $A_3$ on the first coordinate alone: raise an element projecting to a 3-cycle to a sufficiently large power of two. Quotienting this forced subgroup turns that coordinate into one elementary binary coordinate. Its full lift analysis is consequently the same one, with exceptional marker $y/(6\sqrt{2})$. Apply (2.5) at ranks $R$ and $R-1$ and divide their positive weighted sum by $c_R + c_{R-1}/6$. The additional factor $1/\sqrt{2}$ only improves the bound, and the fixed shift changes the constant. This gives the same exponential conclusion. The two options have different actual orbit profiles and are disjoint. □

# 3 Exact terminal attachment

For a critical product $D$, let $N_D(T)$ count subgroups of $D \times T$ full on $T$ and on each displayed critical factor. The group $T$ is the complete exterior projection, with all of its internal couplings. Write $d_2(T) = \dim \mathrm{Hom}(T, C_2)$ and $A_2(T) = T/(T'T^2)$.

**Lemma 3.1** (Binary contraction)**.** *If $O^2(T)$ is the intersection of the normal subgroups of $T$ with 2-group quotient, image and full preimage induce a bijection*

$$N_D(T) = N_D(T/O^2(T)).$$

*Proof.* Goursat's lemma expresses a subgroup above its two projections by a common quotient. The projection inside $D$ is a 2-group, so the common quotient of $T$ is a 2-group. Its kernel contains $O^2(T)$. Every subgroup being counted therefore contains $1 \times O^2(T)$. □

**Lemma 3.2** (Central extension fibre)**.** *Let $1 \to K \to X \to Y \to 1$ be central with $K$ elementary abelian of exponent two, and let $\xi \in H^2(Y, K)$ be its class. Define*

$$\mathrm{Ann}(\xi) = \{\lambda \in K^* : \lambda_* \xi = 0 \text{ in } H^2(Y, \mathbb{F}_2)\}.$$

*The number of subgroups of $X$ mapping onto $Y$ equals*

$$\sum_{L \le \mathrm{Ann}(\xi)} 2^{d_2(Y) \dim L}. \tag{3.1}$$

*Proof.* The annihilator is the image of restriction $\mathrm{Hom}(X, C_2) \to K^*$: a normalized section identifies extension of a character with its scalar defect being a coboundary. Fix $W = H \cap K$. Taking kernels bijects these subgroups with homomorphisms $r : X \to K/W$ restricting to the quotient map on $K$. Such an $r$ exists exactly when every scalar coordinate extends, or $W^\perp \le \mathrm{Ann}(\xi)$. Differences of extensions factor through $Y$, so this fibre is a torsor under $\mathrm{Hom}(Y, K/W)$, of size $2^{d_2(Y) \operatorname{codim} W}$. Different intersections are disjoint. □

Let $V = D/K = \mathbb{F}_2^r$, $c = \dim K$ and $\delta = r/2 - c$. For an arbitrary finite group $T$ define

$$d = d_2(T), \qquad \tau = \dim \ker\big(H^2(A_2(T), \mathbb{F}_2) \longrightarrow H^2(T, \mathbb{F}_2)\big), \qquad X_r(d) = \sum_j \begin{bmatrix} r \\ j \end{bmatrix}_2 2^{dj}. \tag{3.2}$$

The kernel in this definition is retained as a subspace, not merely replaced by the number $\tau$, until a positive upper bound is taken.

**Proposition 3.3** (Exact quotient records)**.** *For $0 \le u \le r$, let $\mathcal{P}_u$ be the linear maps $p : \mathbb{F}_2^u \oplus A_2(T) \to V$ injective on the first summand and surjective onto each displayed critical quotient coordinate. Pull back the critical extension to $C_2^u \times T$, and call its class $\xi_p$. Then*

$$N_D(T) = \sum_{u=0}^{r} \frac{1}{|\operatorname{GL}(u,2)|2^{ud}} \sum_{p \in \mathcal{P}_u} \sum_{L \le \operatorname{Ann}(\xi_p)} 2^{(u+d)\dim L}. \tag{3.3}$$

*Proof.* The image $Y \le V \times T$ of a subgroup being counted is onto $T$. Put $U = Y \cap V$. Its graph map $T \to V/U$ factors through $A_2(T)$. A linear lift of this map identifies $Y$ with $U \times T$. There are exactly $|\operatorname{GL}(u,2)|2^{ud}$ representations by a choice of ordered basis for $U$ and a linear lift. Lemma 3.2 counts every subgroup above $Y$; surjectivity modulo the critical Frattini kernels gives the required full projections. □

**Proposition 3.4** (Uniform terminal bound)**.** *With the notation above,*

$$N_D(T) \le \phi^{-2} \sum_{j=0}^{r} \sum_{\ell=0}^{c} \begin{bmatrix} c \\ \ell \end{bmatrix}_2 72^{\ell} 2^{\tau\ell} 2^{(j-\ell)(r-j+d)}. \tag{3.4}$$

*Consequently*

$$\frac{N_D(T)}{X_r(d)} \le (r+2)^{O(1)} \left( 1 + \sum_{\ell \ge 1} 2^{-\ell(\delta + d/2 - \tau - \log 72) - 3\ell^2/4} \right). \tag{3.5}$$

*In particular, if $\tau = 0$ then $N_D(T) \le C(r+2)^k X_r(d)$ for absolute $C, k$.*

*Proof.* Put $u = r - j$ and $k = u + d$. For an $\ell$-dimensional space of splitting characters in $K^*$ use row-echelon coordinates as in Proposition 2.4. The original product section has a bilinear scalar defect with diagonal $q_\lambda$. If its pullback to $U \times T$ is a coboundary, evaluation on a vertical involution, comparison on commuting vertical and exterior elements, and restriction to $T$ give, respectively,

$$q_\lambda|_U = 0, \qquad \beta_\lambda(U, f(A_2(T))) = 0, \qquad q_\lambda \circ f \in \ker\bigl(H^2(A_2(T)) \to H^2(T)\bigr).$$

Diagonal evaluation $c(a,a) - c(0,0)$ descends to a linear map on actual $H^2(A_2(T), \mathbb{F}_2)$, since it vanishes on coboundaries. The first two conditions force the whole quadratic outcome to be the exterior pullback of the diagonal of a class in the retained inflation kernel. This allowed function space has dimension at most $\tau$. Modulo that space, the joint row relations determine every pivot outcome. Each outcome has at most $2^\tau$ representatives, and Lemma 2.3 leaves at most 72 surjective maps for each representative on the whole $k$-dimensional domain. Dropping the other restrictions leaves at most $72^\ell 2^{\tau\ell} 2^{(r-2\ell)k}$ ordered maps. The record divisor is at least $\phi 2^{uk}$. Multiplication by the fibre size $2^{k\ell}$ and summation over $L$ prove (3.4), even with one fewer factor $\phi^{-1}$.

Use $\begin{bmatrix} c \\ \ell \end{bmatrix}_2 \le \phi^{-1} 2^{\ell(c-\ell)}$ and write $A = r + d$. The exponent apart from this fixed constant is

$$j(A-j) + \ell(c + \tau + \log 72 - \ell - A + j).$$

If $d \le r$, its unconstrained maximum in $j$ is

$$A^2/4 - \ell(\delta + d/2 - \tau - \log 72) - 3\ell^2/4,$$

whereas an integer $j$ nearest $A/2$ gives $X_r(d) \ge 2^{A^2/4 - O(1)}$. If $d \ge r$, the exponent is increasing on $0 \le j \le r$; its maximum is at $j = r$. Divide by the literal term $2^{rd} \le X_r(d)$. The remaining exponent is $\ell(r/2 - \delta + \tau + \log 72 - d - \ell)$, which is smaller than the one in (3.5) by $\ell(d-r)/2 + \ell^2/4$. The $r+1$ choices of $j$ are a polynomial factor. Finally $\delta, d \ge 0$, so at $\tau = 0$ the series is bounded by $\sum_{\ell \ge 0} 72^\ell 2^{-3\ell^2/4} < \infty$. □

*Remark* 3.5. The module-extension annihilator used below is different from the group-extension annihilator in (3.1). No argument identifies the two. In particular a scalar quadratic form which is nonzero on $A_2(T)$ may become zero after inflation to $T$.

# 4 A complete small-support estimate

In even degree write $n = m + b$, with $m, b$ even. Let $M_{n,b}$ count subgroups whose union of noncritical orbits has degree exactly $b$; all other orbits are from Table 1. This definition includes all couplings within the noncritical support, as well as every critical lift.

**Lemma 4.1.** *For $J \le S_b$ one has $\tau_2(J) \le \lfloor b/2 \rfloor$. If $b$ is even and some actual orbit of $J$ is noncritical, then $d_2(J) \le b/2 - 1$.*

*Proof.* Let $U = J'J^2$. In the five-term sequence for $1 \to U \to J \to A_2(J) \to 1$, inflation on $H^1$ is an isomorphism. Thus transgression identifies the inflation kernel in degree two with

$$H^1(U, \mathbb{F}_2)^J \cong \mathrm{Hom}(U/(U'[J,U]U^2), \mathbb{F}_2).$$

The latter is dual to an elementary abelian section of $U \le S_b$, so has dimension at most $b/2$.

The same abelian quotient bound gives $d_2(J) \le b/2$. If equality holds, the moved-support equality theorem of [KP89] applies with the Sylow 2-subgroup moving all $b$ points. Its transitive equality constituents are $C_2$, regular $C_4$, regular $V_4$, $D_8$, the degree-eight central product of two $D_8$ groups, and the stated odd affine constituents. The odd affine constituents have positive moved-point defect; the regular $C_4$ has elementary rank one instead of two. Equality in elementary rank therefore leaves precisely the four actions of Table 1 on every orbit. A noncritical orbit excludes equality, and integrality gives the claimed gap. This is a deduction from the equality classification, not an assertion of independent rank additivity for arbitrary subdirect products. □

**Theorem 4.2** (Square-root support)**.** *Fix $0 < \varepsilon < 2$. Uniformly for even $b \ge 2$ with $b \le (2-\varepsilon)\sqrt{m}$,*

$$\frac{M_{n,b}}{L_n} \le 2^{-\varepsilon(4-\varepsilon)m/16 + o_\varepsilon(m)}.$$

*The sum over this entire range is exponentially small relative to $L_n$.*

*Proof.* Put $R = m/2$, $N = n/2$. For a critical word with multiplicities $(a, v, d, e)$ its central defect is $\delta = a/2 + v + e$, and its weight is

$$W = \frac{1}{2^a a! 24^v v! 8^d d! 384^e e!}.$$

Let $c_R^{(<L)}$ be the sum of these weights with $\delta < L$. Then $a + 2v + 4e < 4L$, so $d > R/2 - 2L$. There are at most $(R+1)^4$ tuples and each weight is at most $1/d!$. Conversely one term in $c_N$ uses $\lfloor N/4 \rfloor$ extraspecial factors and at most one each of $C_2, D_8$ to fill the remainder. Hence

$$c_R^{(<L)} \le \frac{(R+1)^4}{\max(0, \lfloor R/2 - 2L \rfloor)!}, \qquad c_N \ge \frac{1}{16 \cdot 384^{\lfloor N/4 \rfloor} \lfloor N/4 \rfloor!}.$$

Stirling's inequalities imply, for $L = o(R)$ and $N - R = o(R)$,

$$\log \frac{c_R^{(<L)}}{c_N} \le -\frac{R}{4} \log R + O(R + L \log(R+2) + (N-R)\log(R+2)). \tag{4.1}$$

Let $Y_b$ count subgroups on $b$ points all of whose orbit projections are noncritical. Recovering the noncritical support and its complete projection gives

$$\frac{M_{n,b}}{n!} \le \frac{Y_b}{b!} \sum_{\gamma:\,\mathrm{rank}\,\gamma=R} W(\gamma) \max_{J\in\mathcal{Y}_b} N_{D_\gamma}(J). \tag{4.2}$$

Here $Y_b \le s_b \le 2^{b^2/16+O(b^{3/2})}$ by (1.1). There is no extra support-choice factor: it is already in the exact labelled weights of (4.2).

Choose $L = b + \log 72 + \log(R+2)$. For $\delta \ge L$, Lemma 4.1 and Proposition 3.4 give $N_D(J) \le (R+2)^{O(1)} X_R(b/2-1)$. The quantity $X_R(t)$ counts subspaces of $\mathbb{F}_2^{R+t}$ transverse to a fixed $t$-space: choose the projection in $\mathbb{F}_2^R$ and its graph map. Thus

$$\frac{X_R(b/2-1)}{G_N} \le \frac{G_{N-1}}{G_N} \le 2^{-N/2+O(1)}.$$

Also (1.3) gives $c_R/c_N \le (2N)^{b/2}$. The logarithm of the normalized large-defect part is consequently at most

$$-N/2 + b^2/16 + O(b^{3/2} + (b+1)\log(n+2)).$$

Since $b \le (2-\varepsilon)\sqrt{m}$, $N/2 - b^2/16 \ge \varepsilon(4-\varepsilon)m/16 + b/4$. The error is uniformly $o_\varepsilon(m)$.

For $\delta < L$, maximize the $\ell$ exponent in (3.5) using $\tau \le b/2$. The additional factor is

$$2^{b^2/12+O(b+\log(R+2))}.$$

The remaining term satisfies $X_R(b/2-1) \le G_N$. Equation (4.1) bounds its normalized contribution by

$$2^{-R\log R/4+O(R+b^2+b^{3/2}+(b+\log(R+2))\log(R+2))}.$$

For $b = O(\sqrt{R})$ this is $2^{-\Omega(R\log R)}$. Finally only $O(\sqrt{n})$ support sizes occur, which changes the logarithm by $O(\log n)$. □

# 5 Annihilator-aware section capacity

All sections in this section are sections of one specified permutation module. This restriction supplies the capacity bound; odd order alone does not supply it.

## 5.1 Permutation-module bounds

For a transitive group of degree $u$, write $u_2$ for the binary part of $u$, and set $\mathrm{lpp}(1) = 1$. The induced-module theorem of [Tra18] bounds the number of generators of any submodule of the binary permutation module by

$$E(u,2) = \min\left\{ \left\lfloor \sqrt{2/\pi}\,\frac{u}{\sqrt{\log u_2}} \right\rfloor, \frac{u}{\mathrm{lpp}(u/u_2)} \right\}$$

when no soluble transitive subgroup exists. The first entry is infinite if $u_2 = 1$. When such a subgroup exists the applicable bound is

$$E_{\mathrm{sol}}(u,2) = \min\left\{ \frac{u}{2^{K(u)}} \binom{K(u)}{\lfloor K(u)/2 \rfloor}, u_2 \right\}, \qquad K(u) = \sum_{p^a \| u} a(p-1).$$

Both formulas imply the common integer bound $d_T(L) \le E(u,2)$: use $K(u) \ge \log u_2$, the central-binomial estimate, $u_2 \le u/\operatorname{lpp}(u/u_2)$, and integrality of generator number. The sharper soluble formula will be needed as well.

Directly from these formulas,

$$E(u,2) \le u/2 \ (u \ge 2), \qquad E(u,2) \le 3u/8 \ (u \ge 8), \qquad E(u,2) \le 11u/32 \ (u \ge 24 \text{ even}). \quad (5.1)$$

For a nontrivial odd part use $E \le u/3$. For powers of two the degrees $2, 4, 8, 16, 32$ give respectively $1, 2, 3, 6, 11$; from 64 onward the square-root expression is less than $u/3$. The last bound in (5.1) also has a direct formula-level proof. If the odd part is nontrivial, the nonsoluble alternative has largest prime-power divisor at least three, while the soluble alternative has binary part at most $u/3$. If the odd part is one, a Sylow 2-subgroup is transitive; writing $u = 2^a$, one has $a \ge 5$, and Pascal's recurrence propagates $32\binom{a}{\lfloor a/2 \rfloor} \le 11 \cdot 2^a$ from $a = 5$.

**Lemma 5.1** (Orbit filtration)**.** *Let $H \lhd T$ have $k$ orbits of length $u \ge 2$ in a transitive action of degree $s = ku$. Every trivial $H$-section of $\mathbb{F}_2^s$ has dimension at most $kE(u,2)$.*

*Proof.* Filter a submodule by its intersections with successive sums of the actual orbit modules. Each successive section embeds in one orbit module. Right exactness of coinvariants bounds the dimension of the coinvariants of the original submodule by the sum of the corresponding dimensions of these sections. Each is bounded by its module-generator number and hence by $E(u,2)$. A trivial quotient factors through coinvariants. This filtration does not require independent orbit images. □

**Lemma 5.2** (A sharper fixed-section bound)**.** *If $T$ is transitive of even degree $s \ge 24$, every submodule $L$ of $\mathbb{F}_2^s$ satisfies $d_T(L) \le 5s/16$. Consequently every section $A$ satisfies $\dim A^T \le \lfloor 5s/16 \rfloor$.*

*Proof.* Write $s = 2^a o$ with $o$ odd. If $o \notin \{1, 3\}$, then $\operatorname{lpp}(o) \ge 5$, and the two appropriate bounds are respectively $E \le s/5$ and $E_{\mathrm{sol}} \le s/o \le s/5$. If $o = 1$, a Sylow 2-subgroup is transitive: for a point $x$ its stabilizer in a Sylow subgroup has order at most $|T_x|_2 = |T|_2/s$. Here $a \ge 5$. If $o = 3$ and a soluble transitive subgroup exists, then $a \ge 3$ and $K(s) = a + 2 \ge 5$. In both cases the central-binomial proportion is at most $\binom{5}{2}/2^5 = 5/16$; it is nonincreasing.

It remains to consider $s = 3 \cdot 2^a$ without a soluble transitive subgroup. Apply [Tra18, Corollary 3.12] to an actual minimally transitive subgroup. It supplies a soluble subgroup with, for $0 \le j \le e$, exactly $\binom{e}{j}2^{u_1}$ orbits of length $3p^j 2^{u-u_1}$, where $p = 2^h - 1$ is a Mersenne prime, $h \ge 3$, $e \ge 1$, $a = eh + u$, and $u \ge u_1 \ge 0$. Filter by these actual orbit modules. The soluble bound on each is at most its binary part, giving

$$d_T(L) \le 2^{e+u_1} 2^{u-u_1} = 2^{e+u} \le s/12.$$

Restriction can only increase the required generator number, and generator number is subadditive on the filtration. Finally lift $A^T$ to a submodule of $\mathbb{F}_2^s$; its trivial quotient requires $\dim A^T$ generators. □

**Lemma 5.3** (The exact two-step preimage)**.** *Suppose that $T \le S_s$ is faithful and transitive and is not a 2-group. Let $A$ be a section of $\mathbb{F}_2^s$, put $S = A^T$, and set*

$$U = \{a \in A : (g-1)a \in S \text{ for every } g \in T\}.$$

*Then $\dim U \le 3s/8$.*

*Proof.* For $a \in U$ one has $(g-1)a \in S$ and $(g-1)S = 0$. Thus $(g-1)^2 a = 0$, or equivalently $g^2 a = a$ in characteristic two. The image of $T$ on $U$ is therefore an elementary abelian 2-group, so $H = O^2(T)$ fixes $U$ pointwise.

Pull the quotient onto $U$ back to the original permutation submodule and restrict it to the actual $H$-orbits. Since $H \lhd T$ and $T$ is transitive, all these orbits have the same length, say $u$. We have $u \geq 3$: for $u \leq 2$ every orbit image is a 2-group, faithfulness embeds $H$ in their product, and then the extension of $H$ by the 2-group $T/H$ would make $T$ a 2-group. For $u \geq 8$, the exact induced-module formula gives the local $3u/8$ head bound. For $u = 3, 5, 6, 7$, the odd part is nontrivial; in both formula alternatives the local head is at most $u/3$, which is even stronger. Thus only the genuine formula exception $u = 4$ remains.

For $u = 4$, every orbit image is nonbinary. Indeed, the orbit images are conjugate, so if one were binary then faithfulness would again embed $H$ in a product of 2-groups. Suppose a submodule of the four-point permutation module had a trivial quotient of dimension at least two. It then has dimension three. Its one-dimensional quotient is trivial because $\mathbb{F}_2^\times = 1$, so the submodule contains the full permutation-displacement kernel and is exactly the augmentation hyperplane. A nonbinary transitive subgroup of $S_4$ contains an element of order three; since $4 < 2 \cdot 3$, that element is a single 3-cycle. Evaluation at its unique fixed point is injective on the fixed vectors in the augmentation hyperplane, so this fixed space has dimension at most one. The trivial quotient head embeds into that element-fixed space, a contradiction. Hence the local head is at most one and in particular at most $3u/8$. Sum these local bounds over the actual orbit filtration as in Lemma 5.1. □

## 5.2 A coupled socle estimate

**Proposition 5.4.** *For every section $M$ of the binary permutation module of a faithful transitive group $T$ of even degree $s \geq 24$, put $t = \dim M^T$ and $r_{\mathrm{nt}}(M) = \max_{X \neq 1} m_X(\operatorname{Soc} M)/\dim_{\operatorname{End}_T X} X$. Then*

$$2t + 4r_{\mathrm{nt}}(M) \leq 47s/48. \tag{5.2}$$

*No splitting condition on $M^T$ is required.*

*Proof.* We have $t \leq 11s/32$ by (5.1). If there is a nontrivial simple type, choose $X$ attaining the maximum and write $h = \dim_{\mathbb{F}_2} X$, $\operatorname{End}_T X = \mathbb{F}_{2^f}$, $d = h/f$, $m = m_X$, and $H = \ker(T \to \mathrm{GL}(X))$. The mixed section $1^t \oplus X^m$ is the literal sum in $M$ of $M^T$ and the full $X$-isotypic socle component. The sum is direct: a nonzero fixed vector in the latter would generate a trivial simple submodule and hence force $X = 1$. Thus it has dimension $L = t + hm$; no complement to $M^T$ is being chosen.

If $H \neq 1$, it has orbit length at least two, and Lemma 5.1 gives $L \leq s/2$. For $hd \geq 3$,

$$2t + 4m/d = \frac{4}{hd}L + \left(2 - \frac{4}{hd}\right)t \leq \frac{11s}{16} + \frac{5s}{8hd} \leq \frac{43s}{48}.$$

The only remaining nontrivial possibility is $h = 2, d = 1$. The degree identity makes $X$ one-dimensional over its four-element Schur ring. The original action commutes with that ring, so its image lies in $\mathrm{GL}_1(\mathbb{F}_4) = C_3$; simplicity and nontriviality rule out the trivial image. The image is therefore exactly $C_3$. There are one or three $H$-orbits, each of length at least eight. Thus $L \leq 3s/8$, and the required cost is $2L \leq 3s/4$.

If $H = 1$, the group embeds in $\mathrm{GL}_d(2^f)$. Since its degree is even, $d \neq 1$. The values $hd = fd^2 < 9$ are 4 and 8. The first would give $|T| \leq 6$, impossible. In the second case $T \leq \mathrm{GL}_2(4)$, of order 180. Since $s \mid |T|$ and $s \geq 24$, the even possibilities are $s \in \{30, 36, 60, 90, 180\}$, so

the odd part of $s$ has largest prime-power divisor at least five. Thus $t \le s/5$, and $L \le s$ gives $2t + 4m/d \le s/2 + 3t/2 \le 4s/5$. For $hd \ge 9$, the same dimension bound gives

$$2t + 4m/d \le \frac{11s}{16} + \frac{21s}{8hd} \le \frac{47s}{48}.$$

If there is no nontrivial type, the initial bound on $t$ suffices. □

### 5.3 The retained annihilator and the central cut

For a finite $\mathbb{F}_2B$-module $A$, put $S = A^B$ and, for each simple $X$, define the actual module-extension annihilator

$$\mathcal{L}_X = \ker\big(\mathrm{Ext}^1_B(X, S) \longrightarrow \mathrm{Ext}^1_B(X, A)\big).$$

Every $C \le S$ has a module complement in $S$, so $\mathrm{Ext}^1_B(X, C)$ embeds as a specified subspace of $\mathrm{Ext}^1_B(X, S)$. The long exact sequence gives

$$m_X(A/C) = m_X(A) - \mathbf{1}_{X=1} \dim C + \dim_{\mathrm{End}_B X}\big(\mathcal{L}_X \cap \mathrm{Ext}^1_B(X, C)\big). \tag{5.3}$$

Here multiplicity always means socle multiplicity. In particular

$$\dim(A/C)^B = \dim A^B - \dim C + \dim\ker\big(H^1(B, C) \to H^1(B, A)\big). \tag{5.4}$$

These identities explicitly account for newly exposed simple constituents.

*Proof of Theorem 1.2.* Let $t = \dim S$, $\mathcal{L}_1 = \ker(H^1(B, S) \to H^1(B, A))$, and $\ell = \dim \mathcal{L}_1$. The invariants sequence identifies $\mathcal{L}_1$ with $(A/S)^B$. Its full preimage in $A$ is exactly the space $U$ in Lemma 5.3, and therefore has dimension $t + \ell$. Hence

$$t \le \lfloor 5s/16 \rfloor, \qquad t + \ell \le 3s/8 \tag{5.5}$$

by Lemmas 5.2 and 5.3.

Use the natural identification $H^1(B, \mathbb{F}_2) \otimes S \cong \mathrm{Hom}(S^*, H^1(B, \mathbb{F}_2))$ and regard $\mathcal{L}_1$ as an $\ell$-dimensional space of maps out of $S^*$. The elementary evaluation-separator lemma gives a subspace $W \le S^*$ with $\dim W \le \ell$ such that restriction to $W$ is injective on $\mathcal{L}_1$: starting with the whole map space, choose an input on which a nonzero remaining map does not vanish and iterate on the kernel of that evaluation. Put $Z = W^\perp \le S$. Then

$$\dim Z \ge \max(t - \ell, 0), \qquad \mathcal{L}_1 \cap H^1(B, Z) = 0.$$

Indeed, a tensor in $H^1(B, Z)$, viewed as a map on $S^*$, vanishes on $Z^\perp = W$, so injectivity forces every element of the displayed intersection to be zero. Choose $C \le Z$ of dimension

$$c = \min\{\lfloor t/2 \rfloor, \max(t - \ell, 0)\}.$$

Equation (5.4) gives $\dim(A/C)^B = t - c$. Apply Proposition 5.4 to this actual quotient:

$$2c + 4r_{\mathrm{nt}}(A/C) \le 47s/48 - 2t + 4c \le 47s/48.$$

For its trivial type the cost is

$$2c + 4(t - c) = 4t - 2c \le \max\{3t + 1, 2(t + \ell)\}.$$

The inequality follows by considering which entry attains the minimum defining $c$, including $\ell \geq t$. By (5.5),

$$3t + 1 \leq 15s/16 + 1 \leq 47s/48, \qquad 2(t + \ell) \leq 3s/4.$$

The first inequality uses $s \geq 24$. Taking the maximum over the trivial and nontrivial types proves (1.4) with one and the same $C$. □

*Remark* 5.5 (Why the permutation section matters). On $\mathbb{F}_2^t \oplus \mathbb{F}_2^t$ with trivial $C_3$ action, every linear graph from one summand to the other is invariant, giving $2^{t^2}$ graphs. Thus odd order supplies no strict quadratic deficit in an arbitrary trivial representation. The proof above instead uses (5.5), the actual annihilator, and a quotient cut. It does not assert a strict deficit for the ranks of the uncut trivial axes.

# 6 Same-source lifts and original-weight fusion

## 6.1 The binary permutation rank bound

The binary rank input needed below has a direct degree-induction proof. Write $d_2(P) = \dim_{\mathbb{F}_2} \mathrm{Hom}(P, C_2)$; for a finite 2-group this is also the dimension of its Frattini quotient.

**Lemma 6.1.** *If a finite 2-group $P$ acts faithfully on $b$ points, then $d_2(P) \leq \lfloor b/2 \rfloor$.*

*Proof.* Induct on $b$, with degrees zero and one immediate. For an intransitive action, restrict to a nonempty proper orbit of size $a$. Its actual image acts faithfully on that orbit, and the restriction kernel acts faithfully on the original complement. Character restriction and induction give

$$d_2(P) \leq d_2(\text{image}) + d_2(\text{kernel}) \leq \lfloor a/2 \rfloor + \lfloor (b-a)/2 \rfloor \leq \lfloor b/2 \rfloor.$$

For a transitive action, a cover of the original point stabilizer has relative index two. Its actual blocks are pairs, so $b = 2m$. Let $Q$ be the faithful action on the $m$ pairs and $K$ its original kernel. The flip coordinates identify $K$ with a $P$-invariant subspace of $\mathbb{F}_2^m$, retaining all correlations among the flips. Restriction gives

$$d_2(P) \leq d_2(Q) + \dim \mathrm{Hom}(K, \mathbb{F}_2)^P.$$

If $m = 1$, the top is trivial and $\dim K \leq 1$. Otherwise the transitive 2-group $Q$ has a fixed-point-free central involution: any nonidentity central element of order two is semiregular in a faithful transitive action. Choose a lift $g \in P$. Every invariant form on $K$ annihilates $(g-1)K$, hence rank–nullity gives

$$\dim \mathrm{Hom}(K, \mathbb{F}_2)^P \leq \dim K/(g-1)K = \dim \ker((g-1)|_K) \leq \dim \ker((g-1)|_{\mathbb{F}_2^m}) = m/2.$$

Induction bounds $d_2(Q)$ by $\lfloor m/2 \rfloor$, so the total is at most $m = b/2$. No splitting of the original extension is used. □

## 6.2 The fractional Schur cost

The following lemma applies to nonsplit group extensions and to nonsemisimple modules.

**Lemma 6.2** (Conditional cocycle bound)**.** *Let $1 \to A \to Q \to B \to 1$, with $A$ elementary abelian of exponent $p$, and let $r(A) = \max_X m_X(\mathrm{Soc}_B A)/\dim_{\mathrm{End}_B X} X$. For $J \leq S_b$ and a fixed epimorphism $\beta : J \to B$, the number of epimorphisms $J \to Q$ above $\beta$ is at most*

$$|A|\,|H^1(B, A)|\,p^{(b/p)r(A)}. \tag{6.1}$$

*Proof.* If a lift exists, choose one actual lift as origin. All homomorphic lifts are a torsor for $Z^1(J, A_\beta)$; onto lifts are a subset. Let $F = \ker \beta$. Inflation–restriction and the bound on coboundaries give

$$|Z^1(J, A_\beta)| \le |A|\,|H^1(B, A)|\,|\operatorname{Hom}(F, A)^J|.$$

Choose a Sylow $p$-subgroup $P \le F$ and put $D = N_J(P)$. The Frattini argument gives $J = FD$, so $D \to B$ is onto. Restriction embeds

$$\operatorname{Hom}(F, A)^J \hookrightarrow \operatorname{Hom}_D(P/\Phi(P), A).$$

Indeed $P$ maps onto every $p$-group image of $F$. The source here has the actual $D$-action; an action of $B$ on it is not assumed. Filter $W = P/\Phi(P)$ by a composition series. Left exactness of $\operatorname{Hom}(-, A)$ gives

$$\dim \operatorname{Hom}_D(W, A) \le \sum_X [W : X] m_X \dim_{\mathbb{F}_p} \operatorname{End}_D X \le r(A) \dim W.$$

Types not inflated from $B$ contribute zero. Since $D \to B$ is onto, the remaining simple dimensions and endomorphism fields are precisely those used in $r(A)$. Finally $\dim W \le b/p$ by the permutation $p$-group abelianization bound, whose binary case is Lemma 6.1. This proves (6.1). □

**Proposition 6.3** (Central cut and graph moment)**.** *Let $1 \to A \to Q \to B \to 1$ have elementary binary kernel and let $C \le A^B$, $c = \dim C$. Suppose $T \le S_s$ is a specified faithful permutation action with an epimorphism $\pi : T \twoheadrightarrow B$. Put*

$$D_C = |A/C|\,|H^1(B, A/C)|, \quad \lambda = \tfrac{1}{2} r(A/C), \quad v = s + 2c, \quad \Phi(J) = |\operatorname{Epi}(J, B)| 2^{c d_2(J)}.$$

*For $A/C = 0$ take $D_C = 1$. Then, for every $b$ and integer $q \ge 1$,*

$$|\operatorname{Epi}(J, Q)| \le D_C 2^{\lambda b} \Phi(J), \qquad \sum_{J \le S_b} \Phi(J)^q \le s_{b+qv}. \tag{6.2}$$

*Proof.* Since $C \le Z(Q)$, each actual map $J \to Q/C$ has either no lift or a torsor of homomorphic lifts under $\operatorname{Hom}(J, C)$. Apply Lemma 6.2 to the actual extension $1 \to A/C \to Q/C \to B \to 1$ and sum over the same maps to $B$.

A term $\Phi(J)^q$ selects $q$ pairs of literal maps $(\beta_i : J \twoheadrightarrow B, \chi_i : J \to C_2^c)$. Encode them by the subgroup

$$\{(t_1, e_1, \dots, t_q, e_q, j) : \pi(t_i) = \beta_i(j),\ e_i = \chi_i(j) \text{ for all } i\} \le (T \times C_2^c)^q \times J.$$

Realize each $C_2^c$ on $c$ disjoint pairs. Projection onto the fixed coordinate sets recovers $J$ and every selected map. Thus the encoding is injective into the subgroups of $S_{b+qv}$. Repeated maps, dependent characters, and proper joint subdirect images are allowed. □

**Proposition 6.4** (A target-order alternative)**.** *Let $Q$ be a finite 2-group of order $2^a$. For every finite group $J$ and all integers $b, q \ge 0$,*

$$|\operatorname{Hom}(J, Q)| \le 2^{a d_2(J)}, \qquad \sum_{J \le S_b} \left(2^{a d_2(J)}\right)^q \le s_{b+2aq}.$$

*Consequently this supplies the fusion parameters $D = 1$, $\lambda = 0$, $v = 2a$ whenever the original physical width satisfies $2a < w$.*

*Proof.* Every nontrivial finite 2-group has a central subgroup $C$ of order two. Each nonempty fibre of $\mathrm{Hom}(J,Q) \to \mathrm{Hom}(J,Q/C)$ is a torsor under $\mathrm{Hom}(J,C)$: the pointwise difference of two lifts is a homomorphism because $C$ is central. Its cardinality is $2^{d_2(J)}$. Induction on $a$ proves the first inequality, including empty fibres and nonsplit extensions. For the second, encode the $aq$ binary characters together as a graph over the same literal $J$, realizing each binary coordinate on a pair of new points. □

This argument counts actual maps; it takes no quotient by target automorphisms and does not require a faithful action of $Q$ on $2a$ points. Its width condition is distinct from the character criterion used later.

### 6.3 A complete finite-menu estimate

**Theorem 6.5** (Original-weight fusion)**.** *Fix a finite collection of pairs $(U,N)$, where $U \le S_w$ is transitive, $w$ is even, and $N \lhd U$. For each pair suppose there are fixed $D, v, \lambda$ and a nonnegative function $\Phi$ such that*

$$|\operatorname{Epi}(J,U/N)| \le D2^{\lambda b}\Phi(J), \qquad \sum_{J \le S_b} \Phi(J)^q \le s_{b+qv} \quad (q \ge 1), \qquad g = w - v - 8\lambda > 0.$$

*Let $\mathcal{U}_n$ be any family of actual subgroups of $S_n$ each having an orbit of type $U$ whose intersection axis is one of the specified $N$. Then*

$$\frac{|\mathcal{U}_n|}{L_n} \le \eta(n) + \sum_{b<n} K(n,b)\frac{s_b}{L_b}, \qquad \eta(n) = O(2^{-an^2}), \quad \sum_b K(n,b) = O(2^{-a'n}) \tag{6.3}$$

*for some $a, a' > 0$. The weights are those of the original actions. The result is uniform in ambient parity, but its constants may depend on the fixed finite menu.*

*Proof.* For each $U$, first close the selected literal axes under $N_{S_w}(U)$. This finite enlargement leaves the existential physical orbit event unchanged. Choose one certificate on each resulting normalizer orbit and transport it through the induced quotient isomorphisms; thus $D, v, \lambda, \Phi$ can be constant on that orbit. All sums below refer to this closed literal-axis menu.

On a pointed orbit the subgroup has exact Goursat form

$$\{(u,j) \in U \times J : uN = \beta(j)\}$$

for a unique literal epimorphism $\beta : J \to U/N$, once the quotient map is fixed. This counts actual maps, so there is no additional division by $|\operatorname{Aut}(U/N)|$. First apply the pointing factor $n!/(b!a(U))$ to the total normalizer-stable sum over literal axes and complete sources. A single axis need not be fixed by the action normalizer. Only after this division do we split the numerical sum entry by entry; the envelopes need not be conjugacy invariant. For an entry put $e = g/16$, $a_0 = v/8 + e$, and call its complete source hot when $\Phi(J) > 2^{a_0 b}$. If $v = 0$, the moment bound has the same finite right-hand side $s_b$ for every $q$. Hence $\Phi(J) \le 1$ for every $J$, and the strict hot set is empty. We may therefore assume $v > 0$ in the hot estimate below.

For a hot source and any integer $q \ge 1$,

$$\sum_{J \ mhot} |\operatorname{Epi}(J,U/N)| \le D2^{\lambda b - (q-1)a_0 b} s_{b+qv}.$$

Choose $q = \max(1, \lceil 8eb/v^2 \rceil)$. The exact identity

$$\frac{(b+qv)^2}{16} - (q-1)a_0 b + \lambda b - \frac{(b+w)^2}{16} = -\frac{4e^2b^2}{v^2} - eb - \frac{w^2}{16} + \frac{v^2}{16}\left(q - \frac{8eb}{v^2}\right)^2 \tag{6.4}$$

has a strictly negative quadratic term. The all-pairs monomial gives $c_R \geq 1/(2^R R!)$. Thus, writing $R = \lfloor n/2 \rfloor$ and using $b \geq w$ so that $R \leq b$,

$$\frac{n!}{b! L_n} \leq \phi^{-1} 2^{-R^2/4 + R + 1/4} \leq \phi^{-1} 2^{-n^2/16 + 5n/8 + 1/4}.$$

Together with (1.1), this contributes only $O(n^{3/2} + n)$ to the normalized logarithm. No saddle estimate is needed for this normalization. As all entry parameters are fixed, the complete finite hot union is $O(2^{-an^2})$.

On its complement, $|\operatorname{Epi}(J, U/N)| \leq D 2^{(w/8 - g/16)b}$. The cold kernel is bounded by the sum over entries of

$$\Delta(n,b) \frac{D}{a(U)} 2^{(w/8 - g/16)b}, \qquad b = n - w. \tag{6.5}$$

For $w = 2h$, (1.3) and the Gaussian bounds give the explicit estimate

$$\Delta(n,b) \leq \phi^{-2} 2^{(h+1)/4} (n+1)^{h+1} 2^{-hb/4 - h^2/4}.$$

The guarded odd coefficient obeys the same backward-shift bound, including rank zero, so the estimate holds in both parities. Each summand in (6.5) is therefore $2^{-gb/16 + O_w(\log(n+2))}$. The finite sum proves the row estimate.

The hot and cold contributions partition the local numerical sum after pointing. Summing over the finite action menu may overcount, but only by the positive sum already bounded. Restriction by any earlier disjoint family can only reduce it. The actual complement $J$ includes every later orbit and every terminal attachment, as required by (1.6). □

**Proposition 6.6** (Direct first-moment fusion)**.** *Fix a finite original action/normal-axis menu as above. For each entry assume the local envelope and only the first moment*

$$|\operatorname{Epi}(J, U/N)| \leq D 2^{\lambda b} \Phi(J), \qquad \sum_{J \leq S_b} \Phi(J) \leq s_{b+v}.$$

*Suppose $w = 2h$, $v = 2r < w$, and $g = w - v - 8\lambda > 0$. For the same actual orbit family, with any retained survival predicates invariant under the relabellings in the pointing formula, there is a nonnegative forward kernel satisfying*

$$\frac{|\mathcal{U}_n|}{L_n} \leq \sum_{m<n} K_{\mathrm{dir}}(n,m) \frac{s_m}{L_m}, \qquad \sum_{m<n} K_{\mathrm{dir}}(n,m) = O(2^{-\kappa n})$$

*for some $\kappa > 0$. This conclusion does not use the coarse estimate* (1.1)*.*

*Proof.* Apply the original pointing factor to the complete literal-axis sum, as in the preceding proof, and then use the first moment. An entry continues at $m = b + v = n - w + v < n$, with coefficient

$$\Gamma_{U,N}(b) = \frac{(b+w)!}{b!\, a(U)} D 2^{\lambda b} \frac{L_{b+v}}{L_{b+w}}.$$

Define $K_{\mathrm{dir}}(n,m)$ by summing these coefficients over all literal entries with this target degree, taking an entry to be zero when $n < w$. Distinct normals and actions remain separate summands, even when their target degrees agree. The divisor is still $a(U)$, and no additional quotient by target automorphisms or marker choices occurs.

Put $k = h - r > 0$. Since $n = m + 2k$, exact factorial cancellation gives

$$\Gamma_{U,N}(b) = \frac{D}{a(U)} \frac{m!}{b!} 2^{\lambda b} \Delta(n,m).$$

The fixed-width benchmark estimate above applies in either parity, and $m!/b! \le m^v$. Retaining its quadratic width term gives the uniform estimate, valid for every $b \ge 0$,

$$\Gamma_{U,N}(b) \le \phi^{-2} \frac{D}{a(U)} (n+1)^{(w+v)/2+1} 2^{-gb/8-(w^2-v^2)/16+(w-v+2)/8}. \tag{6.6}$$

Indeed the exponent before simplification is $\lambda b - km/4 - k^2/4 + (k+1)/4$. Consequently, for an entry-dependent constant,

$$\Gamma_{U,N}(b) \le C(n+1)^{v+k+1} 2^{\lambda b - km/4} \le C(n+1)^{v+k+1} 2^{-gb/8}.$$

Absorb the fixed polynomial into the exponential and sum the fixed finite menu. This proves the row estimate without estimating any unknown subgroup count $s_m$. □

The order markers $v = 2\log_2 |U/N|$ and the character marker $v = 0$ fit this even-prefix argument whenever their strict width conditions hold. Any class-count input used to establish the character envelope remains necessary. The proposition supplies neither a physical cover by the menu nor uniform control when the widths, menu, or lift constants grow with $n$; those wider regimes remain separate obligations.

**Corollary 6.7** (Nonbinary pair actions)**.** *Let $U$ be a transitive action on $2s$ points with an invariant pair system whose faithful top $T \le S_s$ is nonbinary, with even $s \ge 24$. For every $N \lhd U$, write $K$ for the pair kernel and set*

$$A = KN/N \cong K/(K \cap N), \qquad B = U/(KN).$$

*This entry has the certificate of Theorem 6.5, with $g \ge s/48$. Hence every fixed finite menu of these entries admits* (6.3)*.*

*Proof.* Let $K$ be the pair kernel. For arbitrary $N \lhd U$ put $A = KN/N \cong K/(K \cap N)$ and $B = U/(KN)$. The original quotient $U/N \twoheadrightarrow B$ has kernel $A$, with its literal conjugation action. Moreover the faithful top $T = U/K \le S_s$ maps onto $B$. Inflating the $B$-module $A$ along $T \twoheadrightarrow B$ identifies it with a section of the binary permutation module on the $s$ pairs.

Apply Theorem 1.2 to this faithful nonbinary cover. Surjective inflation preserves the fixed space and the full fixed-quotient displacement dimension. A retained central cut has zero displacement slice before inflation exactly when it does afterwards, and the Schur capacity of the quotient by that cut is unchanged. The selected cut therefore descends to a certificate for the original extension $U/N \twoheadrightarrow B$. Proposition 6.3 uses the same cover $T \twoheadrightarrow B$ for every common-source moment.

With $c = \dim C$,

$$g = 2s - (s + 2c) - 4r(A/C) \ge s/48.$$

If $r_* = s/2 + c$ is the shifted half-degree, then the same retained cut also satisfies

$$32 r_* \le 21 s. \tag{6.7}$$

Indeed $2c \le t = \dim A^B$ and $16t \le 5s$, so $32r_* = 16s+32c \le 16s+16t \le 21s$. For labelled transport, choose one witness on each original-normalizer orbit of literal entries and fixed transporters before counting $J$. The witness is auxiliary certificate data and contributes no additional physical weight. □

*Remark* 6.8 (Arbitrary literal pair axes and faithful-cover descent). Let $K = \ker(U \to T)$ be the literal kernel of the chosen pair action and let $N \lhd U$ be arbitrary. There are canonical epimorphisms

$$q_N : U/N \twoheadrightarrow B_N := U/(KN), \qquad \sigma_N : T = U/K \twoheadrightarrow B_N,$$

and

$$\ker q_N = KN/N \cong K/(K \cap N) =: A_N.$$

The first is the original nonsplit extension attached to the Goursat axis. The second is used only as its faithful permutation cover. Under the literal kernel chart, the $B_N$-module $A_N$, inflated along $\sigma_N$, is the section of the binary permutation module obtained from $K$ by quotienting by $K \cap N$.

Surjectivity of $\sigma_N$ preserves the fixed space and the full fixed-quotient displacement dimension. For every central subspace $C$, the displacement slice for the inflated action is zero if and only if the slice for the $B_N$-action is zero; the Schur capacity of $A_N/C$ is also unchanged by inflation. Hence the cut selected on the faithful nonbinary top descends to a certificate for the actual acting quotient $B_N$. Together with $q_N$ and its kernel chart this gives the complete capacity payload on every literal normal axis. Neither $N \le K$ nor faithfulness or nonbinaryness of $B_N$ is required, and $N$ is never replaced by $K \cap N$ in the original quotient.

For the pair actions of Corollary 6.7, every literal normal axis therefore carries the capacity payload; no earlier-owner alternative is needed to manufacture missing pair data. In a larger ordered residual menu one may nevertheless suppress an axis after proving that every Goursat reconstruction on it belongs to an earlier physical owner. If the residual predicate is disjoint from those owners, its surviving-epimorphism fibre is empty for every complete source $J$, so the axis remains in the original normal-axis sum with coefficient zero. Given a separately proved physical cover, the capacity and zero-fibre branches yield the forward residual row. Faithful-cover descent itself does not prove that cover, and for growing widths it does not prove the uniform original-weight estimate (6.9).

The strengthened prefix bound also makes the growing-width numerical sum transparent. Write $D_{s,i}$ for the lift constant of an entry, $a_{s,i}$ for its original action normalizer, and $e_{s,i} = g_{s,i}/16$. The exact first-moment coefficient at complement degree $b$ obeys

$$\Gamma_{s,i}(b) \le \phi^{-2} \frac{D_{s,i}}{a_{s,i}} (b+2s+1)^{s+r_{s,i}+1} 2^{-sb/384 - 583s^2/4096 + s/4 + 1/4}. \tag{6.8}$$

This follows from (6.6), (6.7), and $e_{s,i} \ge s/768$. Consequently, if the one remaining original-weight input is

$$\sum_i \frac{D_{s,i}}{a_{s,i}} \le 2^{s^2/32+H} \qquad (s \ge 24) \tag{6.9}$$

for a constant $H$, then the complete sum over all $s$ with $2s \le n$ is

$$O_H\left(2^{-3n/512}\right). \tag{6.10}$$

For completeness, eventually $(n+1)^3 2^{1/2} \le 2^{n/512}$. Raising this inequality to the $s$th power absorbs the polynomial in (6.8); the remaining exponent is at most $-sn/2048$. Since $s \ge 24$,

summing at most $n+1$ widths and absorbing that final factor proves (6.10). Thus all module, fusion, and numerical aggregation steps are internal; the physical task is exactly the weighted mass (6.9) together with the literal cover.

*Remark* 6.9 (Scope of the positive enlargement). The original quotient, extension class, and restricted group transgression determine which lifts exist. The cocycle torsor bounds are applied after those data have been fixed. The capacity proof uses the distinct module annihilator in (5.4). The theorem does not grant bounds for arbitrary extra weights on the source groups, and a growing menu requires an additional uniform estimate for its size and the constants $D$. Only $\Phi$, not the cocycle tail $2^{\lambda b}$, is raised to the moment power.

# 7 Complete quotient moments and uniform transfer

For finite $X$ and a complete source $J$ define

$$Z_J(X) = \sum_{N \lhd X} |\operatorname{Epi}(J, X/N)|, \qquad h(w) = w - (w \bmod 2).$$

The normal subgroups here are literal, including $N = X$. Distinct axes with isomorphic quotient groups remain distinct terms.

**Lemma 7.1** (The complete quotient moment). *If $R$ has a specified faithful permutation representation of degree $v$, then for every integer $q \geq 1$,*

$$\sum_{J \leq S_b} Z_J(R)^q \leq s_{b+qv}.$$

*Proof.* $Z_J(R)$ counts full subdirect subgroups of $R \times J$ by their intersection with $R$ and their literal graph map. For an ordered $q$-tuple of such subgroups take their fibre product over the same $J$ inside $R^q \times J$. Projection recovers $J$ and every member of the tuple: each of the other subgroups can lift the same element of $J$. This is an injection on actual counting sets, including coincident maps and proper joint quotient images. □

**Lemma 7.2** (Uniform normalization). *Uniformly for $n = b + w$, $w \geq 1$,*

$$\log \Delta(n, b) \leq -\frac{h(w)b}{8} - \frac{w^2}{16} + O(w \log(n+2)). \tag{7.1}$$

*Proof.* Let $r = \lfloor b/2 \rfloor$ and $R = \lfloor n/2 \rfloor$. The recurrence gives $c_r/c_R \leq (2R)^{R-r}$, and $p_b \leq (1 + b/6)c_r$, whereas $p_n \geq c_R$. These terms cost $O(w \log(n+2))$, since $R - r \leq (w+1)/2$. The Gaussian bounds give $\log(G_r/G_R) = (r^2 - R^2)/4 + O(1)$. For even $w$ this is $-wb/8 - w^2/16 + O(w)$. For odd $w$, the parity correction is at most $b/8 + O(w)$, giving (7.1). The smallest indices are absorbed in the same absolute constant. □

**Theorem 7.3** (A finite complete comparator). *Fix a finite family of actual transitive actions $U$ of degrees $w$. Suppose each has a fixed faithfully represented comparator $R_U$ of degree $0 < v < h(w)$ and constants $B > 0$, $C \geq 0$, $\eta < h(w)/8$ such that for all complete $J \leq S_b$,*

$$Z_J(U) \leq B Z_J(R_U) + C 2^{\eta b}.$$

*Then their complete pointed union has one quadratic-exponentially small scalar and one exponentially small complete forward row, with the original action weights. The comparator need not be a quotient of $U$.*

*Proof.* Put $\alpha = (v+h)/16$, $e = (h-v)/16$ and $\chi = h/8 - \max(\alpha, \eta) > 0$. Split at $Z_J(R) > 2^{\alpha b}$. The cold summand is exactly bounded by

$$\frac{\Delta(n,b)}{a(U)}(B2^{\alpha b} + C2^{\eta b}).$$

Equation (7.1) gives exponential row decay after the finite sum. On the hot part put $d = \max(\eta - \alpha, 0)$, so the whole fibre is at most $(B+C)2^{db}Z_J(R)$. Use Lemma 7.1 at $q = \max(1, \lceil 8eb/v^2 \rceil)$, $q_* = 8eb/v^2$. The main normalized exponent is

$$\frac{(b+qv)^2}{16} - (q-1)\alpha b + db - \frac{(b+w)^2}{16} = -\frac{4e^2b^2}{v^2} - \chi b - \frac{(w-h)b}{8} - \frac{w^2}{16} + \frac{v^2}{16}(q-q_*)^2.$$

Here $0 \le q - q_* \le 1$ and all parameters are fixed. The coarse error in (1.1) and the original factorials cost $O(n^{3/2} + n\log(n+2))$, smaller than the quadratic loss. Take one any-hot union and its cold complement as in Theorem 6.5. □

**Theorem 7.4** (Growing-parameter comparator transfer)**.** *Fix $w_0 \ge 3$ and $0 < \rho \le 1/8$. For each $w \ge w_0$, let $\Sigma_w$ be a finite certificate menu. Each $\sigma \in \Sigma_w$ carries an actual transitive action $U_\sigma$ of degree $w$, a finite comparator $R_\sigma$, a faithful action of $R_\sigma$ on $v_{0,\sigma}$ points, a relabelling-invariant acceptance predicate on its complete physical fibres, and $\eta_\sigma \ge 0$. The same action may occur with several certificate labels. Require*

$$\frac{h(w) - v_{0,\sigma}}{8} - \eta_\sigma \ge \rho w.$$

*Write $Z_{J,\sigma}(U_\sigma)$ for the literal quotient-map weight of the accepted part of the complete source $J$. Suppose that, for every $J \le S_b$,*

$$Z_{J,\sigma}(U_\sigma) \le D_\sigma(b)2^{\eta_\sigma b}Z_J(R_\sigma), \qquad D_\sigma(b) \ge 0. \tag{7.2}$$

*Write $a(U_\sigma)$ for the original action normalizer and assume the direct original-weight menu bound*

$$\text{for every } \gamma > 0, \text{ eventually uniformly for } w_0 \le w \le n, \qquad \sum_{\sigma \in \Sigma_w} \frac{D_\sigma(n-w)}{a(U_\sigma)} \le 2^{\gamma w n}. \tag{7.3}$$

*Then the complete accepted physical union has a forward kernel and scalar with the uniform eventual bounds*

$$K(n, n-w) \le 2^{-\rho w n/8}, \qquad \sum_b K(n,b) \le 2 \cdot 2^{-\rho w_0 n/8}, \qquad \eta(n) \le 2^{-\rho^2 n^2/8}. \tag{7.4}$$

*Proof.* Pad the given representation by fixed points to $v = \max(v_{0,\sigma}, \lfloor 4\rho h(w) \rfloor)$ and put

$$\delta = (h(w) - v)/8 - \eta_\sigma, \qquad a = v/8 + \delta/2.$$

Then $\delta \ge \rho w/2$, $v \le (1-4\rho)w$, and $\eta_\sigma + a = h(w)/8 - \delta/2$. If $R_\sigma \ne 1$, also $v \ge \rho w$: indeed the original margin and $\eta_\sigma \ge 0$ give $v_{0,\sigma} \le h(w) - 8\rho w \le (1-4\rho)w$, while $\lfloor 4\rho h(w) \rfloor \le (1-4\rho)w$. The loss from padding is at most $4\rho w/8$, proving the stated lower bound on $\delta$. Faithfulness gives $v_{0,\sigma} \ge 2$ when $R_\sigma \ne 1$; use this when $\rho w \le 1$, and otherwise $\lfloor 4\rho h(w) \rfloor \ge 4\rho w - 4\rho - 1 > \rho w$. For $R_\sigma = 1$ the hot event below is empty, so no lower bound on $v$ is needed. This controlled padding preserves the original margin.

Split at $Z_J(R_\sigma) > 2^{ab}$. The cold kernel is

$$K(n,b) = \Delta(n,b) \sum_{\sigma \in \Sigma_{n-b}} \frac{D_\sigma(b) 2^{(\eta_\sigma + a)b}}{a(U_\sigma)}.$$

By (7.1) its main exponent is at most $-\rho w b/4 - w^2/16 \le -\rho w n/4 - w^2/32$. The estimate is uniform from $w = 3$: if $r = \lfloor w/2 \rfloor$, then $w \le 3r$ and hence $3w^2 \le 32r^2$; together with $\rho \le 1/8$ this supplies the floor-term comparison used in the last inequality. The number of transitive action classes is $2^{o(w^2)}$ by the transitive-subgroup bound of [LMM00], in the labelled form quoted in the introduction of [RDT25]. Thus the formerly used sufficient conditions

$$D_\sigma(b) \le 2^{\beta(w) + Cw \log^2(w+b+1)}, \qquad \beta(w) = o(w^2), \qquad \max_U |\{\sigma : U_\sigma = U\}| = 2^{o(w^2)}$$

imply (7.3): the logarithms of the class count and the certificate multiplicity, together with $\beta(w)$, are $o(w^2)$, while the logarithmic term is $o(wn)$ uniformly for $w_0 \le w \le n$. In the present formulation the direct menu bound absorbs these terms together with the pointing polynomial. This proves the first bound in (7.4). Sum the geometric tail over $w \ge w_0$ to obtain the second.

For a nonempty hot part use $q = \max(1, \lceil 4\delta b/v^2 \rceil)$, $q_* = 4\delta b/v^2$, and $M = b + qv$. Lemma 7.1 gives $\sum_{J \text{ hot}} Z_J(R_\sigma) \le 2^{-(q-1)ab} s_M$. The exact main exponent, including the lift factor, is

$$\begin{aligned} \frac{M^2}{16} - (q-1)ab + \eta_\sigma b - \frac{n^2}{16} &= -\frac{\delta^2 b^2}{v^2} + (\eta_\sigma + a - w/8)b - \frac{w^2}{16} + \frac{v^2}{16}(q - q_*)^2 \\ &\le -\frac{\rho^2 b^2}{4} - \frac{\rho w b}{4} - (\rho/2 - \rho^2)w^2 \le -\rho^2 n^2/4. \end{aligned}$$

The last comparison uses $\rho \le 1/8$; the adverse odd correction $(h(w) - w)b/8$ is nonpositive. Moreover $M \le (1 + 1/(2\rho))b + w$, so the coarse error is uniformly $O_\rho(n^{3/2})$. The original hot sum is

$$\sum_{w=w_0}^{n} \sum_{\sigma \in \Sigma_w} \frac{D_\sigma(b) 2^{\eta_\sigma b}}{b! a(U_\sigma) p_n G_{\lfloor n/2 \rfloor}} \sum_{J \text{ hot}} Z_J(R_\sigma), \qquad b = n - w.$$

Its remaining normalization costs $O(n \log(n+2))$. Equation (7.3), first at each width and then after the sum over at most $n+1$ widths, absorbs the original weighted menu into an arbitrarily small multiple of $n^2$. Retaining half the quadratic reserve proves the last bound in (7.4). No normalized-count boundedness has been assumed. □

**Lemma 7.5** (Concrete menu-mass certificate)**.** *Suppose $w_0 \ge 1$ and that, for every $\epsilon > 0$, there are $B_\epsilon \ge 0$ and $p_\epsilon \in \mathbb{N}$ such that, uniformly for $w_0 \le w \le n$,*

$$\sum_{\sigma \in \Sigma_w} \frac{D_\sigma(n-w)}{a(U_\sigma)} \le B_\epsilon (n+1)^{p_\epsilon} 2^{\epsilon w^2}.$$

*Then the direct menu bound* (7.3) *holds.*

*Proof.* Given $\gamma > 0$, take $\epsilon = \gamma/3$. Eventually the fixed constant and the fixed polynomial are each at most $2^{\epsilon n}$. Since $1 \le w \le n$,

$$2\epsilon n + \epsilon w^2 \le \gamma w n.$$

The threshold depends on $\gamma$, but is uniform in $w$ on the indicated triangle. □

**Lemma 7.6** (Index–entry menu aggregation)**.** *Suppose that, for every $\epsilon > 0$, there are constants $B_{1,\epsilon}, B_{2,\epsilon} \geq 0$ and $p_\epsilon \in \mathbb{N}$ such that*

$$|\Sigma_w| \leq B_{1,\epsilon} 2^{\epsilon w^2}$$

*and, uniformly for $w_0 \leq w \leq n$ and $\sigma \in \Sigma_w$,*

$$\frac{D_\sigma(n-w)}{a(U_\sigma)} \leq B_{2,\epsilon}(n+1)^{p_\epsilon} 2^{\epsilon w^2}.$$

*Then the hypothesis of Lemma 7.5 holds.*

*Proof.* Apply the two hypotheses with $\epsilon/2$ and sum the common entry bound over $\Sigma_w$. The constants multiply, the polynomial degree is unchanged, and the two width exponents add to $\epsilon w^2$. □

**Lemma 7.7** (Retained-cell incidence)**.** *For a complete source $J$ and finite comparator $R$, put*

$$\mathcal{Q}_J(R) = \coprod_{N \lhd R} \mathrm{Epi}(J, R/N), \qquad |\mathcal{Q}_J(R)| = Z_J(R).$$

*Let $\mathcal{E}$ be an accepted intrinsic carrier-epimorphism family and let $A$ be a finite set of retained invariant flags. If*

$$\kappa : \mathcal{E} \longrightarrow \mathcal{Q}_J(R) \times A \quad \textit{has every fibre of size at most } L,$$

*then*

$$|\mathcal{E}| \leq |A| L Z_J(R). \tag{7.5}$$

*Consequently $|A|L \leq D(b)2^{\eta b}$ proves (7.2). The same conclusion holds on the original literal normal axis whenever the carrier transport reconstructs the original subgroup and acceptance is tested after that inverse reconstruction.*

*Proof.* Partition $\mathcal{E}$ into the fibres of $\kappa$ and sum the bound $L$ over the $Z_J(R)|A|$ possible cells. Reversible transport is injective on the original axis, preserves the complete exterior source $J$, and tests the same original acceptance predicate, so the transported family is a subset of $\mathcal{E}$ with no additional multiplicity.

For the capacity input attached to an epimorphism $\delta : J \twoheadrightarrow Q$, use the literal pullback

$$H_\delta = \{(u, j) : \beta(u) = \delta(j)\} \leq U \times J.$$

Both projections are full and its first axis is exactly $\ker \beta$. Consequently the joint character, derived-normal, invariant-radical and restricted-transgression inequalities apply to this same $H_\delta$, even when it is a nonabelian proper subdirect product. No direct-product replacement or separately maximized axis is introduced before forming the retained cell.

The subgroup $H_\delta$ also recovers $\delta$: surjectivity of $\beta$ lets equality of two pullbacks be tested above every value of the common quotient. Hence, for any preliminary comparator map $\kappa_0$, the exact cell

$$\delta \longmapsto (\kappa_0(\delta), H_\delta)$$

is injective and has fibres of size at most one. A radical or restricted annihilator flag may replace $H_\delta$ only together with a proved bound for the fibres created by that coarsening; this bound is precisely the $L$ in (7.5).

For the comparator–abelian-layer families one may take $Q = U/N$ itself. Fix a top $Q \twoheadrightarrow R/M$ and factor it through a tower of abelian kernel modules $A_\ell$. At layer $\ell$, retain the literal value in the image of the restriction map. Once the lower map and this value are fixed, the remaining lifts form a fibre of size at most $|B^1_\ell||H^1_\ell|$. Thus the flags and fibres are charged jointly by

$$\prod_\ell |\operatorname{im}(\mathrm{res}_\ell)||A_\ell||H^1_\ell|,$$

using $|B^1_\ell| \le |A_\ell|$. A single bound for this product gives the comparator envelope on the original literal quotient, without a carrier change or an owner-precedence hypothesis. □

The repeated action labels in Theorem 7.4 implement the owner-or-capacity dichotomy. Take a finite ordered list of predicates invariant under relabelling of the complete physical subgroup and assign each subgroup to its first eligible predicate. Pair that owner with the single family orbit carrying its certificate whenever such an orbit exists; only the final catch-all branch uses an arbitrary selected bad orbit. First ownership makes these pairs disjoint, while the local action check makes an owner paired with an unrelated orbit an empty cell. Hence the catch-all branch holds precisely when no family orbit exists. Earlier owners and the retained-annihilator branch may use different local certificates $\sigma = (\text{owner}, U)$; only their combined original-weight mass in (7.3) is charged. The complete one-triple family is the first required application of these same local bounds. In its independent finite-owner audit, full projection onto the displayed outer $C_3$ and the retained one-$C_3$ source pattern remove the apparent degree-three ternary-owner label before the selected carrier is forgotten into its canonical cell.

**Proposition 7.8** (Pre-$E_7$ owner-or-cell transfer)**.** *Restrict the first-owner action menu to the nonbinary orbit actions that admit no selected $E_7$ pair frame. On a literal normal axis $N$, suppose either an earlier owner bounds the surviving epimorphisms by the common complete quotient-map weight (as a quotient comparator does), or there is a reversible carrier and a retained-cell map as in Lemma 7.7, with flag set $A_N$, cell-fibre bound $L_N$, and*

$$|A_N|L_N \le C_N(b)2^{\eta b}.$$

*Give an owned axis the extracted coefficient $2^{-\eta b}$ and an unowned axis the coefficient $C_N(b)$. Suppose each resulting complete normal-axis sum, divided by its original action normalizer, satisfies the uniform one-entry hypothesis of Lemma 7.6, and the aggregation parameters satisfy the hypotheses of Theorem 7.4. Then the exact pre-$E_7$ physical family has the forward kernel and scalar bounds* (7.4).

*Proof.* On an owned axis the direct owner envelope bounds the surviving epimorphisms by the complete quotient-map weight. Multiplication by the extracted common factor $2^{\eta b}$ cancels $2^{-\eta b}$, so this contribution has coefficient exactly one. On every other axis reversibility returns the retained-cell bound to the same original source and normal axis, and Lemma 7.7 gives coefficient $C_N(b)2^{\eta b}$. Sum over the literal normal axes before dividing by the original action normalizer. The transitive-subgroup theorem gives $2^{o(w^2)}$ labelled actions. Passing to conjugacy classes and imposing the pre-$E_7$ restriction can only decrease their number, while the fixed first-owner menu adds only a constant factor. Lemma 7.6 therefore supplies Lemma 7.5, which gives (7.3). Theorem 7.4 now applies to the restricted physical cover. □

The owner order in this proposition serves only to make the physical partition disjoint. None of the local estimates needs the assertion that an earlier predicate failed: one may choose any valid

certificate for the retained action class. Formally, the family predicate says that such a certificate exists on a literal orbit. Transporting that orbit under an ambient relabelling proves degree naturality, and the cover displays it before forming the local fusion cell. The residual catch-all may therefore infer the absence of every family orbit without a separate precedence theorem.

The following additive form is used for families whose sharp estimate has a source-independent remainder. It avoids forcing that remainder through the complete-quotient threshold.

**Proposition 7.9** (Additive-tail pre-$E_7$ transfer)**.** *In the setting of Proposition 7.8, suppose the literal normal axis $N$ has the unconditional envelope*

$$Z_{J,N} \le C_N(b)2^{\eta b}Z_J(R) + T_N(b)2^{\theta b}, \qquad C_N(b), T_N(b) \ge 0.$$

*Sum $C_N$ and $T_N$ over the literal normal axes before applying the original action-normalizer divisor. Assume both resulting weighted menus satisfy* (7.3)*, the main parameters satisfy Theorem 7.4, and uniformly*

$$\theta \le \frac{h(w)}{8} - \frac{\rho w}{4}.$$

*Then this family has a complete exponentially contractive forward estimate. The scalar is exactly the scalar of the comparator-weighted main term; the additive remainder contributes only a cold row.*

*Proof.* For fixed $J$, sum the displayed inequality over the original normal axes. Apply the complete quotient moment and its hot/cold split only to the first sum. The second sum is independent of $J$, so summing it over all complete $J \le S_b$ gives

$$s_b\Big(\sum_N T_N(b)\Big)2^{\theta b}.$$

After the exact pointing and benchmark normalization this is the usual cold kernel with exponent $\theta$. Equation (7.1), the displayed gap, and the second menu-mass hypothesis give an exponentially decaying row. Add this cold-only forward estimate to the estimate supplied by Theorem 7.4. No quotient threshold, hot scalar, or independent worst-case value of $Z_J(R)$ is introduced for the tail. □

**Lemma 7.10** (Central nonbinary pair sections)**.** *For a faithful nonbinary transitive action of even degree $s \ge 4$, every trivial section of its binary permutation module has dimension $t \le s/2 - 1$. Consequently a central pair-kernel quotient at original width $2s$ has an effective graph moment degree $s + 2t \le 2s - 2$.*

*Proof.* If an odd prime $p$ divides $s$, its Sylow subgroup has no fixed points. Exactness of invariants bounds a trivial section by the number of its orbits, at most $s/p \le s/2 - 1$. If $s = 2^a \ge 8$, a Sylow 2-subgroup is transitive. The soluble permutation-module bound gives $t \le B_a \le 3s/8 \le s/2 - 1$. At $s = 4$, the 3-cycle fixed-vector argument in Lemma 5.3 gives $t \le 1$. For a central kernel $A$ take $C = A$ in Proposition 6.3. Then the cocycle tail vanishes and the moment degree is $s + 2t$. □

# 8 Faithful character tuples on a common source

All characters in this section are complex irreducible characters. A faithful tuple means that the intersection of its kernels is trivial. Slots may contain the trivial character; this permits harmless padding. Binary-primary linear means every power-of-two order, rather than only order two.

## 8.1 Whole-socle feasibility

Write the abelian part of $\mathrm{Soc}(Q)$ as a direct sum of simple $\mathbb{F}_p Q$-modules over its various primes. For a nontrivial simple type $X$ let $m_X$ be its multiplicity and $e_X = \dim_{\mathrm{End}_Q X} X$. For each prime let $m_p$ be the dimension of the central elementary $p$-socle and let $f_p$ be the dimension of its intersection with $Q'$. Put

$$T = \max\left(\{f_p\}_p, \{\lceil m_X/e_X \rceil\}_{X\neq 1}, \{1 : \mathrm{Soc}(Q) \text{ has a nonabelian factor}\}\right), \qquad M_o = \max_{p \text{ odd}} m_p,$$

with empty maxima zero.

**Theorem 8.1** (Three-category feasibility)**.** *A faithful tuple with $r$ binary-primary linear slots, $l$ unrestricted linear slots, and $t$ general irreducible slots exists if and only if*

$$t \geq T, \qquad l + t \geq M_o, \qquad r + l + t \geq m_2. \tag{8.1}$$

*The trivial group permits the empty tuple.*

*Proof.* Every nontrivial normal subgroup intersects the socle nontrivially. It therefore suffices to detect the entire socle with the joint tuple. For the abelian socle, Clifford theory says that the restriction of an irreducible contains one orbit of linear characters. In a simple isotype $X^{m_X}$, a single orbit spans at most $e_X$ independent multiplicity directions. To see this, identify the image algebra on $X$ with $M_{e_X}(\mathrm{End}_Q X)$; a vector in the isotype supplies at most $e_X$ rows in its multiplicity space. Conversely $t$ vectors generate that isotype whenever $m_X \leq te_X$, by choosing rows to span the whole multiplicity space. The dual module has the same dimensions and multiplicity, so this criterion applies equally to character orbits.

Linear characters kill $Q'$. All nontrivial abelian socle types lie in $Q'$, because their commutator with $Q$ is the whole simple type. Thus they require the general slots. So does each derived-central direction, and a nonabelian socle factor requires at least one general slot. Each irreducible is scalar on a central elementary $p$-group, so a slot detects at most one independent central direction. A binary-primary linear character kills every odd central element. These facts prove necessity of (8.1).

For sufficiency, choose bases of each central $p$-socle adapted to its derived-central subspace. Assign its first $f_p$ dual coordinates to the general slots, and as many remaining coordinates as those slots can hold. At odd $p$, put the remaining coordinates in the $l$ linear slots; at $p = 2$, use the $r + l$ linear slots. The inequalities guarantee room. The functionals assigned to linear slots vanish on the entire intersection of the abelian socle with $Q'$. They extend first to its image in $Q^{\mathrm{ab}}$ and then to complex linear characters of $Q^{\mathrm{ab}}$, since a character of a subgroup of a finite abelian group extends to that group. For a binary-primary slot, extend inside the binary component and take the trivial character on all odd components. Higher binary roots may be necessary and are allowed.

On each nontrivial simple isotype choose $t$ dual vectors whose $Q$-orbits generate its full dual, using the matrix-algebra criterion above. Combine their assignments over all primes with the chosen central functionals to obtain $t$ characters of the abelian socle. If the nonabelian socle is nontrivial, choose a nontrivial irreducible on each of its simple direct factors and take their tensor product. It is faithful: the simple factors are centreless, and a tensor product can have scalar cancellation only from scalar images. Assign this character to one general slot and use the trivial character there in the other slots.

Each slot now specifies an irreducible of the whole socle. Induce it to $Q$ and choose an irreducible constituent. By Clifford theory, its restriction contains every conjugate of the specified

socle character; its kernel on the socle is precisely the corresponding normal core. The chosen orbit spans and central coordinates make the intersection of these cores, together with the extended linear kernels, trivial. The resulting tuple is faithful on $Q$. □

For the source costs below, the cheapest feasible profile is

$$t = T, \qquad l = \max(M_o - T, 0), \qquad r = \max(m_2 - \max(M_o, T), 0). \tag{8.2}$$

Indeed replacing an unrestricted linear slot by a general one increases its coefficient from $(\log 3)/3$ to $(\log 5)/3$, and replacing a binary-primary linear slot by either type also increases its coefficient. For fixed $t$ choose the least feasible $l$ and then the least $r$; increasing $t$ cannot improve that cost.

## 8.2 Joint kernel counting

**Proposition 8.2.** *If $Q$ has a specified faithful tuple with profile $(r, l, t)$, then for every actual $J \le S_b$,*

$$|\operatorname{Epi}(J, Q)| \le |\operatorname{Aut} Q|\, 2^{rb/2} 3^{lb/3} 5^{tb/3}. \tag{8.3}$$

*Proof.* For each onto kernel choose one epimorphism and inflate the fixed tuple. Its joint kernel is that original kernel, so distinct kernels give distinct ordered tuples of characters of the same $J$. There are at most $2^{b/2}$ binary-primary linear characters: a Sylow 2-subgroup of $J$ maps onto the binary component of $J^{\mathrm{ab}}$, and its abelianization has order at most $2^{b/2}$. The total number of linear characters is $|J^{\mathrm{ab}}| \le 3^{b/3}$ by [RDT25, Theorem 2.8] (with the perfect case immediate). The number of all irreducibles is $k(J) \le 5^{b/3}$ by [GM15, Theorem 1.1], including $b \le 3$ by direct inspection. Enlarge the joint tuple set to the product of these three lists. Finally each onto kernel has exactly $|\operatorname{Aut} Q|$ maps. □

**Proposition 8.3** (Several anchors, one preimage)**.** *In a specified faithful tuple of profile $(r, l, t)$ choose an anchor subtuple whose joint kernel $K$ is a $p$-group. Let $u$ be the number of general slots among the anchors, $g = t - u$, and $d = [Q : O_p(Q)]$. Then*

$$|\operatorname{Epi}(J, Q)| \le |\operatorname{Aut} Q| d^g 2^{rb/2} 3^{lb/3} 5^{ub/3} (38/25)^{gb}. \tag{8.4}$$

*If $Q$ itself is a $p$-group the empty anchor is allowed, with $u = 0, d = 1$.*

*Proof.* The inflated anchor tuple determines $M = \varphi^{-1}(K)$. Since $K \le O_p(Q)$ and $O_p(Q/K) = O_p(Q)/K$, it determines the single subgroup $F = \varphi^{-1}(O_p(Q))$ as the preimage of the characteristic $p$-core of $J/M$. Each remaining general character, restricted to $F$, has constituents with $p$-group image. All such constituents factor through the canonical maximal $p$-quotient of $F$. An actual Sylow $p$-subgroup $P \le F$ maps onto that quotient, so its irreducible list has size at most $k(P) \le (38/25)^b$ by the nilpotent class bound [Mar05, Theorem 1.5]. No faithful degree is assigned to the abstract quotient. Frobenius reciprocity bounds the number of irreducibles of $J$ above a fixed constituent by $[J : F] = d$: their dimensions are at least the constituent's dimension, whereas its induction has $d$ times that dimension. The linear slots use the global lists of Proposition 8.2, and only the $u$ general anchors use the general list. Enlarge the joint lists after $F$ has been determined. Faithfulness recovers the onto kernel, and one factor $|\operatorname{Aut} Q|$ counts the maps. For a $p$-group target $F = J$ is already known and no anchor is needed. □

**Corollary 8.4** (Complete finite character entries)**.** *Fix a finite menu of actual actions $U$. For each literal $N \lhd U$ choose one tuple or anchor certificate whose coefficient of $b$ in* (8.3) *or* (8.4) *is at most $h(w)/8 - \delta$, where $w = \deg U$ and $\delta > 0$ is common to the menu. The complete accepted-orbit union has zero scalar and a forward kernel with exponentially small row, in original weights.*

*Proof.* Sum the chosen bounds over the actual normal axes of each $U$, always on the same complete $J$. Their fixed target constants have a finite sum $D_U$, so $Z_J(U) \le D_U 2^{(h(w)/8-\delta)b}$. The pointed cold row is $\Delta(n,b)\sum_{\deg U=n-b} D_U 2^{(h(w)/8-\delta)b}/a(U)$. By Lemma 7.2 its logarithm is at most $-\delta b + O_W(\log(n+2))$, and $b \ge n - W$ for a fixed width cap $W$. The complete finite sum is exponentially small. □

# 9 Relative heads of actual transitive actions

For $N \lhd A$ and a prime $p$, write

$$h_{p,A}(N) = \dim_{\mathbb{F}_p} N/(N^p[N,A]), \qquad \rho_A(N) = h_{2,A}(N), \quad \delta_A(N) = h_{3,A}(N).$$

The distinction between a relative head and the abelianization of an abstract isomorphic group is essential. For $B \le C$ normal in $A$, the quotient word maps onto the corresponding quotient word downstairs; its kernel is an image of the head of $B$. Consequently

$$h_{p,A}(C) \le h_{p,A}(B) + h_{p,A/B}(C/B). \tag{9.1}$$

If $G \le \prod_{i=1}^{t} A_i$ is full on its actual orbit actions, filter $G$ by the kernels of projection onto coordinates $i+1,\dots,t$. The image of the $i$th factor is a normal subgroup $N_i \lhd A_i$. Conjugation on that factor is exactly the action of $A_i$, since the coordinate projection of $G$ is onto. Thus

$$d_p(G) \le \sum_i h_{p,A_i}(N_i). \tag{9.2}$$

This assertion does not add the ranks of the full coordinate images.

**Theorem 9.1** (Binary first head)**.** *For a faithful transitive $U \le S_w$, $w \ge 2$, and $N \lhd U$,*

$$\rho_U(N) \le 3w/8,$$

*except for the whole-action pairs $(U,U)$ of the four critical actions in Table 1. Their heads have dimensions $1,2,2,4$.*

*Proof.* We use strong induction on $w$. Tracey's induced-module theorem [Tra18, Theorem 1.6] gives, for a submodule of an induced binary module with fibre dimension $a$ and index $q \ge 2$, at most $aE(q,2) \le aq/2$ module generators. Indeed, if $q$ has odd part greater than one, the largest-prime-power branch is at most $q/3$; if $q = 2^k$, the other branch is $\lfloor \sqrt{2/\pi}\, q/\sqrt{k} \rfloor$, which is at most $q/2$ for $k \ge 3$ and equals $1,2$ for $k = 1,2$.

Suppose first that $N$ is intransitive, with $q \ge 2$ orbits of size $m$. Let $K$ be the kernel of $U$ on these orbits and let $H$ be its transitive action on one of them. The faithful embedding of $K$ in $H^q$ and its coordinate-kernel filtration give

$$\rho_U(N) \le \rho_K(N) \le \sum_{i=1}^{q} \rho_H(E_i), \qquad E_i \lhd H.$$

If $H$ is not critical, induction suffices. In each critical action every proper normal subgroup has relative head at most $3m/8$: for $C_2, V_4$ this is immediate; in $D_8$, squares and commutators kill its central involution in each proper noncentral normal subgroup; in $E_8$, every noncentral normal subgroup has commutator with $E_8$ equal to its centre and has image of dimension at most three

modulo that centre. These observations also suffice if the common image of $N$ on an orbit is proper in $H$.

The remaining case is a subdirect $N \leq H^q$ with $H$ critical. The block stabilizer induces the actual local automorphisms on every characteristic elementary section of $H$; the sum of its conjugate sections is its induced module of index $q$. For $C_2, V_4$ the preceding module bound gives $q/2, q$. For $D_8, E_8$, use both $N \cap Z(H)^q$ and its image in $(H/Z(H))^q$. Their fibre dimensions are respectively $(1,2)$ and $(1,4)$, so (9.1) gives $3q/2, 5q/2$, respectively. All four bounds are at most $3mq/8$. In particular the central line of a dihedral block has not been omitted.

If $N$ is transitive, then

$$\rho_U(N) \leq d(N) \leq \left\lfloor \frac{\sqrt{3}\, w}{2\sqrt{\log_2 w}} \right\rfloor$$

by [Tra21, Theorem 1.1]. This is at most $3w/8$ for $w \geq 38$, and also for

$$w \in \{16, 19, 24, 27, 30, 32, 33, 35, 36\}.$$

The remaining finite input checks the whole heads in the complete transitive catalogues of degrees

$$2, 3, \ldots, 23, 25, 26, 28, 29, 31, 34, 37.$$

Its only failures are the four critical actions. The catalogue classification is that of [HR20, Hul23]; this slice has 7,259 action classes. A separate enumeration of all 233 transitive normal pairs in degrees $2, 4, 8$ shows that a critical transitive $N$ has a head above $3w/8$ only when $U = N$ is the corresponding critical action. Thus the whole-head test is applied to the actual action of $N$, and the separate pair test handles its possible overgroups. The finite calculation constructs $N/(N^2[N,U])$ and verifies its elementary binary invariants; its precise bounded contract is recorded in Appendix A. Together with the preceding induction this proves the assertion in all degrees. □

We next give the relative ternary bounds needed for the one-triple case. Let $\mathcal{E}_9$ consist of the four transitive actions $9T2, 9T6, 9T7, 9T17$, of orders $9, 27, 27, 81$.

**Lemma 9.2** (Ordered-transversal head bound)**.** *Let $P$ be a finite $p$-group, $H \leq P$, $[P:H] = p^t$, and let $V$ be an $a$-dimensional representation of the original $H$ over a field $k$. For every actual submodule $M \leq \mathrm{Ind}_H^P V$,*

$$\dim_k \mathrm{Hom}_P(M, k) \leq a \ \mathrm{width}(\{0, \ldots, p-1\}^t).$$

*The order defining width is componentwise. For $p = 3$ its width is at most $W(t)$, where*

$$W(0) = W(1) = 1, \qquad W(2) = 3, \qquad W(t) = 7 \cdot 3^{t-3} \quad (t \geq 3).$$

*Proof.* First choose an actual chain $H = H_0 \lhd H_1 \lhd \cdots \lhd H_t = P$ with each index $p$. For existence, a proper subgroup $L < P$ acts on $P/L$. The number of fixed cosets is positive and divisible by $p$, since $[P:L]$ is and all nontrivial orbit sizes are. A fixed coset $gL$ satisfies $g^{-1}Lg \leq L$, hence equality by finiteness. Thus $L$ is properly contained in its normalizer. An element of order $p$ in $N_P(L)/L$ supplies the next subgroup of index $p$. Iteration and index multiplication give exactly $t$ steps. Only consecutive normality is required; $H$ need not be normal in $P$.

Choose $x_i \in H_i$ generating $H_i/H_{i-1}$, and set $T(b) = x_1^{b_1} \cdots x_t^{b_t}$ for $0 \leq b_i < p$. Successively extracting the highest digit proves that these are unique representatives for the original right cosets of $H$. Order the labels by $L(b) = \sum_i b_i p^{i-1}$, so the largest differing coordinate decides the

comparison. If $d < b$ in this order and $b \le c$ componentwise, then the coset of $T(d)T(b)^{-1}T(c)$ has label strictly below $c$. Indeed, at the largest differing coordinate $i$, the equal higher tails of $d, b$ cancel. With $A, B, C \in H_{i-1}$ the remaining expression is

$$Ax_i^{d_i-b_i}B^{-1}Cx_i^{c_i}S_c = Dx_i^{d_i-b_i+c_i}S_c, \qquad D \in H_{i-1}.$$

This uses only $H_{i-1} \lhd H_i$. The exponent satisfies $0 \le d_i - b_i + c_i < c_i < p$, so no carry changes a higher coordinate. The lower coordinates absorb every commutator and stabilizer term.

Use the coinduced model

$$\{f : P \to V : f(hx) = \rho(h)f(x)\}, \qquad (g \cdot f)(x) = f(xg).$$

It is the original induced representation: $g \otimes v$ corresponds to the function supported on $Hg^{-1}$ with value $v$ at $g^{-1}$; the tensor relation gives precisely the displayed $\rho(h)$ twist. Values at the $T(b)$ give one copy of the original $V$ per coset. For $q = T(b)^{-1}T(c)$, act by $q^{-1}$. Then

$$(q^{-1} \cdot f)(T(c)) = f(T(c)q^{-1}) = f(T(b)).$$

Thus the entire leading fibre vector moves unchanged from row $b$ to row $c$. Every lower row moves to a row below $c$ by the preceding calculation. Its original fibre twist is retained; it cannot create a higher row. In particular this argument does not require an invariant basis or a split fibre representation.

Choose an ordered basis of $V$ and order full labels $(b, j)$ by $aL(b) + j$, with coset position first. Work in the intrinsic quotient

$$Q = M/\langle g \cdot u - u : g \in P,\ u \in M\rangle_k.$$

Among lifts in $M$ of all bases of $Q$, minimize the sum of their leading full-label heights. Such lifts exist, and the sum is a natural number. If two different lifts have leading labels $(b, j), (c, j)$ with $b \le c$ componentwise, subtract from the second the scalar multiple of $q^{-1}$ applied to the first which cancels its leading coefficient. The new lift has strictly smaller height: the leading fibre row is unchanged by transport, and every other term stays below $(c, j)$. In $Q$ its class is the old second class minus that multiple of the first, so the classes still form a basis and the new lift is nonzero. This contradicts minimality. The case $b = c$ uses $q = 1$ and also excludes duplicate labels. Thus the leading labels inject into $a$ disjoint copies of the grid, with an antichain in each copy. Invariant forms on $M$ are exactly $Q^*$, proving the dimension bound. The quotient here is intrinsic to $M$, not its image in ambient coinvariants.

For three ternary coordinates the following seven chains partition all 27 labels:

$$\begin{array}{l} 000, 100, 200, 210, 220, 221, 222; \qquad 001, 101, 201, 211, 212; \\ 002, 102, 202; \qquad 010, 110, 120, 121, 122; \qquad 011, 111, 112; \\ 012; \qquad 020, 021, 022. \end{array}$$

Fixing the other $t - 3$ coordinates gives $3^{t-3}$ slices, each met at most seven times by an antichain. For $t = 2$ use the chains $00, 10, 20, 21, 22$; $01, 11, 12$; and $02$. The cases $t = 0, 1$ are immediate. $\square$

For a positive integer $s$, write $s_3 = 3^t$ and define the integer ternary envelope

$$B(s) = \min\{C(s), s/\operatorname{lpp}(s/s_3)\}, \qquad C(s) = \begin{cases} s, & t = 0, \\ s/3, & t = 1, 2, \\ 7s/27, & t \ge 3, \end{cases} \tag{9.3}$$

with $\operatorname{lpp}(1) = 1$. We use this envelope directly in the relative-head recurrence.

**Lemma 9.3** (Primitive inputs and block filtration)**.** *For a primitive action $R$ of degree $r$, let $a_3(R)$ count its composition factors of order three, with multiplicity. Then*

$$a_3(R) \le r/3,$$

*with equality only for the natural $C_3, S_3$ actions and the affine degree-nine actions of orders* $216, 432$*. Outside these four actions, $a_3(R) < 3r/10$. Every primitive normal pair has $\delta_R(N) < 3r/20$, apart from the whole natural $C_3, A_4$ pairs. For a transitive $A$ with a minimal block system of $s$ blocks of size $r$, primitive block component $R$, top $T$, and block kernel $K$,*

$$\delta_A(N) \le a_3(R)B(s) + \delta_T(NK/K). \tag{9.4}$$

*Proof.* We first prove the strict relative-head tail directly from actual group orders. Write $E = \mathrm{Soc}(R)$ and $v_3(m)$ for the exponent of three in $m$. In the nonaffine case, the perfect socle contributes no ternary chief factor, so

$$\delta_R(N) \le v_3(|R/E|), \qquad |R/E| \le a^k k!, \qquad \ell^k \le r,$$

where $E = S^k$, $\ell$ is the least proper subgroup index of $S$ and $a = |\operatorname{Out}(S)|$. The simple-index inequalities give $\ell \ge 5$ and $a \le 3\log_2 \ell$. For $34 \le r < 1024$ there are at most four socle factors. For $k = 1, 3, 4$ the head is at most $3, 8, 12$, respectively. For $k = 2$ it is at most five when $\ell < 27$; otherwise $r \ge 729$ and the bound eight suffices. Each gives $20\delta_R(N) < 3r$.

In the affine case $r = p^d$, the actual translation subgroup and faithful linear complement give

$$\delta_R(N) + v_3(r) \le v_3(|R|), \qquad |R| \mid r \prod_{i=0}^{d-1} (p^d - p^i).$$

For prime $r \ge 34$, this bounds the head by $\lfloor \log_2 r \rfloor$, which is strictly below $3r/20$. For composite $34 \le r < 1024$, necessarily $p < 32$ and $2 \le d < 10$. Integer arithmetic on these prime-power parameters verifies the strict inequality; no action catalogue or normal-subgroup census is involved. For either structure and $r \ge 1024$, putting $L = \lfloor \log_2 r \rfloor + 1$ gives $\delta_R(N) \le L^2 + 2L$. The elementary dyadic inequality $20(L^2 + 2L) < 3r$ completes the tail. Thus no ordinary primitive normal-generator theorem is needed. The chief-weight tail is also obtained from the same original orders. In the nonaffine case only the full-socle quotient contributes ternary weight, and the preceding bound gives $20a_3(R) < 3r$ from degree 34. In the affine case every original chief series satisfies

$$a_3(R) \le v_3\left( p^d \prod_{i=0}^{d-1} (p^d - p^i) \right), \qquad r = p^d.$$

For $45 \le r < 1024$, prime degree is treated directly and the composite cases have $p < 32$, $2 \le d < 10$; integer arithmetic verifies $10a_3(R) \le 3r$. For $r \ge 1024$ the same dyadic quadratic-log estimate settles every original chief series. No primitive composition-length theorem is used.

The alternative finite-row argument below gives a classical proof of the stated bounded-range assertions. The formal T1 route instead proves the strict head estimate away from degrees $10, 12$ and pays their actual semisimple sectors separately; it does not require either catalogue correspondence.

For the bounded strict-head range, put $E = \mathrm{Soc}(R)$. This is a minimal self-centralizing normal subgroup. Its relative ternary evaluation radical is $R$-normal, hence is either 1 or $E$. In the latter case the relative ternary head of $E$ is zero. In the former case evaluation separates $E$; invariance

then centralizes $E$, self-centralization gives $E = R$, and $R$ is a 3-group. A point stabilizer is maximal in a primitive group, so its index is three. Thus this former case has degree three. Outside degree three the relative head of $E$ therefore vanishes.

For any $N \triangleleft R$, minimality gives $N \cap E = 1$ or $E$. The first alternative makes $N$ trivial by self-centralization. In the second, inflation–restriction bounds the relative head of $N$ by $v_3(|R/E|)$. The complete 253-row primitive alphabet in degrees 2–33 records only the exact quotient order $|R/E|$; kernel-checked integer arithmetic verifies

$$20v_3(|R/E|) < 3r$$

outside degrees $3, 4, 18$. Hence no primitive normal-subgroup census is used. Classification completeness and the actual socle correspondence use [RD05, HRDR23].

For the bounded composition-weight statement, a stronger reduction avoids replaying the full composition series. Among all 336 primitive actions through degree 44, the order bound

$$a_3(R) \le v_3(|R|) \le 3r/10$$

settles every nonnatural row in the required range. The failures of the second inequality are exactly the natural $A_r$ and $S_r$ rows in degree 6 and in degrees 10 through 44 (degree 9 is excluded here). Their symbolic composition chains $1 < A_r$ and $1 < A_r < S_r$ show that every chosen chief series has ternary weight zero. This alternative formal receipt is split into five disjoint degree chunks; the kernel checks all 336 exact group orders, their 3-adic valuations, the applicable integer inequalities, the locator uniqueness within each chunk, and the total row count. Classification completeness and the row correspondence again use [RD05, HRDR23]. The weaker $r/3$ assertion follows immediately, including the excluded degree-nine cases handled separately below.

Here is the common ternary head bound used in the block argument. Write $s_3 = 3^t$. Restrict to a Sylow 3-subgroup and decompose the original induced module by its actual double cosets, retaining the twisted fibre actions. Its orbit indices are $3^j$ with $j \ge t$, and their sum is $s$. Filter the given submodule by the kernels of these actual coordinate maps. Head subadditivity and Lemma 9.2 bound each resulting head by $aW(j)$. For $t = 1, 2$, all $W(j) \le 3^j/3$; for $t \ge 3$, all $W(j) = 7 \cdot 3^j/27$. Summation therefore gives $aC(s)$, including mixed orbit sizes. For $t = 0$ the ambient dimension $as$ suffices. Restriction can only increase the invariant character head.

For a prime $q \ne 3$, restrict instead to a Sylow $q$-subgroup. Maschke's theorem makes every submodule of each induced component a direct summand. Its maps to the trivial line therefore extend to the component; Frobenius reciprocity bounds their dimension by $a$. Each actual orbit has size at least $s_q$, so there are at most $s/s_q$ components. Maximizing $s_q$ gives $as/\operatorname{lpp}(s/s_3)$; at $s/s_3 = 1$ use the ambient dimension $as$ instead. Together these give $aB(s)$. All local coefficients and their sums are integers; no floor is moved through the original fibre dimension.

For the block assertion, choose a chief series $R_i$ of $R$ and filter $N \cap K$ by the inverse images of $R_i$ under all original conjugate block evaluations. These maps separate the block kernel. Each elementary layer is a submodule of the actual module induced from a block stabilizer. A ternary layer of fibre dimension $a_i$ therefore contributes at most $a_i B(s)$; a layer at another prime has zero ternary head.

For a nonabelian chief factor $Q$, each coordinate image of the corresponding layer is normal in the original quotient of $R$, hence either trivial or all of $Q$. Minimal normality makes $Q$ perfect. Choose a maximal proper normal subgroup $M \triangleleft Q$. The ambient conjugates of the maps $Q \to Q/M$ are jointly injective, since their common kernel is ambient-normal and proper in $Q$. They give surjections onto simple perfect groups. Thus the original layer, including any proper subdirect

correlations, is a subdirect product of simple perfect groups and is perfect: induct on the number of coordinates, whose successive kernels are either trivial or a full simple factor. Its ternary head is zero.

Sum this single chain using (9.1). The relative second isomorphism theorem identifies the final top section with $NK/K$ under the original ambient conjugation. The ternary fibre dimensions sum to $a_3(R)$, proving (9.4). □

**Theorem 9.4** (Ternary relative stability)**.** *Except for the whole natural $C_3, A_4$ pairs and the four whole $\mathcal{E}_9$ pairs, every transitive normal pair of degree $w$ obeys*

$$\delta_A(N) \le \lfloor 5w/27 \rfloor.$$

*Consequently the bounds $w/4$ and $2w/9$ hold outside, respectively, the whole natural $C_3$ pair and the two whole natural pairs. Positive equality in the former occurs only at natural $A_4$; equality in the latter occurs precisely at the four whole $\mathcal{E}_9$ pairs. For an arbitrary $G \le S_m$ with $c$ actual regular $C_3$ orbits and $a$ actual natural $A_4$ orbits,*

$$9d_3(G) \le 2m + 3c + a. \tag{9.5}$$

*Proof.* The finite primitive input in Lemma 9.3 already implies the assertion in the primitive case. The finite transitive normal-pair input in degree nine says that the only two-ninth equalities are the four whole $\mathcal{E}_9$ pairs and every other pair has head at most one. This is a complete finite catalogue, not an assumption on possible block systems.

From (9.3), $B(s) \le s/3$ for $s \ge 3$ and

$$B(2), B(3), B(4), B(6), B(9), B(18) = 1, 1, 1, 2, 3, 6.$$

For $3 \nmid s$ and $s \ge 3$, the prime-to-three denominator is at least four; the other cases follow from $C(s)$. Use induction in (9.4). For an ordinary top with $s \ge 3$ and $r \ge 3$,

$$\delta_A(N) \le rs/9 + 5s/27 \le 5rs/27.$$

For $r = 2$ there is no ternary layer. At $s = 2$ the top head vanishes and $\delta_A(N) \le a_3(R) \le r/3 \le 10r/27$. For a whole $\mathcal{E}_9$ top the bound is $3a_3(R) + 2 \le r + 2 \le 5r/3$ for $r \ge 3$; $r = 2$ is immediate. A natural $A_4$ top gives $a_3(R) + 1$, which is at most $20r/27$ for $r \ge 3$; again $r = 2$ is immediate. A natural $C_3$ top also gives $a_3(R) + 1$. Use the degree-nine input at $r = 3$, $a_3(R) \le 1$ at $r = 4$, and $r/3 + 1 \le 5r/9$ for $r \ge 5$; the binary-block case is immediate. These cases exhaust the induction. Integrality gives the stated floors.

The coarser conclusions now follow by comparing $5/27 < 2/9 < 1/4$ and inspecting the six stated exceptions. Apply the two-ninth bound to (9.2); a natural $C_3$ layer costs three and a natural $A_4$ layer one above twice its degree. This proves (9.5). In particular a no-$C_3$, no-$A_4$ residual has $d_3(G) \le 2m/9$; neither exclusion can be omitted. □

**Theorem 9.5** (Necessary high ternary actions)**.** *If a transitive normal pair $(A, N)$ of degree $w$ satisfies $20\delta_A(N) > 3w$, its actual action is one of the following:*

1. *natural $C_3$ or natural $A_4$;*
2. *$6T1, 6T4, 6T5, 6T6$;*
3. *$12T20, 12T85, 12T164, 12T194, 12T228, 12T229, 12T265$;*

*4. a transitive* 3*-group of degree* 9 *or* 27*.*

*This is a necessary list of actions, not a claim that all their normal pairs have high relative head.*

*Proof.* Primitive actions are covered by Lemma 9.3. We retain the complete normal-pair catalogues in degrees $6, 12, 18$: 1,300 actions and 20,410 pairs. Their strict high-action list is exactly items two and three, with no degree-eighteen hit. In particular every degree-eighteen pair has head at most two. These remain bounded classification inputs for the actual actions and all their normal subgroups.

Use a minimal block system and put $v = a_3(R)$ in (9.4). For a nonexceptional top with $s \geq 3$, $B(s) \leq s/3$. If $r \geq 5$, then

$$\frac{vs/3 + 5s/27}{rs} \leq \frac{1}{9} + \frac{5}{27r} \leq \frac{4}{27} < \frac{3}{20};$$

at $r = 4$, $v \leq 1$ gives $14s/27 < 3(4s)/20$. At $s = 2$ the top head vanishes and the stronger primitive density bound gives $v \leq 3r/10$ for $r \geq 4$, except possibly $r = 9$; the degree-eighteen input removes that exception.

For a natural $C_3$ top the bound is $v + 1$: the only $r \geq 4$ case not below $9r/20$ is $r = 4$, giving degree twelve. A natural $A_4$ top gives $v+1 \leq 3r/5$ for $r \geq 4$. A whole $\mathcal{E}_9$ top gives $3v+2 \leq 27r/20$ for $r \geq 6$, and $v \leq 1$ deals with $r = 4, 5$. For binary blocks $v = 0$, only the natural $C_3$ top can fail, giving degree six.

It remains to treat ternary blocks, $r = 3$. For a nonexceptional top,

$$B(s) + \lfloor 5s/27 \rfloor \leq \lfloor 9s/20 \rfloor \qquad (s \notin \{2, 6, 18\}).$$

To check all indices, write $s = 3^t m$ with $3 \nmid m$. If $m \notin \{1, 2\}$, use $1/4 + 5/27 < 9/20$. If $m \in \{1, 2\}$ and $t \geq 3$, use $7/27 + 5/27 = 4/9 < 9/20$. For $t \leq 2$ the only remaining indices $s \geq 2$ are $2, 3, 6, 9, 18$; the sums at $s = 3, 9$ are $1, 4$ and satisfy the displayed floors. At $s = 18$, use the actual top bound two from the finite input, giving $B(18) + 2 = 8 < 81/10$. No separate top-degree $27, 54, 81, 162$ refinement is needed.

The only child degrees still to inspect are $6, 9, 12, 18, 27$. Degrees $6, 12, 18$ use the retained finite catalogues. Degree nine is structural. Its minimal block system is $3 \times 3$, and equality in the head recurrence forces both the primitive-fibre contribution and the original normal top head to equal one. A transitive three-point top is $C_3$ or $S_3$. In the second case every permutation is conjugate to its inverse, so invariance sends each ternary character value to its negative; the relative ternary head is zero. The rank-one top is therefore $C_3$. For the literal block kernel, the simultaneous fibre-sign map now gives the same dichotomy used below: a nontrivial sign annihilates its relative ternary head and would give $\delta_A(N) \leq 1$, while trivial signs make the kernel a 3-group. Thus the whole degree-nine action is a 3-group, without a finite order or action-list premise.

At degree 27 the same recurrence removes an apparent choice of block orientation. If $r$ is the primitive block size, $s$ the number of blocks, $d = \delta_A(N)$ and $t$ the relative head of the original normal top image, then the high inequality and the stability envelope give

$$r = 3, \qquad s = 9, \qquad d = 5, \qquad t = 2.$$

Indeed $rs = 27$ leaves $(r, s) = (3, 9)$ or $(9, 3)$. In the second case $v \leq 3$, $B(3) = 1$ and $t \leq 1$, so $d \leq 4$, a contradiction. In the first case $v \leq 1$, $B(9) = 3$ and $t \leq 2$; the high inequality forces equality throughout. The structural degree-nine argument just proved therefore makes the actual top a 3-group.

Let $K$ be the literal block kernel and let $\epsilon : K \to C_2^9$ be the simultaneous vector of its nine fibre signs. Put $K_0 = \ker \epsilon$. Restriction to the original fibres injects $K_0$ into $C_3^9$; this retains every diagonal or proper subdirect correlation. If $\epsilon$ is nontrivial, top transitivity makes each coordinate sign nontrivial. Conjugation through that sign acts by inversion on the corresponding ternary coordinate. Consequently every original normal subgroup of $K_0$, including every proper nonabelian subdirect core, has zero relative ternary head. The quotient $K/K_0$ is binary and has zero ternary head as well. Restriction to the literal kernel followed by the original top map now gives $\delta_A(N) \le t = 2$, contradicting $20d > 81$.

If $\epsilon$ is trivial, $K = K_0$ is a 3-group. Since the actual top is also a 3-group, the original extension $A$ is a 3-group, with no splitting assumption. Thus degrees nine and twenty seven enter the same existing 3-group owner.

The two smallest degrees are structural as well. At degree three, a nonzero relative character forces a quotient of order three; transitivity and the embedding in $S_3$ leave $A_3$ or $S_3$. The latter conjugates every element to its inverse and hence annihilates every invariant ternary character, so only the natural $C_3 = A_3$ action remains. At degree four, the same quotient makes three divide the group order, while transitivity makes four divide it. A subgroup of $S_4$ therefore has order twelve or twenty four. The order-twelve subgroup has index two and is $A_4$; the order-twenty-four subgroup is $S_4$, where conjugacy to inverses again annihilates the relative ternary head. Thus degree four is exactly the natural $A_4$ owner. The only genuinely residual bounded degrees are now 6 and 12. This exhausts the final case and proves the theorem. □

For the separate bounded counting argument below, we retain the finite semiregular witnesses for all 91 nonsoluble degree-eighteen actions: each contains an actual fixed-point-free permutation of order three. Those witnesses are used for the stronger module-generator capacities in Theorem 15.9; they are not needed for the relative-head classification just proved.

## 10 Complete chief-layer fibres on one source

Throughout this section $Z_J(G) = \sum_{N \lhd G} |\operatorname{Epi}(J, G/N)|$ retains every literal normal subgroup of the target. Put $h_b = b \log(b+1)$, with $h_0 = 0$.

**Lemma 10.1** (Semisimple full fibre)**.** *Let $E = S_1 \times \cdots \times S_t$ be a normal subgroup of an arbitrary finite group $G$, with each $S_i$ nonabelian simple. Then, for every actual $J \le S_b$,*

$$Z_J(G) \le Z_J(G/E) \prod_{i=1}^{t} \left(1 + \frac{|\operatorname{Aut} S_i| h_b}{\log |S_i|}\right).$$

*The empty product is one. No permutation representation of $G/E$ or of a simple section is required.*

*Proof.* For an original $N \lhd G$, retain $I = N \cap E$ and $D = NE/E$. The subgroup $I$ is an invariant subproduct. In $G/I$, the group $N/I$ centralizes $E/I$ and intersects it trivially. Projection from $C_{G/I}(E/I)$ to $G/E$ is injective, since $E/I$ is centreless. Thus $I, D$ determine $N$ whenever it exists.

Fix an onto map $\theta : J \to (G/E)/D$ and its actual kernel $F \le S_b$. An onto lift restricts to an onto map $F \to E/I$. After that restriction is fixed, a lift is unique: two lifts have pointwise difference in $E/I$ centralizing the onto image of $F$, and hence have trivial difference.

Distinct kernels of maps $F$ onto one nonabelian simple $S$ give a quotient $S^k$. Indeed, inductively a proper subdirect subgroup of $S^k \times S$ identifies the last coordinate with one of the preceding coordinates, and would give a repeated kernel. Therefore $k \log |S| \le \log |F| \le h_b$. Each kernel

supports exactly $|\operatorname{Aut} S|$ epimorphisms. Bound the maps to $E/I$ by the product of these coordinate bounds, and then sum over the invariant subproducts $I$, enlarging to all subproducts. This gives the displayed product. All $D$ and $\theta$ remain in the single $Z_J(G/E)$; no independent normal-menu multiplier is added. □

Define the composition invariant

$$\kappa(G) = \sum_{C_p \text{ a composition factor of } G} \frac{\log p}{p},$$

with multiplicity. Nonabelian composition factors have weight zero.

**Theorem 10.2** (Fixed-target composition envelope)**.** *For every fixed finite $G$ there are $C_G \geq 1$ and an integer $D_G \geq 0$ such that*

$$Z_J(G) \leq C_G(1+h_b)^{D_G} 2^{\kappa(G)b} \qquad (J \leq S_b).$$

*One may take $D_G$ to be the number of nonabelian composition factors. These constants are uniform on any fixed finite target menu.*

*Proof.* Induct along a chief series, starting with $C_1 = 1, D_1 = 0$. For an elementary abelian minimal normal subgroup $E = \mathbb{F}_p^a$ put $Q = G/E$ and $q = \lfloor \log |Q| \rfloor$. For each original $N \lhd G$, put $I = N \cap E$, $D = NE/E$. A lift of an onto $\theta : J \to Q/D$ restricts on $F = \ker\theta$ to a map $F \to E/I$. There are at most $p^{ad_p(F)} \leq p^{ab/p}$ such restrictions. A fixed generating tuple of $Q/D$ has length at most $q$; choose its preimages in $J$. Their images, together with the restriction to $F$, determine the lift and have at most $|E|^q$ choices. Incompatible assignments contribute zero, so nonsplit extensions require no extra assertion. If $\nu(G)$ is the number of literal normal subgroups of $G$, then

$$Z_J(G) \leq \nu(G)|E|^q p^{ab/p} Z_J(Q).$$

Thus take $C_G = \nu(G)|E|^q C_Q$ and $D_G = D_Q$. For a semisimple minimal normal $E = \prod_i S_i$, use Lemma 10.1, taking $C_G = C_Q \prod_i (1 + |\operatorname{Aut} S_i|)$ and $D_G = D_Q + t$. The abelian exponents add to $\kappa(G)$ by Jordan–Hölder. All permutation rank bounds were applied to the actual $F \leq S_b$. □

**Proposition 10.3** (Irreducible affine last layer)**.** *Let $U = V \rtimes R$ be a fixed primitive affine action of degree $w = p^d$, with $R \neq 1$ faithful and irreducible on $V$. Write $\operatorname{End}_R(V) = \mathbb{F}_{p^f}$ and $A_U = |Z^1(R,V)|$. Then*

$$Z_J(U) \leq (1 + A_U) 2^{fb \log p/(dp)} Z_J(R).$$

*If $R$ is nonabelian, then $f/d \leq 1/2$. In particular this proposition and Theorem 10.2 give a direct contractive row for every fixed finite menu satisfying*

$$\kappa(R) + \frac{f \log p}{dp} < \frac{w - (w \bmod 2)}{8}.$$

*Proof.* Since $C_U(V) = V$, a normal subgroup disjoint from $V$ is trivial; irreducibility makes every other normal subgroup contain $V$. Thus $Z_J(U) = Z_J(R) + |\operatorname{Epi}(J,U)|$ exactly. Fix an onto $\theta : J \to R$ and $F = \ker\theta$. If $V$ has multiplicity $m$ in the semisimple head of $H_1(F, \mathbb{F}_p)$, then $dm \leq d_p(F) \leq b/p$ and the space of equivariant restrictions to $V$ has size $p^{fm}$. Restriction of cocycles has kernel precisely the inflated $Z^1(R,V)$; each nonempty fibre therefore has size $A_U$. This proves the claimed bound, including every obstruction to a lift being onto. The integer $f$ divides $d$. If $f = d$, the action is scalar over its commuting field, so $R$ is abelian. Otherwise $f \leq d/2$. The last assertion follows from original-weight pointing and Lemma 7.2; all target constants are fixed. □

**Theorem 10.4** (Entire affine block kernel)**.** *Let $U \le S_w$ be transitive with $s \ge 2$ blocks of size $r = p^a \ge 2$, exact primitive affine component $L$, and faithful block action $T \le S_s$. Put $w = rs$ and $g(w) = \lceil w/\sqrt{\log w} \rceil$. Choose a chief series $1 = N_0 < \cdots < N_e = L$ and write $K_i = U \cap N_i^s$. For each abelian local chief factor $\mathbb{F}_{p_i}^{a_i}$, let $H_i$ bound the number of module generators of every submodule of its actual induced ambient module and its dual. Then*

$$Z_J(U) \le 2^{\eta_U b + \beta_U} \Psi_U(b) Z_J(T), \tag{10.1}$$

*where*

$$\eta_U = \sum_i H_i \log(p_i)/p_i,$$
$$\beta_U = \sum_i \log(p_i)[a_i s H_i + H_i w/p_i + a_i s(g(w)+1)],$$

*and $\Psi_U$ is the product in Lemma 10.1 over the simple factors of all actual nonabelian intersection layers. With $H_i \le \lceil 4a_i s/\sqrt{\log s} \rceil$ these satisfy, uniformly as $rs \to \infty$,*

$$\eta_U = o(w), \qquad \beta_U = o(w^2), \qquad \log \Psi_U(b) \le C w \log^2(w + b + 1).$$

*All proper, zero and full block kernels, and all original normal quotients, occur in this inequality.*

*Proof.* The endpoint of the local-chief filtration is the block kernel $K = U \cap L^s$, and the final quotient is $T$. Put $G_i = U/K_{i-1}$, $E_i = K_i/K_{i-1}$, $R_i = U/K_i$. For an abelian local factor, the ambient module $V_i = (\mathbb{F}_{p_i}^{a_i})^s$ factors through $R_i$: $K_i$ centralizes the whole local abelian layer. It is induced from one block stabilizer of index $s$. The same is true of its dual. The local-chief construction is that of [Tra18, Lemma 5.8], with this explicit endpoint. For a nonabelian local factor, coordinate projections of $E_i$ are normal in the exact $L$ and hence are trivial or the full local chief factor. Transitivity makes the choice uniform. The subdirect description of products of nonabelian simple groups makes $E_i$ semisimple, possibly trivial.

For one abelian layer abbreviate $G, E, R, p, a, H$ accordingly. Duality gives, for any $R$-module $X$ and every subquotient $M$ of $V_i$,

$$|\operatorname{Hom}_R(X, M)| = |\operatorname{Hom}_R(M^*, X^*)| \le p^{H \dim X}.$$

Indeed a subquotient is a quotient of a submodule, so its dual requires at most $H$ module generators. For $N \lhd G$ retain $I = N \cap E$ and $D = NE/E$. At most $p^{asH}$ invariant submodules $I$ occur, by padded generating tuples in $V_i$. For fixed $I, D$, put $M = E/I$. If a normal complement $N/I$ exists, all others are its graphs of equivariant homomorphisms $D/[D,D]D^p \to M$. Here $D$ acts trivially on $M$: its normal lift centralizes $M$. Thus this is an exact torsor when nonempty. The actual preimage $\widetilde{D} \le U \le S_w$ gives $d_p(D) \le d_p(\widetilde{D}) \le w/p$, so the graph count is at most $p^{Hw/p}$.

Fix now an onto $\theta : J \to R/D$ and its actual kernel $F \le S_b$. Restrictions of compatible lifts to $F$ have at most $p^{Hb/p}$ choices. Once a restriction is fixed, differences descend to cocycles of $R/D$ on $M$. Every quotient of $U$ has at most $g(w)$ generators, by [Tra18, Theorem 1.1(1)]; hence the lift torsor has size at most $|M|^{g(w)} \le p^{asg(w)}$. This argument uses no splitting or cohomology vanishing. Summing $I$, the normal graphs, and the lifts leaves all $D$ and $\theta$ in one $Z_J(R)$. It gives the stated abelian factor, even with the harmless extra $as \log p$ in $\beta_U$. Apply Lemma 10.1 at the other layers and iterate, always on the same literal $J$, to prove (10.1).

For completeness the universal induced-module bound [Tra18, Corollary 4.27(iii)] gives $H_i = \lceil 4a_i s/\sqrt{\log s}\rceil$ for $s \geq 1261$; below that threshold the dimension bound suffices since $\sqrt{\log s} < 4$. Put $L_0 = \log|L|$ and $q_s = \sqrt{\log s}$. The chief-order budget yields

$$\eta_U \leq 2sL_0/q_s + L_0/2,$$
$$\beta_U \leq 4s^2L_0^2/q_s + 2swL_0/q_s + swL_0/\sqrt{\log w} + 3sL_0 + wL_0/2.$$

Since $L$ is affine, $L_0 \leq \log r + (\log r)^2$. Divide the displayed bounds by $rs$ and $r^2s^2$, respectively. For large $r$ their terms tend uniformly to zero; for bounded $r$ they tend uniformly to zero as $s$ increases. This proves both two-variable little-oh estimates, including the ceiling costs.

Every simple factor $S$ in these layers has $\log|\operatorname{Aut} S| \leq (\log|S|)^2$, since a generating tuple of length at most $\log|S|$ determines an automorphism. The total of $\log|S|$ is at most $sL_0$, and each such term is at most $L_0$. Therefore

$$\log \Psi_U(b) \leq sL_0^2 + 2sL_0\log(b+1) \leq Cw\log^2(w+b+1).$$

The last bound follows since $L_0/r$ and $L_0^2/r$ are bounded for all $r \geq 2$. No faithful action was assigned to a simple section in deriving this estimate. □

# 11 Uniform capacity of affine local components

The estimates here apply to the exact primitive local component of an imprimitive action, with arbitrary faithful transitive top.

**Lemma 11.1** (Refined dual capacities)**.** *For a module induced from a block stabilizer of index $s \geq 2$ with local dimension $a$ in characteristic $p$, each of the following available integers bounds the number of generators of every submodule of the ambient module and its dual:*

$$\lfloor as/2\rfloor, \qquad \left\lfloor \frac{as}{\operatorname{lpp}(s/s_p)} \right\rfloor \quad (s/s_p > 1),$$
$$\left\lfloor as\sqrt{\frac{2}{3(p-1)e}} \right\rfloor \quad (s_p = p^e,\ e > 0).$$

*If the block action contains a soluble transitive subgroup, two more bounds are $\lfloor a\widetilde{\omega}(s)\rfloor$ and $as_p$, where*

$$k(s) = \sum_{q|s} \nu_q(s)(q-1), \qquad \widetilde{\omega}(s) = \frac{s}{2^{k(s)}}\binom{k(s)}{\lfloor k(s)/2\rfloor}.$$

*The minimum of these bounds and $\lceil 4as/\sqrt{\log s}\rceil$ may be used in Theorem 10.4.*

*Proof.* The prime-to-$p$ and square-root bounds follow from [Tra18, Theorem 1.6, Corollary 4.27], using $\pi > 3$; the two conditional bounds are its soluble-transitive alternative. The dual is induced from the dual local module. If $s$ is not a pure $p$-power, the prime-to-$p$ denominator is at least two. If $s$ is a $p$-power, a Sylow $p$-subgroup containing a Sylow subgroup of a point stabilizer is transitive, by orbit–stabilizer. Thus the conditional estimate applies. Since $k(s) \geq 1$ and $2^{-k}\binom{k}{\lfloor k/2\rfloor} \leq 1/2$, it gives the same half bound. Integrality gives its floor. Finally the older uniform bound is valid by Theorem 10.4. Taking a minimum preserves every bound and its uniform error estimates. □

**Theorem 11.2** (All but seven affine local degrees)**.** *In Theorem 10.4, suppose*

$$r \notin \{2, 3, 4, 5, 8, 9, 16\}.$$

*Take the refined capacities of Lemma 11.1. With* $h(w) = w - (w \bmod 2)$*, they satisfy*

$$\frac{h(w) - s}{8} - \eta_U \geq \frac{w}{8192}.$$

*Consequently all these actions form one complete transfer family with an exponentially small ordinary kernel row and a quadratically small scalar, uniformly in both the local degree and block count.*

*Proof.* Put $\gamma = (\log 3)/3 < 53/100$, using $3^{100} < 2^{159}$. Retain the actual local chief series and let $c$ be its abelian order charge: each abelian chief factor contributes its number of prime factors with multiplicity, and each nonabelian factor contributes zero. The half capacity gives

$$\eta_U \leq \gamma \frac{s}{2} c < \frac{53}{200} cs.$$

It suffices to prove the integer budget

$$54272c + 38400 \leq 25575r. \tag{11.1}$$

Indeed this gives $(53/200)c + r/8192 + 1/16 \leq (r-1)/8$. Multiplication by $s$ pays the possible parity loss $1/8$, since $s \geq 2$.

For $25 \leq r = p^d < 1024$, $d \geq 2$, $r \neq 32$, the literal embedding gives

$$c \leq \Omega\left(p^d \prod_{i=0}^{d-1} (p^d - p^i)\right).$$

Here $p \leq 31$ and $d \leq 9$, so a bounded integer calculation proves (11.1). This is arithmetic on prime powers, not an action catalogue. At $r = 32$, the original five-dimensional binary complement has abelian chief charge at most four, by the Clifford argument given below; translation adds five, so $c \leq 9$, which satisfies the same budget. Prime degrees use the literal order divisor $r(r-1)$. Finally, for $r \geq 1024$, $c \leq (\lfloor \log r \rfloor + 1)^2$ and the elementary dyadic quadratic-log inequality proves (11.1).

The remaining accepted prime powers are $7, 11, 13, 17, 19, 23$. For prime local degree $p$, the stabilizer is cyclic of order dividing $p - 1$, all chief factors have dimension one, and

$$\eta_U \leq \lfloor s/2 \rfloor \kappa(L), \qquad \kappa(L) \leq \log(p)/p + \kappa(C_{p-1}).$$

At $p = 7$, $\kappa(L) < 3/7 + 1/2 + 17/32 = 327/224$. For even $s$, the margin exceeds $9s/448 > 7s/1024$; for odd $s$, the retained floor gives $(9s + 271)/448$ instead. For a cyclic group, $\kappa(C_m) \leq \log(m)/2$. At $p = 11$, use $10^3 < 2^{10}$ to get $\kappa(L) < 13/6$; the margin is then greater than $s/6 - 1/8 \geq 5s/48$. At $p = 13$, $\kappa(L) < 5/2$ gives margin $s/4 - 1/8 \geq 3s/16$. Both exceed $ps/1024$. For $p \geq 17$, $\log(p-1) \leq (p-1)/4$ and $\log(p)/p < 1/2$, so the margin exceeds

$$s(p-5)/16 - 1/8 \geq s(p-6)/16 > ps/1024.$$

The last inequality holds already at $p = 17$. This exhausts the prime powers outside the seven stated degrees.

The actual quotient $T$ acts on $s \leq h(w)/2$ points and $w \geq 14$. Choose one block and chief-series certificate per action class. Equation (10.1) retains $\beta_U = o(w^2)$ and $\log \Psi_U(b) \leq Cw \log^2(w+b+1)$. Theorem 7.4 applies with $\rho = 1/8192$ and $w_0 = 14$. Its complete ordinary kernel rows are uniformly exponentially small, and its scalar is quadratically small. The original action weight and all quotient and extension data are those already present in (10.1). □

## 12 Explicit affine block counts

Write $s_p = p^{v_p(s)}$, let $\mathrm{lpp}(t)$ denote the largest prime-power divisor of $t > 1$, and put

$$k(s) = \sum_q v_q(s)(q-1), \qquad \omega(s) = s\binom{k(s)}{\lfloor k(s)/2 \rfloor} 2^{-k(s)}.$$

Use Lemma 11.1 for the induced module and its dual. In the notation of that lemma, write

$$H(s,p,a;T) = \min\left\{\bar{H}, \left\lceil \frac{4as}{\sqrt{\log s}} \right\rceil\right\}, \tag{12.1}$$

where $\bar{H}$ is the minimum of all its available integer bounds. The soluble-transitive entries are always available at prime-power block counts.

Always use (12.1) in Theorem 10.4. Its upper cutoff retains the uniform $\beta_U = o(w^2)$ and the semisimple error estimate. Thus a uniform positive margin

$$\frac{h(w) - s}{8} - \eta_U \geq \rho w \tag{12.2}$$

is precisely an application of Theorem 7.4, with the actual top $T$ as quotient. In the following arithmetic $\gamma = \log(3)/3 < 17/32$, using $3^{32} < 2^{51}$.

**Theorem 12.1** (Regular prime and symmetric-three components)**.** *For an actual nonbinary action with exact local $C_p$ on $p$ points,* (12.2) *holds with $\rho = 1/4096$ except possibly at the following block counts:*

| | |
|---|---|
| $p \geq 5$ | $\varnothing$ |
| $p = 3$ | $2, 6$ |
| $p = 2$ | $\{2^a : 1 \leq a \leq 8\} \cup \{3 \cdot 2^a : 0 \leq a \leq 10\}$. |

*For exact natural $S_3$, it holds with $\rho = 1/1024$ outside $\{2, 3, 4, 6, 8, 12, 18\}$.*

*Proof.* For $C_p$ the coefficient is $H \log(p)/p$. If $p \geq 5$, use $H \leq s/2$ and $\log(p)/p \leq 17/32$. Subtracting the desired larger margin $w/64$ leaves at least $((7p-25)s - 8)/64 > 0$.

For $p = 3$, if $s$ has a prime divisor at least five, use $H \leq s/5$; the margin exceeds $23s/160 - 1/8$. If $4 \mid s$, use $H \leq s/4$; the margin exceeds $15s/128$. These cases leave only $s = 3^b$ and $s = 2 \cdot 3^b$. At $s = 3$, $H \leq 1$ gives margin $> 3/32$. For $s = 3^b, b \geq 2$, the soluble bound gives $H \leq 3s/8$, hence margin $> 13s/256 - 1/8$. For $s = 2 \cdot 3^b, b \geq 2$, use $H \leq s/\sqrt{6} < 7s/16$, hence margin $> 9s/512$. All exceed $3s/4096$. Only two and six remain.

For $p = 2$, the margin is $s/8 - H/2$. If the odd part has a prime-power divisor at least five, $H \leq s/5$ leaves $s/40$. For $s = 2^a, a \geq 9$, the central-binomial ratio is at most $63/256$, leaving $s/512$. For $s = 3 \cdot 2^a, a \geq 11$, use $H \leq s\sqrt{2/33} < 127s/512$, leaving more than $s/1024$. This exhausts all counts outside the displayed binary list.

For $S_3$, write $H_p = H(s, p, 1; T)$. Its coefficient is $H_2/2 + \gamma H_3$ and its reserve is $s/4 - (s \bmod 2)/8$. If $s$ has a prime divisor at least five, $H_2, H_3 \leq s/5$ leave more than $7s/160 - 1/8 \geq 3s/160$. For $s = 2^a, a \geq 4$, use $H_2 \leq 3s/8$, $H_3 \leq 1$; the gap is greater than $s/16 - 17/32 > 3s/1024$. At $s = 9$, the integer bounds $H_2 \leq 1$, $H_3 \leq 3$ leave more than $1/32 > 27/1024$. For $s = 3^b, b \geq 3$, use $H_3 \leq 5s/16$, $H_2 \leq 1$; the gap exceeds $43s/512 - 5/8 > 3s/1024$. Finally let $s = 2^a 3^b$, $a, b \geq 1$. For $a \geq 3$, $\eta/s < 1/6 + 17/256$; for $a = 2, b \geq 2$, $\eta/s < 1/18 + 17/128$; for $a = 1, b \geq 3$, $H_2 \leq 2, H_3 \leq s/3$ give $\eta/s < 1/54 + 17/96$. Each is below $1/4 - 3/1024$. The untreated mixed cases are $6, 12, 18$, proving the exact stated sufficient list. □

**Theorem 12.2** (Four-point component capacities)**.** *For exact natural $A_4$,* (12.2) *holds with $\rho = 1/4096$ outside $\{2,3,4,6,8,12,16,24\}$. For exact natural $S_4$ it holds with the same $\rho$ outside*

$$\{2^a : 1 \le a \le 8\} \cup \{3 \cdot 2^a : 0 \le a \le 10\} \cup \{5,10,15,20,30,60\}.$$

*These statements allow all proper, zero and nonsplit block kernels.*

*Proof.* Put $H_{pa} = H(s,p,a;T)$. The respective coefficients are $\eta_A = H_{22}/2 + \gamma H_{31}$ and $\eta_S = \eta_A + H_{21}/2$, and the reserve is $3s/8$.

For $A_4$, a prime divisor at least five gives $\eta_A/s < (1+17/32)/5 = 49/160$. Otherwise $s = 2^a 3^b$. For pure binary powers $a \ge 5$, $H_{22} \le 5s/8, H_{31} \le 1$ leave more than $s/16 - 17/32$. For pure ternary powers $b \ge 2$, $H_{22} \le 2, H_{31} \le 3s/8$ leave more than $45s/256 - 1$. For $a,b \ge 1$ with $a,b \ge 2$ use $\eta_A/s < 1/9 + 17/128$; for $a = 1, b \ge 2$ use $H_{31} \le s/\sqrt{6} < 5s/12$, giving $\eta_A/s < 1/9 + 85/384$. Finally $b = 1, a \ge 4$ gives $\eta_A/s < 1/3 + 17/512$, whose gap is $13/1536$. All displayed gaps are at least $s/1024$, and their complement is exactly the asserted eight counts.

For $S_4$, a prime divisor at least seven gives $\eta_S/s < 65/224$; a factor 25 gives the still smaller bound $65/800$. Write therefore $s = 2^a 3^b 5^c$, $c \in \{0,1\}$. When $c = 1$, the prime-to-characteristic estimates give

$$\eta_S/s \le \frac{3/2}{\max(3^b,5)} + \frac{\gamma}{\max(2^a,5)}.$$

For $a \ge 3$ this is less than $3/10 + 17/256$, leaving $11/1280$; for $b \ge 2$ it is less than $1/6 + 17/160$. The six complementary counts are $5,10,15,20,30,60$. When $c = 0$, pure binary powers $a \ge 9$ give $\eta_S \le 189s/512 + \gamma$, leaving more than $3s/512 - 17/32 \ge s/1024$. Pure ternary powers $b \ge 2$ leave more than $45s/256 - 3/2 \ge s/1024$. For mixed counts, $b \ge 3$ gives $\eta_S/s < 1/18 + 17/64$, and $a,b \ge 2$ gives $\eta_S/s < 1/6 + 17/128$. At the remaining $s = 18$, the integer bounds $(H_{22}, H_{21}, H_{31}) \le (4,2,7)$ leave more than $1/32 > 18/1024$. Finally $b = 1, a \ge 11$ gives

$$H_{22} \le 2s\sqrt{2/33} < 79s/160, \quad H_{21} < 79s/320, \quad H_{31} \le 3;$$

the gap exceeds $3s/640 - 51/32 \ge s/1024$ for $s \ge 6144$. The stated exceptional set is exactly the complement of these cases. □

**Theorem 12.3** (The four remaining affine local degrees)**.** *For $r \in \{5,8,9,16\}$, every action with exact primitive affine local degree $r$ satisfies* (12.2) *with $\rho = 1/4096$, except possibly at*

| $r$ | $s$ |
|---|---|
| 5 | $2,3,4$ |
| 8 | $2,4,6$ |
| 9 | $2,3,4,6,12$ |
| 16 | $2,4.$ |

*Proof.* For $r = 16$, retain the actual correlated translation module inside $(\mathbb{F}_2^4)^s$. The faithful quotient acts on $8s$ original comparator points, using $\mathrm{GL}_4(2) \cong A_8$. The refined capacities give $H(s,2,4) \le 19s/10$ outside $s = 2,4$. Hence the original translation fibre has exponent at most $H/2$ and its margin is

$$\frac{16s - 8s}{8} - \frac{H}{2} \ge \frac{s}{20} > \frac{16s}{4096}.$$

The two remaining block counts use the actual cold-rank sources below. Thus no bound on the number of stabilizer composition factors is needed.

For $r = 5, 8, 9$, let $H(s, p, a)$ be the minimum in Lemma 11.1, using the soluble entries only for prime-power $s$ and omitting the logarithmic cutoff. It has the form $\lfloor ax(s, p) \rfloor$, so $\sum_j H(s, p, a_j) \le H(s, p, \sum_j a_j)$. Floors are taken after multiplying by the dimension. Put $t_2 = 1/2, t_3 = 17/32$, $t_5 = 7/15, t_7 = 3/7$, which upper-bound $\log p/p$. For brevity write $H_{pa} = H(s, p, a)$. The following upper bounds retain the true translation dimension:

$$E_5 = t_5 H_{51} + \tfrac{1}{2} H_{22},$$
$$E_8 = \tfrac{1}{2} H_{23} + t_3 H_{31} + t_7 H_{71},$$
$$E_9 = \tfrac{1}{2} H_{24} + t_3 (H_{32} + H_{31}).$$

Indeed the stabilizers at degrees five and nine divide $C_4$ and $\mathrm{GL}_2(3)$, respectively. For degree eight, a proper subgroup of the simple group $\mathrm{GL}_3(2)$ has index at least seven, since an index below seven would embed the group in $S_6$. Its order divides 168 and is at most 24, and therefore has at most two prime divisors and is soluble. An irreducible soluble stabilizer has trivial normal 2-subgroup. A minimal normal subgroup is then $C_3$ or $C_7$; $C_3$ has a nonzero proper fixed space on $\mathbb{F}_2^3$, contradicting irreducibility. A normal $C_7$ puts the stabilizer between $C_7$ and its order-21 Singer normalizer. Thus the only possibilities are $C_7, C_7 : C_3, \mathrm{GL}_3(2)$, proving the envelope $E_8$. Nonabelian local factors in these arguments stay in $\Psi_U$.

Here is an exhaustive arithmetic verification for all $s$, without a group census. Independent upper bounds on weighted dimensions and a binary exponent cutoff are

| $r$ | $A_2$ | $A_3$ | $A_5$ | $A_7$ | $a_0$ |
|---|---|---|---|---|---|
| 5 | 1 | 0 | 7/15 | 0 | 5 |
| 8 | 3/2 | 17/32 | 0 | 3/7 | 3 |
| 9 | 2 | 51/32 | 0 | 0 | 4 |

The reserve per $s$ is at least $B_r = (r-1)/8 - 1/16$ for odd $r$, and $B_r = (r-1)/8$ for even $r$. If $5 \mid s$, the capacity per dimension at $p = 5$ is less than $5s/12$, and at the other primes at most $s/5$. If $7 \mid s$, use $s/3$ at seven and $s/7$ elsewhere. A prime divisor at least eleven gives $s/11$ everywhere. If $9 \mid s$, use $5s/12$ at three and $s/9$ elsewhere. If $2^{a_0} \mid s$, use $3s/8$ at two for $r = 5$, $s/2$ for $r = 8$, and $5s/12$ for $r = 9$; other primes get $s/2^{a_0}$. Substitution gives these lower gaps per $s$, respectively:

| $r$ | $5 \mid s$ | $7 \mid s$ | $q \ge 11$ | $9 \mid s$ | $2^{a_0} \mid s$ |
|---|---|---|---|---|---|
| 5 | 31/720 | 383/1680 | 73/240 | 593/2160 | 23/480 |
| 8 | 429/1120 | 99/224 | 1605/2464 | 1181/2688 | 9/1792 |
| 9 | 7/32 | 95/224 | 215/352 | 59/1152 | 7/1536 |

Every entry exceeds $r/4096$. The complement is exactly $s = 2^a 3^b$, $0 \le a < a_0$, $b \in \{0, 1\}$, $s \ge 2$. The accepted cells there, with exact gaps $(h(rs) - s)/8 - E_r$, are

| | |
|---|---|
| 5 | $(6, 1/15), (8, 8/15), (12, 3/5), (16, 23/15), (24, 13/5), (48, 33/5)$ |
| 8 | $(3, 37/224), (12, 363/224)$ |
| 9 | $(8, 13/32), (24, 103/32)$ |

Direct substitution into the displayed integer formula proves each entry; each exceeds $rs/4096$. Together with the two degree-sixteen cells, the eleven remaining cells are exactly the exceptions in the theorem. Taking the logarithmic cutoff again restores the uniform error estimates and cannot worsen any margin. □

**Theorem 12.4** (Zero ternary kernel at local degree four)**.** *Let $U$ have exact local $A_4$ or $S_4$ on $s \geq 2$ blocks, put $M = (\mathbb{F}_2^2)^s$, $E = U \cap M$, and view $R = U/E$ inside $R_0 \wr T$, where $R_0 = C_3$ or $S_3$. If $R \cap C_3^s = 1$, then $R = T$, counts two and three are impossible, and every count outside $\{4, 8, 16\}$ satisfies* (12.2) *with $\rho = 1/96$.*

*Proof.* For local $A_4$, the linear block kernel is already $R \cap C_3^s$. For local $S_4$, it embeds in $C_2^s$. Each of its coordinate images is a normal 2-subgroup of the exact $S_3$, and is trivial. Thus $R = T$ in both cases. At counts two and three its point stabilizer cannot surject onto the required local $C_3$ or $S_3$.

Only the translation intersection layer remains. With $H = H(s, 2, 2; T)$ its entire fibre is bounded by

$$Z_J(U) \leq 2^{Hb/2+4sH+2s(g(4s)+1)} Z_J(T).$$

If $s$ is not a power of two, $H \leq 2s/3$ leaves gap $3s/8 - H/2 \geq s/24 = w/96$. If $s = 2^a, a \geq 5$, $H \leq 2\binom{a}{\lfloor a/2 \rfloor} \leq 5s/8$ leaves $s/16$. The other powers are two, four, eight and sixteen, of which two is impossible. The cutoff in (12.1) supplies the uniform quadratic error, so Theorem 7.4 applies. □

# 13 Nonaffine primitive components and compressed quotients

**Lemma 13.1** (Degree forms of a primitive socle)**.** *Let a nonaffine primitive group of degree $r$ have full socle $S^a$, where $S$ is nonabelian simple and $\ell$ is its least proper subgroup index. Then either $r = [S : H]^a$ for a proper subgroup $H < S$, or $r = |S|^d$ with $a \leq 2d$. In particular, $\ell \geq 5$ and $r \geq \ell^a$.*

*Proof.* Use the structural cases in [LPS88, §1 and the theorem on p. 392]. Almost simple and their product actions give the first form. Diagonal and compound diagonal actions give $a = mt$, $d = (m-1)t$, with $m \geq 2$; this also includes the cases with two regular minimal normal subgroups. Twisted wreath actions give $d = a$. Thus the full socle, including both minimal normal subgroups when present, satisfies the stated alternatives.

An index below five would embed $S$ in the soluble group $S_4$, so $\ell \geq 5$. The first form gives $r \geq \ell^a$ directly. In the second, use $\ell^2 < |S|$ from [BBERJS22, Theorem A(4)] to obtain $\ell^a \leq \ell^{2d} \leq |S|^d = r$. □

**Theorem 13.2** (Primitive compression)**.** *If $L \leq S_r$ is nonaffine primitive, its socle $E$ is semisimple and $L/E$ has a specified faithful action of positive degree $u$ with $2u \leq h(r)$. Beyond the structural O'Nan–Scott theorem, the finite primitive catalogue is used only in actual degrees $5 \leq r \leq 29$.*

*Proof.* Every nontrivial normal subgroup of a primitive group is transitive. An abelian minimal normal subgroup is regular and gives the affine case. Distinct minimal normal subgroups commute; since the centralizer of a transitive group is semiregular, two such subgroups must be regular mutual centralizers. They are isomorphic, and a third is impossible. Consequently $E = S^a$ for a nonabelian simple $S$. Also $C_L(E) = 1$: a nontrivial centralizer would contain a minimal normal subgroup of $L$, whereas its intersection with the centreless $E$ is trivial. Thus

$$L/E \leq \mathrm{Out}(S) \wr S_a.$$

Put $q = |\mathrm{Out}(S)|$. Its regular action gives the wreath product a faithful action of degree $aq$, including $q = 1$.

Theorem A(4) of the cited paper gives $q \leq 3 \log \ell$, where $\ell \geq 5$ is the least proper subgroup index of $S$. Lemma 13.1 gives $r \geq \ell^a$. For $a \geq 2$, $2aq \leq \ell^a$: at $a = 2, \ell = 5$, integrality gives $q \leq 6$

and $4q \le 24 < 25$; for $\ell \ge 6$ use $12\log\ell \le \ell^2$. Increasing $a$ multiplies the right side by at least five and the left side by at most $3/2$. For $a = 1, \ell \ge 30$, use $6\log\ell \le \ell$ instead.

In the remaining almost simple case, failure of the regular action of the actual quotient to have $2|L/E| \le r$ would imply $r < 2|L/E| \le 6\log\ell < 30$. The finite primitive certificate therefore checks precisely the required bounded range. Its complete degree-5–29 primitive alphabet has 116 simple nonabelian socles; each records the exact action and socle orders, checks their factorization $|L| = q|E|$, and verifies $2q \le r$. The published primitive-catalogue correspondence identifies the literal action and its actual socle with one of these rows. Lagrange's theorem then gives $q = |L/E|$ inside the formal proof. The structural primitive profile has already retained the literal socle, its semisimple chart, and a faithful action of $L/E$ of degree $q$, so the finite bounded step does not construct a second quotient map or reidentify its kernel. Completeness is relative to the stated primitive classification [RD05, HRDR23]; the four nonsimple nonaffine socles in this range are covered by the preceding argument. Thus $2u \le r$, and parity gives $2u \le h(r)$. □

**Theorem 13.3** (Complete semisimple compression)**.** *Suppose an actual transitive $U \le S_w$, $w \ge 5$, has a normal semisimple subgroup $E$, possibly trivial, such that $U/E$ has a faithful action of degree $v_0 \le h(w)/2$. The complete family of these actions satisfies Theorem 7.4, with $\eta_U = 0$ and $\rho = 1/20$. Every action with a nonaffine primitive exact local component has such a certificate.*

*Proof.* Write $E = \prod_i S_i$. Each $S_i$ is a literal subgroup of $S_w$. The retained simple-index inequalities give $|\operatorname{Out}(S_i)| \le |S_i|$: if $\ell_i \ge 5$ is the least proper subgroup index, then $|\operatorname{Out}(S_i)| \le 3\log\ell_i < \ell_i^2 < |S_i|$. Since a nonabelian simple group is centerless,

$$|\operatorname{Aut}(S_i)| = |S_i|\,|\operatorname{Out}(S_i)| \le |S_i|^2.$$

This supplies the required automorphism count without a simple-group generation theorem. Since $\sum_i \log|S_i| \le \log(w!) \le w\log w$, the product in Lemma 10.1 has logarithm at most $Cw\log^2(w+b+1)$ for an absolute $C$. The reserve is $(h(w) - v_0)/8 \ge h(w)/16 \ge w/20$ for $w \ge 5$. Theorem 7.4 applies with $\beta = 0$. Here Lemma 10.1 is applied to the complete sum over all literal normal axes before the physical transfer. Its single outer-factor product is therefore paid once. Assigning that complete bound to every axis and summing again would introduce a spurious normal-subgroup multiplicity and is not part of the estimate.

For a block system of size $r$ and count $s$, let $L$ be its exact primitive component and put $E_0 = \operatorname{Soc}(L)$, $E = U \cap E_0^s$. Each coordinate projection of $E$ is normal in the exact $L$, and hence is a normal subproduct of $E_0$. The subdirect-product description of products of nonabelian simple groups makes $E$ semisimple, allowing proper diagonal and zero intersections. Moreover $U/E$ embeds into $(L/E_0)\wr T$, whose faithful action has degree $v_0 = us$. Theorem 13.2 gives $2v_0 \le h(r)s \le h(rs)$. This proves the last assertion, including primitive actions by taking $s = 1$. □

## 14 Complete primitive-action consumers

Throughout this section $Z_J(U) = \sum_{N \lhd U} |\operatorname{Epi}(J, U/N)|$. Every quotient is literal, and $J$ is the same complete source throughout.

**Lemma 14.1** (One derived source for soluble affine targets)**.** *Let $U = V \rtimes R$, where $V = \mathbb{F}_p^d$ and $R \ne 1$ is soluble, faithful and irreducible. Put $\operatorname{End}_R(V) = \mathbb{F}_{p^f}$, $c_R = |N_{\mathrm{GL}_d(p)}(R)|/|C_{\mathrm{GL}_d(p)}(R)|$, and $\eta = f\log(p)/(pd)$. For every $J \le S_b$,*

$$Z_J(U) = Z_J(R) + |\operatorname{Epi}(J, U)|, \qquad |\operatorname{Epi}(J, U)| \le |V| c_R 2^{\eta b}.$$

*More precisely, if $t$ is the derived length of $R$ and $a_X$ are the multiplicities in the head of $H_1(J^{(t)}, \mathbb{F}_p)$ of the distinct pulled-back natural $R$-modules, then*

$$|\operatorname{Epi}(J, U)| = |V| c_R \sum_X (p^{f a_X} - 1). \tag{14.1}$$

*Proof.* Put $F = J^{(t)}$, $Q = J/F$. The nontrivial last derived group $R^{(t-1)}$ is abelian and has order prime to $p$, because $O_p(R) = 1$ for a faithful irreducible linear group. It has no fixed vector in $V$. For an onto $\theta : Q \to R$, the prime-to-$p$ part of the abelian group $Q^{(t-1)}$ maps onto $R^{(t-1)}$. Averaging and inflation–restriction therefore give $H^i(Q, V_\theta) = 0$ for all $i \geq 0$. Restriction identifies $H^1(J, V_\theta)$ with $\operatorname{Hom}_Q(H_1(F, \mathbb{F}_p), V_\theta)$, and each restriction has exactly $|V|$ literal cocycles. A nonzero restriction is onto $V$ by irreducibility, and is exactly the condition that the lift to $U$ is onto: $U^{(t)} = V$.

Two top maps give isomorphic modules precisely when they differ by an automorphism of $R$ induced by $N_{\mathrm{GL}_d(p)}(R)$. Each such orbit has $c_R$ members, proving (14.1). All types lie in one head, so $d \sum_X a_X \leq d_p(F) \leq b/p$ and $\sum_X (p^{f a_X} - 1) \leq p^{f \sum_X a_X} - 1$. Finally, an original normal subgroup of $U$ either contains $V$ or intersects it trivially. In the latter case it centralizes $V$; since $C_U(V) = V$, it is trivial. This proves the exact full-menu identity and the bound. □

**Theorem 14.2** (All sufficiently small-order actual actions)**.** *All transitive nonbinary actions of degree $w \geq 64$ with $\log |U| \leq w/8$ have one complete ordinary forward kernel*

$$K_{\mathrm{so}}(n, b) = \Delta(n, b) \sum_{\substack{[U]:\deg U = n-b \geq 64 \\ \log |U| \leq (n-b)/8}} \frac{2^{(\log |U|)^2 + (\log |U|)(b/2+1)}}{a(U)},$$

*with row at most $2 \cdot 2^{-n}$ eventually and zero scalar. In particular this covers every primitive affine action of degree at least* 1024.

*Proof.* The ordinary permutation-generator bound [HRD13, Proposition 2.1] gives $d(J) \leq b/2 + 1$. For $\ell = \log |U|$, the number of target normal subgroups is at most $2^{\ell^2}$, by padding a generating tuple of each subgroup to length $\lfloor \ell \rfloor$. The images of source generators then prove the displayed fibre envelope. Original pointing gives this kernel directly.

The uniform coefficient/Gaussian comparison gives

$$\log \Delta(n, b) \leq -\frac{(w - (w \bmod 2))b}{8} - \frac{w^2}{16} + O(w \log(n + 2)).$$

The envelope costs at most $w^2/64 + wb/16 + w/8$. The action-class entropy is $o(w^2)$ by the transitive enumeration theorem. For $w \geq 64$, the negative terms are at most $-wn/32 - w^2/64$; uniformly absorb class entropy into $w^2/128 + O(w)$ and the remaining terms into $wn/64$. Hence $K_{\mathrm{so}}(n, n - w) \leq 2^{-wn/64}$. Sum over all $w \geq 64$. For affine degree $w = p^d$, $\log |U| < (d + 1) \log w \leq (\log w)^2 + \log w \leq w/8$ when $w \geq 1024$. The last inequality follows at $\log w = 10$ and thereafter by differentiating twice. A primitive affine group of that degree cannot be binary: in characteristic two an irreducible binary linear group has only the trivial simple module. □

**Theorem 14.3** (The soluble primitive affine range)**.** *All soluble primitive affine actions of degrees $5 \leq w < 1024$, except the natural action $C_5 \rtimes C_4$ of degree five, have a complete ordinary forward row $O(2^{-an})$ and a complete scalar $O(2^{-a' n^2})$, for fixed $a, a' > 0$.*

*Proof.* For each actual $U = \mathbb{F}_p^d \rtimes R$, use Lemma 14.1. A regular prime action has $Z_J(C_p) = |\operatorname{Hom}(J, C_p)| \le p^{b/p}$; for $p \ge 5$, $\log(p)/p < (p-1)/8$. Suppose $R \ne 1$. If $p = 2$, $w \ge 8$, use the faithful action of $R$ on the $w-1$ nonzero vectors. It has degree $v = w-1 < w$ and $\eta \le 1/2 < w/8$, so Theorem 7.3 applies with comparator $R$.

For odd $p$ and abelian $R$, irreducibility makes $R$ cyclic in the multiplicative group of a field. Thus $Z_J(R) = |\operatorname{Hom}(J, R)| \le |J/J'| \le 3^{b/3}$. At $w \ge 7$, its exponent and $\eta$ are smaller than $(w-1)/8$. At degree five the complements are $1, C_2, C_4$. For $C_2$ use its degree-two comparator; the $C_4$ case is the specified exception.

At degree nine let $Z = R \cap \{\pm I\}$ and $P = R/Z$, acting faithfully on the four projective points. For each original normal subgroup of $R$, a fixed top map has at most $|\operatorname{Hom}(J, C_2)|$ lifts through its central kernel. If $h_2(J) = |\operatorname{Hom}(J, C_2)|$ and $e = d_2(P)$, then

$$h_2(J) Z_J(P) \le 2^e Z_J(P \times C_2).$$

Indeed, for one quotient $P/N$ with $c = |\operatorname{Hom}(P/N, C_2)|$, the two product kernels contribute $[1 + h_2(J) - c]|\operatorname{Epi}(J, P/N)|$, and $h_2(J) \le 2^e[1 + h_2(J) - c]$ whenever the epimorphism count is nonzero. These kernels are distinct for each $N$. Consequently $Z_J(R) \le \nu(R) 2^{d_2(P)} Z_J(P \times C_2)$. The comparator has faithful degree six, smaller than $h(9) = 8$; the affine exponent is at most $\log(3)/3 < 1$.

At degree twenty-five the scalar subgroup of $R \le \mathrm{GL}_2(5)$ has order dividing four. Its projective image is a soluble subgroup of $\mathrm{PGL}_2(5) \cong S_5$ [Adl92, Lemma 13], of order at most twenty-four. Indeed, a subgroup $P \le S_5$ not containing $A_5$ has index at least five: otherwise its coset kernel is trivial and would embed $S_5$ in $S_4$. A soluble $P$ cannot contain $A_5$, so $|P| \le 24$. Each central quotient lift costs at most $|\operatorname{Hom}(J, C_4)| \le 2^{b/2}$. Counting all fixed target normals and using $d(J) \le b/2 + 1$ gives $Z_J(R) \le D_R 2^{(\log 48) b/2}$. Its exponent is strictly below $h(25)/8 = 3$.

At degree twenty-seven the actual faithful irreducible soluble complement in $\mathrm{GL}_3(3)$ has order at most seventy-eight. The direct linear proof separates a monomial action from the homogeneous restriction to a minimal normal subgroup; the latter gives the field $\mathbb{F}_{27}$ and its scalar normalizer of order $3(27-1) = 78$. Thus $Z_J(R) \le D_R 2^{(\log 78) b/2}$, with $(\log 78)/2 < 26/8$. No primitive-catalogue correspondence is needed.

For the remaining odd prime-power degrees, elementary linear-map counting gives $|R| \le p^{d^2}$ when $w = p^d$. At $w = 49$, this gives $|R| \le 49^2$. At $w = 81$, use instead the actual order divisor $|R| \mid |\mathrm{GL}_4(3)| = 2^9 3^6 \cdot 5 \cdot 13$. Every chosen chief series therefore has abelian charge at most seventeen. The complete bottom fibre costs at most one additional $\gamma = (\log 3)/3$ unit, even without solubility. Consequently the complete source exponent is at most $18\gamma < 9.6 < h(81)/8 = 10$. This proves the needed source directly; a soluble-complement cube bound is unnecessary. For all other remaining pairs $(p, d)$ below the large-degree cutoff, the elementary bound $p^{d^2}$ gives the gap directly; the previously proved large-degree capacity estimates handle the infinite range. The fixed-target ordinary generator envelope applies as before.

These cases exhaust the finite set of action classes. Every pointwise exponent has a strict physical gap; every comparator has $0 < v < h(w)$ and $\eta < h(w)/8$. Their original kernels are $\Delta(n, n-w)$ times their respective fibre envelopes divided by $a(U)$. Apply Theorem 7.3 and take finite positive minima of these gaps. The pointwise classes have scalar zero. This proves complete row and scalar rates without assuming any normalized count is bounded. □

**Proposition 14.4** (Natural symmetric-four and Frobenius-five actions)**.** *The natural $S_4$ and $C_5 \rtimes C_4$ actions have complete ordinary forward rows $O(2^{-an})$ and complete scalars $O(2^{-a'n^2})$, for positive constants.*

*Proof.* For $S_4 = \mathbb{F}_2^2 \rtimes S_3$, Lemma 14.1 gives $Z_J(S_4) \le Z_J(S_3) + 24 \cdot 2^{b/4}$. The comparator $S_3$ has faithful degree three, less than the original width four, and $1/4 < 4/8$. The finite comparator theorem applies, using the original normalizer 24.

For the natural degree-five action put $z(J) = Z_J(C_4) = |\operatorname{Hom}(J, C_4)|$ and $\eta = \log(5)/5$. The same lemma gives $Z_J(U) \le z(J) + 5 \cdot 2^{\eta b}$. Declare $z(J) \le 2^{3b/8}$ cold. Its exact kernel is

$$\frac{\Delta(n, n-5)}{20} \left(2^{3(n-5)/8} + 5 \cdot 2^{\eta(n-5)}\right),$$

whose gap is positive since $\eta < 15/32 < 1/2$. On the hot event, the whole original fibre is at most $6 \cdot 2^{(\eta - 3/8)b} z(J)$. Use Theorem 26.1 with $r = \max(1, \lceil b/8 \rceil)$. Its weighted first moment, including the hot cutoff, has quadratic exponent

$$b^2/16 + br/4 + r^2/2 - (r-1)3b/8 + (\eta - 3/8)b - n^2/16 = -b^2/128 + (r - b/8)^2/2 + (\eta - 5/8)b - 25/16.$$

The actual scalar includes exactly the original factor $1/(20b! p_n G_{\lfloor n/2 \rfloor})$ on the hot source sum. Its logarithmic normalization costs $O(n \log(n+2))$; the moment error is $o(n^2)$. Thus it is at most $2^{-n^2/512}$ eventually. The hot event is empty at $b = 0$. □

**Lemma 14.5** (A nonabelian order saving)**.** *Write $\Omega(m)$ for the number of prime factors of $m$, counted with multiplicity, and let $c_{\mathrm{ab}}(G)$ count the abelian composition factors of a finite group $G$. If $E$ is a nonabelian simple normal subgroup of $G$, then*

$$c_{\mathrm{ab}}(G) + \Omega(|E|) \le \Omega(|G|).$$

*The same inequality holds if $G$ maps onto a finite group $S$ with a nonabelian normal subgroup $E$ contained in every nontrivial normal subgroup of $S$. In this second case $E$ is a subgroup of the target $S$; no uniqueness condition is required in $G$.*

*Proof.* Fix an actual chief series of $G$. In the first case, take the first term whose intersection with $E$ is nontrivial. Simplicity makes that intersection equal to $E$, so $E$ injects into the corresponding chief factor. In the second case, take the first term with nontrivial image in $S$. The preceding term maps trivially, and the image of the chosen term contains $E$. Thus that chief factor maps onto a group containing $E$. In either case the factor is nonabelian and its order is divisible by $|E|$. Additivity of $\Omega$ along the series proves the claim: abelian chief factors pay their full order charge, while this nonabelian factor saves at least $\Omega(|E|)$. □

**Theorem 14.6** (The nonsoluble primitive affine range)**.** *All primitive affine actions $U = \mathbb{F}_p^d \rtimes R$ of degrees $5 \le w = p^d \le 1024$ with nonsoluble $R$ have one complete pointwise ordinary forward row at most $2^{-n/8}$ eventually and scalar zero.*

*Proof.* By Theorem 10.2 and Proposition 10.3, the complete fibre has the bound

$$Z_J(U) \le (1 + A_U) C_R (1 + b \log(b+1))^{D_R} 2^{(\kappa(R) + \eta_U) b},$$

where $A_U = |Z^1(R, \mathbb{F}_p^d)|$, $\eta_U = f \log(p)/(dp)$, and $\mathbb{F}_{p^f} = \operatorname{End}_R(\mathbb{F}_p^d)$. Because $R$ is nonabelian, $f$ is a proper divisor of $d$; otherwise the faithful image would be scalar over its commuting field. Thus $d \ge 2$ and $\eta_U \le \log(p)/(2p)$. Write $\gamma = \log(3)/3 < 17/32$.

The actual chief traces give the following abelian charge budgets:

| $w$ | 32 | 49 | 64 | 128 | 256 |
|---|---|---|---|---|---|
| $c_{\mathrm{ab}}(R)$ | 4 | 8 | 14 | 29 | 40. |

The entries at 49 and 256 follow from the literal full-linear orders. At 128 the full order has thirty prime factors with multiplicity, and a nonabelian chief factor saves at least one unit.

For 32, Clifford restriction to a nontrivial normal subgroup is irreducible: constituent dimension divides five, and dimension one over $\mathbb{F}_2$ would give a trivial restriction. An abelian normal subgroup has cyclic scalar centralizer of order thirty-one, and the conjugation quotient embeds in the cyclic automorphism group of that normal subgroup. It would therefore make the whole group soluble. A nonabelian minimal normal subgroup is simple; the other product factors would lie in its cyclic linear centralizer. Its ambient centralizer is trivial. The actual quotient embeds in the outer automorphism group of this simple subgroup, whose order is at most thirty by the retained index inequalities. The quotient therefore spends at most four abelian order units.

For 64, nonhomogeneous Clifford restriction gives actual block patterns $(2,3),(3,2),(6,1)$. The corresponding order divisors are $|\operatorname{GL}_3(2)|^2\,2!$, $|\operatorname{GL}_2(2)|^3\,3!$, and $6!$; their prime-factor budgets are eleven, eight and seven. A homogeneous abelian normal subgroup gives an actual scalar field of degree $e = 2, 3, 6$ on the same module. The scalar-linear kernels give order divisors $|\operatorname{GL}_3(4)|\,2$, $|\operatorname{GL}_2(8)|\,3$ and $|\operatorname{GL}_1(64)|\,6$, with budgets thirteen, eight and five. If there is no abelian normal subgroup, every soluble normal subgroup is trivial. A nonabelian minimal normal subgroup is a product of simple groups. Each faithful homogeneous simple constituent has dimension three or six. Its linear centralizer is soluble: the irreducible cases have order dividing seven or sixty-three, and the repeated three-dimensional case has centralizer order dividing six. Thus there is only one simple factor and its ambient centralizer is trivial. The original quotient embeds in its outer automorphism group, whose order is at most thirty-six, costing at most five units. These cases prove the displayed uniform bound fourteen while retaining the original chief series.

Using $\gamma < 8/15$ and $\eta_U \le 1/4$ in the binary cases, and $\eta_U \le \gamma/2$ otherwise, all five displayed budgets leave a physical gap greater than $1/4$. Degree 81 uses the complete $18\gamma$ source above. For $343 \le w < 1024$, the elementary original affine-order ceiling has exponent $(\lfloor \log w \rfloor + 1)^2$; its half-log slope is at most $81/2$ below 512 and 50 thereafter, again leaving more than $1/4$. The remaining medium odd degrees use their actual linear-order ceilings.

At $w = 25$, quotient the scalar subgroup of order dividing four. The projective image is a nonsoluble subgroup of $\operatorname{PGL}_2(5) \cong S_5$, hence $A_5$ or $S_5$. To see this uniformly, a subgroup not containing $A_5$ has order at most twenty-four, as above. Its order divides 120 and hence has at most two prime divisors, so Burnside's theorem [Bur04] makes it soluble. Thus a nonsoluble projective image contains $A_5$ and is $A_5$ or $S_5$. Now $|R|$ divides $|\operatorname{GL}_2(5)| = 480$, so $\Omega(|R|) \le 7$, while $\Omega(|A_5|) = 4$. Lemma 14.5, applied in the actual projective quotient, gives at most three abelian factors of $R$. Its exponent is therefore at most $3\gamma + 17/64 < 119/64 < 3 - 1/4$. The argument also covers nonsplit scalar extensions.

The only remaining prime-power degrees below sixty-four are $8, 16, 27$: prime-degree stabilizers are cyclic, and $\operatorname{GL}_2(2), \operatorname{GL}_2(3)$ are soluble. The complete primitive action catalogue at these three degrees gives the following finite composition-factor data for the literal stabilizers:

| degree | primitive indices | abelian factors of $R$ | bound on $\kappa(R)$ |
|---|---|---|---|
| 8 | 3 | none | 0 |
| 16 | 11–20 | at most two | 17/16 |
| 27 | 10,11 | at most one | 17/32 |

The affine classification is that of [RDU03]; the representative conventions and cross-catalogue indexing are those of [HRDR23, RD05]. These counts also follow from Lemma 14.5: the degree-sixteen complements have alternating simple cores or actual projective quotients with alternating

cores, and their exact orders leave at most two prime-factor charges. In particular the scalar $C_3$ in $\mathrm{GL}_2(4)$ is retained; no unique-minimal-normal claim is made there. At degree twenty-seven the determinant-one subgroup of $\mathrm{GL}_3(3)$ is simple of index two. The displayed bounds supply respectively exponent bounds $1/4, 21/16, 51/64$, all with physical gap greater than $1/4$. For the formal proof, degree eight is covered by checked subgroup and linear-intertwiner certificates in $\mathrm{GL}_3(2)$, degree sixteen keeps the original translation kernel and uses the proved $\mathrm{GL}_4(\mathbf{F}_2) \cong A_8$ geometry, and degree twenty-seven uses proved linear structure. The bounded classifiers in the displayed table are an alternative verification; the final source construction needs no primitive composition-length premise. These are sufficient actual-source consumers; the stronger small-affine carrier correspondence and the primitive-sixteen locator are unused. The other ranges were covered analytically.

Finally use the original pointed kernel

$$\Delta(n,b) \sum_{\substack{[U] \text{ in this family} \\ w=n-b}} \frac{(1+A_U)C_R(1+b\log(b+1))^{D_R}2^{(\kappa(R)+\eta_U)b}}{a(U)}.$$

The family is fixed and finite, so all target constants and polynomial exponents are fixed. Its complete row is $O(n^D)2^{-n/4} \le 2^{-n/8}$ eventually. The full cocycle factor $A_U$ remains in this argument; it is not set equal to $|\mathbb{F}_p^d|$ for nonsoluble stabilizers. □

## 15 The bounded ternary block counts

Put $\gamma = (\log_2 3)/3 < 17/32$. All fibres in this section use the complete source $J \le S_b$, without any restriction on its orbits.

**Lemma 15.1** (Finite multiplicative comparison)**.** *Fix finitely many actual actions $U$ of widths $w$, and fixed finite comparators $Q_U$ with faithful actions of degrees $0 < v_U < h(w)$. Suppose*

$$Z_J(U) \le C_U 2^{\eta_U b} Z_J(Q_U), \qquad \delta_U = (h(w) - v_U)/8 - \eta_U > 0.$$

*Their complete pointed union has an exponentially small forward row and a quadratically small scalar, with original normalizers $a(U)$. The comparators need not be quotients of $U$.*

*Proof.* Set $\alpha = v/8 + \delta/2$, and call $J$ hot when $Z_J(Q) > 2^{\alpha b}$. The cold coefficient is exactly $\Delta(n,b)C_U 2^{(\eta_U+\alpha)b}/a(U)$. Its exponent is $-\delta b/2 - w^2/16 + O_w(\log(n+2))$. For the full hot fibre use Lemma 7.1 with $q = \max(1, \lceil 4\delta b/v^2 \rceil)$ and $q_* = 4\delta b/v^2$. The normalized leading exponent is

$$\frac{(b+qv)^2}{16} - (q-1)\alpha b + \eta b - \frac{(b+w)^2}{16} = -\frac{\delta^2 b^2}{v^2} - \frac{\delta b}{2} - \frac{(w-h(w))b}{8} - \frac{w^2}{16} + \frac{v^2(q-q_*)^2}{16}.$$

The rounding term is bounded and the coarse counting error is $O(n^{3/2} + n\log(n+2))$. The original hot weight is $(b!a(U)p_n G_{\lfloor n/2 \rfloor})^{-1}$ times the complete hot sum of $Z_J(U)$. A finite union therefore preserves both strict losses. Choose certificates before the any-hot/cold partition. □

**Lemma 15.2** (Central binary comparison)**.** *If $X$ has central subgroup $C_2$ and quotient $T$, then*

$$Z_J(X) \le \nu(X)|\operatorname{Hom}(J, C_2)|Z_J(T), \qquad |\operatorname{Hom}(J, C_2)|Z_J(T) \le 2^{d_2(T)} Z_J(T \times C_2),$$

*where $\nu(X)$ is its number of literal normal subgroups.*

*Proof.* For each $N \lhd X$, $X/N$ is a central extension of a quotient of $T$ by a group of order at most two. A fixed map to that quotient has no lifts or at most $|\operatorname{Hom}(J, C_2)|$ lifts. Summing over all $N$ proves the first inequality, including repetitions of the same normal subgroup of $T$. For the second put $h = |\operatorname{Hom}(J, C_2)|$. For each $S \lhd T$ put $Q = T/S$, $a = |\operatorname{Hom}(Q, C_2)|$. An onto map $J \to Q$ and a binary character are jointly onto $Q \times C_2$ precisely when that character does not factor through the map. Thus the distinct product normals $S \times 1, S \times C_2$ contribute $(h - a + 1)|\operatorname{Epi}(J, Q)|$. Whenever the latter epimorphism set is nonempty, $h \geq a$ and $h \leq a(h - a + 1) \leq 2^{d_2(T)}(h - a + 1)$. All remaining product normals add nonnegative terms. □

**Proposition 15.3** (Regular ternary exceptions)**.** *Every actual transitive action with exact local regular $C_3$ and two or six blocks has a complete exponentially small forward row and quadratically small scalar.*

*Proof.* Write $U \leq C_3 \wr T$, $E = U \cap C_3^s$. At $s = 6$, every transitive $T \leq S_6$ contains a semiregular $C_3$: a Sylow three-subgroup of order three has trivial point stabilizers, while one of order nine is a Sylow subgroup of $S_6$ and contains two disjoint three-cycles. The ambient module and its dual restrict to two regular $\mathbb{F}_3[C_3]$ modules. Since $\mathbb{F}_3[C_3] \cong \mathbb{F}_3[t]/(t^3)$, every submodule has at most two generators: its number of Jordan blocks is its socle dimension, at most two. Quotients retain this bound. The full-chief proof of Theorem 10.4, with this capacity, gives $Z_J(U) \leq C_U 2^{2\gamma b} Z_J(T)$. Its physical width is eighteen, its comparator degree six, and $3/2 - 2\gamma > 0$, so Lemma 15.1 applies.

At $s = 2$, exact local $C_3$ forces $E$ to project onto each coordinate. Its only $C_2$-invariant possibilities in $\mathbb{F}_3^2$ are the diagonal, the antidiagonal, and the full plane. Coprimality splits the extension, giving $C_6, S_3, C_3 \times S_3$, respectively, all in their original degree-six actions. The first has full fibre $|\operatorname{Hom}(J, C_6)| \leq 3^{b/3}$. For the other two use the common source $F = J'J^2$, $B = J/F$. The semisimple module $H_1(F, \mathbb{F}_3)$ has trivial multiplicity $d = d_3(J)$ and nontrivial sign multiplicities $a_\theta$, with $d + \sum a_\theta \leq b/3$. Coprime inflation–restriction gives

$$|\operatorname{Epi}(J, S_3)| = 3 \sum_{\theta \neq 0} (3^{a_\theta} - 1), \qquad |\operatorname{Epi}(J, C_3 \times S_3)| = (3^d - 1) 3 \sum_{\theta \neq 0} (3^{a_\theta} - 1).$$

The second joint map is onto because $C_3, S_3$ have no common nontrivial quotient. Each expression is at most $3^{b/3+1}$. The complete normal quotients of $C_3 \times S_3$ are $1, C_2, C_3, C_6, S_3, C_3 \times S_3$; the cyclic terms have the same abelianization bound. Thus $Z_J(U) \leq C_U 2^{\gamma b}$. Its slope is less than $h(6)/8 = 3/4$, so direct normalization already gives an exponentially small row and zero scalar. □

## 15.1 Signed modules on a bounded number of triples

For exact natural $S_3$ on $s$ blocks write

$$E = U \cap C_3^s, \quad R = U/E \leq C_2 \wr T, \quad D = \ker(R \longrightarrow T) \leq C_2^s.$$

These are the actual kernel and actual top of the chosen block system. If $E = 1$, the whole block kernel is trivial: a nontrivial normal local image is $C_3$ or $S_3$, and in the latter case its derived subgroup projects onto $C_3$. Hence $U \cong T$ has faithful degree $s$ and satisfies Theorem 7.3. If an ambient signed ternary module and its dual have capacity $H_3$, the full-chief argument gives

$$Z_J(U) \leq C_U 2^{\gamma H_3 b} Z_J(R). \tag{15.1}$$

Adding a binary capacity $H_2$ gives the same bound with $2^{(\gamma H_3 + H_2/2)b} Z_J(T)$. These bounds include all proper kernels, normal graphs and nonsplit extensions.

**Lemma 15.4** (Signed isotypes)**.** *Assume $D \neq 1$. Its coordinate characters are all nonzero and have $q$ distinct values, each repeated $a = s/q$ times. They span $D^*$; thus $q \mid s$, $d = \dim D \leq q \leq 2^d - 1$, and $q = 1, 2$ forces $d = 1, 2$, respectively. If $a$ is a power of two or $a = 3$, every submodule of the signed ambient module and of its dual is cyclic.*

*Proof.* Transitivity makes the character multiplicities equal, and their common kernel is zero because $D$ is its actual coordinate image. The idempotents of $\mathbb{F}_3[D]$ decompose a submodule into isotypes. The stabilizer of one isotype acts transitively on its $a$ coordinates by signed permutations. If $a$ is a power of two, a transitive Sylow two-subgroup of its coordinate image lifts to a two-group of signed permutations. The coordinate module is cyclic and semisimple over this group; an equivariant projection of a cyclic generator generates any submodule. If $a = 3$, a Sylow three-subgroup maps isomorphically onto a coordinate three-cycle. Its signs have product one, so it untwists to the regular $\mathbb{F}_3[C_3]$ module, whose ideals are cyclic. Transport a generator of the chosen isotype by $R$ to generate all isotypes. Signed permutation matrices identify the dual with the original module. Quotients of cyclic modules are cyclic as well. □

**Lemma 15.5** (Two common-source onto bounds)**.** *(i) Let $V$ be elementary abelian of order $3^e$, and $R$ a two-group acting faithfully with cyclic dual $V^*$. Then*

$$|\operatorname{Epi}(J, V \rtimes R)| \leq 3^e |\mathrm{GL}_e(3)| 3^{b/3}.$$

*The same conclusion holds for $R = C_6$ acting faithfully on a single unipotent ternary block, with its involution acting as $-1$.*

*(ii) If $X$ has a normal Sylow $p$-subgroup $P$ with cyclic center and $C_X(P) = Z(P)$, then*

$$|\operatorname{Epi}(J, X)| \leq |Z(P)| \, |\operatorname{Aut}(P)| p^{b/(p-1)}.$$

*Proof.* For (i), fix $F = O^2(J)$ in the two-group case and $F = J'$ in the $C_6$ case. Formation residuals commute with onto maps: the image of the source residual has the required quotient type, and the preimage of the target residual has that type, proving both inclusions. Consequently every map under consideration sends this same $F$ onto $V$. Put $B = J/F$, $A = H_1(F, \mathbb{F}_3)$. The restriction is an onto $B$-map $A \to V_\theta$ and dualizes to an injection $\iota : V_\theta^* \to A^*$. Fix a cyclic vector $v_0 \in V^*$. Its image $f$ determines the whole submodule $\langle Bf \rangle$. For each $f$, at most $|\mathrm{GL}_e(3)|$ identifications with $V^*$ are possible, and faithfulness determines the quotient map $\theta$ uniquely from each identification. Thus all quotient maps and onto restrictions together have at most $|\mathrm{GL}_e(3)| 3^{\dim A}$ choices. For fixed restriction the lift fibre is empty or a cocycle torsor on $B$. Averaging gives $H^1(B, V_\theta) = 0$ when $B$ is a two-group. In the second case $B$ is abelian and its two-primary Hall subgroup has zero invariants because it maps onto the scalar involution; averaging and inflation–restriction give the same conclusion. The torsor has size at most $|V|$. Finally $\dim A = d_3(F) \leq b/3$.

For (ii), fix $F = O^{p'}(J)$ and its largest $p$-group quotient $Q = F/O^p(F)$. Every onto map to $X$ sends $F$ onto $P$ and factors there through $Q$. Since $F \leq S_b$, $|Q| \leq p^{v_p(b!)} \leq p^{b/(p-1)}$. A faithful linear character of $Z(P)$ induces an irreducible constituent faithful on $P$: every nontrivial normal subgroup of a $p$-group meets its center. Fix its faithful character $\chi$. Pullback along onto maps $Q \to P$ gives irreducible characters of $Q$; one pulled-back character determines the kernel and therefore at most $|\operatorname{Aut}(P)|$ maps. The degree-square identity bounds their number by $|\operatorname{Aut}(P)||Q|$. For a fixed onto restriction, two extensions to $J$ differ in $C_X(P) = Z(P)$, so their quotient maps coincide. Their differences form a cocycle torsor on the $p'$-group $J/F$ with coefficients $Z(P)$. Coprime averaging bounds its size by $|Z(P)|$. □

**Proposition 15.6** (Two symmetric triples)**.** *All exact-local-$S_3$ actions on two blocks satisfy a complete finite forward estimate.*

*Proof.* The block stabilizer is its kernel, so its binary image is either $C_2^2$ or diagonal $C_2$. In the first case distinct signs force $E = \mathbb{F}_3^2$ and $U = S_3 \wr C_2$. The ternary kernel is faithful irreducible for $R = D_8$, and Lemma 14.1 gives $Z_J(U) \le Z_J(D_8) + 216\,3^{b/6}$. The comparator degree is four. In the diagonal case $R$ is $C_4$ or $V_4$. Coprimality splits the extension. For $C_4$, its square is $-I$, so $E = \mathbb{F}_3^2$ is irreducible and the derived-source bound gives $Z_J(U) \le Z_J(C_4) + 18\,3^{b/3} \le 19\,3^{b/3}$. For $V_4$, the ambient module has two distinct sign lines; choosing one or both gives $S_3 \times C_2$ or $S_3^2$. The former has a faithful degree-five action.

For the latter use the same $F = J'J^2$ and sign multiplicities $a_\theta$ as above. An ordered pair of onto maps to $S_3$ is onto $S_3^2$ precisely when its binary characters differ: a proper subdirect has common quotient $C_2$ or $S_3$, forcing equality. Hence

$$|\operatorname{Epi}(J, S_3^2)| = 9 \sum_{\theta \neq \psi} (3^{a_\theta} - 1)(3^{a_\psi} - 1) \le 18\,3^{b/3}.$$

The last inequality expands $\prod_\theta (1 + (3^{a_\theta} - 1))$ and uses $\sum a_\theta \le b/3$. Its complete normal menu is

$$Z_J(S_3^2) = |\operatorname{Epi}(J, S_3^2)| + 2|\operatorname{Epi}(J, S_3 \times C_2)| + 2|\operatorname{Epi}(J, S_3)| + |\operatorname{Epi}(J, V_4)| + 3|\operatorname{Epi}(J, C_2)| + 1.$$

Indeed intersection with $C_3^2$ is zero, one of its two sign lines, or the full plane; self-centralization resolves the zero case, and the other cases give exactly these quotients. Thus $Z_J(S_3^2) \le 2Z_J(S_3 \times C_2) + 18\,3^{b/3}$. Every slope is below $h(6)/8$ and every comparator degree is below six, so Theorem 7.3 finishes all cases. □

**Proposition 15.7** (Four blocks with two signs)**.** *All exact-local-$S_3$ four-block actions whose binary kernel has two distinct coordinate characters have a complete finite forward estimate.*

*Proof.* Their equal character classes have size two, $D = C_2^2$, and the top preserves the two-class partition. Thus $T \le D_8$ has order four or eight, and $R$ is a two-group of order sixteen or thirty-two. Coprimality splits $U = E \rtimes R$. The two isotypes have equal dimension in every submodule, so nonzero $E$ has dimension two or four. Every two-dimensional section has monomial image exactly $D_8 \le \mathrm{GL}_2(3)$: its diagonal signs are independent and its two lines are interchanged. Its action kernel $K$ in $R$ has order two or four. Such a normal kernel has a central series of at most two order-two factors, first by meeting $Z(R)$ and then because a normal subgroup of order two is central.

Put $P = \mathbb{F}_3^2 \rtimes D_8 = S_3 \wr C_2$ in its faithful degree-six action and $Q_* = P \times C_2^2$ in degree ten. The central series, Lemma 15.2, and the literal quotient-menu inclusion imply $Z_J(V \rtimes R) \le A_V Z_J(Q_*)$ for every two-dimensional section $V$, with a fixed finite $A_V$. They also give $Z_J(R) \le A_R Z_J(Q_*)$ whenever such a section exists. No splitting of the central extensions is required.

If $E$ is the full four-dimensional module, it is faithful and its dual is cyclic: the coordinate module is cyclic and semisimple for the two-group $R$. Lemma 15.5(i) bounds its onto maps by $C_0 3^{b/3}$, where $C_0 = 81|\mathrm{GL}_4(3)|$. A normal subgroup disjoint from $E$ centralizes $E$, so is trivial. All other proper intersections are two-dimensional sections. Consequently

$$Z_J(U) \le |\operatorname{Epi}(J, U)| + Z_J(R) + \sum_{\substack{I < E \\ \dim I = 2,\, I \ R\text{-stable}}} Z_J(U/I).$$

If $E$ is reducible, the preceding comparisons bound this by $B_U Z_J(Q_*) + C_0 3^{b/3}$. If it is irreducible, the sum is empty and use $R$ in faithful degree eight instead. If $\dim E = 2$, the central comparison

already bounds the entire fibre. These are all normal strata. Comparator degrees at most ten and slope $\gamma$ are strictly below the required width-twelve thresholds. □

**Proposition 15.8** (Three blocks with one sign)**.** *All exact-local-$S_3$ three-block actions with diagonal binary kernel have a complete finite forward estimate.*

*Proof.* The top is $C_3$ or $S_3$. A lifted coordinate three-cycle of order three untwists its signs. In the first case $R = C_2 \times C_3$. In the second the determinant is nontrivial on the scalar binary kernel and splits $R \to S_3$; a transposition conjugating the three-cycle to its inverse has constant signs. Thus after diagonal rescaling $R = C_2 \times S_3$ acts by scalar signs and ordinary coordinate permutations. Its nonzero invariant ternary submodules are exactly the three ideals in $\mathbb{F}_3[t]/(t^3)$, of lengths $e = 1, 2, 3$. All are stable under $S_3$ as well. The scalar involution has zero invariants on the module and on every quotient. Averaging and inflation–restriction give $H^1(R, \mathbb{F}_3^3/E) = 0$; a translation therefore removes the cocycle of the actual embedded subgroup. Hence $U_e = E_e \rtimes R$.

For top $C_3$, $U_1 = S_3 \times C_3 =: Q$, faithfully of degree six. At $e = 2, 3$, the action is faithful, $U_e' = E_e$, and the dual is cyclic. Lemma 15.5(i) gives $|\operatorname{Epi}(J, U_e)| \le C_e 3^{b/3}$. Normal intersections with $E_e$ form its uniserial chain. At surviving length at least two, faithfulness forces a normal subgroup disjoint from the ternary kernel to be trivial. At length one the quotient is $Q$, whose normals disjoint from its first ternary line are 1 and its central $C_3$. There are no mixed normal graphs, because the involution inverts one line and fixes the other. Thus, exactly,

$$\begin{aligned} Z_J(U_1) &= Z_J(Q), \\ Z_J(U_2) &= Z_J(Q) + |\operatorname{Epi}(J, U_2)|, \\ Z_J(U_3) &= Z_J(Q) + |\operatorname{Epi}(J, U_2)| + |\operatorname{Epi}(J, U_3)|. \end{aligned}$$

For top $S_3$, $U_1 = S_3^2 =: Q_*$, again faithfully of degree six. At lengths two and three the action of $R$ is faithful. The normal Sylow three-subgroup $P_e = E_e \rtimes C_3$ has center the first ideal $C$, of order three, and $C_{U_e}(P_e) = C$. This follows since centralizing $E_e$ forces an element into $E_e$, where centralizing the top three-cycle forces it into $C$. Lemma 15.5(ii) gives $|\operatorname{Epi}(J, U_e)| \le 3|\operatorname{Aut}(P_e)|3^{b/2}$. Every nontrivial normal subgroup contains $C$, by the uniserial chain and faithfulness. Moreover $U_2/C \cong Q_*$ and $U_3/C \cong U_2$. To see the necessary twist, the length-one quotient is the sign representation of $S_3$, while $(\mathbb{F}_3^3/C)^*$ is its augmentation module. The dual of a two-dimensional module is its determinant-inverse twist; here the determinant is the sign of the $S_3$ factor. Multiplying the scalar $C_2$ coordinate of $R$ by this sign is an automorphism of $R$ and supplies both asserted abstract identifications. Consequently the same three displayed identities hold with $Q_*$ in place of $Q$ and these new onto tails. Both comparators have degree six, physical $h(9) = 8$, and the largest tail slope $(\log_2 3)/2$ is less than one. Apply Theorem 7.3. □

**Theorem 15.9** (All bounded symmetric-three counts)**.** *Every actual action with exact local natural $S_3$ and $s \in \{2, 3, 4, 6, 8, 12, 18\}$ blocks has a complete exponentially small forward row and a quadratically small scalar. Together with Theorem 12.1, this covers every imprimitive exact-local-$S_3$ action.*

*Proof.* The $E = 1$ case was disposed of above. When $D = 1$, use (15.1) with comparator $T$ and the respective capacities

| $s$ | 3 | 4 | 6 | 8 | 12 | 18 |
|---|---|---|---|---|---|---|
| $H_3$ | 1 | 1 | 2 | 1 | 3 | 7. |

These follow from Lemma 11.1; at six use the semiregular three-cycle argument of Proposition 15.3, lifting through a binary kernel by a Sylow three-subgroup. Every margin $(h(3s) - s)/8 - \gamma H_3$ is

positive. At $s = 2$, $D = 1$ contradicts the exact local $S_3$ image; Proposition 15.6 covers that count completely.

At $s = 8$, irrespective of $D$, the prime-to-three bound is $H_3 = 1$; retain $R$ in degree sixteen, with margin $1 - \gamma > 0$. At $s = 18$, a soluble transitive subgroup of $T$ gives $H_3 \le 5$ and $H_2 \le 2$, so the degree-eighteen comparator $T$ has margin $7/2 - 5\gamma > 0$. Otherwise use the actual semiregular $C_3$ witness for every nonsoluble degree-eighteen action recorded in the finite input to Theorem 9.5. Its lift through $D$ has order three. The signed ternary module and its dual untwist to six regular modules over $\mathbb{F}_3[C_3]$, so $H_3 \le 6$. The same comparator has margin $7/2 - 6\gamma > 0$.

Now assume $D \neq 1$ and use the actual isotype count $q$. At $s = 12$, if $q \le 2$, then $d = q \le 2$ and $H_3 \le 3$. For each literal normal quotient of $R$, a fixed map to its top quotient has at most $2^{db/2}$ restrictions into its binary kernel, times a fixed cocycle constant. Thus $Z_J(R) \le C_R 2^{db/2} Z_J(T)$, including all normal graphs. The margin is at least $2 - 3\gamma > 0$. If $q \ge 3$, its isotype size is $4, 3, 2$ or one; Lemma 15.4 gives $H_3 = 1$ and the degree-twenty-four comparator $R$ has margin $3/2 - \gamma > 0$. At $s = 6$, $q \ge 2$ similarly gives isotype sizes $3, 2, 1$ and margin $3/4 - \gamma > 0$ with degree-twelve comparator $R$.

At $s = 4, 6$, the remaining $q = 1$ means that $D$ is central of order two in $R$. Combine Lemma 15.2 with the capacities $H_3 = 1, 2$. The comparator $T \times C_2$ has degrees six and eight; its margins are respectively $3/4 - \gamma > 0$ and $5/4 - 2\gamma > 0$. The $s = 4, q = 2$ branch is Proposition 15.7. For $s = 4, q = 4$ or $s = 3, q = 3$, every coordinate sign space is distinct. Transitivity makes $E$ the whole ambient module, irreducible and faithful for $R$. Since $R$ is soluble, the common-derived estimate Lemma 14.1 gives $Z_J(U) \le Z_J(R) + C_U 2^{\gamma b}$; the comparator degrees $2s < h(3s)$ and tail slopes meet Theorem 7.3. The only remaining case is $s = 3, q = 1$, covered by Proposition 15.8.

The listed cases exhaust all divisors $q \mid s$ and the zero kernels. Each multiplicative bound has strictly positive margin, so Lemma 15.1 applies. There are finitely many actual actions in these widths. Choose one block system and the first applicable certificate for each action, then one any-hot/cold partition on the complete accepted pointing union. The cold coefficients are summed with their original $\Delta(n, n-w)/a(U)$, and the hot sums with their original $((n-w)!a(U)p_n G_{\lfloor n/2 \rfloor})^{-1}$. The finite sum preserves exponential and quadratic decay. There is no additional action weight for isotypes, quotient representations, or alternative block systems. □

## 16 The bounded affine exceptions

We retain the complete quotient killing translations as a comparator. Only the other normal quotients need character covers. This distinction allows the following arguments to include arbitrary nonsplit extensions.

**Lemma 16.1** (Homogeneous sections of a block module)**.** *Suppose $G$ acts linearly on $M = \bigoplus_{i=1}^{s} V_i$, with $\dim_F V_i = d$ and transitive permutation of the $s \ge 2$ summands. If $X^m$ is a homogeneous semisimple section, then either $\dim_F X \ge 2$ and $m \le ds/2$, or $\dim_F X = 1$ and $m \le d\lfloor s/2 \rfloor$. In particular a transitive monomial module on $s$ lines has homogeneous multiplicity at most $\lfloor s/2 \rfloor$.*

*Proof.* The first assertion is dimension. For the second choose a derangement in the transitive block action: the average fixed-point count is one, whereas the identity fixes $s > 1$ points. A lift $g$ has at most $\lfloor s/2 \rfloor$ block cycles. One basis of a starting block generates its whole cycle over $F[x]$, with $x$ acting as $g$ and the return twist retained. Thus $M$ has at most $d\lfloor s/2 \rfloor$ generators over the PID $F[x]$. Every subquotient has at most that many: take its preimage in a free module of that rank, then use the submodule theorem over a PID. On $X^m$, the element $g$ is a single scalar, so

its polynomial-module generator number is $m$. This proof does not require semisimplicity of the ambient module. □

**Lemma 16.2** (One orbit of constituent types)**.** *Let $R$ act faithfully on $M = \bigoplus_i V_i$ and transitively on the summands, with each $V_i$ irreducible under its stabilizer. Let $D \lhd R$ fix every summand, with nontrivial coordinate images $D_i$. Suppose each $D_i$ is normal in the exact local stabilizer. Then $M \downarrow_D$ is semisimple, has no trivial constituent, and its simple types form one $R$-orbit. Every nonzero $R$-section of $M$ is faithful on $D$.*

*Proof.* A simple $D_i$-submodule of $V_i$ and its stabilizer translates span $V_i$ by irreducibility; their sum is semisimple. The isotypic summands form one orbit, since the sum over a proper orbit would be invariant under the stabilizer. The fixed space is invariant under that stabilizer, and is zero by faithfulness and $D_i \neq 1$. Transitivity on blocks joins the local orbits of types into one $R$-orbit. A nonzero $R$-section has nonempty invariant type support, so it contains every type in that orbit. The intersection of their kernels is the kernel of $D$ on the faithful $M$, which is trivial. □

We repeatedly use the following elementary section observation. An elementary abelian $p$-section of an actual $H \leq S_d$ has rank at most $\lfloor d/p \rfloor$: lift it to a subgroup of $H$ and apply the abelian $p$-quotient bound there. This does not assign degree $d$ to an arbitrary quotient of $H$.

**Theorem 16.3** (Binary affine local degrees eight and sixteen)**.** *Let $U$ have exact primitive affine local degree $r = 2^d$ on $s$ blocks, where $(r, s)$ is one of*

$$(8, 2), (8, 4), (8, 6), (16, 2), (16, 4).$$

*Put $E = U \cap (\mathbb{F}_2^d)^s$ and $R = U/E$. For every literal $N \lhd U$ with $E \not\leq N$, the quotient $U/N$ has a faithful tuple of at most $t = s$ irreducible complex characters when $r = 8$, and at most $t = 2s$ when $r = 16$. Consequently all these actions have complete contractive forward rows and negligible scalars.*

*Proof.* Write the exact linear stabilizer as $R_0 \leq \mathrm{GL}_d(2)$, and let $K = R \cap R_0^s$. Its coordinate image $K_i$ is normal in $R_0$. If it were trivial, the onto local map from a block stabilizer would factor through $T_i \leq S_{s-1}$. For $d = 3$, the three possibilities $C_7, C_7 : C_3, \mathrm{GL}_3(2)$ proved in Theorem 12.3 all have order divisible by seven, impossible for $s \leq 6$. For $d = 4, s \leq 4$, it would make $R_0$ a quotient of a subgroup of $S_3$. The groups of order dividing six, namely $1, C_2, C_3, C_6, S_3$, have no irreducible binary module of dimension four. For the last assertion, a normal 2-subgroup acts trivially on a simple module. On a nontrivial simple binary $S_3$-module, restriction to its normal $C_3$ supplies scalar multiplication by $\mathbb{F}_4$, and the inverting involution acts semilinearly. Choose a nonzero vector fixed by that involution, which exists in characteristic two. Its $\mathbb{F}_4$-line is invariant under both generators, so irreducibility forces binary dimension two. Thus $K_i \neq 1$ in all cases.

Every normal 2-subgroup of a faithful irreducible binary linear group is trivial, since its fixed space is nonzero and invariant. Therefore $O_2(K_i) = 1$ and $O_2(K) = 1$. By Lemma 16.2, the physical module is semisimple on $K$, with no fixed constituent, and each nonzero translation section is faithful on $K$. Its simple $K$-constituents have dimension at least two. At degree eight they have dimension three: a nontrivial normal subgroup of $C_7 : C_3$ contains $C_7$, and the other two local possibilities are $C_7$ and the simple $\mathrm{GL}_3(2)$.

Fix a retained quotient $Q = U/N$, write $\bar{E}$ for its nonzero translation image, and let $B$ be the preimage of $K$ in $U$. Faithfulness gives $N \cap B = N \cap E$, so $\bar{B}/\bar{E} = K$. A binary minimal normal subgroup outside $\bar{E}$ meets it trivially; its intersection with $\bar{B}$ maps to a normal 2-subgroup of $K$

and is trivial. Such a subgroup therefore centralizes $\bar{B}$. Every nontrivial-core binary type in the whole socle lies in $\bar{E}$, and its multiplicity is at most $s$ for $d = 3$ and $2s$ for $d = 4$.

Keep the whole sum $A_2^0$ of the core-trivial binary socle types. It meets $\bar{E}$ trivially, because $\bar{E}^{\bar{B}} = 0$; its intersection with $\bar{B}$ again maps to $O_2(K) = 1$. Thus it embeds in a quotient of the actual top $T$, and has total rank at most $\lfloor s/2 \rfloor$. These are different simple types from those in $\bar{E}$, so the bounds take a maximum.

For an odd prime $p$, the whole $p$-socle intersects $\bar{B}$ in a normal $p$-subgroup. That subgroup centralizes $\bar{E}$, and faithfulness of $K$ on $\bar{E}$ forces its image in $K$ to be trivial. Hence the whole odd socle injects a quotient of $T$ and has rank at most $\lfloor s/p \rfloor$. The whole-socle criterion of Theorem 8.1 therefore supplies the stated common tuple length, including any nonabelian socle in the same slots.

Normals containing $E$ contribute exactly $Z_J(R)$. Fixing tuples on all remaining target quotients before inspecting $J$ gives

$$Z_J(U) \le Z_J(R) + C_U 5^{tb/3}, \qquad C_U = \sum_{N \lhd U,\ E \not\le N} |\operatorname{Aut}(U/N)|.$$

The action on the nonzero vectors of each local space gives $R$ a concrete faithful degree $(r-1)s < rs$. The exponents satisfy $s \log 5/3 < s$ and $2s \log 5/3 < 2s$, respectively. Theorem 7.3 now applies to this finite set of physical degrees. If $E = 1$, its retained sum is empty and the same comparison is exact. □

**Theorem 16.4** (All exceptional affine-nine counts)**.** *Every action with exact primitive affine local degree nine and $s \in \{2, 3, 4, 6, 12\}$ has a complete contractive forward row and negligible scalar, on arbitrary complete sources.*

*Proof.* Write $M = (\mathbb{F}_3^2)^s$, $E = U \cap M$, $R = U/E \le R_0 \wr T$, $B_0 = O_2(R_0)$ and $D = R \cap B_0^s$. The complete local matrix classification gives

| $R_0$ | $B_0$ | $R_0/B_0$ |
|---|---|---|
| $C_4, C_8, D_8, Q_8, QD_{16}$ | $R_0$ | 1 |
| $\mathrm{SL}_2(3)$ | $Q_8$ | $C_3$ |
| $\mathrm{GL}_2(3)$ | $Q_8$ | $S_3$. |

This is the seven-local input of the finite affine-nine structural certificate: adjoining all elements to the retained sixteen subgroup classes of the literal $\mathrm{GL}_2(3)$ proves subgroup coverage; irreducibility is checked against its four lines. The displayed normal 2-cores and quotients are the indicated elementary matrix groups, not a classification of any imprimitive degree.

Suppose first $D \ne 1$. Its coordinate images are nontrivial normal subgroups of the exact $R_0$. The physical module is semisimple on $D$, and Lemma 16.2 shows every nonzero translation section is faithful on $D$. In a retained quotient $Q = U/N$, let $\bar{B}$ be the image of the preimage of $D$. Then $\bar{B}/\bar{E} = D$ and $O_2(\bar{B}) = 1$: a normal 2-subgroup centralizes the normal ternary $\bar{E}$ and its image in $D$ would act trivially.

The entire binary socle therefore injects into a quotient of $R/D \le S_3 \wr T$. Removing the normal ternary base embeds it further in a quotient of $C_2 \wr T \le S_{2s}$, so its rank is at most $s$. Ternary simple types nontrivial on $\bar{B}$ lie in $\bar{E}$; Lemma 16.1, with local dimension two, bounds their multiplicities by $s$. The whole $\bar{B}$-trivial ternary type sum meets $\bar{E}$ and then $\bar{B}$ trivially, and embeds in a quotient of $R/D \le S_{3s}$, again giving rank at most $s$. The two sets of types are distinct. For $p \ge 5$, the whole $p$-socle injects a quotient of $T$, since both local kernels have order divisible only by two and

three. The whole-socle criterion gives a faithful tuple of at most $s$ characters for every retained quotient. Thus

$$Z_J(U) \le Z_J(R) + C_U 5^{sb/3}.$$

The actual action on nonzero local vectors represents $R$ faithfully in degree $8s$. For each of the five counts, $8s < h(9s)$ and $s \log 5/3 < h(9s)/8$. The finite comparator theorem applies, including the odd physical width 27 with $h(27) = 26$.

If $D = 1$, the linear block kernel $K = R \cap R_0^s$ is trivial. For binary $R_0$ this is immediate. Otherwise $K$ embeds in $S_3^s$ and every binary subgroup of $K$, and hence every binary subgroup of its quotients, is elementary. A normal subgroup of $\mathrm{SL}_2(3)$ or $\mathrm{GL}_2(3)$ outside $B_0$ contains $Q_8$ and cannot have this property. Every coordinate image of $K$ therefore lies in $B_0$, giving $K = D = 1$. So $R = T$. For $s \le 4$, a point stabilizer of $T$ cannot surject onto any displayed $R_0$, whose orders are $4, 8, 16, 24, 48$; these counts are impossible in this branch.

For $s = 6, 12$, any quotient has binary and other nonternary socles embedding in a quotient of $T$. Filter its whole ternary socle by its intersection with $\bar{E}$. The intersection is an actual section of $M$ with type multiplicity at most $s$, and the outer elementary section of $T$ has rank at most $\lfloor s/3 \rfloor$. The semisimple socle filtration gives a common cover of length $s + \lfloor s/3 \rfloor = 8, 16$. Its exponent is strictly less than $9s/8$, since $5^{64(s+\lfloor s/3 \rfloor)} < 2^{216s-3}$. Corollary 8.4 consumes these entire normal menus. Every argument retained the original source and action weight. □

**Theorem 16.5** (Nonzero ternary core at local degree four)**.** *Let $U$ have exact natural $A_4$ or $S_4$ local action on $s$ blocks, $2 \le s \le 3072$, and put $E = U \cap (\mathbb{F}_2^2)^s$, $R = U/E \le S_3 \wr T$, $D = R \cap C_3^s$. If $D \ne 1$, the complete action is accepted by Theorem 7.3.*

*Proof.* Every coordinate image of $D$ is $C_3$. Its binary simple types in $M = (\mathbb{F}_2^2)^s$ are the inverse pairs of the nonzero coordinate characters. They form one $R$-orbit, allowing semilinear inversion. Each nonzero $R$-section of $M$ is therefore faithful on $D$ and has no fixed vector, as in Lemma 16.2. For a retained quotient, write $\bar{E}, \bar{B}$ for translations and the preimage of $D$. Then $\bar{B}/\bar{E} = D$ and $O_3(\bar{B}) = 1$.

Consider a homogeneous binary $R$-section $X^m$ of $M$ and put $e = \dim_{\mathrm{End}_R(X)} X$. Its binary dimension is even. If it is at least four, then $m \le \lfloor s/2 \rfloor$. In dimension two with endomorphism field $\mathbb{F}_2$, $e = 2$ and $\lceil m/e \rceil \le \lceil s/2 \rceil$. In dimension two with endomorphism field $\mathbb{F}_4$, the image on $X$ is $C_3$. Faithfulness on $D$ makes $D = C_3$ central in $R$. Changing the field convention on each coordinate by Frobenius aligns its generator with the global scalar $\omega I$. Every $R$-map then commutes with this scalar and is $\mathbb{F}_4$-linear; the frame remains monomial. Binary submodules and maps are now genuinely $\mathbb{F}_4$-linear. Lemma 16.1 over $\mathbb{F}_4$ gives $m \le \lfloor s/2 \rfloor$. This use explicitly fixes the field and makes no assertion about arbitrary restriction of scalars.

At $s = 3$ improve the remaining endomorphism-$\mathbb{F}_2$ case. The image on $X$ is $S_3$ and faithful $D$ is its $C_3$. Choose a 3-power lift $z$ of a physical three-cycle. It centralizes $D$. Multiply it by $d \in D$ cancelling its image on $X$; the resulting $y = zd$ still has odd order and induces the physical three-cycle. Its fixed space on $M$ has dimension at most two, since a fixed tuple is determined by one coordinate. Odd-order invariants are exact in characteristic two. All of $X^m$ is fixed by $y$, so $2m \le 2$. Thus the translation-type Schur capacity is

$$k(s) = 1 \quad (s = 2, 3), \qquad k(s) = \lceil s/2 \rceil \quad (s \ge 4).$$

Every binary socle type nontrivial on $\bar{B}$ lies in $\bar{E}$. For the whole trivial-type sum $A_2^0$, put $L = NE/E \lhd R$. Faithfulness gives $L \cap D = 1$, so $[L, D] = 1$. Its image in $R/L$ is injective; the full preimage $H \le R$ centralizes $D$ because $[H, D] \le L \cap D = 1$. It also meets $D$ trivially. The

actual frame gives $C_R(D) \cap S_3^s = D$: every coordinate centralizer of the projected $C_3$ is $C_3$. Hence $H$ embeds in the actual top $T$, and $A_2^0 = H/L$ has rank at most $\lfloor s/2 \rfloor$.

For odd $p$, the whole $p$-socle intersects $\bar{B}$ trivially: for $p = 3$ use $O_3(\bar{B}) = 1$, and otherwise use orders. It injects a quotient of $R/D \leq C_2 \wr T$, then a quotient of $T$, so has rank at most $\lfloor s/p \rfloor$. The whole-socle criterion supplies $k(s)$ characters on every retained quotient, including the nonabelian socle in the same slots.

Thus $Z_J(U) \leq Z_J(R) + C_U 5^{k(s)b/3}$, with all killed translation normals exactly in $Z_J(R)$. The comparator has its concrete faithful degree $3s < 4s$. The retained exponent is less than $4s/8$: for even $s$ use $5^{32} < 2^{75}$, and for odd $s \geq 5$ use $75(s+1) \leq 96s - 3$; counts two and three are immediate. This is a fixed finite band of actual actions, so Theorem 7.3 applies once to their union. □

# 17 The exceptional small pair actions

Put $\beta = \log_2(38/25)$ and $\gamma_5 = \frac{1}{3}\log_2 5$. Throughout this section, a pair action means an actual transitive group $U \leq C_2 \wr S_s$ in its specified action on $2s$ points. Write $E = U \cap C_2^s$, $T = U/E$, and $M_s = \mathbb{F}_2^s$ with its actual $T$ action. Every normal subgroup mentioned below is a normal subgroup of this original $U$, unless another group is specified.

**Lemma 17.1** (Relative socles and fixed prefixes)**.** *Let $F \lhd U$ be a binary group and $S = U/F$. Suppose that, for every $N \lhd U$, the multiplicity $m_X$ of each simple binary type $X$ in $\mathrm{Soc}(U/N) \cap FN/N$ satisfies $m_X \leq qe_X$, where $e_X = \dim_{\mathrm{End}_U X} X$. Suppose every quotient of $S$ has a faithful tuple of at most $k$ irreducible characters. Then every $Q = U/N$ has a faithful tuple consisting of a $k$-character prefix with joint binary kernel, followed by $q$ general characters. Consequently*

$$|\operatorname{Epi}(J, Q)| \leq |\operatorname{Aut} Q|[Q : O_2(Q)]^q 5^{kb/3}(38/25)^{qb} \qquad (J \leq S_b).$$

*If $Q$ is binary, the whole tuple can instead be counted in the known binary-source mode. If a specified top quotient has a faithful tuple of $r$ binary-primary linear characters, one may use those $r$ positions and the same $q$ relative general positions. For $F = E$, the relative hypothesis follows whenever every homogeneous section $X^m$ of $M_s$ satisfies $m \leq qe_X$.*

*Proof.* Put $A = FN/N$. A minimal normal subgroup of $Q$ contained in $A$ is central in $A$ and elementary abelian. Thus $V = \mathrm{Soc}(Q) \cap A$ is a semisimple binary $Q/A$-module. The dual $V^*$ has $q$ module generators: for $X^m$ the required number is $\lceil m/e_X \rceil$, and distinct simple types share positions. Induce each corresponding linear character of $V$ to $Q$ and choose an irreducible constituent. Its restriction to $V$ consists of conjugates of that linear character; hence the intersection of these $q$ kernels with $V$ is trivial. Inflate a faithful tuple of $Q/A$. The joint kernel of the resulting tuple is normal in $Q$, lies in $A$, and meets $\mathrm{Soc}(Q)$ trivially, so it is trivial. This does not extend a character to a nonabelian $A$ or split the original extension. Apply Proposition 8.3 to the fixed prefix. When $F = E$, $V$ is a section of the original $M_s$. All choices depend on $Q$, not on the later complete source $J$. □

**Lemma 17.2** (A point-orbit generator bound)**.** *Let a group $T$ act on a finite set $\Omega$, and let $g \in T$ have at most $c$ cycles on $\Omega$. Over any field $k$, every $T$-invariant subspace of $k^\Omega$ is generated by at most $c$ vectors as a $k[T]$-module. The same bound holds for its quotient modules. There is no restriction on the characteristic or the order of $g$.*

*Proof.* Let $t$ act as $g$. One delta function from each cycle generates $k^\Omega$ over $k[t]$. Pull an invariant subspace back along the resulting surjection $k[t]^c \twoheadrightarrow k^\Omega$. The pullback is free of rank at most $c$, since $k[t]$ is a principal ideal domain. Mapping a basis down gives at most $c$ generators for the subspace, first over $k[t]$ and hence over $k[T]$. Their images generate every quotient. □

The same argument applies to any element $a \in k[T]$: if $k^\Omega$ is generated by $c$ vectors as a $k[t]$-module with $t$ acting as $a$, then every $T$-invariant subquotient needs at most $c$ generators. A finite certificate can give the seed vectors, successive applications of $a$, and coefficients reconstructing every coordinate vector. These identities prove generation directly and require no invariant-subspace list. For the first zero-core sixteen-point action, one such group-algebra operator gives the required four-generator bound.

**Lemma 17.3** (Sharing generators across a full pair kernel)**.** *Suppose a group acts transitively on a set of pairs and contains every individual pair flip. If every invariant subspace of the binary module on the pairs has at most $c$ module generators, the same bound holds for every invariant subspace of the binary module on the points.*

*Proof.* Let $q$ sum the coordinates in each pair. Its kernel $K$ consists of the pair-constant vectors and is isomorphic to the module on the pairs. If an invariant subspace $W$ is not contained in $K$, some $v \in W$ has unequal coordinates in a pair. Subtracting its image under that pair flip gives the constant vector supported on the pair. Transitivity then gives all of $K$, so $K \le W$. When $W \le K$ the asserted bound is immediate. Otherwise lift at most $c$ generators of $q(W)$ to $W$ and let $S$ be their generated submodule. Since $q(W) \neq 0$, also $S \not\le K$; the same argument gives $K \le S$. The equality $q(S) = q(W)$ now implies $S = W$. No splitting of the characteristic-two extension is required. □

The full pair kernel can be weakened to the even pair flips, provided the group contains an element with odd total pair offset. Here an element sends $(i, b)$ to $(\pi(i), b + \epsilon_i)$, and its total offset is $\sum_i \epsilon_i \in \mathbb{F}_2$. To prove the same conclusion, choose $v \in W$ with $q(v) \neq 0$. If $q(v)$ is nonconstant, flipping one pair where its value is one and one where its value is zero gives an individual pair constant. Otherwise $q(v)$ is the all-one vector, and the double flips give every even pair-constant vector. For an element of odd total offset, its displacement of $v$ is pair-constant with coordinates

$$a_i = v(\pi(i), 0) + \epsilon_i - v(i, 0), \qquad \sum_i a_i = 1.$$

This supplies the missing odd direction, hence all of $K$. The lifting argument in Lemma 17.3 again shares the same $c$ generators across the kernel and quotient. This criterion uses only original group elements and avoids listing invariant subspaces.

**Lemma 17.4** (Normal generators for product and affine tops)**.** *Suppose every normal subgroup of finite groups $G$ and $H$ is normally generated by at most $a$ and $b$ elements, respectively. Then every normal subgroup of $G \times H$ is normally generated by at most $a + b$ elements. Suppose also that $V \neq 0$ is a finite elementary abelian group and $R \le \mathrm{GL}(V)$ acts irreducibly. If every normal subgroup of $R$ is normally generated by at most $t \ge 1$ elements, the same bound holds for every normal subgroup of $V \rtimes R$. More generally, if $G \twoheadrightarrow H$ has cyclic kernel and every normal subgroup of $H$ has at most $t$ normal generators, then every normal subgroup of $G$ has at most $t + 1$ normal generators. In each conclusion the bound also holds for the socle of every quotient.*

*Proof.* For $N \lhd G \times H$, use normal generators of $N \cap (G \times 1)$ and lifts in $N$ of normal generators of its image in $H$. Their normal closure has the same intersection and image as $N$, so it is $N$; this does not require $N$ to split as a product.

Every nontrivial normal subgroup $N$ of $V \rtimes R$ contains $V$. Indeed, irreducibility makes $N \cap V$ either 0 or $V$; in the first case $[N, V] = 1$, and faithfulness gives $N \leq C_{V \rtimes R}(V) = V$, hence $N = 1$. If $N = V$, any nonzero vector normally generates it. Otherwise lift normal generators of the nontrivial image of $N$ in $R$. Their normal closure is nontrivial, hence contains $V$, and therefore is $N$. For a cyclic kernel $K$, choose a generator of $N \cap K$ and lifts of normal generators of the image of $N$ in $H$. Their normal closure is contained in $N$, contains $N \cap K$, and has the same image as $N$; it is therefore $N$. Neither a splitting nor a central kernel is needed. Finally, normal generators descend under surjections; apply this to the preimage of each quotient socle. □

When the normal subgroups of a finite group form a chain, each is normally cyclic: a largest normal closure of one of its elements contains every other such closure. Thus the lemma also supplies a two-generator bound for products of two finite groups with normal chains. These structural arguments replace normal-subgroup enumeration whenever the literal top has the displayed affine or product identification.

**Lemma 17.5** (Diagonal binary block pairs)**.** *Let $V \neq 0$ be a finite binary vector space and let $R \leq \mathrm{GL}(V)$ act irreducibly. Let $W = (V \oplus V) \rtimes (R \times C_2)$, where $R$ acts diagonally and $C_2$ interchanges the two summands. If every normal subgroup of $V \rtimes R$ has at most $t$ ambient normal generators, then every normal subgroup of $W$, and every quotient socle of $W$, has at most $t + 1$ ambient normal generators. In particular the bound is two when the normal subgroups of $V \rtimes R$ form a chain.*

*Proof.* Write $T = V \oplus V$ and $D = \{(v, v) : v \in V\}$. The map

$$((v, w), (r, e)) \longmapsto ((v + w, r), e)$$

is a surjection $W \twoheadrightarrow (V \rtimes R) \times C_2$ with kernel $D$. Every nontrivial normal subgroup $N$ of $W$ contains $D$. Indeed, $C_W(T) = T$ by faithfulness, so $N \cap T = 0$ would give $[N, T] = 1$ and hence $N = 1$. Choose a nonzero $(v, w) \in N \cap T$. If $v = w$ it already lies in $D$; otherwise its sum with its swapped vector is the nonzero vector $(v + w, v + w) \in N \cap D$. Irreducibility now gives $D \leq N$.

The product bound in Lemma 17.4 supplies $t + 1$ normal generators for each normal subgroup of $(V \rtimes R) \times C_2$. If $N = D$, one nonzero diagonal vector normally generates $N$. Otherwise lift normal generators of its nontrivial image. Their normal closure is nontrivial, contains $D$, and has the same image as $N$, so it equals $N$. The quotient-socle claim follows by taking preimages and descending the generators. □

**Lemma 17.6** (Coprime cyclic quotients)**.** *Let $E \lhd G$ be a normal $p$-subgroup of a finite group, with $G/E$ cyclic of order prime to $p$. Suppose $t \geq 1$ and every $G$-normal subgroup contained in $E$ has at most $t$ ambient normal generators. Then the socle of every quotient of $G$ has at most $t$ ambient normal generators. If $E$ is elementary abelian, it suffices that its actual conjugation module over $\mathbb{F}_p[G/E]$ has at most $t$ generators.*

*Proof.* The $p$-primary part of $\mathrm{Soc}(G)$ lies in $E$, since its image in $G/E$ is trivial. It therefore has at most $t$ normal generators. The complementary part of the socle intersects $E$ trivially and embeds in the cyclic quotient, so it is cyclic. These two parts commute and have coprime orders. Their generators can be combined in $t$ positions, recovering the two components by taking powers of coprime orders. This proves the bound for $G$.

For a surjection $\pi : G \twoheadrightarrow Q$, retain the actual image $\pi(E)$. It is a normal $p$-subgroup, and $Q/\pi(E)$ is a cyclic quotient of $G/E$. If $H \lhd Q$ lies in $\pi(E)$, then $E \cap \pi^{-1}(H)$ is normal in $G$ and maps onto $H$; its normal generators descend. The same argument thus applies to every $Q$. Finally, in the elementary abelian case Maschke's theorem makes every invariant subspace a direct summand. Projecting $t$ module generators onto that summand gives $t$ ambient normal generators for the corresponding subgroup of $E$. □

**Lemma 17.7** (Coprime quotient transfer)**.** *Let $E \lhd G$ be a normal $p$-subgroup of a finite group, with $|G/E|$ prime to $p$. Suppose every $G$-normal subgroup contained in $E$ has at most $t$ ambient normal generators, and every normal subgroup of $G/E$ has at most $a$ ambient normal generators. Then the socle of every quotient of $G$ has at most $\max(t, a)$ ambient normal generators. In particular, for an elementary abelian kernel, the first hypothesis follows from a $t$-generator actual conjugation module. Neither cyclicity nor solubility of $G/E$ is needed.*

*Proof.* Each minimal normal factor which is not a $p$-group intersects $E$ trivially, hence embeds in $G/E$ and has order prime to $p$. Their join also has order prime to $p$: on adjoining a minimal normal factor to a normal join, the factor is either already contained in the join or has trivial intersection with it, in which case the orders multiply. Thus the entire non-$p$ socle part is disjoint from $E$. It embeds as a normal subgroup of $G/E$, so its ambient normal generators lift to at most $a$ generators in $G$. The $p$-primary socle part lies in $E$ and uses at most $t$ generators. Coprime orders let the two lists share $\max(t, a)$ positions.

For each surjection $\pi : G \twoheadrightarrow Q$, use the actual kernel image $\pi(E)$. The kernel-contained bound descends as in Lemma 17.6. The uniform normal bound on $G/E$ descends to its quotient $Q/\pi(E)$ by taking preimages and mapping normal generators. The argument therefore applies to $Q$. The elementary abelian assertion again follows by Maschke's theorem. □

**Lemma 17.8** (A quaternary triple with a ternary cycle)**.** *Let $G = B \rtimes \langle c \rangle$, where $B = (\mathbb{Z}/4\mathbb{Z})^3$ and $c$ cyclically permutes the three coordinates. Every normal subgroup of $G$ has one ambient normal generator. Consequently every quotient of $G$ has a socle with at most one ambient normal generator.*

*Proof.* Write $B = B_0 \oplus B_1$, where $B_0$ is the diagonal and $B_1$ is the coordinate-sum-zero plane. The operator $3(1 + c + c^2)$ projects onto $B_0$, so it splits every invariant subgroup of $B$. The rotation on $B_1/2B_1$ has irreducible polynomial $X^2 + X + 1$ over $\mathbb{F}_2$. Its only invariant subgroups in $B_1$ are therefore 0, $2B_1$, and $B_1$: a subgroup not contained in $2B_1$ surjects modulo $2B_1$ and equals $B_1$ by Nakayama's lemma. Each of these is generated by one rotation orbit. The diagonal part is cyclic, and the two projection operators combine its generator with the plane generator in one position. This proves the assertion for normal subgroups contained in $B$.

If $N \lhd G$ is not contained in $B$, an element of $N$ acts on $B_1$ as $c$ or $c^2$. Both $1 - c$ and $1 - c^2$ are invertible there, so commutators give $B_1 \leq N$. The group $G/B_1 \cong C_4 \times C_3$ is cyclic. Lift a generator of $N/B_1$ to $g \in N$. Its normal closure again contains $B_1$ by the same commutator argument, and hence equals $N$. Finally, the preimage of any normal subgroup of a quotient is normal in $G$; its normal generator maps to a normal generator of that subgroup. Apply this to the quotient socle. □

**Lemma 17.9** (Paired forms and a block swap)**.** *Let $R \leq \mathrm{GL}(V)$ act irreducibly on a nonzero finite binary vector space, and suppose every normal subgroup of $R$ has at most one ambient normal generator. Let $K \leq R \times R$ be stable under interchanging its factors, with first projection $K \to R$*

*an isomorphism. Let $H = K \rtimes C_2$ act on $V \oplus V$ by the two coordinates and their swap. Suppose some $s \in H$ satisfies*

$$s^4(v, w) + s^2(v, w) + s(v, w) = (0, w) \qquad (v, w \in V).$$

*Then every normal subgroup of $(V \oplus V) \rtimes H$, and every quotient socle of this group, has at most two ambient normal generators.*

*Proof.* The kernel $K \cong R$ and quotient $C_2$ each have uniform normal generator bound one. For a normal subgroup of $H$, first generate its intersection with $K$, then lift generators of its image in $C_2$. This gives the bound two in $H$.

The displayed polynomial preserves every invariant subspace and projects onto the second block. Together with the swap and the surjective local action, it shows that the $H$-module $V \oplus V$ is irreducible. The action is faithful. Consequently every nontrivial normal subgroup of the affine group contains its translation group: its intersection with the translations is nontrivial, since otherwise it centralizes them, and irreducibility gives the whole translation group. Lift generators of the normal image in $H$. When that image is nontrivial their normal closure already contains the translations; when it is trivial, one nonzero translation suffices. Finally descend the uniform bound through each original normal quotient. □

In the twisted two-block models, $K$ is the group preserving the specified nondegenerate pairing between the blocks. The pairing gives injectivity of the first projection; the displayed generator partners give surjectivity. The two distinct cubic Singer actions give the projection polynomial above. Thus the same argument treats the three literal complements together.

**Lemma 17.10** (Matrix units and uniform cyclicity)**.** *Let $k$ be a field and $R$ a $k$-algebra acting on a module identified $k$-linearly with $k^{n\times m}$, where $1 \le n$ and $m \le n$. Suppose that, for every $i, j$, an element of $R$ acts by putting row $j$ in row $i$ and zeroing all other rows. Then every $R$-submodule is cyclic.*

*Proof.* Fix an $R$-submodule $W$. Let $U \le k^m$ be its first-row image. The matrix units show that every row of a member of $W$ lies in $U$, and that every vector of $U$ can be placed alone in any row of a member of $W$. Choose a basis $u_1, \dots, u_d$ of $U$, where $d \le m \le n$, and form $w \in W$ by putting $u_j$ in row $j$ for $1 \le j \le d$. The matrix units recover each $u_j$ in each row from $w$. These vectors span $W$ over $k$, so $Rw = W$. The argument also covers $W = 0$, with the empty basis. No semisimplicity assumption is needed. □

**Lemma 17.11** (Normal covers in a two-block wreath product)**.** *Let $A$ be a nontrivial centerless group, and let $D \le A$ be contained in every nontrivial normal subgroup of $A$. Then $D \times D$ is contained in every nontrivial normal subgroup of $A \wr C_2$. Consequently, if $\pi : A \twoheadrightarrow Q$ has nontrivial kernel $D$ and every normal subgroup of $Q \wr C_2$ has at most $t \ge 1$ ambient normal generators, the same bound holds for every normal subgroup and every quotient socle of $A \wr C_2$. If $A$ is centerless and simple, these bounds are one.*

*Proof.* The centralizer of $A \times A$ in $A \wr C_2$ is trivial: a base element centralizing both factors has both coordinates in $Z(A)$, whereas an element exchanging the factors cannot centralize the first factor. Thus a nontrivial normal subgroup $N$ meets the base nontrivially. After interchanging factors, choose $(a, b) \in N$ with $a \ne 1$. Since $A$ is centerless, some commutator of this element with $(x, 1)$ is nontrivial. Hence $N \cap (A \times 1)$ is a nontrivial normal subgroup of the first factor and contains $D \times 1$. The block swap also gives $1 \times D \le N$.

The induced epimorphism $A \wr C_2 \twoheadrightarrow Q \wr C_2$ has kernel exactly $D \times D$. Lift generators of a nontrivial normal image. Their normal closure contains this kernel by the first assertion, and therefore equals the original normal subgroup. The kernel itself is normally cyclic, since any of its nonidentity elements has a nontrivial normal closure contained in the kernel. This proves the positive bound $t$; taking preimages gives the quotient-socle assertion. For simple $A$, take $D = A$ and the block quotient $C_2$. □

**Lemma 17.12** (Centralizers in an abelian affine model)**.** *Let $V$ be an abelian group and $\varphi : H \to \mathrm{Aut}(V)$. Projection onto $H$ induces an isomorphism*

$$C_{V \rtimes_\varphi H}(V)/V \cong \ker \varphi.$$

*Proof.* An element $(v, h)$ commutes with every translation $(w, 1)$ exactly when $\varphi(h)w = w$ for every $w \in V$. The projection of the centralizer is therefore $\ker \varphi$, and its kernel is $V$. □

For a literal affine identification, this reduces the centralizer-quotient certificate to the action kernel in the complement. The complement action on $V$ may have a nontrivial kernel; the permutation action on the original points is retained separately. In the four-ternary models the complement has order at most 64, even when the full top has order 5184.

**Lemma 17.13** (Two elementary module bounds)**.** *A transitive monomial module of dimension $r > 1$ over a field has no homogeneous section $X^m$ with $m > \lfloor r/2 \rfloor$. For the regular binary $V_4$ module $W = \mathbb{F}_2[x, y]/(x^2, y^2)$, every submodule has head dimension at most two; the unique submodule attaining two is the augmentation ideal $I = (x, y)$, of dimension three. These statements also hold after a field extension.*

*Proof.* A transitive permutation group has a derangement: the average number of fixed points is one, and the identity fixes more than one point. Choose a monomial element whose permutation is a derangement. Its module over the principal ideal domain $k[t]$ is a sum of at most $\lfloor r/2 \rfloor$ cyclic modules, one per cycle. The pullback argument of Lemma 17.2 gives the same generator bound for every subquotient. On the other hand $X^m$, restricted to this element, requires at least $m$ generators, by any nonzero primary component and the invariant-factor theorem. For the second assertion a proper ideal lies in $I$, whose radical is $I^2 = \langle xy \rangle$. A one-dimensional image in $I/I^2$ generates a cyclic ideal, whereas a two-dimensional image forces the entire $I$. The whole ring has cyclic head. This proves the claim and its equality assertion over any field of characteristic two. □

**Lemma 17.14** (An alternating four-point cyclic operator)**.** *Let $k$ have characteristic two. Every invariant subquotient of the natural permutation module $k^4$ for a group containing $A_4$ is cyclic. The conclusion is preserved by any simultaneous change of coordinates intertwining the actual operators used below.*

*Proof.* Index the coordinates by $0, 1, 2, 3$, and put $a = (1\,2\,3)$, $t = (0\,3)(1\,2)$ and $E = 1 + a + a^2$. The group-algebra operator $M = EtE + 1 + a^2$ has matrix

$$M = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}.$$

The vectors

$$e_3, \quad Me_3 = (1, 0, 1, 1), \quad M^2 e_3 = e_2, \quad M^3 e_3 = (1, 1, 1, 0)$$

form a basis. Thus $k^4$ is cyclic as a $k[M]$-module. A submodule of a cyclic module over the principal ideal domain $k[x]$ is cyclic, as is every quotient of that submodule. Each group-invariant subquotient is $M$-invariant, and an $M$-cyclic generator is also a group-cyclic generator because $M \in kA_4$. □

## 17.1 The finite inputs and what they certify

We use the following finite input on *top actions*, separately from the universal lifting arguments that follow. For a simple binary $T$-module $X$, write $\delta_X = \dim_{\mathbb{F}_2} X$. Define

$$\tau(T) = \max_{B \le M_s,\ X} \left\lceil \frac{\dim_{\mathbb{F}_2} \mathrm{Hom}_T(B, X)}{\delta_X} \right\rceil, \qquad \kappa(T) = \max_{L \lhd T} c(T/L),$$

where $c$ denotes the least size of a faithful irreducible character tuple. The formula for $\tau$ is exactly the largest Schur demand of a homogeneous section: the multiplicity of $X$ in the head of $B$ is $\dim \mathrm{Hom}_T(B, X) / \dim \mathrm{End}_T(X)$.

**Lemma 17.15** (Certified top data)**.** *The top-action certificates have the following properties.*

*(i) Among the* 301 *transitive degree-twelve actions,* 283 *admit upper bounds for $(\tau, \kappa)$ belonging to*

$$(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1),$$

*Only upper bounds are needed: a top may use any certified pair in this list, without attaining either bound or agreeing with a previously recorded profile. Both bounds must hold for the same original top action. The remaining eighteen consist of eight binary-core actions with quotient $C_3$ or $S_3$ as in Lemma 17.30; eight actions with an elementary normal ternary group on four triples as in Lemma 17.31; and two paired actions with six-point tops as in Lemma 17.32.*

*(ii) For the two six-point groups*

$$S_{36} = \langle (124), (24)(56), (13)(25)(46) \rangle, \quad S_{72} = \langle (124), (24), (13)(25)(46) \rangle,$$

*all homogeneous sections of $M_6$ have Schur demand at most one. Their only simple types have dimensions one and four, with commuting field $\mathbb{F}_2$. Every nontrivial quotient has a faithful irreducible character except $C_2^2$, which has two faithful binary linear positions.*

*(iii) For all* 24 *nonbinary transitive degree-eight groups* $S$*, every nontrivial homogeneous type in* $M_8$ *has Schur demand at most one, every trivial section has dimension at most two, and* $\kappa(S) \le 2$*. If an invariant submodule has trivial head dimension two, it is a unique five-dimensional submodule* $B$*. Every invariant* $D \ge B$ *satisfies* $\dim D/(B + [D, S]) \le 1$*.*

*(iv) The* 29 *zero-ternary minimal-four degree-sixteen tops, the* 15 *degree-sixteen tops with two primitive affine eight-point blocks, and the* 20 *primitive affine degree-sixteen tops all satisfy* $\tau \le 4$ *and* $\kappa \le 2$*. The two inequivalent nontrivial binary composition factors of the projective eight-point* $\mathrm{PSL}_2(7)$ *module have dimension three, commuting field* $\mathbb{F}_2$*, and each occurs once; the trivial factor occurs twice.*

*Proof.* For the sixty-four sixteen-point module bounds in (iv), use Lemma 17.2 and its group-algebra variant. Sixty-three literal actions have a displayed element with at most four point orbits; the remaining action has the four-seed polynomial certificate described above. These give $\tau \le 4$ for every invariant subquotient directly.

The other finite calculations are recorded by the pair-top certificate contract in Appendix A. We specify their mathematical coverage, since a list of tested module examples would not suffice. Starting with every vector, form its cyclic submodule and close the resulting list under sums. Every invariant subspace is the sum of cyclic submodules of its vectors, so this constructs the complete invariant-subspace lattice. Maximal intervals give the simple modules; intertwining linear equations on the same generators identify their isomorphism types, their endomorphism fields, and the displayed Hom spaces. A composition chain gives the separate composition bounds. Thus every numerator and every homogeneous quotient is covered. One may also certify the nontrivial socle bounds without identifying all simple types. For invariant $A \le C \le B$ with $[B, S] \le C$, the quotient $B/C$ is pointwise fixed. Every map from a nontrivial simple module $X$ into $B/A$ therefore lands in $C/A$, so

$$\mathrm{Hom}_S(X, B/A) = \mathrm{Hom}_S(X, C/A)$$

under the natural inclusion. One can also remove a fixed lower layer: if $[C, S] \le A$, the projection $B/A \to B/C$ has fixed kernel $C/A$. Postcomposition induces an injection

$$\mathrm{Hom}_S(X, B/A) \hookrightarrow \mathrm{Hom}_S(X, B/C).$$

Indeed, a nonzero map from the nontrivial simple module $X$ is injective, so its image cannot lie in a pointwise fixed kernel. Thus removing either kind of fixed layer does not decrease the bound on the nontrivial socle. A remaining quotient with cyclic dual has Schur demand at most one; the same holds for a binary quotient of dimension at most two when $X$ is nontrivial. These steps commute with surjective pullback of the acting group. This certificate allows several trivial generators in the whole dual while retaining the required nontrivial bound. In the finite registry, it suffices to record for every nested pair a reversed cyclic edge, a dimension jump at most two, or an intermediate state removing a nonzero fixed upper or lower layer. Induction on $\dim B - \dim A$ verifies every such descent with one uniform argument. Vector-orbit completeness can be checked without enumerating the acting group. For chosen representatives $v_i$, record for every vector $v$ an index $i$ and a word $g$ in the retained generators such that $gv_i = v$. An invariant submodule containing $v$ also contains $v_i = g^{-1}v$, so the cyclic-span and join tables recover every invariant submodule. For $M_8$ this uses only 256 vector witnesses, including for the natural $A_8$ and $S_8$ actions. For normal subgroups start with kernels of all irreducible characters and close under intersections. Every normal subgroup occurs, since the regular representation of its quotient is faithful and decomposes into irreducibles. Literal kernels, not quotient isomorphism types, are

retained. Searching these kernel intersections proves the asserted covers for all quotients. These quotient checks share one finite-table proof. For each normal kernel $N$, the preimage of the quotient socle is the least normal subgroup $C$ containing $N$ and every normal subgroup minimal strictly above $N$. A tuple $x_1, \dots, x_t \in C$ normally generates $C/N$ in the ambient quotient precisely when every normal subgroup containing $N$ and all the $x_i$ also contains $C$. Completeness of the normal registry makes both assertions finite inclusion tests. The same argument then covers every quotient, with only the inclusion and tuple-membership tables varying between actions. Another shortcut is a uniform bound of $t$ ambient normal generators for every normal subgroup of the top. Apply it to the preimage of a quotient socle and map the tuple into the quotient. Such a uniform bound can be checked by reaching every normal-registry state from the identity in at most $t$ steps, each adjoining the normal closure of one class representative.

For this route, completeness requires only a conjugacy cover inside the original group. Let $g_i$ list every element of $G$, let $r_j$ be selected representatives, and supply original elements $h_i$ and labels $\ell(i)$ with

$$h_i g_i = r_{\ell(i)} h_i.$$

Suppose the normal-state table contains 1 and records every join $N\langle r_j^G \rangle$. It then contains every normal subgroup $K$: start at 1 and adjoin representatives conjugate to successive elements of $K$. Each join stays inside $K$, and after all its elements have been included the resulting state is $K$. Neither distinct conjugacy classes nor exact centralizer orders are needed. The displayed equations can be checked pointwise in a faithful permutation action, while every representative and conjugator remains an element of the original group. A cyclic top needs no normal registry: each of its quotient socles is cyclic. If its point action is regular, the polynomial-module argument above also gives one generator for every invariant submodule over any field. For the first twelve-point top, the two retained generators are the third and fourth powers of the cycle $x \mapsto x + 1 \pmod{12}$; the cycle itself is their quotient. These two identities and its single point orbit therefore prove both one-position bounds directly. There is a useful shortcut when the complete normal registry is a chain. Every quotient again has a chain of normal subgroups, hence at most one minimal nontrivial normal subgroup. If present, that subgroup is the socle and any nonidentity element normally generates it; otherwise the socle is trivial. Thus all quotient socles have a one-position normal-generating tuple, without separate generator searches for the quotients. For a faithful irreducible affine group $V \rtimes R$ with nontrivial simple complement $R$, this chain is $1 < V < V \rtimes R$. Indeed, a normal subgroup meeting $V$ trivially centralizes $V$ and is therefore trivial; any other normal subgroup contains $V$, and projection to $R$ gives the two remaining possibilities. The literal degree-eight affine actions can thus be handled by explicit binary coordinates and certificates for their complements of orders 7 and 168, without enumerating the full affine groups.

The same argument works whenever the normal subgroups of $R$ form a chain: every nontrivial normal subgroup of the faithful irreducible affine group contains $V$, and the subgroups above $V$ correspond to those of $R$. In fact, the chain hypothesis can be removed if every quotient socle of $R$ already has a $t$-position normal-generating tuple. Every quotient socle of $V \rtimes R$ then has a $\max\{1, t\}$-position tuple: the quotient by the trivial kernel has socle $V$, while every nontrivial kernel contains $V$ and hence gives an actual quotient of $R$. For the ternary augmentation module on four labels, the Klein four group splits the module into its three distinct nontrivial character spaces. The alternating group permutes these spaces transitively, so the module is irreducible under $A_4$ and $S_4$. Twisting the $S_4$ action by its sign leaves the $A_4$ restriction unchanged and remains faithful. The normal subgroups of both complements form chains. Consequently, all three affine models have one-position quotient-socle bounds. Literal generator identities identify the original twelve-point actions with the full images of these models, so the bounds apply to their

original normal quotients.

The complement generators in a literal action may carry translation offsets. These offsets must be retained. If $s : R \to \mathbb{F}_3^\times$ is the scalar character and $d : R \to \mathbb{F}_3$ satisfies $d(gh) = d(g) + s(g)d(h)$, the map

$$(v, r) \longmapsto (v + d(r)\mathbf{1}, r)$$

embeds the augmentation affine model into the full four-coordinate model. The cocycle identity verifies multiplication. To prove that the displayed lifts generate the full image, it suffices that their projections generate $R$ and that they contain a nonzero augmentation translation. Indeed, the intersection with the translations is a nonzero $R$-invariant subspace, hence the whole augmentation module. Before applying the embedding, the augmentation component of each lift can therefore be cancelled inside the group. This establishes the required full-image equality while retaining the cocycle offsets of the displayed generators. This also handles the scalar extension with complement $A_4 \times C_2$, whose central involution negates the augmentation coordinates. Every normal subgroup of this complement either lies in the central $C_2$ or contains the Klein four kernel of its cyclic quotient $C_3 \times C_2$. In the latter case, choose a lift of a generator of the normal subgroup's image and, if necessary, multiply it by a kernel element to leave the central axis. Its normal closure then contains the kernel and has the required image, so it is the whole normal subgroup. Thus every normal subgroup is normally cyclic, every quotient socle has a one-position tuple, and the affine transfer applies without asserting a normal chain for $A_4 \times C_2$.

The full ternary permutation module on four labels has the equivariant splitting $\mathbb{F}_3^4 = V \oplus \mathbb{F}_3\mathbf{1}$, where $V$ is its augmentation module; the projection onto the constant coordinate is the coordinate sum, since $4 = 1$ in $\mathbb{F}_3$. Thus, for $R = A_4$ or $S_4$, the full affine model is $(V \rtimes R) \times C_3$. Every normal subgroup missing $V \times 1$ lies in the central $C_3$ axis: its commutators with $V$ vanish, and self-centralization of $V$ followed by $V^R = 0$ removes its first coordinate. Such a subgroup is either trivial or the whole central axis. The original socle $V \times C_3$ has one normal generator, because the nontrivial $R$-module $V$ and the trivial central factor can share a position. A kernel containing $V$ instead leaves a quotient of $R \times C_3$. Hence the all-quotient bound reduces to that smaller complement product. For $A_4 \times C_3$, two positions suffice. For $S_4 \times C_3$, one suffices: reducing successively through the Klein four and rotation kernels leaves the cyclic quotient $C_2 \times C_3$, with the same central-axis argument at each step. This proves bounds two and one for the two full four-coordinate models and all their original normal quotients, without enumerating their normal subgroups.

If the block action also has a common scalar character with values in $\{1, -1\}$, projection from the full four-coordinate module onto its augmentation module is still equivariant. The induced map of semidirect products is onto and has cyclic kernel, the constant translations; that kernel need not be central. Whenever the augmentation semidirect product has a normal chain, the cyclic-kernel argument above therefore gives two normal generators for every normal subgroup of the full model, and hence a two-position bound for every original quotient socle. In particular, this applies to the sign-twisted $S_4$ model already identified above, with its actual scalar action retained.

An independent global sign gives a second common reduction. For $G = \mathbb{F}_3^4 \rtimes (H \times C_2)$, with $H = A_4$ or $S_4$, let $V$ be the augmentation subgroup and $L$ the constant translations. They are distinct minimal normal subgroups. Every nontrivial normal subgroup of $G$ contains one of them: a normal subgroup disjoint from the translation group centralizes it and is trivial by faithfulness, while an invariant subgroup of $V \oplus L$ contains a simple summand. The two summands have different $H$ types, so their normal generators share one position. The two quotient maps are the actual augmentation projection $G \twoheadrightarrow V \rtimes (H \times C_2)$ and the coordinate-sum map $G \twoheadrightarrow H \times D_6$.

Thus every nontrivial quotient kernel reduces to one of these two smaller models, and the trivial kernel has a normally cyclic socle. Their quotient-socle bounds are one when $H = A_4$ and two when $H = S_4$. For the complement products this follows by reducing through the Klein four and rotation kernels to $A_4 \times C_2$ and $C_3 \times D_6$, or to the corresponding symmetric products. The scalar action on $L$ and both literal quotient maps remain part of this argument; no direct-product decomposition of $G$ is assumed.

The same complement arguments cover the actions whose ternary translations are restricted to the constant line. Their literal groups are $C_3 \times A_4$, $A_4 \times D_6$, $C_3 \rtimes_{\mathrm{sgn}} S_4$, $C_3 \times S_4$, and $S_4 \times D_6$; the respective bounds for every quotient socle are $2, 1, 2, 1, 2$. In the sign-twisted case, the cyclic translation kernel and the normal chain of $S_4$ give the two-position bound. Each identification follows from the actual translation and block generators, so it transfers to every normal quotient of the displayed action.

More generally, a cover of the normal registry by $t$ chains gives a $t$-position tuple for every quotient. The quotient correspondence restricts the original cover to the normal subgroups above the quotient kernel. Distinct minimal normal subgroups are incomparable, so each chain contains at most one of them. Choose a nonidentity element in each occupied chain, and the identity in each empty chain. The chosen elements normally generate all minimal normal subgroups and hence the socle. Thus a single finite chain-cover certificate can replace all quotient-specific socle-generator searches. These structural bounds also supply the finite-registry certificates: lift a normally generating tuple for the quotient socle to the original group and take its normal closure together with the quotient kernel. By the subgroup correspondence this is exactly the relative socle carrier $C$. Thus the same registry interface accepts either an explicit finite tuple certificate or a structural proof, without changing any quotient kernel or adding a counting assumption.

There is also a common reduction for coprime affine subgroups. Let $T = V \rtimes H$, where $V$ is a vector space in characteristic $p$ and $p \nmid |H|$. If $K \leq T$ contains the displayed complement $H$, then $K = W \rtimes H$ for the actual invariant subspace $W = K \cap V$. Maschke's theorem gives an $H$-equivariant projection $P : V \to W$ whose restriction to $W$ is the identity. Hence

$$V \rtimes H \longrightarrow K, \qquad (v, h) \longmapsto (P(v), h)$$

is a retraction. Composing with any original quotient map of $K$ shows that every such quotient is a quotient of the full affine model. Thus a bound valid for the socles of all quotients of that model transfers to $K$ without listing $W$ or its normal subgroups. The complement inclusion must be checked in the literal action; an abstract subgroup inclusion alone does not supply this retraction.

Completeness of the degree-twelve and primitive action lists is the published classification input [Hul05, HR20, RD05, Hul23, HRDR23]. The imprimitive degree-eight and two-affine-block constructions use all normal quotients of the local group, all graph isomorphisms between these quotients, and all block-swap cosets in their actual normalizers; Goursat's lemma and the order-two coset condition prove completeness. The four-pair constructions use the full invariant-kernel list and all lifts of fixed top generators. The zero-ternary list similarly uses all degree-four local subgroups, both induced binary linear frames, all invariant kernels, and every solution of the generator-relator cocycle equations. The computed cocycle-space dimension, modulo the coboundary-space dimension, verifies that every complement class occurs. Transporters identify actions in the stated lists. These procedures certify only the finite top data. In particular they make no assertion about unlisted normal subgroups of a subsequent pair lift. □

**Lemma 17.16** (Natural binary modules)**.** *Let $n \geq 4$ be even and let $A_n \leq T \leq S_n$ act on $M = \mathbb{F}_2^n$ by permuting coordinates. Every $T$-invariant subspace of $M$ is generated by the orbit of one vector.*

*Proof.* If a vector $v$ is nonconstant, choose three coordinates on which its values are not all equal. Adding $v$ to its image under the three-cycle on these coordinates gives a vector of weight two. The alternating group is transitive on pairs, so the orbit span of $v$ contains every vector $e_i + e_j$, and hence the even-sum hyperplane $I$.

Now let $W \leq M$ be invariant. If $W$ contains an odd-sum vector $v$, then $v$ is nonconstant because $n$ is even. Its orbit span contains $I$ and $v$, and is therefore all of $M = W$. If every vector in $W$ has even sum but some vector is nonconstant, the same argument gives $W = I$ generated by that vector. Otherwise $W$ consists of constant vectors and has dimension at most one. This proves the assertion, including $W = 0$. In particular the natural $A_{16}$ and $S_{16}$ rows need no invariant-vector census. □

**Lemma 17.17** (Matrix rows and polynomial fibres)**.** *Let $k$ be a field, let $R$ be a $k$-algebra, and identify an $R$-module $M$ with $V^r$ as a $k$-vector space, where $r \geq 1$. Suppose elements of $R$ act as all the matrix units on these rows, and one element acts as the same operator $T$ on every row. If $V$ is generated by at most $r$ elements over $k[X]$, with $X$ acting as $T$, every $R$-submodule of $M$ is cyclic.*

*Proof.* Let $W \leq M$ be an actual $R$-submodule and let $U \leq V$ be its first-row image. The matrix units identify every row image with $U$ and place each element of $U$ in any single row, so $W = U^r$. The common operator makes $U$ a $k[X]$-submodule of $V$. The submodule theorem over the PID $k[X]$ supplies generators $u_1, \ldots, u_r$ of $U$, allowing zero entries. The vector $(u_1, \ldots, u_r)$ lies in $W$. Applying the matrix units recovers each $u_j$ in each row; applying polynomials in the common operator recovers all of $U$ there. This vector therefore generates $W$ as an $R$-module. □

For the regular twelve-point action of $C_3 \rtimes C_4$, with the order-four generator inverting $C_3$, the ternary average separates four block constants from an eight-dimensional augmentation module. The constants form a cyclic polynomial module. On the augmentation, actual group-algebra matrix units give two rows, and the square of the order-four generator gives two polynomial generators of the first-row fibre. Lemma 17.17 makes every augmentation submodule cyclic. The averaging projection then combines the constant and augmentation generators in one position. Literal embeddings transfer this bound to the larger signed dihedral actions; no list of invariant subspaces is required.

## 17.2 A uniform source for six-pair actions

Write $s(b) = |\operatorname{Sub}(S_b)|$, $r_b = \lceil b/100 \rceil$, and $\mathcal{Z}_U(J) = \sum_{N \lhd U} |\operatorname{Epi}(J, U/N)|$. The next estimates treat all original normal denominators at once.

**Lemma 17.18.** *Let $T \leq S_6$ be transitive and imprimitive. Every nonbinary quotient of $T$ has a socle with one ambient normal generator. Moreover, for every transitive $T \leq S_6$, every invariant subspace of its binary natural permutation module has two module generators.*

*Proof.* A minimal block system gives either three pairs or two triples. In the three-pair case the actual block kernel is a submodule of $\mathbb{F}_2^3$, and the transitive three-point top is $C_3$ or $S_3$. Restriction to the actual cubic rotation is semisimple; its three-point permutation module is cyclic, so every kernel submodule is cyclic. For the $C_3$ top, the coprime cyclic quotient argument applies to every original normal denominator. For the $S_3$ top, a nonbinary quotient must retain the whole top: otherwise both its surviving kernel and its top are binary. Its binary socle lies in the cyclic kernel section, since $S_3$ has no nontrivial normal binary subgroup. Its odd socle uses one position, since the whole group has order $2^a 3$. The two prime parts share that position.

In the two-triple case retain the actual ternary kernel inside $\mathbb{F}_3^2$ and its actual binary signed-permutation top. The latter exchanges the two coordinates. Its order is prime to three, and the monomial two-coordinate module, and every invariant section, is cyclic. In a nonbinary original quotient the retained ternary section is nonzero. The kernel of the actual binary top's action on this section injects into the permutation group of the two coordinates: a nontrivial diagonal sign cannot act trivially on such a section, by semisimplicity and coordinate transitivity. Its order is therefore at most two. The binary socle lies in this action kernel, while the ternary socle uses one module generator. Coprime sharing again gives one position for the whole socle.

For the natural binary module, an actual Sylow-three subgroup has at most two point orbits. Indeed each orbit has size divisible by three, by the transitive stabilizer-index formula. Its order is invertible in $\mathbb{F}_2$, so every invariant subspace is a direct summand on restriction. The permutation module on the two or fewer orbits has two generators; project them onto that summand. They also generate under $T$. □

**Proposition 17.19.** *Let $U \le S_{12}$ be transitive with a specified system of six pairs, and suppose its actual six-point top is imprimitive. Then*

$$\sum_{J \le S_b} \mathcal{Z}_U(J) \le 2^{490} 2^{43b/30} s(b) + 2^{256} (b!)^{736} 2^{-51 b r_b/200} s(b + 2r_b).$$

*No quotient map or original normal denominator is omitted.*

*Proof.* Let $K$ be the actual binary pair kernel and $T = U/K$. For each original $N \lhd U$ put $Q = U/N$, $V = KN/N$ and $R = U/KN$. A binary $Q$ has order at most 32. To see this, an actual Sylow-three subgroup lies in $N$, so odd averaging bounds the dimension of $V$ by its at most two point orbits, giving $|V| \le 4$. A binary quotient of a transitive six-point group has order at most eight. The only larger possibility allowed by $|T| \mid 6!$ would have order sixteen and an odd normal kernel; that kernel has two three-point orbits, embedding $T$ in $S_3 \wr S_2$, of order 72, a contradiction. Thus $|Q| \le 4 \cdot 8$. The binary-target character-rank bound gives cold slope $5 \cdot 51/200 < 43/30$ and the usual binary rank tail.

If $Q$ is nonbinary, so is its exact residual top $R$. Lemma 17.18 gives socle capacity one for $R$ and dual-module capacity two for $V$; the latter follows by self-duality and passage to the actual invariant subquotient. For each top epimorphism $\theta : J \twoheadrightarrow R$, with kernel $F$, the full lift fibre has at most

$$|Q|^{\lfloor \log_2 |R| \rfloor} 2^{2d_2(F)} \le 2^{144} 2^{2d_2(F)}$$

elements: $|Q| \le 2^{16}$, $|R| \le 720$ and $\lfloor \log_2 |R| \rfloor \le 9$. On cold sources $d_2(F) \le 51b/200$ for every top map. The ordinary one-character bound contributes $\frac{1}{3} \log_2 5$, and $\frac{1}{3} \log_2 5 + 51/100 < 43/30$. The actual automorphism factor is at most $2^{90}$, giving coefficient $2^{234}$ per original axis.

On hot sources retain all target maps and all coset representatives of $F$ in $J$. Their factorial cost is at most $(b!)^{16+|R|} \le (b!)^{736}$, and the rank-tail moment gives $2^{-51 b r_b/200} s(b + 2r_b)$. Finally $|U| \le 2^{16}$ bounds the number of original normal subgroups by $2^{256}$. Summing all axes gives the stated coefficients. □

The cold coefficient fits the original width-twelve menu, and the hot term is paid by the proved rank-tail deficit. The primitive six-point top uses the earlier primitive source. Together these give the complete six-pair source for every actual representative, without a separate 182-action source table or a normal-subgroup census.

### 17.3 A uniform source for eight-pair actions

**Proposition 17.20.** *Let $U \le S_{16}$ be transitive and nonbinary, with a specified system of eight two-point blocks. Then*

$$\sum_{J \le S_b} \mathcal{Z}_U(J) \le 2^{1216} 2^{31b/16} s(b) + 2^{576} (b!)^{65560} 2^{-51 b r_b/200} s(b + 2r_b).$$

*The estimate retains every epimorphism to every original normal quotient.*

*Proof.* Let $K \lhd U$ be the actual binary block kernel and $T = U/K \le S_8$. This top is transitive and nonbinary. The published degree-eight classification, together with the checked point conjugacies, identifies its actual permutation action with one of the twenty-four retained tops. The checked natural dual-module bound is $H = 1$ for eighteen of them and $H = 2$ for the other six. The latter six have orders $24, 24, 48, 24, 96, 192$, respectively, so their odd part is exactly three. For every original $N \lhd U$, put

$$Q = U/N, \qquad V = KN/N, \qquad R = U/(KN).$$

The exact $R$-module $V$ is a subquotient of the original binary permutation module. Its dual therefore has at most $H$ generators. No condition $N \le K$ is imposed.

Fix an onto top map $\theta : J \twoheadrightarrow R$, with kernel $F$. A lift to $Q$ is determined by its restriction to $F$ and by its values on lifts of a generating set of $R$. Evaluation on the $H$ dual generators bounds the equivariant restrictions by $2^{Hd_2(F)}$. Thus the complete fibre has at most

$$|Q|^{\lfloor \log_2 |R| \rfloor}\, 2^{Hd_2(F)} \le 2^{384} 2^{Hd_2(F)}$$

elements. This uses $|Q| \le 2^{24}$ and $|R| \le 2^{16}$. On cold sources every such $F$ has $d_2(F) \le 51b/200$.

Every quotient of every retained top has socle capacity at most two. Put $\gamma = \frac{1}{3}\log_2 5$ and $\beta = \log_2(38/25)$. For binary $R$, the known-binary-source character certificate costs $2\beta b$. For nonbinary $R$ among the six $H = 2$ tops, its odd part is three, so two faithful positions give one ordinary anchor with binary kernel and one relative position, costing $(\gamma + \beta)b$. The other eighteen tops use two ordinary positions. The three combined cold slopes are therefore

$$2\beta + \frac{51}{100}, \qquad \gamma + \beta + \frac{51}{100}, \qquad 2\gamma + \frac{51}{200}.$$

Each is at most $31/16$: it suffices to use $\gamma \le 7/9$ and $\beta \le 5/8$. The actual automorphism factor is at most $2^{256}$, giving cold coefficient $2^{640}$ on each original axis. There are at most $2^{576}$ original normal axes, since $|U| \le 2^{24}$.

On a hot source retain a top map whose kernel $F$ has $d_2(F) > 51b/200$. All its original coset representatives are counted. They cost at most $(b!)^{|R|}$, while the unrestricted target fibre costs at most $(b!)^{24}$. The binary rank-tail moment bounds the hot kernels by $2^{-51 b r_b/200} s(b + 2r_b)$. Since $|R| \le 2^{16}$, the factorial power is at most 65560. Summing the original axes gives the displayed hot coefficient and completes the estimate. □

The width-sixteen numerical consumer pays the cold coefficient inside the original menu envelope and the hot term through the proved rank-tail deficit, with the original pointing and normalizer divisor. This supplies the complete eight-pair source endpoint uniformly. A separate enumeration of degree-sixteen source certificates is unnecessary for this endpoint.

## 17.4 A source-summed cubic central extension

The following bound treats a central extension before choosing any complete source. It applies to nonsplit extensions and retains every normal subgroup of the original target.

**Proposition 17.21.** *Suppose that a finite group $U$ maps onto $C_3$ with kernel $K$, and there is a homomorphism $a : K \to (\mathbb{F}_4, +)^d$ whose kernel $Z$ is central in $U$ and embeds in $C_2^c$. Assume $c, d \le 3$, and conjugation by a lift of a fixed generator of $C_3$ acts on $a(K)$ as multiplication by a primitive cube root in $\mathbb{F}_4$. With the notation above,*

$$\sum_{J \le S_b} \mathcal{Z}_U(J) \le 2^{512} 2^{381b/192} s(b) + 2^{256} (b!)^{19} 2^{-51 b r_b/200} s(b + 2r_b).$$

*For an original action of width sixteen, the second term, with its original pointing and normalizer divisor, is $O(2^{-b^2/50000})$.*

*Proof.* Fix one original $N \lhd U$ and write $Q = U/N$. If $N \le K$, the quotient retains its $C_3$ top. The image of $a(N)$ is an $\mathbb{F}_4$ subspace, since it is invariant under the actual cubic scalar. Taking its quotient gives an active rank at most $d$. The kernel of this descended active map is exactly the image of the original $Z$, so its binary rank is at most $c$; this statement does not require the extension to split.

If $N \not\le K$, it contains an element whose image generates $C_3$. Killing that element identifies every element of $K$ with an element of the original $Z$: on the active image the difference of the second and first cubic conjugates is the identity operator, because $\omega^2 - \omega = 1$ in $\mathbb{F}_4$. The same commutator difference has trivial image in $Q$. The whole $Q$ is therefore a quotient of $Z$, with binary rank at most $c$.

For a retained top map $\theta : J \twoheadrightarrow C_3$, keep its actual kernel $F$. Invariant central restrictions and scalar-twisted active restrictions share their maximum, rather than their sum, against $|\operatorname{Hom}(F, C_2)|$. This bounds each fixed-top fibre, including nonsplit lift choices, by a fixed target coefficient times $2^{\max(c,d) d_2(F)}$. Count all top maps. On the cold sources $d_2(F) \le 51b/200$ for every such $\theta$, the resulting slope is at most $153/200 + (\log_2 5)/3 < 381/192$. The killed-top quotients have the unconditional slope $3/2$, which satisfies the same bound.

On a hot source, retain every target epimorphism. Its original kernel $F \le S_b$ and three actual coset representatives recover $J$, so the finite-index multiplicity is at most $(b!)^3$. The binary marker moment at $r_b$ bounds the number of hot kernels by $2^{-51 b r_b/200} s(b + 2r_b)$. Since $|U| \le 3 \cdot 2^c 4^d \le 2^{16}$, all epimorphisms to one target cost at most $(b!)^{16}$. There are at most $2^{256}$ original normal subgroups. The fixed cold coefficient per quotient is at most $2^{64}$, so summing the retained and killed alternatives is bounded by the displayed $2^{512}$. This proves the complete raw sum, without division by target automorphisms. Finally the binary rank-tail square completion and the ordinary coarse subgroup bound give the stated decay after the original width-sixteen pointing. □

**Corollary 17.22** (A symmetric top)**.** *Proposition 17.21 also holds with top $S_3$ and the stated scalar action of a selected element of order three, provided $(b!)^{19}$ is replaced by $(b!)^{22}$.*

*Proof.* Every nontrivial normal subgroup of $S_3$ contains its cubic subgroup. Thus an original quotient either retains the actual $S_3$ top, with smaller central and active ranks, or kills a cubic lift. In the latter case the image of $K$ is again the image of the original $Z$, and its index is at most two. The quotient is a binary group of order at most $2^{c+1} \le 16$; it need not be elementary abelian and may contain $C_4$.

For each such binary target $H$, retain all its epimorphisms and apply the cutoff $d_2(J) \le 51b/200$ directly to the same source $J$. The binary target-order bound $|\operatorname{Hom}(J,H)| \le 2^{(\log_2|H|)d_2(J)}$ gives cold slope at most $51/50$. Its complementary source sum costs at most $(b!)^4$ times the same binary marker tail. For retained tops, the original kernel and six coset representatives cost $(b!)^6$, so the full target factor is $(b!)^{22}$. The fixed cold coefficient per normal quotient is at most $2^{128}$, still absorbed by $2^{512}$ after summing all original normals. Apply the split separately for each original normal and commute the finite sums. The original pointing gives the same quadratic decay. □

**Proposition 17.23** (A central cyclic-four kernel)**.** *Let the top in Proposition 17.21 or Corollary 17.22 be $T = C_3$ or $S_3$. Replace the embedding $Z \hookrightarrow C_2^c$ by $Z \hookrightarrow C_4^c$, keeping $c, d \le 3$ and the actual cubic action. The group $K$ may be nonabelian. Put $t_b = \lceil b/200 \rceil$. Then*

$$\begin{aligned}\sum_{J \le S_b} \mathcal{Z}_U(J) \le{}& 2^{512}2^{381b/192}s(b)\\ &+ 2^{256}(b!)^7 2^{-51br_b/200}s(b+2r_b)\\ &+ 2^{256}(b!)^{22}2^{-51bt_b/200}M_4(b,t_b).\end{aligned}$$

*For a width-sixteen action, the two exceptional terms together are $O(2^{-b^2/100000})$ after the original pointing and normalizer divisor.*

*Proof.* For one retained top map $\theta : J \twoheadrightarrow T$, set $F = \ker\theta$ and retain the images of actual coset representatives. The remaining restrictions map $F$ to $K$. Projecting to the active coordinates gives cubic-twisted characters $F \to \mathbb{F}_4$. Inside each literal projection fibre, two restrictions differ by invariant central characters $F \to Z$, since $Z$ is central. This argument works for nonabelian $K$ and does not split the extension.

One invariant $C_4$ character and one twisted $\mathbb{F}_4$ character together inject into $\operatorname{Hom}(F, C_4)$. To see this, embed the first binary coordinate of $\mathbb{F}_4$ in the subgroup of order two of $C_4$ and multiply the two characters. The product of its three successive cubic translates is the cube of the invariant character. Cubing is invertible on $C_4$; after recovering that character, its twisted partner is recovered from two successive binary coordinates. Consequently the restriction count is at most $|\operatorname{Hom}(F,C_4)|^{\max(c,d)}$. The complete fixed-top fibre has the additional factor $|U|^{|T|}$.

Cut off the *literal* count $|\operatorname{Hom}(F,C_4)|$ at $2^{51b/200}$, for every onto top map. The cold slope is again at most $153/200 + (\log_2 5)/3 < 381/192$. Hot kernels are counted by the full moment $M_4(b,t_b)$ and at most six original coset representatives; the target epimorphism count costs $(b!)^{16}$. This gives the last displayed term and keeps all top maps.

Every original normal quotient either retains descended central and active coordinates or kills the selected cubic. In the retained case, the central image still embeds in at most $c$ copies of $C_4$. Indeed, a quotient of an abelian exponent-four group with at most $c$ generators again has that property, so its decomposition has at most $c$ cyclic factors of orders two or four. In the killed case, the image of $K$ is the image of $Z$ and has index at most two; hence the whole target is a binary group of order at most $2^{2c+1} \le 128$. It need not have exponent four. The direct binary-rank cutoff on $J$ gives cold slope at most $357/200$ and the second displayed term. Since $|U| \le 6 \cdot 4^{c+d} \le 2^{16}$, summing every original normal subgroup costs at most $2^{256}$; the fixed cold coefficients are absorbed in $2^{512}$. Finally Theorem 26.1, the binary marker bound, and the two respective square completions give the normalized deficits $b^2/100000$ and $b^2/50000$. □

**Proposition 17.24** (Three full pair kernels)**.** *Let $U$ be a transitive action on three four-point blocks, each equipped with two retained pairs. Suppose its kernel on the three blocks is the full*

*product $D_8^3$ of the pair-preserving local groups. Then*

$$\sum_{J\le S_b} \mathcal{Z}_U(J) \le 2^{289}2^{43b/30}s(b) + 2^{81}(b!)^{15}2^{-51br_b/200}s(b+2r_b).$$

*This hypothesis can equivalently be certified by order* 512 *of the actual block kernel, using its faithful local-coordinate embedding. It does not assert coverage of proper or correlated block kernels.*

*Proof.* Let $K = D_8^3$ and $Z = Z(K) = C_2^3$. For every original normal subgroup $N$ of $U$, either $N \le Z$ or $Z \le N$. Indeed, if $N \cap K$ contains a noncentral local coordinate, its commutator with that local $D_8$ contains the corresponding central involution. Transitivity on the three blocks then gives $Z \le N$. Otherwise $N \cap K \le Z$. Commuting $N$ with $K$ shows that its induced top centralizes the faithful action on $K/Z$, hence is trivial; thus $N \le K$, and then $N \le Z$.

If $N \le Z$ but $Z \not\le N$, the socle of $U/N$ lies in the image of $Z$. A minimal normal subgroup centralizes the normal class-two subgroup $K/N$, and the actual local-coordinate centralizer calculation leaves precisely that central image. The module $Z$ is the three-coordinate binary permutation module; a cubic cycle makes every invariant subquotient cyclic by the polynomial-module argument. Thus the socle of $U/N$ has one ambient normal generator. The bound $|U| \le 3072 \le 2^{12}$ and the one-position character count give $2^{144}2^{43b/30}$ per quotient. There are at most $2^{144}$ original normal subgroups, giving the first contribution $2^{288}2^{43b/30}$.

For $Z \le N$, the literal quotient correspondence injects these axes into the normal axes of $U/Z$, with identical epimorphism counts; there is no further normal-menu factor. Here $U/Z$ embeds in $(C_2^2)^3 \rtimes S_3$. Its two constant coordinates are central, and each augmentation coordinate is the natural binary plane for the selected cubic. The rank-two versions of the complete cubic and symmetric source bounds therefore give cold coefficient $2^{256}$, slope $43/30$, and the displayed hot term. Their normal-menu factor is $2^{81}$ because $|U/Z| \le 384 \le 2^9$. Adding the proper-central contribution uses $2^{288} + 2^{256} \le 2^{289}$. Every normal axis and every source epimorphism has thus been retained. □

**Proposition 17.25** (A small three-block kernel)**.** *In the same physical three-block setting, suppose instead that the actual block kernel has order at most eight. Then*

$$\sum_{J\le S_b} \mathcal{Z}_U(J) \le 2^{100}2^{43b/30}s(b) + 2^{36}(b!)^{12}2^{-51br_b/200}s(b+2r_b).$$

*The kernel need not be elementary abelian, and the extension need not split.*

*Proof.* The actual top $T$ is $C_3$ or $S_3$, so $|U| \le 48 \le 2^6$ and there are at most $2^{36}$ original normal subgroups. If $N \le K$, the quotient has the same top and its kernel is a quotient of the original $K$, of binary order at most eight. For each onto top map $\theta : J \twoheadrightarrow T$, retain $F = \ker\theta$ and all coset representatives. Maps in that fibre are determined by their restrictions $F \to K/(K \cap N)$ and these representatives. Their number is at most $|U/N|^{|T|}2^{3d_2(F)}$. The number of onto top maps is at most a fixed constant times $3^{b/3}$. For $C_3$ this follows from the abelianization bound. For $S_3$, the derived cyclic dual restricts each onto map to a separating character on $J'$; the same abelianization bound applies to its faithful degree-$b$ action, with the fixed automorphism factor of $S_3$. The cold rank cutoff therefore gives slope at most $153/200 + (\log_2 3)/3 < 43/30$. The hot sum costs at most $(b!)^{6+|T|} \le (b!)^{12}$ times the binary kernel-marker tail.

If $N \not\le K$, its nontrivial normal top image contains the cubic subgroup. The residual top has order at most two. The same original quotient $U/N$ is therefore binary of order at most sixteen. Its direct binary-rank cutoff has cold slope at most $204/200$ and hot factorial power at most four.

Uniformly, the cold coefficient per quotient is at most $2^{64}$. Summing over all original normals gives the displayed coefficients $2^{100}$ and $2^{36}$. □

**Proposition 17.26** (Monolithic quotient reduction)**.** *Let $Z \lhd G$ be contained in every nontrivial normal subgroup of the finite group $G$. For every finite source $J$,*

$$\mathcal{Z}_G(J) \le |\operatorname{Epi}(J,G)| + \mathcal{Z}_{G/Z}(J).$$

*If $M \lhd G/Z$ is likewise contained in every nontrivial normal subgroup of $G/Z$, this gives*

$$\mathcal{Z}_G(J) \le |\operatorname{Epi}(J,G)| + |\operatorname{Epi}(J,G/Z)| + \mathcal{Z}_{(G/Z)/M}(J).$$

*In particular, no multiplicative normal-menu factor is needed at either step.*

*Proof.* Separate the original zero normal subgroup, whose contribution is $|\operatorname{Epi}(J,G)|$. Every remaining original normal contains $Z$. The quotient correspondence injects those axes into the normal subgroups of $G/Z$, preserving each actual quotient and its raw epimorphism count. Their sum is therefore at most $\mathcal{Z}_{G/Z}(J)$. Apply the same argument to $G/Z$ for the second inequality. This reduction requires the stated normal-subgroup containments; it does not infer them from a numerical quotient profile. □

**Corollary 17.27** (The literal correlated cubic carrier)**.** *For the literal degree-sixteen action $G = 16T1542$, every original normal axis is included in the bound*

$$\sum_{J \le S_b} \mathcal{Z}_G(J) \le 2^{322} 2^{381b/192} s(b) + 2^{256} (b!)^{19} 2^{-51 b r_b/200} s(b + 2r_b).$$

*No normal-subgroup census is required for this estimate.*

*Proof.* The original word certificates give a binary kernel $K \lhd G$ of index three, a central subgroup $Z$ of order two, and an original normal subgroup $N$ of order eight containing $Z$. They prove

$$|G| = 3072, \qquad C_G(K) = Z, \qquad Z(K/Z) = N/Z \cong C_2^2,$$

and the actual cubic acts transitively on the three nonidentity elements of $N/Z$. The centralizer certificate tests the sixteen possible images of one point, using actual transitive words in $K$. For the second center, the checked quotient graph and the original kernel of order 64 cover $K$ by a fixed coset grid; each element outside $N$ has an explicit commutation witness modulo $Z$. These certificates establish containments for every actual group element, rather than presupposing a complete list of normal subgroups.

Every nontrivial normal subgroup of $G$ contains $Z$. Indeed, a nontrivial intersection with the binary group $K$ meets its center, while a trivial intersection would force that normal subgroup into $C_G(K) = Z$. In $G/Z$, the plane $N/Z$ is minimal normal. The center of its binary kernel lies in this plane, and an element outside that kernel cannot centralize it: its nontrivial cubic image would force the whole group to centralize the plane. The same binary-center intersection argument therefore puts $N/Z$ inside every nontrivial normal subgroup of $G/Z$. Both groups have socle capacity one.

Proposition 17.26 now reduces the complete weight to the two faithful targets and the same original quotient $G/N$. The latter has actual central binary rank three and two synchronized $\mathbb{F}_4$ coordinates over $C_3$. Retaining the constant before the final enlargement in the cubic complete-source bound gives cold coefficient $2^{321}$ and hot coefficient $2^{256}$, with factorial power 19. Each faithful target has order at most $2^{12}$; the one-position character estimate bounds their combined contribution by $2^{145} 2^{43b/30} s(b)$. Since $43/30 < 381/192$ and $2^{145} + 2^{321} \le 2^{322}$, the asserted complete bound follows. □

## 17.5 The twelve-pair actions

**Lemma 17.28** (A binary-primary prefix)**.** *Let $Q$ be finite and put $V = \Omega_1(Z(Q))_2$. If $\dim V \le r+1$ and $\dim(V \cap Q') \le r$, there is a linear character $\chi$ of binary-power order with $\dim(V \cap \ker \chi) \le r$. If $\psi$ is a linear character of odd order with binary kernel, then $\chi\psi$ has binary kernel and the same residual central binary subspace as $\chi$.*

*Proof.* If $\dim V \le r$, take the trivial character. Otherwise choose $z \in V \setminus Q'$. Characters of the finite abelianization separate points, so some linear character detects $z$. Raising that character to the odd part of $|Q|$ gives binary-power order and still detects $z$. Its kernel on $V$ therefore has codimension one. This uses full-order characters: a sign character need not detect the square of an element of order four.

Binary-power and odd-order roots of unity cannot cancel, so $\ker(\chi\psi) = \ker\chi \cap \ker\psi$. Moreover $\psi$ is trivial on every involution. This proves both assertions about the twist. For either prefix, the ambient-fixed part of the kernel's central binary socle injects into the corresponding residual subspace of $V$. Thus the same bound applies to its trivial simple type; the nontrivial simple types must still be bounded on that actual kernel. □

**Lemma 17.29** (An anchor when the odd part is three)**.** *Let $|Q| = 2^a 3$. Every jointly faithful tuple of irreducible characters of $Q$ contains a character with binary kernel. Thus a faithful tuple of $r+1$ characters supplies one ordinary prefix and $r$ relative positions on its binary kernel, using the same whole-group characters.*

*Proof.* A nonbinary normal subgroup contains a Sylow three subgroup of $Q$, and normality makes it contain every conjugate. If every character kernel were nonbinary, their intersection would therefore contain a fixed nontrivial Sylow three subgroup, contradicting faithfulness. Choose a binary-kernel member as the prefix and retain the other $r$ characters as the tail. For each tail character, Maschke supplies an irreducible constituent on the actual prefix kernel; the normal restriction argument in Lemma 17.1 applies. □

**Lemma 17.30** (The binary-core twelve-point tops)**.** *Let $T$ be one of the eight binary-core actions in Lemma 17.15(i). Put $P = O_2(T)$. For every pair lift $U$ and every $N \lhd U$, put $Q = U/N$. Every nonbinary $Q$ has a socle normally generated by at most four elements and admits four general faithful positions, one of which has binary kernel. Thus it has one ordinary anchor and three relative general positions. Every binary $Q$ admits one binary-primary linear and four relative general positions.*

*Proof.* In all eight actions the original group $U \le S_{24}$ has order $2^a 3$, is solvable, and has only semiregular nonidentity elements of order three. The literal geometry supplies these facts directly. The unique $C_3$ case has

$$T = P \rtimes \langle h \rangle, \qquad P = \{(a, b, a+b) : a, b \in V_4\} \le V_4^3,$$

where $P$ acts by translations on three four-point blocks and $h$ cycles the blocks. In particular $|T| = 48$. For the $S_3$ cases, the literal product chart or an equivariant tower $\Omega_{12} \twoheadrightarrow \Omega_6 \twoheadrightarrow \Omega_3$ with two-point fibres gives a full $S_3$ image with binary kernel. A nonidentity element of order three survives on the three blocks and fixes no point. The pair lift has binary kernel as well, so this semiregularity holds on the original twenty-four points. Generator identities and short covering words certify the displayed actions; no normal-subgroup list of the pair lift is used.

First let $Q$ be nonbinary, and write $\pi : U \twoheadrightarrow Q$ for the original quotient map. Then $N$ is binary and $|Q| = 2^b 3$. Fix $h \in U$ of order three. For any central elementary binary subgroup $A \leq Q$, coprime centralizer lifting maps $C_U(h) \cap \pi^{-1}(A)$ onto $A$. The kernel of the action on the eight $h$-orbits lies in the odd group $C_3^8$, so it is killed by the map to $A$. Thus $A$ is a binary quotient of a subgroup of $S_8$, and has rank at most four by the permutation prime-rank bound. Every elementary binary section of $Q$ has rank at most twelve, using the original faithful action of $U$ on twenty-four points. Put $V = \mathrm{Soc}(Q)_2$. The fixed type in $V$ therefore has multiplicity at most four. A nonfixed simple type $X$, with binary dimension $\delta_X$ and Schur dimension $e_X$, satisfies $\delta_X m_X \leq 12$. If its Schur product degree $\delta_X e_X$ is at least three, then $m_X \leq 4e_X$. Only product degree two needs further attention; then $\delta_X = 2$, $\mathrm{End}_Q(X) = \mathbb{F}_4$, and $e_X = 1$.

When $T/P = S_3$, this last type cannot occur. Indeed, $U$, and hence $Q$, has no quotient of order three: such a map kills the binary kernel of $U \twoheadrightarrow S_3$ and would descend to $S_3$. A simple binary type of product degree two gives precisely such a scalar quotient of order three.

For the $C_3$ action, we prove the needed bound on the actual module $M_{12}$. Let $B \leq M_{12}$ be $T$-invariant and put $d = \dim B/[B,P]$. Each four-point coordinate is a regular $V_4$ module. By Lemma 17.13, every local submodule $S$ has head dimension at most two, with equality only for its three-dimensional augmentation ideal. In particular

$$3 \dim S/[S,P] \leq \dim S + 3.$$

Applying the head inequality along the three-coordinate filtration gives $d \leq 6$ and $3d \leq \dim B + 9$: head dimension is subadditive in exact sequences, while ordinary dimension is additive. If $d = 6$, the kernel of each coordinate projection has head dimension at most four, so every coordinate image is its augmentation ideal $I_i$. Hence $B \leq I_1 \oplus I_2 \oplus I_3$ and $\dim B \leq 9$, while the weighted inequality gives $\dim B \geq 9$. Thus the original numerator is exactly this augmentation sum. The $T$-orbits of two augmentation generators supported on one block span this sum.

Now let $X$ be the actual simple target of product degree two. The normal binary subgroup $P$ fixes $X$, so every $T$-map $B \to X$ factors through the original coinvariant module $B/[B,P]$. This is a semisimple $C_3$-module. Its Hom space to $X$ has even binary dimension, by postcomposition with $\mathrm{End}_T(X) = \mathbb{F}_4$, and has dimension at most $d$. Thus $d \leq 5$ gives $\dim \mathrm{Hom}_T(B,X) \leq 4$. When $d = 6$, the two generators just constructed give the same bound $2 \dim X = 4$. For any homogeneous section $X^m$, pull its numerator back to the original $M_{12}$. Precomposition embeds $\mathrm{Hom}_T(X^m, X)$ into the Hom space of that numerator, so $2m \leq 4$. This proves $m \leq 2$ for every such section, with its original action.

Filter $V$ by $V \cap EN/N$ and its image in $Q/(EN/N)$. The first term embeds in the actual pair section $E/(E \cap N)$ of $M_{12}$. Pulling the same simple $Q$-type back to $U$ and descending along $U \twoheadrightarrow T$ gives multiplicity at most two there. The second term is elementary binary in a quotient of the literal top of order 48, so has dimension at most four and contains at most two copies of that same type. Multiplicities add along this filtration, giving $m_X \leq 4 = 4e_X$. No quotient is replaced by an abstract isomorphic group.

In both top cases every binary socle type now has demand at most four. The odd socle has order at most three and shares the binary generator positions; solvability excludes nonabelian minimal normal subgroups. Thus the whole socle needs at most four ambient normal generators. The whole-socle case of Theorem 8.1 gives four faithful general positions, and Lemma 17.29 selects a binary-kernel member of this same tuple. This is profile $(0,0,1,3)$ for both $C_3$ and $S_3$ tops.

Finally let $Q$ be binary, for either top case. A Sylow three subgroup $\langle h \rangle \leq N$ has eight three-point orbits. Frattini gives $U = NN_U(\langle h \rangle)$. In the ambient normalizer $C_3^8 : (C_2 \times S_8)$, the common inversion factor $C_2$ must be retained. The odd base of the actual normalizer is killed by

$N$; its actual joint image $H \le C_2 \times S_8$ surjects onto $Q$. Retaining this joint image preserves the correlations between inversion and orbit permutation. Its faithful ten-point action bounds every elementary binary section by five. Since the inversion factor is abelian, $H'$ acts faithfully on the eight orbit points, so every elementary binary section of $H'$, and hence of $Q'$, has rank at most four. Thus the central binary rank of $Q$ is at most five and its derived central rank at most four. Lemma 17.28 gives a binary-primary linear character $\chi$ with $\dim(\Omega_1(Z(Q)) \cap \ker \chi) \le 4$. Since all simple binary $Q$-types are trivial, the $Q$-socle of $\Omega_1(Z(\ker \chi))$ is precisely this ambient-fixed subspace. The relative-socle construction supplies four general positions on the actual kernel. This is profile $(1, 0, 0, 4)$ in the known binary-source mode. □

**Lemma 17.31** (Four ternary orbits)**.** *Suppose $T$ has an elementary normal ternary subgroup $C$ of rank at least two with four orbits of size three, $T/C$ binary, and $|O_2(T)| \le 2$. Every pair lift $U$ has a quotient $R$ with a specified faithful twenty-point action and a constant $D_U$ such that*

$$Z_J(U) \le Z_J(R) + D_U 5^b \qquad (J \le S_b).$$

*Proof.* Averaging over $C$ gives the $T$-equivariant decomposition $M_{12} = M_0 \oplus W$, with $\dim M_0 = 4$, $\dim W = 8$. The map $(v, t) \mapsto (\mathrm{avg}_C(v), t)$ on the ambient semidirect product has kernel $W$. Restricted to $U$, its kernel is $E_1 = E \cap W = [E, C]$; put $R = U/E_1$. The ambient $M_0 : T$ acts faithfully on the disjoint union of the twelve pair labels and the eight $C$-orbit labels on the original twenty-four points. The first action detects $T$ and the second detects $M_0$, proving faithfulness of this twenty-point representation after restriction to $R$.

Fix an original $N$ not containing $E_1$, and choose a minimal normal $S$ in the nonzero image of $E_1$ in $Q = U/N$. On restriction to $C$, $W$ consists of four nontrivial two-dimensional local types. Their kernels have trivial intersection, and they form one $T$-orbit. There are at least two distinct types, since $C$ has rank at least two; each has multiplicity at most two. Every nonzero $T$-section of $W$ therefore acts faithfully on $C$. In particular if $L$ is the kernel of the action of $T$ on $S$, then $L \cap C = 1$, so $L$ is normal binary and $L \le O_2(T)$.

Put $A = EN/N$. Since $A$ is abelian, there is an exact sequence

$$1 \longrightarrow A \longrightarrow C_Q(S) \longrightarrow L/\pi(N) \longrightarrow 1.$$

Every minimal normal subgroup of $Q$ commutes with $S$, including $S$ itself. Thus the entire socle lies in this binary centralizer. Its image modulo $A$ has dimension at most one and is trivial. The trivial part in $A$ is a section of $M_0$, the ordinary permutation module on four blocks, so has rank at most two by Lemma 17.13. Every nontrivial type in $A$ lies in $W$ and has multiplicity at most two by restriction to $C$. The whole socle therefore has Schur demand at most three, and Theorem 8.1 supplies three general positions. Its epimorphisms cost at most $|\operatorname{Aut} Q| 5^b$. Finally the interval $N \ge E_1$ is exactly the normal-subgroup lattice of $R$, with the same quotients and epimorphism sets. Summing that interval gives exactly $Z_J(R)$; summing the other normals gives the asserted constant $D_U = \sum_{N \not\supseteq E_1} |\operatorname{Aut}(U/N)|$. □

**Lemma 17.32** (The paired six-point tops)**.** *Suppose $U \to T \to S$ has degrees $24, 12, 6$, with both successive systems consisting of pairs, and $S = S_{36}$ or $S_{72}$ from Lemma 17.15. Every original quotient admits either one general anchor and three relative general positions, or two binary-primary linears and three relative general positions in the known binary-source mode. A binary quotient with no exceptional $C_2^2$ top needs at most four general positions.*

*Proof.* In both literal six-point tops, the first cubic generator followed by the displayed block swap is a six-cycle. The polynomial-module argument above makes every invariant subquotient

of $M_6$, and its dual, cyclic. Thus each such section has Schur demand at most one, without an invariant-subspace enumeration. Let $F$ be the kernel of $U \to S$. On the six original four-point blocks, $F \le D_8^6$, and $F/E$ is a submodule of $M_6$. Let $C_0 \le M_{12}$ be the vectors constant on the six pairs of pair labels. Both $C_0$ and $M_{12}/C_0$ are $M_6$. Fix $N$, put $A = FN/N$, and take a homogeneous component $X^m$ of $\mathrm{Soc}(U/N) \cap A$. Its preimage $D \le F$ has the three successive factors obtained from

$$D \cap E \cap C_0, \qquad (D \cap E + C_0)/C_0, \qquad \pi(D) \le F/E.$$

The denominators are the respective images of $D \cap N$; the last is $\pi(F \cap N)$, since $D$ contains $F \cap N$. Thus all three factors are homogeneous sections of the same $M_6$ for the same original $S$ type. Their multiplicities add, and the preceding cyclic bound gives $m \le 3e_X$. Apply Lemma 17.1. Except for top $C_2^2$, a nontrivial top quotient has a one-character prefix with binary joint kernel. For top $C_2^2$, $Q$ itself is binary and the prefix consists of two full-order binary linear characters. A trivial top needs only the three relative positions. No part of this proof assumes that $F$ or $A$ is abelian. □

**Theorem 17.33** (Complete twelve-pair estimate)**.** *The family of all actual pair actions with twelve pairs and nonbinary top has a complete exponentially small forward row and a quadratic-exponentially small scalar, with its original action weights.*

*Proof.* For the 283 generic tops, Lemma 17.1 gives slope $\kappa\gamma_5 + \tau\beta$ and target-dependent finite constants. For each of the six pairs in Lemma 17.15(i),

$$5^{64\kappa}38^{192\tau} < 2^{573}25^{192\tau},$$

so the slope is less than $3 - 1/64$. The binary-core slopes are $\frac{1}{2} + 4\beta$ and $\gamma_5 + 3\beta$, both less than $3 - 1/64$. The paired-six slopes are $\gamma_5 + 3\beta$, $1 + 3\beta$, and $4\beta$, with the same strict margin. These inequalities follow, without rounding logarithms, by raising to the 192nd power; the corresponding numerators are $2^{96}38^{768}$, $5^{64}38^{576}$, $2^{192}38^{576}$, and $38^{768}$, against $2^{573}$ times the matching power of 25. Proposition 8.3 and Corollary 8.4 sum every literal normal in these finite original width-twenty-four actions. Constants such as $[Q : O_2(Q)]^q |\operatorname{Aut} Q|$ are fixed before choosing $J$. For the other eight tops, Lemma 17.31 and Theorem 7.3 apply with $v = 20 < 24$ and $\eta = \log_2 5 < 3$. This last assertion includes the entire normal interval of the quotient, rather than only its faithful target. There are finitely many subgroups of $S_{24}$, so the complete union is covered by one fusion, retaining the factor $1/(b!a(U)p_nG_{\lfloor n/2 \rfloor})$ in each original hot sum. □

## 17.6 The sixteen-pair actions

**Lemma 17.34** (Paired eight-point tops)**.** *Suppose $U \to T \to S$ has degrees $32, 16, 8$, with both successive systems consisting of pairs and $S$ nonbinary. For every original normal subgroup, the relative socle in the direct quartet kernel has Schur demand at most four. Every quotient of $U$ therefore admits a two-character general prefix with binary joint kernel and four relative general positions.*

*Proof.* Write $F = \ker(U \to S) \le D_8^8$. The three layers used in Lemma 17.32 are now sections of $M_8$. A nontrivial simple type has demand at most three by Lemma 17.15(iii). We must improve the possible sum of three trivial heads from six to four using the original commutator denominator.

For an invariant binary module $B$ put $d_S(B) = \dim B/[B,S]$. For invariant $K, W, L \le M_8$ such that their coordinatewise product span $KW$ lies in $L$, we claim

$$d_S(K) + d_S(W) + \dim L/([L,S] + KW) \le 4. \tag{17.1}$$

A Sylow binary subgroup of the actual transitive $S \le S_8$ is transitive. Hence every nonzero invariant submodule contains the constant vector $\mathbf{1}$: a binary group has a nonzero fixed vector on every nonzero binary module, and $M_8^P$ consists of constants. Dually, every proper invariant submodule lies in the augmentation hyperplane $I$. If $K$ or $W$ is zero, the two remaining heads are at most two each. If both heads are at most one, the result is again immediate. If $d_S(K) = 2$, $d_S(W) = 1$, then $KW \supseteq K$ and the last term is at most one by the five-dimensional equality assertion in Lemma 17.15. If both heads are two, $K = W$ is that five-dimensional submodule. It cannot be self-orthogonal in $\mathbb{F}_2^8$, so some product has odd coordinate sum. Thus $KW$ is an invariant submodule not in $I$, whence $KW = M_8$ and the last term vanishes. This proves (17.1).

Fix $N$ and put $A = FN/N$. Lift the trivial part of $\mathrm{Soc}(U/N) \cap A$ to $D \le F$. Set $B = D \cap E$ and let $C \le M_{16}$ be constant on the eight pairs of pair labels. The three binary coordinate modules are

$$K = \pi(D) \le F/E \le M_8, \quad W = (B + C)/C \le M_8, \quad L = B \cap C \le M_8.$$

Inside a single quartet the commutator of an internal pair flip and a pair interchange is their coordinate product. Therefore the original commutator relation is $[B, D] = KW \le L \cap N$. Also $[L, S] \le L \cap N$, since the lifted section is central in $U/N$. The exact three-step socle filtration consequently has total rank at most the left side of (17.1), hence at most four. This uses the literal intersection with $N$ in the bottom layer; replacing it by an unrelated denominator would lose the improvement. The top cover is two by Lemma 17.15(iii), so Lemma 17.1 completes the proof, even when $A$ is nonabelian. □

**Lemma 17.35** (Quotients of an actual degree-eight group)**.** *Let $G \le S_8$ be transitive. Every quotient which is nonbinary or nonabelian has faithful character cover at most two. An abelian binary quotient has at most four faithful binary linear positions.*

*Proof.* We use the following precise consequence of the excessive binary normal-pair classification [RDT25, Definition 5.1 and Proposition 5.3(i)]: if $P$ is transitive binary of degree eight and $S \lhd P$, then

$$d_P(S) > 2 \implies [P : S] \le 2, \qquad d(P) \le 4.$$

Here $d_P(S) = \dim S/(S^2[S,P])$ by [RDT25, Lemma 2.5(i)]. Every Sylow binary subgroup of an actual transitive group of binary degree is transitive, by the orbit–stabilizer inequality.

We first show that $\dim A/(A^2[A,G]) \le 2$ whenever $G/A$ is nonbinary. Otherwise, with $S = A \cap P$, the displayed result gives $[G : A]_2 \le 2$. The Frattini normalizer $L = N_G(S)$ satisfies $G = AL$ and is transitive on the original eight points: $S$ is transitive on each original $A$-orbit, and $L$ is transitive on their labels. The group $L/S$ is nonbinary with binary part of order at most two. It has a nontrivial normal odd subgroup. Indeed this is immediate for odd order; for order twice an odd number use the kernel of the sign of its regular permutation action. This subgroup survives in $L/O_2(L)$.

Here is the needed compression fact, with its original action specified. If a faithful transitive $L$ of degree $n$ has a nontrivial normal odd subgroup modulo $P_0 = O_2(L)$, take an odd complement $K$ in its full preimage. Every $L$-trivial elementary section of $P_0$ is an image of a subgroup of $C_{P_0}(K)$, by coprime fixed lifting. This centralizer acts faithfully on the $K$-orbits: the kernel on each odd orbit centralizes a transitive odd group, hence is semiregular of odd order. If $K$ moves

the $P_0$-orbit labels, its orbits cover at least three such labels, giving at most $n/3$ orbits. Otherwise its local action on every $P_0$-orbit is nontrivial, by transitivity and conjugacy of complements. On a binary orbit of size $d$, a prime-order odd element has at most $d/4$ fixed points. To verify this, its fixed points form one orbit of its binary centralizer, by complement conjugacy in the binary point stabilizer; thus their number is $2^c < d = 2^a$. If its prime order is $q$, then $q \mid 2^a - 2^c$, so $a - c \geq \mathrm{ord}_q(2) \geq 2$. The element has at most $(d + (q-1)d/4)/q \leq d/2$ cycles. In either case $K$ has at most $n/2$ orbits. Applying the actual abelianization bound [RDT25, Theorem 2.8] to the faithful centralizer action bounds every such binary section by $n/4$. Apply this with $n = 8$ to the image of $S$ in $A/(A^2[A, G])$, a trivial section of $O_2(L)$, to obtain the contradiction.

For a nonbinary $Q = G/N$, take the preimage of $\Omega_1(Z(Q))_2$. The marked bound just proved gives central binary rank at most two. Every elementary binary socle section has rank at most four by the actual Sylow-preimage form of Theorem 2.8; a nontrivial simple binary type has dimension at least two and therefore multiplicity at most two. All odd primary ranks are at most $\lfloor 8/p \rfloor \leq 2$, again using actual preimages, so Theorem 8.1 gives cover two. For a nonabelian binary $Q$, the actual $P$ surjects onto $Q$. The preimage of $\Omega_1(Z(Q))$ has index at least four, since a group cyclic modulo its center is abelian. The excessive-index bound gives central rank at most two. Finally an abelian binary quotient needs $d(Q) \leq d(P) \leq 4$ full-order linear characters. □

**Lemma 17.36** (Four ternary blocks and their top quotients)**.** *Let $R \leq S_3 \wr B$ on twelve points, where $B$ is transitive of degree four, and $H = R \cap C_3^4 \neq 1$. Put $L = R \cap S_3^4$ and $A = L/H$. If $A = 1$, every quotient of $R$ has cover at most two. In general every quotient retaining a nontrivial image of $H$ has cover at most two. In the exact-local-$S_3$ case, $R/H$ has a specified faithful transitive degree-eight sign action.*

*Proof.* The coordinates of $H$ are all nonzero, so $R$ is transitive on the original twelve points. Suppose $A \neq 1$. Its nontrivial coordinate inversion characters form one $R$-orbit and have trivial common kernel. Maschke's theorem on this binary $A$ shows that every nonzero $R$-section of $H$ is faithful on $A$. Thus for a quotient retaining $\bar{H} \neq 1$, $N \cap L \leq H$ and $C_{\bar{L}}(\bar{H}) = \bar{H}$. The whole binary socle, and every $p > 3$ socle, consequently injects into a quotient of the actual four-point $B$. Their ranks are at most two and $\lfloor 4/p \rfloor$. The total ternary socle rank is at most four from the original twelve-point Sylow preimage. A nonscalar ternary type therefore has Schur demand at most two. For a scalar type $\chi$, if $\chi|_A \neq 1$ its top image is zero, since a normal ternary subgroup of the quotient by $\bar{H}$ centralizes the normal binary $A$. Its entire contribution is a homogeneous section of the transitive monomial $\mathbb{F}_3^4$, of dimension at most two by Lemma 17.13. If $\chi|_A = 1$, its $H$ piece is zero by the inversion-character support, and it injects into a quotient of $B$, giving rank at most one.

If $A = 1$, all nonternary socles inject past $\bar{H}$ into a quotient of $B$, while nonscalar ternary types still have demand at most two. For a scalar ternary socle $X^m$, let $D$ be its preimage in $R$. For the trivial scalar, $D^3[D, R] \leq N$ and the degree-twelve relative bound of Theorem 9.5 gives $m \leq 2$. For a nontrivial scalar put $K = \ker \chi$. Then $D, N, H \leq K$. The actual $K$ action is transitive of degree twelve or has two six-point orbits. Filtering the same relative head by those orbit kernels gives respectively the bounds 2 or $1 + 1 = 2$ from that theorem. This includes every original normal subgroup, whether or not it retains $H$. The simultaneous socle criterion proves the claimed cover bounds. Finally the sign map on each local $S_3$ has kernel exactly $H$ on $R$. The induced group in $C_2 \wr B$ is transitive on its eight points because each block stabilizer contains local inversion. This is the asserted faithful action. □

**Lemma 17.37** (Nonzero ternary four-point frames)**.** *Suppose $T \le L \wr B$ has four blocks of size four with exact primitive local $L = A_4$ or $S_4$. Put*

$$E_0 = T \cap V_4^4, \qquad R = T/E_0, \qquad H = R \cap C_3^4 \neq 1.$$

*Let $M_0$ be the full preimage of $E_0$ in any original pair lift $U$. For every original $N$, the relative socle in $M_0N/N$ has Schur demand at most four. In the $S_4$ case, its $H$-nontrivial types have demand at most two. If moreover $E_0 = V_4^4$, the whole relative demand is at most two.*

*Proof.* The preimage $T_0$ of $H$ in $T$ has local image $A_4$: that image is normal in $L$ and maps onto $C_3$. Write a local binary permutation module as $W = \mathbb{F}_2[V_4]$, with flag $0 < C = I^2 < I < W$. The natural translation module $V_4$ is canonically $I/C$, via $v \mapsto v - 1$ modulo $I^2$. Consequently, for each $H$-nontrivial simple type in the relative socle, its pair piece and its $E_0$ piece are sections of the *same* natural module $V = \mathbb{F}_4^4$, with the same $R$ action and endomorphism field. This remains true for correlated $E_0$ and nonsplit extensions. Every nonzero $R$-section of $V$ is faithful on $H$: on restriction to $H$, the local nontrivial types form one orbit with trivial intersection of their kernels.

In the $A_4$ case, a type of binary dimension at least four has multiplicity at most two in each copy of $V$; a type of dimension two with Schur dimension at least two has total demand at most $\lceil 8/2 \rceil = 4$. In the remaining case, dimension two and Schur dimension one, faithfulness on $H$ implies $H = C_3$ and $[R, H] = 1$. Orient the coordinates so that $H$ is the same scalar of $\mathbb{F}_4^*$ on all four coordinates. The actual $R$ action is now transitive monomial over $\mathbb{F}_4$. Lemma 17.13 bounds the multiplicity in each copy by two, again giving total Schur demand four. This orientation is applied to the same frame; it does not merge distinct binary types.

For $L = S_4$ no natural type has Schur dimension one: such an image would be scalar and abelian, whereas faithfulness on $H$ would then kill $[R, H]$, contrary to local inversion. A type of dimension at least four consequently has total demand at most two. For dimension two, faithfulness again gives $H = C_3$. After orienting $H$ globally, $R^+ = C_R(H)$ is an index-two transitive monomial group over $\mathbb{F}_4$. It is transitive because every coordinate stabilizer contains an inverting element outside $R^+$. Restricting the homogeneous section to $R^+$ and applying Lemma 17.13 bounds its multiplicity in each copy by two; division by Schur dimension two gives total demand at most two.

An $H$-trivial type has no $E_0$ piece and lies entirely in the pair layer. It is trivial under $T_0$. The local $A_4$ module has trivial homogeneous sections of dimension at most one: its invariant submodules are $0, C, I, W$, and only $C$ and $W/I$ are trivial. Filter by the four coordinate kernels, with denominator the intersection of the original denominator with each kernel. Every factor is such a local section, so the whole dimension is at most four. This proves the general relative bound.

If $E_0 = V_4^4$, let $D/F$ be an $H$-trivial homogeneous section in the pair layer. The coordinate projections of $D$ are uniformly $0, C, I$, or $W$, by transitivity and the same local flag. If they are $W$, independent translations imply $I^4 \le [D, E_0] \le F$; the section is then a section of the ordinary four-coordinate module. If they are $I$, independent translations give $C^4 \le F$, and the remaining natural layer has no $H$-trivial constituent, so the section vanishes. If they are $C$, it is again a section of the ordinary four-coordinate module. Its homogeneous multiplicity is at most two by Lemma 17.13. Together with the natural-type bound this proves the final assertion. □

**Lemma 17.38** (The rank-four sign boundary)**.** *In the exact-local-$S_4$ situation of Lemma 17.37, if the actual sign group $K = R/H \le C_2 \wr B$ has an abelian binary quotient needing four generators, then $E_0 = V_4^4$.*

*Proof.* Let $A = K \cap \mathbb{F}_2^4$. The elementary abelianization sequence gives $d_2(K) \le \dim A/[A, B] + d_2(B)$. The five transitive groups of degree four are $C_4, V_4, D_8, A_4, S_4$. For $C_4, D_8, S_4$, an actual

regular four-cycle makes every invariant base submodule have head at most one; their binary abelianizations have ranks $1, 2, 1$. For $A_4$ the corresponding bounds are one and zero. Thus only $B = V_4$ can reach four. Lemma 17.13 then forces $A = I$, the regular augmentation ideal, and $B$ has rank two. Modulo $I$, the actual $K$ is the graph of a homomorphism $\phi : B \to C_2$. For a lift $(v, b)$ the square has image

$$(1 + b)v = \phi(b)(b - 1) \pmod{I^2}.$$

If $\phi \neq 0$ this kills a nonzero additional direction of the base head in the abelianization, giving at most three generators. Hence $\phi = 0$ and $K = I : B$.

The four coordinate inversion characters of $I$ are nontrivial and pairwise distinct. Over $\mathbb{F}_3$ their weight spaces are the four coordinate lines. The nonzero invariant $H \leq \mathbb{F}_3^4$, with full coordinate projections, must therefore be $C_3^4$. Its four natural binary coordinate types on $V_4^4$ are distinct. Each projection of $E_0$ is the full $V_4$, since the local derived subgroup of $T_0$ is $V_4$. Semisimplicity under this $H$ now forces $E_0 = V_4^4$. □

**Lemma 17.39** (The two primitive eight-point blocks)**.** *If the sixteen-point top $T$ has two minimal eight-point blocks, the socle of every quotient of $T$ has at most three normal generators. Every invariant subquotient of its natural binary permutation module has at most four module generators.*

*Proof.* The primitive degree-eight list is

$$2^3 : C_7, \quad 2^3 : (C_7 : C_3), \quad \mathrm{AGL}_3(2), \quad \mathrm{PSL}_2(7), \quad \mathrm{PGL}_2(7), \quad A_8, \quad S_8.$$

This is the actual action list supplied by the primitive classification cited in Lemma 17.15. In each local group every normal subgroup has one normal generator. The two-block normal-generation argument uses one position for each coordinate kernel section and one for the image in the cyclic block top. It therefore gives three normal generators for every normal subgroup of $T$. This bound passes to every quotient, in particular to its socle.

For the module assertion, let $P$ be an actual Sylow-seven subgroup of $T$. It fixes the two block labels. Its local image in each primitive component is a Sylow-seven subgroup: the block stabilizer maps onto that component, and its index in $T$ is two. Seven divides each of the seven listed component orders exactly once. A faithful group of order seven on eight points has a seven-point orbit and one fixed point. Hence $P$ has at most four orbits on the original sixteen points. Restriction to this odd group is semisimple, and one orbit seed per orbit generates the permutation module. Submodules and quotients inherit the same four-generator bound by semisimplicity; enlarging the acting group to $T$ preserves it. This argument covers all seven local types and keeps correlated local Sylow images. □

**Lemma 17.40** (Natural alternating and symmetric tops)**.** *If $T = A_s$ or $S_s$ in its natural action, $s \geq 5$, then every quotient of an original pair lift has either one faithful general character or at most two faithful full-order linear characters.*

*Proof.* The invariant submodules of $M_s$ are $0, C, I, M_s$, where $C$ is constant and $I$ is augmentation. The three-cycle argument in Lemma 17.16 shows that the orbit span of any nonconstant vector contains $I$; this step does not require even degree. For odd $s$, $M_s = C \oplus I$ with $I$ simple; for even $s$, $0 < C < I < M_s$ and $I/C$ is simple. In either case the socle of every subquotient has multiplicity at most one in each type. If $\pi(N) = 1$, then $N \leq E$, and every abelian minimal normal of $Q = U/N$ lies in $E/N$, since its image in $T$ would be abelian normal. Their multiplicities are at most one; the nonabelian socle shares the same one position. Theorem 8.1 gives one faithful character. If $\pi(N) \neq 1$, it contains $A_s$ and $[E, A_s] \leq E \cap N$, using actual lifts from $N$ in the

commutator. From the displayed submodule list, $[E, A_s]$ has codimension at most one in $E$. Since $T/\pi(N)$ has order at most two, $|Q| \leq 4$. This is abelian binary and needs at most two full-order linear characters. The original extension was never assumed split. □

**Lemma 17.41** (The actual quartet source)**.** *Let $T \leq S_{16}$ be nonbinary and transitive, preserving four specified quartets. Either $T$ has an actual system of eight pairs, or every nonbinary quotient of $T$ has socle normal-generator bound two. The natural binary module has uniform generator bound eight. In the unpaired alternative, for any original pair lift $U \twoheadrightarrow T$, every binary quotient of $U$ has order at most $2^{15}$; in the zero-odd no-pair case the stronger bound $2^{11}$ holds.*

*Proof.* Take the paired alternative first. Otherwise identify each quartet with $V = \mathbb{F}_2^2$ using the actual point chart. The original group embeds in

$$V^4 \rtimes (\mathrm{GL}_2(2) \wr S_4).$$

Write $A = T \cap V^4$, let $C = T/A$ be its actual linear image, and let $B \leq S_4$ be its actual coarse image. Put $O = \ker(C \to C_2 \wr S_4)$, an elementary ternary group. No independence of the local coordinates and no splitting of $T$ is asserted. For each original $N \lhd T$ retain

$$K_N = AN/N \cong A/(A \cap N), \qquad B_N = T/(AN).$$

Suppose first $O \neq 1$. Each local projection of $O$ is nontrivial by transitivity, so $V^4$ and all its retained sections are fixed-free under the whole $O$. For cyclic coarse top the orbit argument gives translation-module bound two. For a regular Klein top, kernel-averaging projectors separate the local cubic types into classes of size at most two; the repeated-kernel case is the rank-one cubic chart. A dihedral top restricts to its regular Klein subgroup. For $A_4$ or $S_4$, the local-kernel partition is constant or discrete and the same projectors give the stronger bound one. The corresponding actual linear quotients have socle bound two whenever needed on a nonbinary axis. These are the cyclic, Klein, dihedral and alternating cases of the retained ternary kernel argument in Lemma 17.36; binary quotients are not included in the Klein and dihedral assertion.

Let $D$ be the image of the preimage of $O$ in $T/N$. Its moving socle row lies in $K_N$ and costs two positions by the translation bound. The fixed row meets $K_N$ trivially and injects into the socle of $B_N$, so also costs two. These rows have no common nontrivial normal quotient: $D$ acts by full commutator on the moving row and trivially on the fixed row. The subdirect-product criterion therefore shares their two positions by a maximum, rather than adding them. This proves the asserted whole socle bound. The possible coarse orders $4, 8, 12, 24$ exhaust the actual transitive four-point action, so this dispatch needs no sixteen-point catalogue.

If $O = 1$, a binary $B$ would make $C$ binary by its injective sign map, and then make $T$ binary, a contradiction. Hence $A_4 \leq B$. Either $C \to B$ is injective or its binary kernel supplies an invariant nonzero line in each actual quartet. Translation along those four lines is an equivariant fixed-point-free involution, yielding the excluded pair system. Thus $C \cong A_4$ or $S_4$. The actual regular Klein lift and a cubic lift give the twisted quartet chart. On the same $K_N$, put $e = 1+a+a^2$ and $f = a + a^2$ for its cubic operator. If the socle lies in $K_N$, the uniform two-generator chart suffices. Otherwise the actual Klein subgroup acts trivially on $K_N$; its nontrivial normal image is forced into the centralizer by a socle constituent outside $K_N$. The twisted chart then gives $\dim eK_N \leq 2$ and $\dim fK_N \leq 2$. The binary socle image in $B_N$ adds zero fixed dimension and at most two active dimensions. Over $\mathbb{F}_2[C_3] \cong \mathbb{F}_2 \times \mathbb{F}_4$ the whole binary socle consequently needs at most two generators. The odd-primary socle needs one and shares positions with the binary part,

again giving two. The no-pair hypothesis also excludes the untwisted local cubic chart; otherwise its invariant line gives the original pair system.

For the natural-module bound, choose a derangement of the actual coarse top and lift it to $T$. It has at most two block orbits. Four seeds above each orbit give a polynomial-module cover with eight generators, so Lemma 17.2 applies to all invariant sections. For binary quotients use odd averaging on the original pair kernel. If $O \neq 1$, an actual ternary preimage subgroup has at most two orbits per quartet, hence eight orbits on the sixteen pair labels. The binary image of $T$ has order at most $2^7$, through its actual sign action; the binary quotient of $U$ therefore has order at most $2^{8+7}$. If $O = 1$, a cubic coarse lift has at most eight label orbits and every binary quotient of the actual $T$ has order at most eight by the same fixed projector. This gives $2^{8+3}$. In both calculations the retained top quotient and kernel section use the same original normal denominator. □

**Theorem 17.42** (Complete sixteen-pair estimate)**.** *All actual pair actions with sixteen pairs and nonbinary top have a complete exponentially small forward row and a quadratic-exponentially small scalar, with their original action weights.*

*Proof.* The actual minimal-block geometry of $T$ is exhaustive: $T$ is primitive, has eight pairs, has four quartets, or has two primitive eight-point blocks. The paired case is Lemma 17.34; the primitive case uses the twenty affine rows of Lemma 17.15(iv) and natural $A_{16}, S_{16}$ from Lemma 17.40. These statements are used only within their displayed geometric scopes. The remaining two branches use Lemmas 17.39 and 17.41.

Here is their uniform counting step. Set $a = 51/200$, $r = \lceil b/100 \rceil$ and $\theta = 39/10$. A quotient of a two-eight-block top has socle bound three and its original binary kernel section has dual generator bound four. A nonbinary quotient of a quartet top has the respective bounds two and eight. Their cold slopes satisfy

$$3\gamma_5 + 4a < \theta, \qquad 2\gamma_5 + 8a < \theta, \qquad 15a < \theta < 4 - 32/32768.$$

The last left inequality pays all binary original targets separately; no two-generator claim for their socles is used.

Fix an original normal denominator and the actual residual top map. The dual-kernel lift count charges its top generators before the remaining binary kernel maps. Sources with any retained finite-index kernel of binary rank exceeding $ab$ are charged by the complete binary rank-tail moment. Keeping those hot sources through the sum, all original normals in either branch satisfy, for fixed constants $C, k$,

$$\sum_{J \le S_b} Z_J(U) \le C\, 2^{\theta b} s_b + C(b!)^k 2^{-arb} s_{b+2r}.$$

One may take the cold constant $2^{4864}$, hot constant $2^{2304}$ and $k = 2^{32} + 48$: the actual top order is at most $2^{32}$, the original pair-action order at most $2^{48}$, and there are at most $2^{2304}$ normal axes. The unchanged menu envelope pays the cold constant since $16 \cdot 32 \log^2(34 + b) \ge 5000$. The rank-tail square completion and the published coarse subgroup estimate absorb every fixed factorial power, leaving a quadratic deficit on the hot sum. Thus the same original width-thirty-two normalization gives the stated row and scalar. The finitely many paired and primitive branches are combined with these two uniform sources. Point transports retain $a(U) = |N_{S_{32}}(U)|$ and every original pointing weight. The cover uses the actual block geometry and the bare primitive classifications. □

## 18 Repeated natural heads and the smaller pair counts

For a finite group $J$, write $d_p(J) = \dim_{\mathbb{F}_p} J/(J'J^p)$. All head modules below come from an actual subgroup of the same complete $J \le S_b$.

**Proposition 18.1** (Repeated natural heads with central binary lifts)**.** *Let*

$$R_d^A = \mathbb{F}_4^d : C_3, \qquad R_d^S = (\mathbb{F}_2^2)^d : S_3,$$

*where the same scalar, respectively natural, action is used on all $d$ copies. Suppose $Q$ has a specified central elementary binary subgroup $C$ of rank $c$ and $Q/C \cong R_d^A$ or $R_d^S$, with $d \ge 1$. For every complete $J \le S_b$,*

$$|\operatorname{Epi}(J,Q)| \le B_T(d) 2^{\max(c,d)b/2}, \qquad B_A(d) = 2 \cdot 4^d, \quad B_S(d) = 6 \cdot 4^d.$$

*No factor $|\operatorname{Aut} Q|$ is required in this bound.*

*Proof.* Put $r = d_2(J)$. In the $A$ case let $F = J'$, and in the $S$ case let $F = J''$. In either case set $B = J/F$ and $D = H_1(F, \mathbb{F}_2)$, with its honest $B$ action. For every onto top map $\theta$ to $C_3$ or $S_3$, let $V_\theta$ be the natural two-dimensional binary module. In the first case the odd part of the abelian $B$ maps onto $C_3$; in the second case the odd part of the abelian $B'$ does so. This is a normal odd subgroup with zero invariants on $V_\theta$. Averaging and inflation–restriction therefore give

$$H^i(B, V_\theta) = 0 \quad (i \ge 0), \qquad H^1(J, V_\theta) \cong \operatorname{Hom}_B(D, V_\theta).$$

The vanishing in degree two is what retains all restrictions even when the source extension is nonsplit.

Let $a_I$ be the multiplicities of represented natural simple types in the one semisimple head of $D$, and put $A = \sum_I a_I$. Their commuting fields are $\mathbb{F}_4$ in the $A$ case and $\mathbb{F}_2$ in the $S$ case. Each type corresponds to two, respectively six, concrete top maps. Onto restrictions onto $d$ identical copies are ordered independent rows over that field. Every cohomology class has exactly $4^d$ literal cocycles, since the module has no fixed vectors. The target's first, respectively second, derived group is precisely its repeated binary module. Thus a map is onto exactly when this restriction is onto, giving the exact formulas

$$|\operatorname{Epi}(J, R_d^A)| = 2 \cdot 4^d \sum_I \prod_{j=0}^{d-1} (4^{a_I} - 4^j), \tag{18.1}$$

$$|\operatorname{Epi}(J, R_d^S)| = 6 \cdot 4^d \sum_I \prod_{j=0}^{d-1} (2^{a_I} - 2^j). \tag{18.2}$$

A term is zero when $a_I < d$. No independent copy of the source is assigned to a repeated constituent.

We need the simultaneous constraints

$$r + 2A \le \lfloor b/2 \rfloor \quad (A \text{ case}), \qquad r + A \le \lfloor b/2 \rfloor \quad (S \text{ case}). \tag{18.3}$$

Choose bases of all the Hom spaces at once. Their evaluations give an onto map from $D$ to $V_{\text{all}} = \bigoplus_I V_I^{a_I}$. Lift all its coordinates by the displayed cohomology isomorphism. The joint image is exactly $V_{\text{all}} : R$, where $R$ is the joint image of the chosen top maps: the restriction of the same

$F$ is onto $V_{\mathrm{all}}$, so the full inverse image of $R$ is obtained. In the $A$ case $R \le C_3^I$ is odd and acts without trivial constituents on the binary module. This target has no binary quotient. Pair its quotient map with the maximal elementary binary quotient $C_2^r$ of $J$; Goursat's lemma gives the full product. The preimage of $V_{\mathrm{all}} \times C_2^r$ is an actual subgroup of $S_b$ with an elementary binary quotient of rank $2A + r$. The actual subgroup abelianization bound [RDT25, Theorem 2.8] proves the first inequality.

In the $S$ case put $N = R \cap C_3^I$ and $K \cong R/N \le C_2^I$. The coordinate sign characters on $N$ are nontrivial and have no trivial constituent; hence $[N, K] = N$. A complement to the normal odd Hall subgroup gives $R = N : K$ with elementary binary $K$. The maximal binary quotient of $V_{\mathrm{all}} : R$ is exactly $K$. Pairing with $C_2^r$ therefore gives $(V_{\mathrm{all}} : R) \times C_2^{r-u}$, where $u = \dim K$; it is the full fibre product over all $u$ binary characters. Pull back $(V_{\mathrm{all}} : K) \times C_2^{r-u}$. Each natural type restricted to $K$ is a nontrivial transvection module, so its coinvariant dimension is one. This actual subgroup has binary abelian quotient rank $A + u + r - u = A + r$, proving the second inequality.

For $q = 4$ or 2,

$$\sum_I \prod_{j<d} (q^{a_I} - q^j) \le \sum_I (q^{a_I} - 1)^d \le \left( \sum_I (q^{a_I} - 1) \right)^d \le (q^A - 1)^d < q^{dA}$$

when the sum is nonempty, and it is zero otherwise. Above one fixed map to $Q/C$, homomorphic lifts to $Q$ are absent or a torsor under $\mathrm{Hom}(J, C)$, of size $2^{cr}$. Onto lifts are only a subset. Consequently the two exponents are bounded by $cr + 2dA$ and $cr + dA$, respectively. Equations (18.3) bound both by $\max(c, d)\lfloor b/2 \rfloor$. This proves the assertion while retaining every lift obstruction before the positive enlargement. □

**Corollary 18.2** (Finite menus with repeated-head entries)**.** *Fix a finite menu of original transitive actions $U$ of width $w$. For every original $N \lhd U$, suppose its quotient has either a fixed mixed character tuple with slope less than $h(w)/8$, or a specified central repeated-head certificate as in Proposition 18.1 with $\max(c, d)/2 < h(w)/8$. Then the complete menu has one exponentially small forward row and zero scalar, with original action weights.*

*Proof.* Fix one choice at every original normal subgroup. The finite set has a common positive strict gap; sum the estimates of Propositions 8.2 and 18.1 on the same $J$, and apply Lemma 7.2 to $\Delta(n, b)Z_J(U)/a(U)$. Different original normals remain different summands. Repeated-head entries already count literal epimorphisms and do not receive an additional automorphism factor. □

**Lemma 18.3** (A binary rank tail on complete sources)**.** *For each fixed $\epsilon > 0$,*

$$\#\{J \le S_b : d_2(J) > (1/4 + \epsilon)b\} \le 2^{b^2/16 - \epsilon^2 b^2 + O_\epsilon(b^{3/2})}.$$

*The same estimate, with an additional $O(b \log(b + 2))$ in the exponent, counts complete $J$ having a normal subgroup $F$ of index three with $d_2(F) > (1/4 + \epsilon)b$.*

*Proof.* For any integer $t \ge 0$, the graphs of all maps $J \to C_2^t$, with the latter on $t$ fixed disjoint pairs, are subgroups of $S_{b+2t}$. Projection recovers $J$ and its graph. There are $2^{td_2(J)}$ maps. Apply the coarse bound (1.1) and take $t$ within one of $2\epsilon b$. The exact main exponent is

$$\frac{(b + 2t)^2}{16} - (1/4 + \epsilon)bt = \frac{b^2}{16} - \epsilon^2 b^2 + \frac{(t - 2\epsilon b)^2}{4}.$$

For the second assertion, every index-three overgroup of a fixed $F \le S_b$ is generated by $F$ and one permutation, so there are at most $b!$ possibilities. This is a positive union bound over all such actual kernels; no eligible kernel multiplicity is divided out. □

**Lemma 18.4** (Binary quotients of six-pair actions)**.** *Every binary quotient of a transitive action on twelve points preserving six pairs has order at most* 32*.*

*Proof.* Write $E \le \mathbb{F}_2^6$ for the pair kernel and $T \le S_6$ for the transitive pair top. Let $U/N$ be the original binary quotient. A Sylow three subgroup $P$ of $U$ lies in $N$. Its image is Sylow in $T$ and has no fixed pair, since the point-stabilizer index six is divisible by three. Thus it has two orbits on the six pairs. Odd-order averaging embeds every $P$-trivial quotient of $E$ into the $P$-fixed part of its ambient permutation module, of dimension two. In particular $|EN/N| \le 4$.

Every binary quotient of a transitive $T \le S_6$ has order at most eight. Otherwise its order is sixteen, since $|T|$ divides 6!, and its kernel is odd. This normal kernel is nontrivial, as a binary group cannot be transitive in degree six. Its orbits have a common odd length dividing six; faithfulness excludes length one, so they are two triples. Consequently $T$ embeds in $S_3 \wr S_2$, of order 72, contradicting a quotient of order sixteen. Apply this to $U/EN$ and multiply its order by $|EN/N| \le 4$. All kernels and block actions here are the actual ones of $U$. □

**Proposition 18.5** (All six-pair actions)**.** *Every transitive action on twelve points with six pairs has a complete exponentially small forward row and a quadratic-exponentially small scalar, with its original action weight.*

*Proof.* The six-block certificates in Appendix A give 182 physical action classes and 2476 literal normal records. Here is the constructive coverage behind this finite premise. The 56 subgroup representatives in $S_6$ contain the identity and are closed, up to explicit conjugacy, under adjoining every double-coset representative. If $x = h_1 g h_2$, then $\langle H, x\rangle = \langle H, g\rangle$; induction on generators therefore proves completeness. Sixteen are transitive. The 2825 distinct subspaces of $\mathbb{F}_2^6$, the exact Gaussian sum, give all 112 invariant-kernel interfaces. On each quotient module, evaluating a presentation on corrected generator lifts gives the linear cocycle equations. The number of base-conjugacy classes is exactly $2^{\dim Z^1 - \dim M + \dim M^T}$, proving completeness of the listed complement graphs and their full preimages, including nonsplit original lifts. The 224 transitive block certificates are identified by actual degree-twelve conjugacy into the 182 physical actions, with their original symmetric normalizers.

Every original normal list is closed under adjoining the normal closure of each conjugacy-class representative and contains the identity. Since a normal subgroup is generated by its conjugacy classes, this proves exhaustive normal coverage. The certificate at each normal is either a faithful irreducible tuple consisting of at most one linear and one general character (two linears also allowed), a binary quotient, or a literal onto map with exactly that kernel to one of the three fixed models

$$C_2^e \times R_2^A \quad (0 \le e \le 2).$$

Every binary quotient has order at most 32 by Lemma 18.4, uniformly in the original normal denominator. Kernel intersections of the character tuples and the explicit repeated-head model maps verify the other alternatives.

Call the complete $J$ cold if $r = d_2(J) \le 13b/50$. Character entries cost at most $|\operatorname{Aut} Q| 15^{b/3}$. Every map to a binary target factors through the canonical maximal binary quotient of $J$. That quotient has generator rank $r$, so each binary quotient has at most $2^{5r} \le 2^{13b/10}$ maps. Proposition 18.1 bounds each repeated-head model by $32 \cdot 2^b$. As $15 < 16$, summing every original normal gives a constant $C_U$ with

$$Z_J(U) \le C_U 2^{4b/3} \qquad (J \text{ cold}).$$

This has strict margin $3/2-4/3=1/6$ at original width twelve. For arbitrary $J$, its generator number is at most $\log_2(b!)$; the fixed targets have orders at most $46080<2^{16}$. Thus $Z_J(U)\le \nu(U)(b!)^{16}$, where $\nu(U)$ is its normal count. Lemma 18.3 with $\epsilon=1/100$ bounds the hot complete-source count by $2^{b^2/16-b^2/10000+O(b^{3/2})}$. The full hot fibre and original normalization are consequently at most

$$\sum_U \frac{\nu(U)(b!)^{16}}{b!a(U)p_nG_{\lfloor n/2\rfloor}}2^{b^2/16-b^2/10000+O(b^{3/2})}=2^{-\Omega(n^2)},\qquad n=b+12.$$

The factorials cost only $O(n\log(n+2))$. The cold sum is $\Delta(n,b)\sum_U C_U2^{4b/3}/a(U)$ and is exponentially small by Lemma 7.2. Equivalently, it is the ordinary additive cold row of slope $4/3$; only the square-completed hot rank tail enters the exceptional scalar. Fuse the whole family once, using the existence of any hot accepted pointing and its cold complement. There is one complete terminal on the cold row. □

**Lemma 18.6** (A cyclic dual on a common derived source)**.** *Let $X=V:R$, where $V$ is a binary vector space of dimension $d$, $R$ acts faithfully and $V^*$ is cyclic. Suppose, for $\ell=1$ or $2$, that $R^{(\ell)}=1$ and the odd part of $R^{(\ell-1)}$ has zero invariants on $V$. Then*

$$|\operatorname{Epi}(J,X)|\le 2^d|\operatorname{GL}_d(2)|2^{b/2}.$$

*Proof.* The hypotheses give $X^{(\ell)}=V$. Every map therefore sends the same $F=J^{(\ell)}$ onto $V$. Put $B=J/F$ and $D=H_1(F,\mathbb{F}_2)$. The odd part of $B^{(\ell-1)}$ maps onto the stipulated odd group under every onto top map, so averaging gives $H^i(B,V_\theta)=0$ for all $i$. Onto restrictions $D\to V_\theta$ dualize to injections $V_\theta^*\to D^*$. Fix one cyclic vector in $V^*$. Its image $f$ determines its whole image module $\mathbb{F}_2[B]f$. For a fixed $f$, there are at most $|\operatorname{GL}_d(2)|$ linear identifications; faithfulness determines the top map from each identification. Thus every top map and onto restriction together has at most $|\operatorname{GL}_d(2)|2^{\dim D}$ possibilities. Each restriction has at most $|V|$ cocycle lifts, and the actual $F\le S_b$ has $d_2(F)\le b/2$. This proves the bound for reducible and nonsemisimple modules as well. □

**Proposition 18.7** (All four-pair actions)**.** *Every nonbinary transitive action on eight points with four pairs has a complete exponentially small forward row and a quadratic-exponentially small scalar, with its original action weight.*

*Proof.* Its top is $A_4$ or $S_4$. In $M=\mathbb{F}_2^4$, the invariant modules are exactly $0,C,I,M$, with $C$ constant and $I$ augmentation: weight-two vectors span $I$, weight-one or weight-three orbits span $M$, and weight four is constant. For each kernel, try all 256 pairs of binary corrections to two fixed top generators, and require that the resulting whole kernel be the selected one. Every projecting subgroup contains such lifts, so this is exhaustive. The four-block certificates retain the ten transitive classes, their original normalizers and all 72 original normals. Normal completeness follows from intersection closure of irreducible character kernels, as in Lemma 17.15. Five classes are

$$C_2\times A_4,\quad \operatorname{SL}_2(3),\quad S_4 \text{ in degree eight},\quad C_2\times S_4,\quad \operatorname{GL}_2(3).$$

The other five, denoted here $Y_1,\dots,Y_5$, have binary kernels $I,M,I,I,M$, respectively, with $A_4$ top in the first two cases and $S_4$ top in the last three. The two augmentation lifts over $S_4$ are the ordinary lift and the graph whose base parity equals the top sign; the latter need not split over $I$.

For these five, $C$ is the unique minimal normal subgroup. The only minimal submodule of their binary kernel is $C$, and any normal subgroup disjoint from that kernel centralizes it. The top action on $I$ and $M$ is faithful, so such a subgroup is trivial. A character nontrivial on $C$, induced and then

restricted to an irreducible constituent, is therefore faithful on $Y_i$. Thus $|\operatorname{Epi}(J, Y_i)| \le |\operatorname{Aut} Y_i| 5^{b/3}$. The exact central quotients recorded by the literal maps are

$$Y_1/C = R_2^A, \quad Y_2/C = X_c, \quad Y_3/C = Y_4/C = X_g, \quad Y_5/C = X_s,$$

where, on $V = \mathbb{F}_4^2$, $a$ is common multiplication by $\omega$, $t$ common Frobenius and $s$ coordinate swap, and

$$X_c = V : \langle a, s \rangle, \quad X_g = V : \langle a, ts \rangle, \quad X_s = V : \langle a, t, s \rangle.$$

Because every nontrivial normal contains $C$, in each case $Z_J(Y_i) = |\operatorname{Epi}(J, Y_i)| + Z_J(Y_i/C)$ exactly.

For $X_c, X_s$, the module is cyclic over $\mathbb{F}_4[s]$ with unique proper nonzero submodule the diagonal line. The invariant trace form $\operatorname{Tr}(x_1 y_1^2 + x_2 y_2^2)$ identifies it with its binary dual. For $X_g$, the three invariant $\mathbb{F}_4$ lines are $y = cx$ with $c^3 = 1$; any two are complementary natural $S_3$ modules. Their commuting field is $\mathbb{F}_2$, and the matrix algebra $\operatorname{Mat}_2(\mathbb{F}_2)$ shows that two natural copies have a cyclic dual (take components forming a basis). Thus Lemma 18.6 applies with $\ell = 1$ for $X_c$ and $\ell = 2$ for the other two, giving onto bounds $C_0 2^{b/2}$, $C_0 = 16|\operatorname{GL}_4(2)|$. Their exact full normal menus are

$$\begin{aligned} Z_J(X_c) &= |\operatorname{Epi}(J, X_c)| + Z_J(C_2 \times A_4), \\ Z_J(X_g) &= |\operatorname{Epi}(J, X_g)| + 3|\operatorname{Epi}(J, S_4)| + Z_J(S_3), \\ Z_J(X_s) &= |\operatorname{Epi}(J, X_s)| + Z_J(C_2 \times S_4). \end{aligned}$$

Indeed in the first and third case every nontrivial normal contains the diagonal line. In the second case the three line kernels give three $S_4$ quotients, and all other proper normals contain $V$. These yield faithful degree-six comparators $C_2 \times A_4, S_4, C_2 \times S_4$ (padding $S_4$), with multipliers $1, 3, 1$. Since both $1/2$ and $\gamma_5$ are less than one, Theorem 7.3 handles $Y_2, \dots, Y_5$.

For $C_2 \times T$, $T = A_4, S_4$, normals not containing the natural $V_4$ lie in the central $C_2$. Put $z = 2^{d_2(J)}$. The complete remaining normal sum is $z|\operatorname{Epi}(J, A_4)|$ in the alternating case and $(z-1)|\operatorname{Epi}(J, S_4)|$ in the symmetric case: in the latter, the central binary row must be independent of the existing sign. The joint facets in Proposition 18.1, with $d = 1$, bound these sums by $8 \cdot 2^{b/2}$ and $24 \cdot 2^{b/2}$ respectively. Their complete quotient intervals are exactly $Z_J(C_2 \times C_3)$ or $Z_J(C_2 \times S_3)$, with faithful five-point representations. This proves the finite comparator estimate for these two classes. For $\operatorname{SL}_2(3)$ and $\operatorname{GL}_2(3)$ the center $C_2$ is unique minimal normal, so the same faithful-character argument and exact normal interval give comparators $A_4, S_4$ of degree four, with slope $\gamma_5 < 1$. The nonnatural $S_4$ itself has that comparator with zero error. This handles all classes except $Y_1$.

For $Y = Y_1$, the repeated quotient has five distinct $\mathbb{F}_4$-line kernels and therefore

$$Z_J(Y) = |\operatorname{Epi}(J, Y)| + |\operatorname{Epi}(J, R_2^A)| + 5|\operatorname{Epi}(J, A_4)| + Z_J(C_3). \tag{18.4}$$

The first-derived proof of Proposition 18.1 gives one common head with natural multiplicities $a_I$ and $r = d_2(J)$. For each corresponding $\theta : J \to C_3$, put $H = \ker \theta$. Averaging on $J/H = C_3$ gives

$$H^1(J, \mathbb{F}_2) = H^1(H, \mathbb{F}_2)^{C_3}, \qquad H^1(J, V_\theta) = \operatorname{Hom}_{C_3}(H_1(H, \mathbb{F}_2), V_\theta).$$

Comparing the latter with the same common-head Hom space shows that $H_1(H, \mathbb{F}_2)$ has trivial multiplicity $r$ and natural multiplicity $a_I$. Thus

$$d_2(\ker \theta) = r + 2a_I. \tag{18.5}$$

There are at most $b/4$ represented types, since $2\sum_I a_I \le d_2(J') \le b/2$.

Declare $J$ hot when *some* normal subgroup of index three has rank $> 51b/200$; test all such subgroups, even those whose natural type is absent. On the cold set, (18.5) and (18.1) give

$$5|\operatorname{Epi}(J, A_4)| \le 10b\, 2^{51b/200}, \qquad |\operatorname{Epi}(J, R_2^A)| \le 8b\, 2^{51b/100}.$$

Together with $Z_J(C_3) \le 3^{b/3}$ and the faithful-character bound on $Y$, the exact full sum (18.4) has slope at most $\gamma_5 < 1$, up to a fixed linear factor in $b$. For all sources that full sum is at most $C(1+b)2^b$: use $a_I \le b/4$ in the repeated-head formula. Lemma 18.3, now with $\epsilon = 1/200$ and its index-three clause, gives the complete hot count $2^{b^2/16 - b^2/40000 + O(b^{3/2} + b\log(b+2))}$. Multiplying by the whole fibre and by the original factor $1/(384b!p_nG_{\lfloor n/2\rfloor})$, $n = b+8$, leaves a quadratic-exponentially small scalar. The cold row is exponentially small by Lemma 7.2. The divisor 384 is the original $S_8$ normalizer: $Y$ is normal in $C_2 \wr S_4$, and its characteristic $C$ forces every normalizer to preserve those four pairs. The finite union of all ten classes is fused once, with all original weights and complete cold terminals retained. □

## 18.1 The three exceptional five-point block counts

**Theorem 18.8** (Exact affine-five exceptions)**.** *Every action with exact primitive affine local degree five and $s \in \{2,3,4\}$ has a complete exponentially small forward row and a quadratic-exponentially small scalar.*

*Proof.* Write $M = \mathbb{F}_5^s$, $E = U \cap M$, $R = U/E \le C_4 \wr T$. Since $s \le 4$, the latter group has order prime to five. Transitivity forces $E \ne 1$, since otherwise $5 \nmid |U|$. Schur–Zassenhaus first splits $U = E : R$, then conjugates its complement into the actual monomial complement in $M : R$. This does not split $R \to T$. The semisimple monomial modules $M, M^*$ are cyclic: the orbit of one coordinate vector spans the module. Every section $W = E/I$ and its dual are therefore cyclic and can be realized as actual submodules of $M$.

Let $D = \ker(R \to T)$ and $K = \ker(R \to \mathrm{GL}(W))$ for $W \ne 0$. Every coordinate projection of this realized $W$ is nonzero by transitivity. A diagonal element trivial on $W$ is therefore one, so $K \cap D = 1$ and $K$ embeds normally in $T$. Its coordinate orbits have one common size $q \mid s$; a fixed vector on a monomial orbit is determined by one coordinate, giving $\dim W \le s/q$. Thus $K = 1$ if $\dim W \ge 2$ at $s = 2, 3$, or if $\dim W \ge 3$ at $s = 4$. The only further possibility is $s = 4, \dim W = 2, q = 2$. Then $K \le C_2^2$ and $T \le D_8$, so $R$ is binary. The normal subgroup $K$ has a series of at most two central $C_2$ factors in $R$: choose an involution in $K \cap Z(R)$ and then use centrality of a normal order-two group in the quotient. The same holds after quotienting any normal $K_0 \le K$.

We need a joint-source onto estimate. If $X = W : S$, with $S$ a faithful $5'$-group on $W = \mathbb{F}_5^d$ and $W^*$ cyclic, then

$$|\operatorname{Epi}(J, X)| \le 5^d |\mathrm{GL}_d(5)| 5^{b/5} \qquad (J \le S_b). \tag{18.6}$$

Indeed fix the single characteristic source subgroup $F = O^{5'}(J)$ and $B = J/F$. Residuals commute with epimorphisms, so every map onto $X$ sends $F$ onto $W$ and induces a map $B \to S$. Put $A = H_1(F, \mathbb{F}_5)$. A surjective equivariant restriction dualizes to an injection $W^* \hookrightarrow A^*$. The image of one fixed cyclic vector determines its whole cyclic $B$-submodule. For each image there are at most $|\mathrm{GL}_d(5)|$ identifications; faithfulness determines the top map from each identification. Thus this bounds all top maps and restrictions jointly by $|\mathrm{GL}_d(5)||A^*|$. The remaining lift fibre is empty or a torsor for $Z^1(B, W)$; coprime averaging gives $|Z^1(B, W)| \le |W|$. Finally $d_5(F) \le b/5$ in its actual permutation action, proving (18.6). A central $C_2$ lift has at most $|\operatorname{Hom}(J, C_2)| \le 2^{b/2}$

choices over a fixed map, whether split or not. The two central steps above therefore cost at most $2^b$ in total.

Fix now $N \lhd U$ and $I = N \cap E$. If $I = E$, these terms sum exactly to $Z_J(R)$. If $\dim(E/I) = 1$, retain the whole comparator $U/I$. For this line let $\chi : R \to \mathbb{F}_5^*$ be its action. The map $(\chi, \pi_T)$ is injective since $K \cap D = 1$, so $U/I \hookrightarrow \mathrm{AGL}_1(5) \times T$ has a faithful action of degree $5 + s \le 4s$. For $\dim(E/I) \ge 2$, the normal $N/I$ is disjoint from $W$ and centralizes it. It lies in $W \times K$ and projects injectively to the $5'$-group $K$; its graph into $W$ is trivial. Hence $N/I = K_0 \lhd R$. Apply (18.6) if $K = 1$, or apply it after the at most two central binary steps otherwise.

This proves, with a fixed list $\mathcal{Q}_U$ consisting of $R$ and all literal line quotients $U/I$, that

$$Z_J(U) \le \sum_{Q \in \mathcal{Q}_U} Z_J(Q) + C_U 2^{\eta_s b}, \qquad C_U = \nu(U) 5^s |\mathrm{GL}_s(5)|,$$

where $\eta_2 = \eta_3 = \log 5/5$ and $\eta_4 = 1 + \log 5/5$. Equal abstract quotients at different $I$ remain different entries of this fixed list. Pad every comparator to faithful degree $v = 4s$.

For clarity the finite comparator argument extends directly to this fixed list. Let $m = |\mathcal{Q}_U|$ and $S_U(J) = m^{-1} \sum_{Q \in \mathcal{Q}_U} Z_J(Q)$. Jensen and Lemma 7.1 give, for every integer $q \ge 1$,

$$\sum_{J \le S_b} S_U(J)^q \le m^{-1} \sum_Q \sum_J Z_J(Q)^q \le s_{b+qv}.$$

The proof of Theorem 7.3 uses precisely this moment inequality and positivity, so applies with statistic $S_U$ and multiplier $m$. Here $h(5s) = 10, 14, 20$ and $v = 8, 12, 16$, respectively; $v < h(5s)$ and $\eta_s < h(5s)/8$, since $\log 5/5 < 7/15$. The finite union therefore has the claimed bounds with its original physical weights and one complete residual terminal. □

## 18.2 Three pairs and the complete degree-sixteen slice

**Proposition 18.9** (Three original pairs)**.** *Every nonbinary transitive action on three pairs has a complete contractive forward row and negligible scalar.*

*Proof.* The full pair wreath group has the decomposition

$$C_2 \wr S_3 \cong C_2 \times (A \rtimes S_3) \cong C_2 \times S_4,$$

where $\mathbb{F}_2^3 = C \oplus A$ is the invariant sum of the constant line and the augmentation plane. The affine action of $A \rtimes S_3$ on the four vectors of $A$ is faithful and has order 24. Let $H$ be the image of the original action in this product, and let $L \le S_4$ be its first projection. Transitivity gives $6 \mid |H|$, hence $3 \mid |L|$. Thus $|L|$ is $3, 6, 12$ or $24$. An order-six subgroup has a normal cubic subgroup, whose unique fixed point is fixed by the whole subgroup; restriction to the other three points identifies it with $S_3$. The order-twelve subgroup is $A_4$, and the order-twenty-four subgroup is $S_4$.

The kernel of $H \to L$ is either trivial or the central $C_2$. In the first case $H \cong L$; in the second $H = L \times C_2$. The order-three case is not transitive on six points. Consequently every transitive actual group is isomorphic to one of

$$C_6, \quad S_3, \quad A_4, \quad S_4, \quad C_2 \times S_3, \quad C_2 \times A_4, \quad C_2 \times S_4.$$

These are abstract charts on the original groups. Distinct permutation actions, including the two $S_4$ actions, retain their original pointings.

The targets $C_6, A_4, S_3, C_2 \times S_3, S_4$ have specified faithful representations of degrees at most five, less than six. Use the target itself in Theorem 7.3, with $C = 0$. For $U = C_2 \times L$, $L = A_4$ or $S_4$, let

$V$ be the standard $V_4$. Any normal subgroup not containing $V$ centralizes it; normality then forces that subgroup into the central $C_2$. Normals containing $V$ contribute exactly $Z_J(C_2 \times (L/V))$, whose comparator has faithful degree five. The two other quotients are $L$ and $C_2 \times L$. Proposition 18.1, with repeated rank one and central rank at most one, bounds both their onto-map counts by a fixed constant times $2^{b/2}$. Thus

$$Z_J(U) \le Z_J(C_2 \times (L/V)) + C_U 2^{b/2}.$$

Its exponent $1/2 < 6/8$ is strict. The finite comparator theorem applies with the original normalizer for each original action. □

**Theorem 18.10** (Every nonbinary action of degree sixteen)**.** *The complete union of nonbinary transitive degree-sixteen actions has a contractive forward row and negligible scalar. In particular this covers every eight-pair action and every zero-ternary action on four four-point blocks.*

*Proof.* The finite action certificate contains the full nonbinary slice of the degree-sixteen transitive classification, with 527 original permutation representatives and their original normalizers. Its completeness premise is the stated transitive classification [Hul05, Hul23]. Each of its 9249 literal normal entries is retained separately. To justify normal completeness, the certificate lists all conjugacy classes, their exact sizes and normal closures. The supplied normals contain one and are closed under adjoining every such normal closure. Every normal subgroup is generated by the closures of the classes it contains, so this join closure proves that no original normal is omitted.

For 518 actions all quotients have a fixed mixed character tuple, and for all but a specified interval or head quotient of each remaining action the same is true. A mixed tuple has $l$ linear and $t$ general slots with literal kernel intersection equal to the original normal, and satisfies

$$3^{64l} 5^{64t} \le 2^{381}.$$

Proposition 8.2 gives coefficient at most $381/192 = 2 - 1/64$, strictly below the physical reserve $16/8$. All target automorphism factors are fixed finite constants.

Two actions have a specified original derived subgroup with quotient $C_2^4$, concretely represented on eight points. Precisely the normal interval above that subgroup is retained as the whole $Z_J(C_2^4)$; every other original normal has a mixed tuple. Theorem 7.3 applies with $v = 8$ and $\eta = 2 - 1/64$.

Seven actions have, among their original normals, eight exceptional quotients. Each is a central elementary binary extension of a repeated $A_4$ or $S_4$ model: the certified whole centre has rank $c$, and the quotient has repeated rank $d$, with $32 \max(c, d) \le 127$. The maps and central preimages are literal; no splitting is inferred. Proposition 18.1 bounds each onto fibre by a fixed constant times $2^{\max(c,d)b/2} \le 2^{3b/2}$. All other normals are mixed. Original-weight pointing and Lemma 7.2 give their direct exponential row. The finite union, with the two comparator scalars, proves the theorem. No certificate is selected after inspecting the complete source $J$. □

## 19 The two zero-ternary frame exceptions

Put

$$z_2(G) = \max_{L \lhd G} \dim_{\mathbb{F}_2} L/[L, G]L^2, \qquad \mu_T(M) = \max_{B \le M \text{ invariant}} \dim_{\mathbb{F}_2} B/[B, T].$$

The second quantity is the maximum dimension of a trivial homogeneous section of the one actual module $M$.

**Lemma 19.1** (A fixed-frame bound)**.** *If $1 \to E \to U \to T \to 1$ has elementary binary kernel $E \le M$ as a $T$-module, then $z_2(U) \le \mu_T(M) + z_2(T)$. Moreover*

$$z_2(U) = \max_{N \lhd U} \operatorname{rank} \Omega_1(Z(U/N))_2.$$

*Proof.* For $K \lhd U$, let $L$ be its image in $T$, put $N = [K,U]K^2$ and $I = K \cap E$. The image of $N$ is exactly $[L,T]L^2$, so there is an exact sequence

$$0 \longrightarrow I/(I \cap N) \longrightarrow K/N \longrightarrow L/[L,T]L^2 \longrightarrow 0.$$

Its first term is a trivial quotient of the actual submodule $I$ of $M$, proving the bound. Each $K/[K,U]K^2$ is central elementary binary in its corresponding quotient of $U$. Conversely the preimage of any central elementary binary subgroup in $U/N$ has relative head surjecting onto that subgroup. Taking maxima proves the identity. □

**Lemma 19.2** (Noncentral socle types in the two frames)**.** *Suppose $U$ has exact local $A_4$ or $S_4$ on $s \in \{8,16\}$ blocks and zero ternary linear kernel. Let $M = (\mathbb{F}_2^2)^s$, $E = U \cap M$ and $T = U/E$, its actual faithful block top. Every nontrivial binary simple type in the socle of every quotient has Schur demand at most $5s/8$, and every odd-primary type has demand at most $\lfloor s/3 \rfloor$. Thus every quotient has a faithful tuple of at most $5s/8$ characters if $z_2(U) \le 5s/8$.*

*Proof.* In $Q = U/N$, the whole binary socle $A$ is a semisimple $T$-module: the normal binary $\bar{E}$ acts trivially on each simple type. Filter $A$ by $A \cap \bar{E}$. For a type $X$ of binary dimension $d$ and Schur dimension $e$, its multiplicities satisfy

$$m_E \le \lfloor 2s/d \rfloor, \qquad m_T \le \lfloor s/(2d) \rfloor.$$

The second bound uses the whole elementary outer section of $T$. These multiplicities add, since the types on the two sides may agree. For $d \ge 4$, their sum is at most $5s/8$. For $d = 3, e = 3$, dividing the sum by three and rounding up also suffices. For $d = 3, e = 1$, the image is $C_7$; its normal kernel is transitive, since its orbit number divides both seven and $s$. Restricted to that kernel, $X^{m_E}$ is trivial of dimension $3m_E$. Lemma 16.1 bounds this dimension by $s$, giving $m_E \le \lfloor s/3 \rfloor$ and $m_T \le \lfloor s/6 \rfloor$.

For $d = 2, e = 2$ the demand is at most $\lceil (s + s/4)/2 \rceil = 5s/8$. If $d = 2, e = 1$, the image is $C_3$ and its normal kernel $K$ is transitive. At a block stabilizer, restriction to $K$ either keeps a nontrivial local $C_3$ or $S_3$, or kills the local $C_3$ in the $A_4$ case. In the first case take a Sylow 3-subgroup $P$ of $K$. At a $P$-fixed block it is Sylow in the stabilizer and its local image is $C_3$, with no fixed vector. Every nonfixed orbit contributes at most two invariant dimensions. Exactness of odd-order invariants on sections gives $2m_E \le 2\lfloor s/3 \rfloor$. Adding $m_T \le s/4$ suffices at both counts. In the second case the induced frame restricted to $K$ is two copies of the honest permutation module $\mathbb{F}_2^s$. Its transitive Sylow 2-subgroup gives the scalar submodule capacity $\binom{a}{\lfloor a/2 \rfloor} = 3s/8$ for $s = 2^a \in \{8,16\}$. Filtering a trivial section through the two copies gives $m_E \le 3s/8$, and again $m_E + m_T \le 5s/8$.

For odd $p$, the entire $p$-socle meets $\bar{E}$ trivially and embeds in a quotient of $T$, so its rank is at most $\lfloor s/p \rfloor$. The only remaining binary type is the trivial type, whose maximum rank over the normal quotients is exactly $z_2(U)$ by Lemma 19.1. Apply the whole-socle criterion. □

The following bounds concern tops and induced linear frames, and do not enumerate all extensions $U$ above a frame. The four-by-four case has a structural proof; the remaining finite inputs have literal permutation and module witnesses.

**Lemma 19.3** (Small frame bounds)**.** *The zero-ternary frames have the following properties.*

*(i) Every nonbinary transitive top $T$ of degree eight satisfies $z_2(T) \le 2$. Every frame induced from an onto map of a point stabilizer to $C_3$ or $S_3$ satisfies $\mu_T(M) \le 2$.*

*(ii) For each degree-sixteen top having four four-point blocks, exact local $A_4$ or $S_4$, and zero internal ternary kernel, every frame $M$ induced from an onto point-stabilizer map to $C_3$ or $S_3$ satisfies $z_2(T) + \mu_T(M) \le 10$.*

*(iii) Every primitive degree-sixteen top satisfies $z_2(T) \le 2$ and has an odd subgroup $A$ with at most four point orbits. Consequently every frame induced from a two-dimensional binary local representation satisfies $\mu_T(M) \le 8$ and $z_2(T) + \mu_T(M) \le 10$.*

*(iv) The primitive degree-eight actions are the three affine groups $2^3 : C_7$, $2^3 : (C_7 : C_3)$, $\mathrm{AGL}_3(2)$ and $\mathrm{PSL}_2(7)$, $\mathrm{PGL}_2(7)$, $A_8, S_8$. Their normal subgroup orders are respectively*

$$(1, 8, 56),\ (1, 8, 56, 168),\ (1, 8, 1344),\ (1, 168),\ (1, 168, 336),\ (1, 20160),\ (1, 20160, 40320).$$

*Except for the first group, a Sylow 3-subgroup has orbit lengths $1, 1, 3, 3$ in each actual action.*

*Proof.* The degree-eight top alphabet consists of the seventeen imprimitive classes constructed by the four-pair and two-four-block Goursat and lift certificates, and the seven primitive actions of the stated primitive classification. All normal subgroups are retained for the relative-rank check; equivalently, their checked quotient-socle bounds imply $z_2(T) \le 2$. Four point stabilizers admit no quotient $C_3$ or $S_3$, as follows from their exact affine or natural stabilizer structures. For each of the twenty remaining literal actions, a Sylow 3-subgroup $P$ and one $g \in N_T(P)$ have the following point property: every point is fixed by $P$ or belongs to one displayed $\langle P, g\rangle$-orbit. Short words on the original eight physical points check this property. Only the selected Sylow subgroup, of order three or nine, needs a Cayley certificate.

This single witness covers every original local epimorphism and its twist. At a $P$-fixed point, $P$ is Sylow in the point stabilizer; its local image is the full $C_3$, which has no fixed vector on the faithful binary plane. Thus every such fibre of the induced module vanishes after taking $P$-invariants. Evaluation at the displayed moving-orbit representative injects $(M^P)^g$ into the original two-dimensional local space. For every invariant $B \le M$, odd-order averaging and restriction to $g$ give

$$\dim B/[B, T] \le \dim(M^P)^g \le 2.$$

This proves the complete local-frame bound without an invariant-submodule list or a classification of the point-stabilizer kernels.

For (ii), write $E$ for the actual binary translation kernel and $J = T/E \in \{A_4, S_4\}$ for the faithful four-point top. We first bound the whole binary socle of every original quotient $T/N$ by two ambient generators, including its central trivial constituents. Here is the common-envelope calculation used for this step. Put $k = \mathbb{F}_2$, and let $W$ be the irreducible binary plane for $C_3$. The two envelopes are

$$M_6 = k^2 \oplus W_m \oplus W_t, \qquad M_7 = (P_2 \oplus k) \oplus W_m \oplus W_t.$$

The $A_4$ action on $M_6$ factors through $C_3$; the $S_4$ action on $M_7$ factors through $S_3$. Here $P_2$ is the two-point permutation module for the reflection, while $W_m, W_t$ are isomorphic natural binary $S_3$-planes. The reflection on $P_2$ is retained. Every submodule of either envelope is two-generated: the trace $1 + c + c^2$ for a cubic generator projects onto the fixed part and its complement projects

onto $W_m \oplus W_t$. The latter needs at most two generators under $C_3$. A submodule of $k^2$ needs at most two; a submodule of $P_2 \oplus k$ either lies in its two-dimensional fixed space, or contains the two-dimensional cyclic span of a moved vector and needs at most one more generator. Adding the two pairs of generators componentwise and applying the trace projections gives two generators for their common submodule.

To apply this to the actual binary socle $A$ of $T/N$, put $A_0 = A \cap EN/N$. The binary preimage of the Klein group acts trivially on $A$. The four-point translation coordinates lie in $k[V_4] \otimes W$, of dimension eight. Filtering the regular factor by its bottom and augmentation layers gives a source $L \twoheadrightarrow A_0$ and an injection $L \hookrightarrow M_j$. This uses the bottom map when the source lies in the bottom layer, the whole-source map for the whole translation module, and the induced middle-section map for a proper intermediate source. In coordinates $(a_0, a_1, a_2; w_m, w_t)$, the resulting source has $a_2 = w_t = 0$, whereas the injection of $A/A_0$ into the same envelope has $a_0 = a_1 = w_m = 0$. In the alternating case its remaining scalar $a_2$ also vanishes, giving $M_6$. These coordinate maps use the original $N$ throughout; no containment of a bottom layer in $N$ or splitting of the group extension is required. Their images are disjoint. Since $A$ is semisimple, $L \oplus A/A_0$ surjects onto $A$ and embeds into this one envelope. Thus $A$ is two-generated. The other prime-primary socle parts have bound two and combine by a maximum, so every original quotient has whole-socle bound two. Lemma 19.1 gives $z_2(T) \leq 2$.

Choose an actual Sylow three subgroup $S = \langle s \rangle$. Its point action has one fixed point and five moving triples. Every fibre at an $S$-fixed point contributes zero to $M^S$: the original local epimorphism maps $S$ onto the local $C_3$. If $C_E(S) \neq 1$, choose a nonidentity centralizing translation $e$. It is trivial on the fixed four-point block and nonzero on the three moving blocks. It pairs their four moving $S$-orbits, so there are at most three active $\langle S, e \rangle$-orbits in all. Twisted-fibre evaluation bounds the head of every invariant submodule by six. Hence $z_2(T) + \mu_T(M) \leq 2 + 6$.

If $C_E(S) = 1$ and $J = A_4$, odd-order averaging shows that every $S$-stable section of $E$ has zero trivial head. The remaining relative binary head comes from a normal subgroup of $A_4$ and is also zero. Thus $z_2(T) = 0$, and the five moving triples give $\mu_T(M) \leq 10$. If instead $J = S_4$, Sylow conjugacy in its preimage lifts a reflection through $N_T(S)$. Its square belongs to $E \cap N_T(S) = C_E(S)$, so the actual lift is an involution. On a moving block which it stabilizes, the actual block stabilizer $S_3$ maps isomorphically onto its local linear quotient, so its local action is nontrivial modulo translations. A fixed-point-free involution on four points is a translation, so this local involution has a fixed point. The reflection therefore has at most three orbits on the four cross-block $S$-orbits. Including the triple in the fixed block leaves at most four active orbits, giving $\mu_T(M) \leq 8$ and $z_2(T) + \mu_T(M) \leq 2 + 8$. This proves (ii) for every original local map and twist.

For (iii), the quotient-socle certificates give $z_2(T) \leq 2$ for each of the twenty affine primitive tops. The displayed cyclic subgroups of orders five, seven or fifteen, or the displayed $C_3 \times C_3$, have at most four orbits on the original sixteen points. In any induced two-dimensional local frame, an invariant vector on one such orbit is determined by its value at a single point, so $\dim M^A \leq 8$. Exactness of odd-order invariants then gives $\mu_T(M) \leq 8$. For the natural $A_{16}$ and $S_{16}$ actions, the normal chains give $z_2(T) \leq 2$, and an even fifteen-cycle has just two point orbits. Thus the same argument covers all twenty-two tops and every two-dimensional local frame; no classification of point-stabilizer kernels is needed here. The primitive completeness premise is [RD05, HRDR23]; part (i) uses the separate constructive eight-point completeness argument above, whereas part (ii) requires no four-by-four catalogue. The normal orders and actual Sylow orbits in (iv) are direct checks on the same seven primitive representatives. The finite completeness certificates identify actual actions, rather than inferring coverage from their cardinalities. □

**Theorem 19.4** (Complete zero-ternary eight and sixteen frames)**.** *Every exact-local $A_4/S_4$ action with zero ternary kernel and eight or sixteen blocks has a faithful character cover on every normal quotient of length five or ten, respectively. Their entire original-normal menus satisfy Corollary 8.4.*

*Proof.* By Lemmas 19.1 and 19.2, it suffices to bound $\mu_T(M) + z_2(T)$ by five or ten. At degree eight Lemma 19.3(i) gives four.

We record two tools for degree sixteen. Odd-order invariants are exact, so for an odd subgroup $P$, $\mu_T(M) \le \mu_{N_T(P)}(M^P)$. If $g$ is a 2-element normalizing $P$, the binary cyclic module $M^P \downarrow_{\langle g \rangle}$ is a sum of modules over a chain ring. The head and socle have equal dimension, and submodules cannot have larger socle. Hence

$$\mu_T(M) \le \dim(M^P)^g.$$

Also, if $P$ is Sylow three in a transitive top of power-of-two degree, a $P$-fixed block contributes zero to $M^P$: the local onto $C_3/S_3$ map has nontrivial $C_3$ image there. Each nonfixed $P$-orbit contributes at most two invariant dimensions.

Suppose first $T$ preserves eight pairs, with kernel $K$ and top $S \le S_8$. If $K \ne 1$, it projects onto every pair. The local map from a point stabilizer kills its intersection with $K$, because $C_3, S_3$ have no nontrivial normal 2-subgroup. It descends onto the stabilizer of a pair in $S$, giving one of the degree-eight frames $M_0$. Induction in stages and the length-two filtration of the binary two-point permutation module give a filtration of $M$ with two inflated copies of $M_0$. Thus $\mu_T(M) \le 2\mu_S(M_0) \le 4$. Furthermore Lemma 19.1 on $K \le \mathbb{F}_2^8$ gives $z_2(T) \le 3 + z_2(S) \le 5$, using the scalar induced capacity three in degree eight. The sum is at most nine. If $K = 1$, then $T \cong S$ and $z_2(T) \le 2$. A Sylow three has at most two nonfixed orbits on the eight pairs, each lifting to two orbits on the sixteen points, since an odd pair stabilizer fixes its pair pointwise. Hence $\mu_T(M) \le 8$, giving ten.

Otherwise a minimal nontrivial block of $T$ has size four or eight, or $T$ is primitive. For size four let $F$ be its internal binary translation kernel, let $D$ be its internal ternary kernel and let $S \le S_4$ be the top. If $D \ne 1$, retained-$F$ quotients have character cover two by Theorem 16.5. For quotients killing $F$, the central elementary binary socle injects after removing the odd $D$ into a quotient of $C_2 \wr S \le S_8$. Therefore $z_2(T) \le 4$. If $S = A_4$ or $S_4$, a Sylow three has at most three nonfixed orbits on the sixteen points: over its three cycled four-blocks the moving local triples give one orbit of length nine and their fixed points one orbit of length three; the fourth block contributes at most one further orbit. Thus $\mu_T(M) \le 6$. If $S$ is binary, let $P$ be Sylow three of the full block kernel. It is also Sylow in $T$, and Frattini gives a normalizer transitive on the four blocks. Each block contributes the same dimension $d \le 2$ to $M^P$. The normalizer modulo $P$ is binary, so the local $d$-space has a series of trivial factors. Induction gives $d$ copies of the four-point permutation module in a filtration of $M^P$. Their scalar capacity is two, giving $\mu_T(M) \le 2d \le 4$. If instead the internal $D = 1$, use the joint bound in Lemma 19.3(ii).

For a minimal block of size eight, the two components are conjugate primitive actions. The fixed degree-eight catalogue leaves the seven locals in Lemma 19.3(iv): each of the other seventeen actions preserves one of the displayed nontrivial partitions. Every normal subgroup of each of these seven locals is normally generated by one element. The original two-block kernel is a Goursat subdirect product of the two components. Projecting an ambient normal subgroup to one component and retaining its kernel in the other gives a normal generating set of size at most two. The cyclic block top adds at most one generator, so $z_2(T) \le 3$.

For the coinduced binary-plane frame, choose a Sylow three-subgroup $P$ of the block kernel, hence of $T$. Frattini's argument makes $N_T(P)$ transitive on the two blocks. At a point fixed by $P$, the stabilizer epimorphism maps $P$ onto a Sylow three of the local $C_3$ or $S_3$, and the faithful binary

plane has no fixed vector. Thus only moving $P$-orbits contribute. Every normalizer orbit meeting them has at least three points in each block, hence at least six points. There are at most two on sixteen points; the two-dimensional fibres give $\mu_T(M) \le 4$. Consequently $z_2(T) + \mu_T(M) \le 7$. This argument retains both actual projections and their common kernel; the bound is uniform in the Goursat twists and the local frame maps. The primitive degree-sixteen case is Lemma 19.3(iii).

These cases exhaust the top: a primitive binary group has prime degree, so no omitted binary primitive local occurs at sizes four or eight. The original extension $U$ is arbitrary throughout. Finally $5^{64\cdot5} < 2^{24\cdot32-3}$ and $5^{64\cdot10} < 2^{24\cdot64-3}$ give the character-forward margin at the two physical widths. □

## 20 Exhaustion of the nonbinary orbit actions

**Theorem 20.1** (Complete nonbinary action reduction)**.** *Let $\mathcal{X}_n$ consist of the actual subgroups of $S_n$ with an orbit action outside the following list:*

$$\textit{binary transitive actions}, \qquad \textit{natural } C_3, A_4, S_3, \qquad \textit{fixed points}.$$

*There are positive constants $c, c'$ and complete kernels $K(n,b)$, with $K(n,b) = 0$ for $b \ge n$, such that eventually*

$$\frac{|\mathcal{X}_n|}{L_n} \le \eta(n) + \sum_{b<n} K(n,b)\frac{s_b}{L_b}, \qquad \sum_{b<n} K(n,b) = O(2^{-cn}), \qquad \eta(n) = O(2^{-c'n^2}).$$

*Every forward term retains the complete complementary subgroup and the original permutation-action weight. Thus every subgroup outside this counted union has only the displayed orbit actions.*

*Proof.* We first prove that every transitive action outside the list meets one of the proved consumers on its entire original normal menu. This is an action-by-action exhaustion, preceding the counting union.

For a primitive action, a nonaffine socle is covered by Theorem 13.3. An affine action of degree at least 1024 is covered by Theorem 14.2. At smaller degrees at least five, the soluble and nonsoluble cases are Theorems 14.3 and 14.6, with the exceptional degree-five Frobenius action covered by Proposition 14.4. At degree four the only primitive groups are the natural $A_4, S_4$, the latter again covered by that proposition. Degrees three and two leave precisely the allowed natural actions and the binary group.

Now let $U$ be imprimitive. Choose a minimal nontrivial block of size $r$, with $s$ blocks and faithful top $T \le S_s$. The full block stabilizer induces an exact primitive local group $L \le S_r$: any smaller block for that induced action and all its translates would give a smaller block for $U$. Transport coordinates along representatives of the block action to obtain the actual embedding $U \le L \wr T$. This does not impose a splitting or full block kernel.

If $L$ is nonaffine, use Theorem 13.3. If it is affine with $r \notin \{2,3,4,5,8,9,16\}$, use Theorem 11.2. For the four degrees $5, 8, 9, 16$, Theorem 12.3 covers every count except its explicit finite lists; these lists are consumed, respectively, by Theorems 18.8, 16.3, 16.4 and 16.3. Each is an entire-normal statement on arbitrary complete sources.

For $r = 3$, the exact local group is $C_3$ or $S_3$. Theorem 12.1 and Proposition 15.3 cover all $C_3$ counts. The same capacity theorem and Theorem 15.9 cover all $S_3$ counts.

For $r = 4$, the local group is natural $A_4$ or $S_4$. Theorem 12.2 leaves only its displayed finite count sets, all at most 3072. In the actual affine frame put $E = U \cap (\mathbb{F}_2^2)^s$, $R = U/E$ and

$D = R \cap C_3^s$. For $D \neq 1$, Theorem 16.5 applies. For $D = 1$, Theorem 12.4 covers all counts except $4, 8, 16$, with counts two and three impossible. At four blocks use Theorem 18.10; at eight and sixteen use Theorem 19.4. This disposes of every exact four-point component.

Only $r = 2$, $L = C_2$ remains. Its pair kernel is binary, so its top is nonbinary whenever $U$ is nonbinary. The prime-capacity theorem leaves exactly

$$\{2^a : 1 \leq a \leq 8\} \cup \{3 \cdot 2^a : 0 \leq a \leq 10\}.$$

At count two the whole action would be binary. Counts three, four and six are Proposition 18.9 and Propositions 18.7, 18.5. Count eight is Theorem 18.10, and counts twelve and sixteen are Theorems 17.33 and 17.42. The remaining necessary counts are

$$24, 32, 48, 64, 96, 128, 192, 256, 384, 768, 1536, 3072.$$

Every one is even and at least 24. For each literal $N \lhd U$, put $K$ for the pair kernel, $A_N = KN/N \cong K/(K \cap N)$ and $B_N = U/(KN)$. Corollary 6.7 applies through the canonical faithful cover $T = U/K \twoheadrightarrow B_N$, even when $B_N$ is a 2-group. It supplies the complete payload for the original extension $U/N \twoheadrightarrow B_N$ without assuming $N \leq K$. The displayed physical degrees contain only finitely many labelled action classes and literal normalizer orbits, so these payloads form a fixed finite menu. Theorem 6.5 gives its strict forward row. This proves the required exhaustion.

For the final insertion one may sharpen the source before applying the numerical aggregate. Partition the outside family according as every actual orbit is binary, natural $S_3$, or one of the selected pair-frame actions. In the complementary, pre-insertion family choose one orbit witnessing failure of this alphabet and transport its action to the fixed conjugacy representative. That representative cannot admit a selected pair frame: transporting such a frame back to the chosen orbit would contradict its defining failure. Hence the earlier-owner aggregate is required only on the restricted menu of nonbinary action classes with no selected pair frame. The post-alphabet family is charged to the selected pair owner separately, so the two forward rows have disjoint physical sources. The numerical insertion of the restricted row is Proposition 7.8: earlier-owned axes keep coefficient one, while every rejected axis retains its radical/transgression flag together with its Yoneda fibre, and the complete original-weight sum is certified by Lemma 7.5.

The complete degree-three packet of Theorem 29.6 may likewise be removed before this last aggregate. Partition the exact pre-$E_7$ residual according as every violating orbit has degree three. Relabelling is a bijection, so the covered piece has exactly the normalized ratio bounded by that theorem. In the complementary piece choose the certified non-three violation; its degree is at least four. Thus the terminal owner–cell aggregate begins at width four, where the only retained actions are the natural $A_4$ and $S_4$ actions and their checked rows. Width three never re-enters as an unused numerical cell.

Finally choose one of the proved certificates on each original action class, before inspecting its complementary group, and give the finitely many consumer families a fixed order. For a subgroup in $\mathcal{X}_n$, at least one orbit and its actual Goursat normal entry belong to the first eligible family. Apply that family's complete incidence sum. For a family with a hot/cold split, the existence of any hot eligible occurrence defines its hot part; all eligible occurrences in its remaining part are cold. Thus a chosen cold pointing cannot hide another hot one.

The incidence factor for deleting the original $w$-point orbit is $n!/(b!a(U))$, with $b = n - w$. It changes the occurrence factorial once. None of the block systems, chief factors, comparator degrees or character tuples acquires a physical support or normalizer. The full-fibre theorems are evaluated first on the same actual complete $J \leq S_b$, so the cold bounds multiply $s_b/L_b$ and the hot bounds

are complete scalars at $n$. Every finite family has strict positive margin, and every unbounded family has its proved uniform margin and subquadratic error. A finite sum of their exponential rows and quadratic scalars gives the asserted $K, \eta, c, c'$. No bound on $s_b/L_b$ was assumed. □

## 21 Binary sections and joint character incidence

The nonbinary hypothesis in Lemma 5.3 is essential. For binary tops it is replaced by a joint budget indexed by actual character kernels. Write $B_a = \binom{a}{\lfloor a/2 \rfloor}$, including $B_0 = 1$.

**Lemma 21.1** (Locally dependent maps over $\mathbb{F}_2$)**.** *Let $\mathcal{M} \le \mathrm{Hom}(V, W)$ have dimension at least two, and suppose that $\dim \mathcal{M}(v) \le 1$ for every $v \in V$. All maps in $\mathcal{M}$ have image in one common line of $W$.*

*Proof.* If $f \in \mathcal{M}$ has rank at least two, then for every $g \in \mathcal{M}$ one has $\ker f \le \ker g$. To see this, take $u \in \ker f$ and two vectors with independent $f$-images. Applying pointwise dependence to each vector and its translate by $u$ forces $g(u)$ to lie in both independent lines. Thus $g$ induces a map on $\mathrm{im}\, f$ taking each $x$ to either 0 or $x$. Linearity on independent pairs forces this map to be zero or the identity. Hence $g = 0$ or $f$, contradicting $\dim \mathcal{M} \ge 2$. All nonzero maps therefore have rank one. Two maps with distinct image lines have the form $u\alpha$ and $v\beta$ for independent $u, v$. There is an input on which both nonzero functionals $\alpha, \beta$ are one, contradicting pointwise dependence. □

**Lemma 21.2** (A deterministic separator)**.** *Let $\mathcal{L} \le H \otimes S$ have dimension $\ell$, and put*

$$j_0 = \max_{0 \ne \lambda \in H} \dim(\mathcal{L} \cap (\lambda \otimes S)),$$

*with empty maximum zero. There are $q$ independent functionals on $S$ whose simultaneous evaluation is injective on $\mathcal{L}$, where*

$$q \le \left\lfloor \frac{\ell + \max(1, j_0)}{2} \right\rfloor. \tag{21.1}$$

*For $\mathcal{L} = 0$ take $q = 0$.*

*Proof.* View tensors as maps $S^* \to H$. On the current common kernel, select an evaluation of rank at least two whenever one exists. Each step removes at least two dimensions, and its functional is independent of those already selected. If a remainder of dimension $r \ge 2$ has only evaluations of rank at most one, Lemma 21.1 puts it in one character line, so $r \le j_0$. Separate its support with $r$ more functionals. A remainder of dimension one requires only one functional. If $k$ first-stage steps occurred, then $\ell \ge 2k + r$ and $q = k + r$, proving (21.1) in all cases. The final functionals remain independent: earlier ones vanish on the remaining support. □

Let $T \le S_{2^a}$ be transitive and binary and $A$ a section of its permutation module whose action factors through $B$. Set

$$S = A^B, \quad t = \dim S, \quad H = H^1(B, \mathbb{F}_2), \quad \mathcal{L} = \ker(H^1(B, S) \to H^1(B, A)), \quad \ell = \dim \mathcal{L}.$$

The soluble module bound gives $t, \ell \le B_a$, by applying it to $A^B$ and the actual section $(A/S)^B \cong \mathcal{L}$.

**Lemma 21.3** (Joint character-kernel budget)**.** *Inflate $H$ into $\mathrm{Hom}(T, C_2)$ and fix a point $x$. Let $H_0 = \{\chi \in H : \chi(T_x) = 0\}$. For $W \le H$ put $j = \dim(W \cap H_0)$. Then*

$$t + \dim(\mathcal{L} \cap (W \otimes S)) \le 2^j B_{a-j}. \tag{21.2}$$

*Proof.* Let $K_W = \bigcap_{\chi \in W} \ker \chi$ in $T$. Its number of orbits is $[T : K_W T_x] = 2^j$, since the annihilator of the image of $T_x$ in $T/K_W$ is $W \cap H_0$. The orbit lengths are $2^{a-j}$. The preimage in $A$ of the connecting-map slice $\mathcal{L} \cap (W \otimes S)$ has dimension equal to the left side of (21.2). The group $K_W$ fixes it: the corresponding cocycles into the trivial module $S$ vanish on $K_W$. Filter this actual trivial section by the $K_W$-orbit modules and apply the soluble bound $B_{a-j}$ to each. This proves the inequality. □

**Theorem 21.4** (Binary capacity from degree 64)**.** *For $a \geq 5$ there is $C \leq S$ with*

$$2 \dim C + 4 \dim (A/C)^B \leq 2B_a + 2B_{a-1}. \tag{21.3}$$

*For $a \geq 6$ the right side is at most $15 \cdot 2^a/16$.*

*Proof.* A nonzero character has one or two kernel orbits, so Lemma 21.3 gives $j_0 \leq J = 2B_{a-1} - t$, using $B_a \leq 2B_{a-1}$. Apply Lemma 21.2 and let $C$ be the common kernel in $S$ of the selected functionals. Then $\dim C = t - q$ and $\mathcal{L} \cap (H \otimes C) = 0$. Equation (5.4) gives $\dim (A/C)^B = q$, so the cost is $2t + 2q$. If $J \geq 1$, this is at most $2t + \ell + J = t + \ell + 2B_{a-1} \leq 2B_a + 2B_{a-1}$. If $J = 0$, then $t = 2B_{a-1} \leq B_a$. The binomial identities imply $a$ even and $t = B_a = 2B_{a-1}$; this central binomial coefficient is even. Therefore $q \leq \lceil \ell/2 \rceil \leq B_a/2$, giving the same result. Finally, for $a \geq 6$, both central-binomial proportions $B_a/2^a$ and $B_{a-1}/2^{a-1}$ are at most $5/16$. □

**Theorem 21.5** (The degree-32 boundary)**.** *If $a = 5$, there is $C \leq S$ with $2 \dim C + 4 \dim (A/C)^B \leq 30$.*

*Proof.* Here $t, \ell \leq 10$ and $j_0 \leq 12 - t$. The separator gives cost at most $t + \ell + 12 \leq 32$. Unless $t = \ell = 10$, it is at most 31 and hence at most 30, since the cost is even.

Assume $t = \ell = 10$. Put $H_0$ as in Lemma 21.3 and choose a complement $H = H_0 \oplus H_1$. For

$$(M_0, M_1, M_2, M_3, M_4, M_5) = (10, 12, 12, 16, 16, 32)$$

the joint budget is $\dim (\mathcal{L} \cap (W \otimes S)) \leq M_j - 10$. In particular $\mathcal{L} \cap (H_1 \otimes S) = 0$, so projection embeds $\mathcal{L}$ into a ten-dimensional space $\overline{\mathcal{L}} \leq H_0 \otimes S$. For $W_0 \leq H_0$,

$$\dim (\overline{\mathcal{L}} \cap (W_0 \otimes S)) \leq M_{\dim W_0} - 10.$$

We have $\dim H_0 \leq 5$ from the orbit count; taking $W_0 = H_0$ forces equality, because $M_j - 10 \leq 6$ for $j \leq 4$.

Let $R_q$ be the number of nonzero tensors of rank $q$ in $\overline{\mathcal{L}}$, $1 \leq q \leq 5$. Double count containment of their row support in $r$-spaces of $H_0$. The instances $r = 2, 4$ give the simultaneous constraints

$$\begin{aligned} 15R_1 + R_2 &\leq 465, \\ 15R_1 + 7R_2 + 3R_3 + R_4 &\leq 1953, \\ R_1 + R_2 + R_3 + R_4 + R_5 &= 1023. \end{aligned} \tag{21.4}$$

Indeed the exact double count is $\sum_{\dim W = r} (2^{\dim (\overline{\mathcal{L}} \cap (W \otimes S))} - 1) = \sum_q R_q \begin{bmatrix} 5-q \\ r-q \end{bmatrix}_2$.

For a uniform five-space $C \leq S = \mathbb{F}_2^{10}$, the probability of containing a fixed $q$-dimensional image is

$$p_q = \frac{\begin{bmatrix} 10-q \\ 5-q \end{bmatrix}_2}{\begin{bmatrix} 10 \\ 5 \end{bmatrix}_2}, \qquad (p_1, \ldots, p_5) = \left( \frac{1}{33}, \frac{5}{5621}, \frac{1}{40953}, \frac{1}{1733677}, \frac{1}{109221651} \right).$$

Set $z = p_5$, $y = (p_3 - p_5)/3$, and $x = (p_1 - 15y - z)/15$. These are positive, and the coefficient vector of $x$ times the first row of (21.4), $y$ times the second, and $z$ times the third dominates $(p_1, \dots, p_5)$ coordinatewise. The slacks in coordinates two and four are respectively 1932416/1638324765 and 2480/327664953; the other three are equalities. Consequently

$$\sum_q p_q R_q \le 465x + 1953y + 1023z = \frac{103925485}{109221651} < 1. \tag{21.5}$$

Projection from $H$ to $H_0$ never increases tensor rank, and is injective on $\mathcal{L}$. Since containment probability decreases with rank, the expected number of nonzero tensors of the original $\mathcal{L}$ contained in $H \otimes C$ is also less than one. This number is an integer, so some $C$ has zero intersection. Equation (5.4) then gives $\dim(A/C)^B = 10 - 5 = 5$, of cost $10 + 20 = 30$. □

Every simple binary module for a 2-group is trivial. Hence these costs are precisely $2 \dim C + 4r(A/C)$. A binary transitive pair action of width $2s \ge 64$ therefore has an entry certificate for every normal axis, with $g \ge s/16$, by Proposition 6.3. To sum over all widths one must also bound the number of actions, axes, pair systems, cuts, and constants $D_C$ uniformly. The finite-menu theorem alone does not perform that unbounded sum.

## 22 Reversible simultaneous quotient transport

The construction below permits nonabelian common quotients and proper subdirect replacement groups. These are actual group maps, rather than identifications of abstract quotient types.

**Proposition 22.1** (Simultaneous reconstruction)**.** *Let $H \le \prod_i U_i$ be full subdirect. For selected coordinates let $N_i = H \cap U_i$ be the exact embedded coordinate axis, and specify maps*

$$\alpha_i : U_i \twoheadrightarrow Q_i = U_i/N_i, \qquad \beta_i : P_i \twoheadrightarrow Q_i, \qquad P_i \le \prod_{V \in W_i} V$$

*with $P_i$ full on the displayed word $W_i$. Use identity maps on the untouched coordinates. Set $\alpha = \prod_i \alpha_i$, $\beta = \prod_i \beta_i$, and*

$$\overline{H} = \alpha(H), \qquad H_{\mathrm{new}} = \beta^{-1}(\overline{H}). \tag{22.1}$$

*Then $H_{\mathrm{new}}$ is full on every replacement factor. The decorated map is injective, with inverse*

$$H = \alpha^{-1}(\beta(H_{\mathrm{new}})). \tag{22.2}$$

*Proof.* The embedded $N_i$ commute and each is contained in $H$. Hence $\ker \alpha = \prod_i N_i \le H$, proving $H = \alpha^{-1}(\overline{H})$. The projection of $\overline{H}$ onto each $Q_i$ is full. Any element of $P_i$ has an image appearing in some element of $\overline{H}$; surjectivity of the other $\beta_j$ permits a simultaneous lift. Thus the projection of $H_{\mathrm{new}}$ onto the whole $P_i$ is full, which implies fullness on its displayed factors. Surjectivity of $\beta$ gives $\beta(H_{\mathrm{new}}) = \overline{H}$, proving the inverse formula. No step replaces $P_i$ by the full product of its projections. □

**Proposition 22.2** (Bounded decoration fibre)**.** *Suppose the original actions, words, and maps range over fixed finite menus of bounded degree, each replacement has the same physical degree as its original coordinate, and original critical coordinates are untouched. If the original noncritical support has degree $b$, the number of decorated preimages of any target subgroup on $n$ points is at most $(n+2)^{Kb}$ for a menu-dependent constant $K$.*

*Proof.* Realize each replacement on the same labelled points as its original orbit. To reconstruct the decorated block structure, select and group the target orbits into the changed bounded blocks, and record the finitely many local action types, coordinate bijections, normal axes, and map choices. Once this complete decoration is fixed, the fixed-chart physical realization is injective and recovers the simultaneous carrier subgroup. Equation (22.2), followed by the recorded complete point chart, then recovers the original labelled subgroup.

There are $O(b)$ selected orbits, grouping slots, and finite labels. Allowing $n+2$ choices at each slot bounds all these possibilities by $(n+2)^{Kb}$. This counts labelled groups directly, so division by $n!$ retains the original action-normalizer weights and occurrence factorials. □

For the degree-eight/degree-sixteen application, the local construction removes most of this generic decoration before counting. Pulling a certificate to the canonical orbit axis preserves its underlying slot and therefore all of its numerical support data. The retained route also gives an exact chart from that slot's displayed output points to the corresponding original local action. These charts assemble along the canonical orbit chart to a single fusion-natural chart on the original ambient labels, and the resulting supplied-chart physical realization is injective. Thus the proposed mixed record needs only nineteen symbols per changed block: degree tag, least point, route tag, and labelled point representatives. Since at most $b/2$ records occur, twenty available symbols per record would give $K = 10$.

This numerical packing is separate from the variable-word reconstruction step. Equality of the mixed table aligns all changed routes, and the retained route-output charts transport equality of the fusion-natural physical targets back to the native point sets. Indexed full-preimage reflection then recovers the complete correlated word, including quotient-identity slots, while the slot-preserving source reindex and faithfulness recover the two canonical sources. Thus the application below has fibre one without recording an extra point chart.

The original denominator can also be retained directly when only an upper bound is needed; uniqueness of the orbit presentation is unnecessary.

**Lemma 22.3** (Upper assembly with original normalizers)**.** *Fix nonempty finite sets $\Omega_i$, actions $U_i \le S(\Omega_i)$, and multiplicities $m_i$, and let $X$ be the disjoint union of these occurrences, of total size $n$. Put*

$$W = \prod_i (N_{S(\Omega_i)}(U_i) \wr S_{m_i}) \le S(X).$$

*If $\mathcal{P}$ is any family of subgroups of $S(X)$ invariant under conjugation by $W$, then its assembled family $\mathcal{A} = \{gKg^{-1} : g \in S(X),\ K \in \mathcal{P}\}$ satisfies*

$$|\mathcal{A}| \le \frac{n!\,|\mathcal{P}|}{\prod_i |N_{S(\Omega_i)}(U_i)|^{m_i} m_i!}.$$

*Proof.* For every pair $(g, K)$ and every $w \in W$, the pair $(gw, w^{-1}Kw)$ has the same assembled subgroup. It remains in $S(X) \times \mathcal{P}$ by invariance, and distinct $w$ give distinct first coordinates. Every nonempty assembly fibre therefore has at least $|W|$ elements. Summing the fibre sizes gives $|\mathcal{A}||W| \le n!|\mathcal{P}|$. The displayed denominator is the order of the original internal-symmetry group $W$. □

No transitivity, separation of the action types, or fullness of the local subgroups is required for this upper bound. Invariance is a condition on the whole family; individual subgroups need not be fixed.

Here is the quantitative form needed to use such a transport. Suppose a target binary word has critical rank $R$, $a$ regular $C_4$ factors, and noncritical carrier support of half-degree $4T$, so $N = R+2a+4T$. Put $C_{\mathrm{new}} = 2a+4T$ and write the original noncritical support as $2C_{\mathrm{old}}$. Assume each replacement retains at least a fixed proportion $\delta > 0$ of noncritical points. Then

$$\delta C_{\mathrm{old}} \le C_{\mathrm{new}} \le C_{\mathrm{old}}. \tag{22.3}$$

**Proposition 22.4** (Absorbing decorations)**.** *Assume the complete normalized target mass at each profile satisfies the two uniform estimates*

$$\begin{aligned} T \le a/140, \quad a \ge 1: &\quad 2^{-aN/25+O(a\log(N+2))}, \\ T > a/140: &\quad 2^{-(29/656)TN+O(N\log(N+2))}. \end{aligned} \tag{22.4}$$

*With the finite-menu assumptions of Proposition 22.2 and* (22.3)*, the full original family with a noncritical orbit has normalized mass* $2^{-\Omega(N)}$*.*

*Proof.* First consider the target range $0 < C_{\mathrm{new}} \le \sqrt{N}/4$. Theorem 4.2 gives its normalized completed-mixture mass a factor $2^{-N/4}$. By (22.3), $C_{\mathrm{old}} \le C_{\mathrm{new}}/\delta$, so Proposition 22.2 contributes only $2^{O_\delta(\sqrt{N}\log(N+2))}$. Thus this entire decorated range is still exponentially small. In the remaining range $C_{\mathrm{new}} > \sqrt{N}/4$. After summing over original supports the additional decoration factor is $2^{O_\delta(C_{\mathrm{new}}\log(N+2))}$.

If $T \le a/140$, then $C_{\mathrm{new}} < 3a$. This factor is absorbed by the first error in (22.4), and the negative term is at most $-N/25$ before the smaller error. If $T > a/140$, then $C_{\mathrm{new}} < 284T$, so $T > \sqrt{N}/1136$. The negative term in the second envelope has order at least $N^{3/2}$ and absorbs both the displayed error and the decoration cost. There are only polynomially many profiles and support sizes. Their sum retains the exponential gain. □

The inequalities (22.4) are the stronger profile bounds proved for the full twelve-carrier mixture in Theorem 23.15. Multiplying only a final $2^{-\gamma N}$ mixture bound by the decoration factor would be insufficient. An all-critical replacement also fails (22.3), and is outside the negligible-mass conclusion even though the reconstruction map remains valid.

# 23 The complete critical, cyclic-four, and carrier mixture

All groups and all direct-product coordinates in this section are labelled. Write $\mathrm{SD}(G_1, \dots, G_t)$ for the number of subgroups full on every displayed coordinate, and put

$$\begin{aligned} \mathcal{K}_8 &= \{8T18, 8T26, 8T27, 8T28, 8T29, 8T31, 8T35\}, \\ \mathcal{K}_{16} &= \{16T1082, 16T1083, 16T1084, 16T1332, 16T1547\}. \end{aligned}$$

The names specify the literal permutation actions in the finite data. A carrier of degree eight has weight one, and one of degree sixteen has weight two. A carrier word of weight $T$ has degree $8T$. Let $D$ be a critical word of elementary quotient rank $R$.

## 23.1 Three simultaneous parameters on a full projection

For a finite 2-group $H$ set

$$d(H) = \dim H/\Phi(H), \qquad \rho(H) = \max_{M \lhd H,\, M \le H'} d_H(M), \qquad d_H(M) = \log|M : M^2[M,H]|.$$

The invariant $\tau(H)$ is the one in (3.2); the five-term sequence identifies it with $d_H(\Phi(H))$.

**Lemma 23.1** (Joint transition)**.** *Let $K \le L \times G$ be full on both factors and let $N = K \cap G$. Put*

$$k = d_G(N), \quad n = \log|N|, \quad m = \max_{M \lhd G,\, M \le N \cap G'} d_G(M), \quad R_N = N^2[N,G], \quad a_2 = d_G(R_N).$$

*Then $\delta = d(K) - d(L)$ and $\epsilon = \max(0, \rho(K) - \rho(L))$ satisfy*

$$0 \le \delta \le k, \qquad 0 \le \epsilon \le m, \qquad \delta + \epsilon \le n, \qquad \tau(K) - \tau(L) \le k - \delta + a_2. \tag{23.1}$$

*Proof.* The map on Frattini quotients is onto, and $\delta = \log|N : N \cap \Phi(K)|$. Fullness gives $[N,K] = [N,G]$, so $R_N \le N \cap \Phi(K)$ and $\delta \le k$. For $M \lhd K$ with $M \le K'$, let $M_0$ be its projection to $L$ and $M_1 = M \cap N$. Lifting normal generators gives

$$d_K(M) \le d_L(M_0) + d_K(M_1).$$

Here $M_1 \lhd G$, $M_1 \le N \cap G'$, and $d_K(M_1) = d_G(M_1) \le m$. Also $M_1 \le N \cap \Phi(K)$, so $d_K(M_1) \le n - \delta$. Maximization proves the two bounds on $\epsilon$.

The map $\Phi(K) \to \Phi(L)$ is onto. After quotienting by squares and ambient commutators, its kernel is generated by $N \cap \Phi(K)$. Indeed products of squares and commutators in $\Phi(L)$ lift to such products in $\Phi(K)$. Consequently

$$\tau(K) \le \tau(L) + d_K(N \cap \Phi(K)).$$

The quotient $(N \cap \Phi(K))/R_N$ is trivial elementary of dimension $k - \delta$, and the image of $R_N$ has dimension at most $a_2$. This proves the last inequality. □

For a row $r = (k, n, m, a_2, c, g)$ define

$$h_r(x,y) = \max\{x\delta + y\epsilon : 0 \le \delta \le k,\ 0 \le \epsilon \le m, \delta + \epsilon \le n\}, \qquad q_r = \max(m, a_2).$$

The quantity $a_2$ cannot in general be replaced by $m$. The following criterion makes that replacement available for an entire ambient group and reduces the radical certificates to generator tests.

**Lemma 23.2** (Finite radical certificates)**.** *Let $G = \langle s_1, \dots, s_u \rangle$ be a finite 2-group and $N = \langle x_1, \dots, x_v \rangle \lhd G$. Then*

$$R_N = \langle x_i^2,\ [x_i, s_j] : 1 \le i \le v,\ 1 \le j \le u \rangle^G.$$

*If $\Phi(G) \le G'$, then $R_N \le N \cap G'$ and $a_2 \le m$ for every such $N$. If also $G' \le N$, then $m = \rho(G)$ and $G/N$ is abelian.*

*Proof.* Write $U$ for the displayed normal closure. Its generators lie in $R_N$, and $R_N \lhd G$, so $U \le R_N$. In $G/U$, each $x_iU$ has square one and commutes with every $s_jU$, hence with the whole quotient. The image of $N$ is therefore central elementary abelian, which gives $N^2[N,G] \le U$ and proves equality. An ordinary closure can replace this normal closure when its normality in the original $G$ has been certified.

Every square and ambient commutator from $N$ lies in $\Phi(G)$. Under the stated containment this puts the same $G$-normal subgroup $R_N$ inside $N \cap G'$. Its relative head is one of the terms in the maximum defining $m$, so $a_2 \le m$. Finally $G' \le N$ gives $N \cap G' = G'$, proving the remaining assertions. □

**Lemma 23.3** (One centralizer controls all normal heads)**.** *Let $G$ be a finite group, $p$ a prime, $N \lhd G$, and $u \in G$. Write $R = N^p[N, G]$ and $d_{G,p}(N) = \log_p |N : R|$. Then*

$$p^{d_{G,p}(N)} \le |C_N(u)|.$$

*Consequently, for a finite 2-group $G$ and any subgroup $S \le G$,*

$$2^{\max_{M \lhd G,\, M \le S} d_G(M)} \le |C_S(u)|, \qquad 2^{\rho(G)} \le |C_{G'}(u)|.$$

*Neither $N$ nor $G'$ is required to be abelian.*

*Proof.* The set map $f : N \to R$ given by $f(x) = [x, u]$ need not be a homomorphism. If $f(x) = f(y)$, then $y^{-1}x \in C_N(u)$. Choose one representative $y_r$ of each nonempty fibre. The map

$$x \longmapsto (f(x), y_{f(x)}^{-1} x)$$

injects $N$ into $R \times C_N(u)$. Thus $|N| \le |R|\,|C_N(u)|$, proving the first bound after division by $|R|$. Apply it to every original $G$-normal subgroup $M \le S$ and use $C_M(u) \le C_S(u)$. Taking the maximum gives the second bound; putting $S = G'$ gives the last one. □

**Lemma 23.4** (Joint commutator radicals)**.** *Let $G$ be a finite 2-group with $\Phi(G) = D = G'$, and put*

$$V = G/D, \qquad E = D/(D^2[D, G]).$$

*For $\lambda \in E^*$ let $b_\lambda$ be the alternating form on $V$ given by $b_\lambda(xD, yD) = \lambda([x, y])$. For every original normal subgroup $N \ge D$, write $W = N/D$ and*

$$J_W = \{\lambda \in E^* : W \le \operatorname{rad}(b_\lambda)\}.$$

*Then*

$$d_G(N) \le \dim W + \dim J_W. \tag{23.2}$$

*Let $q, a$ be nonnegative integers. If the forms associated with every $q + 1$ independent characters in $E^*$ have common radical of dimension at most $a$, then*

$$\dim W > a \quad \Longrightarrow \quad d_G(N) \le \dim W + q.$$

*In particular, zero common radicals for independent character pairs give $d_G(N) \le \dim W + 1$ for $W \ne 0$. If each nonzero form has radical dimension at most $a$, then $d_G(N) \le \dim W$ whenever $\dim W > a$. The subgroup $D$ need not be abelian.*

*Proof.* Elements of $E^*$ are exactly the $G$-invariant characters of $D$. The commutator identities and this invariance make $\lambda([x, y])$ additive in both original variables. The resulting maps vanish on $G'$ in each variable, so they descend to the displayed alternating forms on $V$.

Restriction of a $G$-invariant character of $N$ to $D$ has kernel the characters of $N/D$, of dimension $\dim W$. Every character in the restriction image annihilates $[N, G]$, so that image lies in $J_W$. Rank–nullity proves (23.2). This uses only a necessary condition on the restriction image; no assertion that every member of $J_W$ extends is needed.

If $\dim J_W > q$, choose $q + 1$ independent characters in $J_W$. The same $W$ lies in all their radicals, contradicting $\dim W > a$. Thus $\dim J_W \le q$. The two special cases follow by taking $(q, a) = (1, 0)$ and $q = 0$, respectively. □

**Corollary 23.5** (Star commutator family)**.** *In the setting of Lemma 23.4, suppose that there are linear identifications $V = U \oplus \mathbb{F}_2$ and $E^* = U^*$ under which the actual commutator forms are*

$$b_\lambda((u,t),(v,s)) = \lambda(u)s + t\lambda(v).$$

*Then every original normal subgroup $N \geq D$ satisfies*

$$d_G(N) \leq \max\{\dim U, \dim(N/D)\}.$$

*Proof.* Put $W = N/D$. If $W$ contains $(u,t)$ with $t \neq 0$, then every $\lambda \in J_W$ satisfies $t\lambda(v) = 0$ for all $v \in U$, by testing against $(v,0)$. Thus $J_W = 0$. Otherwise $W \leq U \oplus 0$, and first projection identifies $W$ with a subspace $S \leq U$. Testing against $(0,1)$ shows that $J_W = S^\perp \leq U^*$, so $\dim W + \dim J_W = \dim U$. Both cases give the result through (23.2). The required identifications concern the complete original character space and the actual quotient; ranks of selected numerical forms alone do not supply them. □

**Lemma 23.6** (Derived-intersection directions)**.** *Keep $G, D, V, E$ as in Lemma 23.4, but let $N \lhd G$ be arbitrary. Put*

$$B = N \cap D, \qquad W = ND/D, \qquad L_B = \{\lambda \in E^* : \lambda|_B = 0\}.$$

*Then $L_B \leq J_W$ and $|N| = |B|2^{\dim W}$. If $B < D$, then $L_B \neq 0$. Consequently, a radical bound $a$ for every nonzero $b_\lambda$ gives $\dim W \leq a$. If independent pairs have zero common radical, then $\dim L_B \geq 2$ implies $N \leq D$. For the star family of Corollary 23.5, $L_B \neq 0$ gives the sharper bound*

$$\dim W + \dim L_B \leq \dim U.$$

*Proof.* Original normality gives $[N,G] \leq N \cap D = B$, so every character in $L_B$ annihilates all commutators from $W$. The second isomorphism theorem identifies $W$ with $N/B$, proving the order identity. If $B < D$, choose a maximal proper $G$-normal subgroup $J$ of $D$ containing $B$. Then $D/J$ is a minimal nontrivial normal subgroup of the finite 2-group $G/J$, and hence has order two and is central. Its nonzero character gives a member of $L_B$.

The single-form bound follows by choosing a nonzero member of $L_B$. Two independent members force $W = 0$ under the stated pair hypothesis. For the star family, a nonzero member of $L_B$ forces every vector of $W$ to have last coordinate zero. First projection identifies $W$ with a subspace of the common annihilator of $L_B$ in $U$, whose dimension is $\dim U - \dim L_B$. This proves the last inequality. These are constraints from characters of $D$ vanishing on $B$; the extendibility and power constraints on characters of $B$ remain separate. □

**Lemma 23.7** (Exact annihilator and center detection)**.** *Keep $G, D, V, E$ as above and write $R = D^2[D,G]$. For every subgroup $B \leq D$, the common kernel in $D$ of $L_B$ is $BR$. If $R \leq B$, put*

$$C_B = \{x \in G : [x,G] \leq B\}, \qquad U_B = \bigcap_{\lambda \in L_B} \operatorname{rad}(b_\lambda).$$

*Then $B \lhd G$, $D \leq C_B$, and the original quotient map induces*

$$C_B/D \cong U_B, \qquad |C_B| = |D|2^{\dim U_B}.$$

*Suppose in addition that independent character pairs have zero common radical and that $N \lhd G$ satisfies neither $N \leq D$ nor $D \leq N$. For $B = N \cap D$, the space $L_B$ has a unique nonzero member $\lambda$. If $R \leq B$, then, with $w = \dim(ND/D)$ and $r = \dim \operatorname{rad}(b_\lambda)$,*

$$|D : B| = 2, \qquad 1 \leq w \leq r, \qquad |Z(G/N)| = 2^{r+1-w}, \qquad |(G/N)'| = 2. \tag{23.3}$$

*Proof.* The invariant characters of $D$ are the complete dual of $D/R$. The ordinary double-annihilator identity in this vector space says that their common kernel on $B$ is the inverse image of $BR/R$, which is $BR$. This does not identify $B$ itself unless $R \le B$.

Under that containment, $B/R$ lies in the central elementary subgroup $D/R$ of $G/R$, so $B$ is normal in $G$. For every original $x, y \in G$, detection gives $[x, y] \in B$ exactly when $\lambda([x, y]) = 0$ for every $\lambda \in L_B$. Thus $x \in C_B$ exactly when $xD \in U_B$. In particular $D \le C_B$, and the asserted equivalence and order formula follow from the original quotient map.

In the crossing case, Lemma 23.6 gives $L_B \neq 0$ and $W = ND/D \neq 0$. Two independent members of $L_B$ would force $W = 0$, so $\dim L_B = 1$. Its unique nonzero member detects $BR$; when $R \le B$, this is its literal kernel $B$ in $D$, of index two. Also $W \le \mathrm{rad}(b_\lambda)$, so $1 \le w \le r$. Since $|N| = |B|2^w$ and $Z(G/N) = C_B/N$, the center formula follows. Finally the original derived group maps onto $(G/N)'$ with kernel $B$, proving its order. □

**Lemma 23.8** (Saturation and exact crossing heads)**.** *Keep the notation of Lemma 23.7. Suppose* $\dim E \ge 2$*, independent character pairs have zero common radical, and every original $G$-normal subgroup $M \le D$ satisfies*

$$M \le R \quad \text{or} \quad R \le [M, G].$$

*Then every crossing normal $N$ satisfies, with $B = N \cap D$,*

$$[N, G] = B = N^2[N, G], \qquad d_G(N) = \dim(ND/D).$$

*For every original normal $B \le D$ with $B \not\le R$,*

$$B^2[B, G] = R, \qquad \max_{M \lhd G,\, M \le B} d_G(M) = \max\left\{ \max_{M \lhd G,\, M \le R} d_G(M), d_G(B) \right\},$$

*where $d_G(M) = \dim \mathrm{Hom}(M, \mathbb{F}_2)^G$.*

*Proof.* Put $H = [N, G] \le B$. If $H \le R$, every character of $D$ annihilates $H$, so every form annihilates $W = ND/D$ against $V$. Two independent characters would then force $W = 0$, a contradiction. Thus saturation gives $R \le H$. The full space $L_H$ is the joint annihilator of $W$, and the pair hypothesis makes its dimension at most one. Since $0 \neq L_B \le L_H$, these spaces are equal. Exact detection now gives $H = B$. Every original square in $N$ lies in $N \cap D$, since $G/D$ has exponent two, proving the relative radical identity and the head formula.

If $M \le D$ and $M \not\le R$, saturation and monotonicity of the relative radical give $R \le [M, G] \le M^2[M, G] \le R$. For $M \le B$ outside $R$, the resulting quotient inclusion $M/R \le B/R$ gives $d_G(M) \le d_G(B)$. Subgroups inside $R$ contribute the other maximum. Both terms are attained within $B$, which proves equality without enumerating its normal subgroups. □

For the original pair-family masters $16T1082$, $16T1083$, $16T1084$ and $16T1547$, the saturation hypothesis is certified by original commutator words on their 16 or 64 derived rows. Every word has positive commutator depth, so an element of a normal subgroup outside $R$ generates all of $R$ through its original ambient commutators. The same finite certificates give saturation for $16T1332$, whose star family is treated separately from the pair hypothesis. For the four pair-family masters, every nonzero form has radical dimension two, $d_G(D) = 3$, and $\max_{M \lhd G,\, M \le R} d_G(M) \le 2$. Consequently every crossing normal has exactly one of the following rows, in the order $(k, n, m, a_2, c, g)$:

| $G$ | $w = 1$ | $w = 2$ |
|---|---|---|
| $16T1082$, $16T1083$, $16T1084$ | $(1, 4, 2, 2, 2, 1)$ | $(2, 5, 2, 2, 1, 1)$ |
| $16T1547$ | $(1, 6, 2, 2, 2, 1)$ | $(2, 7, 2, 2, 1, 1)$. |

Indeed $B$ has index two in $D$, its relative radical is $R$, and $d_G(B) = 2$; hence $m = a_2 = 2$. The remaining coordinates follow from Lemma 23.7 and $|N| = |B|2^w$. These are exhaustive possibilities for crossing normals; no realization of every listed row or identification of different normals is required.

For $16T1547$, the remaining normals inside $R$ also admit a uniform argument. Here $|R| = 8$, and the original commutator tests on its eight rows give a central subgroup $Z$ of order two with $R^2[R, G] = [R, G] = Z$. Every original normal $M \le R$ either lies in $Z$ or contains $Z$; if $M \not\le Z$, the same saturation words give $M^2[M, G] = Z$. Thus $M$ is 1, $Z$, an order-four subgroup above $Z$, or $R$. This argument requires no list or count of the order-four subgroups. Their exact rows are

| $M$ | $(k, n, m, a_2, c, g)$ |
|---|---|
| $1$ | $(0, 0, 0, 0, 1, 6)$ |
| $Z$ | $(1, 1, 1, 0, 2, 5)$ |
| $Z < M < R$ | $(1, 2, 1, 1, 1, 4)$ |
| $R$ | $(2, 3, 2, 1, 3, 3)$. |

For completeness, the original form separation first gives $C_M \le D$. If $M < R$, saturation strengthens this to $C_M \le R$. The identity $[R, G] = Z$ then gives $C_M = R$ for $Z \le M < R$; the eight-row fixed tests give $C_1 = Z$, and $C_R = D$. These are the center orders in the table. The derived orders are $|D|/|M|$, and the maximum defining $m$ ranges over every original normal below $M$, using the same relative-radical identities.

**Lemma 23.9** (Crossing capacities for the full star family)**.** *Keep $G, D, V, E$ as above, and suppose the complete original form family is the star family on $V = U \oplus \mathbb{F}_2$, with $E^* = U^*$ and $d = \dim U$. Assume every original normal subgroup of $D$ is comparable with $R$. For a crossing normal $N$, put $B = N \cap D$, $H = [N, G]$, $w = \dim(ND/D)$ and $g = \dim L_B$. Then*

$$R \le H \le B, \qquad d_G(N) + g \le d, \qquad |(G/N)'| = 2^g, \qquad |Z(G/N)| = 2^{d-w}.$$

*Proof.* The complete space $L_H$ is exactly the joint annihilator of $W = ND/D$. Since it contains the nonzero space $L_B$, the star formula forces $W \le U \oplus 0$. Its first projection identifies $L_H$ with the annihilator of a $w$-dimensional subspace of $U$, so $\dim L_H = d - w$. If $H \le R$, then $L_H = E^*$, forcing $w = 0$; comparability therefore gives $R \le H$. Exact character detection and the order identity now give

$$|D| = 2^{d-w}|H| = 2^g|B|, \qquad |N| = 2^w|B|.$$

All original powers remain in the relative radical $N^2[N, G]$, which contains $H$. Hence $2^{d_G(N)}|H| \le |N|$, and the displayed identities yield $d_G(N) + g \le d$.

The full common radical of $L_B \ne 0$ is its annihilator in $U \oplus 0$, of dimension $d - g$. Thus $|C_B| = |D|2^{d-g}$ and $|Z(G/N)| = |C_B|/|N| = 2^{d-w}$. The image of $D$ in $G/N$ has order $|D : B| = 2^g$. This proof uses neither a one-dimensional parameter space nor an equality between $H$ and the full relative radical of $N$. □

**Lemma 23.10** (The remaining original normal branches)**.** *For each of the five degree-sixteen masters put $D = G'$, $R = D^2[D, G]$, $d = \log|D|$, $s = \log|G/D|$ and $e = d_G(D)$. Their parameters are*

| $G$ | $d$ | $s$ | $e$ | $\log\lvert R\rvert$ |
|---|---|---|---|---|
| $16T1082$, $16T1083$, $16T1084$ | 4 | 6 | 3 | 1 |
| $16T1332$ | 6 | 5 | 4 | 2 |
| $16T1547$ | 6 | 6 | 3 | 3. |

*If $D \le N$, with $w = \dim(N/D)$, then $0 \le w \le s$ and the six invariants are bounded coordinatewise by*

$$(\max(e, w), d + w, e, e, s - w, 0).$$

*Here the order, the maximum $m = e$, and both quotient slopes are exact. For $R < N < D$ the exhaustive exact alternatives are*

| $G$ | $(k, n, m, a_2, c, g)$ |
|---|---|
| $16T1082$, $16T1083$, $16T1084$ | $(1, 2, 1, 1, 2, 2)$, $(2, 3, 2, 1, 3, 1)$ |
| $16T1547$ | $(1, 4, 2, 2, 2, 2)$, $(2, 5, 2, 2, 3, 1)$ |
| $16T1332$ | $(t, 2 + t, t, 1, 4, 4 - t)$, $\quad t \in \{1, 2, 3\}$. |

*Inside $R$, the first three masters have the two exact rows $(0, 0, 0, 0, 1, 4)$ and $(1, 1, 1, 0, 3, 3)$. For $16T1332$ there is a central subgroup $Z$ of order two, and every original normal inside $R$ is $1, Z$ or $R$, with respective rows*

$$(0, 0, 0, 0, 1, 6), \qquad (1, 1, 1, 0, 1, 5), \qquad (1, 2, 1, 1, 4, 4).$$

*Together with the four small-radical rows of $16T1547$ and the crossing results above, these alternatives cover every original normal subgroup.*

*Proof.* For $D \le N$, the image dimension gives $|N| = 2^{d+w}$ and the abelian quotient has order $2^{s-w}$. The original derived-normal maximum is $e$: its upper bound follows from the checked original centralizer ranks, and $D$ itself attains the lower bound. The full-form head estimate gives $d_G(N) \le \max(e, w)$, while $N^2[N, G] \le N \cap D$ gives $a_2 \le m = e$.

For $R < N < D$, nonempty original commutator saturation gives $N^2[N, G] = R$. Put $t = d_G(N)$ and $\ell = \dim L_N$. Then $t, \ell > 0$, $t + \ell = e$, $|N| = |R|2^t$, and $m = \max(m(R), t)$. In a pair family, $e = 3$; the full common radical of $L_N$ has dimension zero when $\ell = 2$, and dimension two when $\ell = 1$. Lemma 23.7 gives $c = \ell + \dim U_N$ and $g = \ell$. In the complete star family the nonzero full parameter space $L_N$ has common radical dimension $4 - \ell = t$, so $c = 4$ and $g = 4 - t$. The excluded case $\ell = 0$ is $N = D$ and belongs to the first branch.

For the first three masters, $|R| = 2$ makes $R$ central, its relative radical trivial, and its head and normal-head maximum both one. Full pair-form separation and saturation give $C_1 = R$ and $C_R = D$, proving the two small rows. For $16T1332$, the four original radical rows, with all five original generators acting, give $R^2[R, G] = [R, G] = Z$, with $|R| = 4$ and $|Z| = 2$. Every original normal $M \le R$ lies in $Z$ or contains $Z$; outside $Z$ it has relative radical $Z$ and head at most one. Cardinality therefore forces $M = R$. The same tests and full star separation give $C_1 = Z$, $C_Z = R$, and $C_R = D$, yielding the three rows and $m(R) = d_G(R) = 1$.

Finally split an arbitrary $N \lhd G$ first by $D \le N$, then by $N \le D$. In the latter case saturation makes $N$ comparable with $R$, giving either $N \le R$ or $R < N < D$. The remaining case is crossing. This proof never counts normals by counting their possible rows and does not assume that every displayed possibility is realized. □

There is a useful simplification when comparing these rows numerically. The feasible pairs satisfy $\delta \le k$, $\epsilon \le m$ and $\delta + \epsilon \le n$, so their polygon depends on $n$ only through $\min(n, k + m)$. Thus a comparison row $(K, N_0, M, A, C, G_0)$ dominates all pair supports and both marks whenever

$$k \le K, \quad m \le M, \quad a_2 \le A, \quad \min(n, k + m) \le N_0, \quad c \le C, \quad g \le G_0.$$

This comparison preserves the original order $n$ in the group-theoretic identities. It simply includes its entire feasible polygon in the comparison polygon. In particular it remains valid for a negative first mark, with a nonnegative terminal multiplier.

For $Q = G/N$ take $(c, g) = (\log|Q|, 0)$ if $Q$ is abelian, and $(c, g) = (\log|Z(Q)|, \log|Q'|)$ otherwise. For a full subdirect product $L$ of the bounded carrier factors,

$$|\operatorname{Epi}(L, Q)| \le C(d(L)+2)^C 2^{cd(L)+g\rho(L)}. \tag{23.4}$$

The following elementary bound proves (23.4) for every finite 2-group source, without a restriction on its direct-product factors. Its constant is absolute on the fixed target alphabet.

**Proposition 23.11** (Structured epimorphisms)**.** *Let $G, Q$ be finite 2-groups, and put $d = d(G)$, $\rho = \rho(G)$ and $q = \log|Q|$. Then*

$$|\operatorname{Epi}(G, Q)| \le (d+1)^q\,|Q'|^\rho\,|Q|^{|Q'|q}\,|Z(Q)|^d. \tag{23.5}$$

*In particular,* (23.4) *holds with*

$$C = C_Q = \max\{q, |Q|^{|Q'|q}\} \ge 1, \qquad c = \log|Z(Q)|, \qquad g = \log|Q'|.$$

*These choices also give the stated abelian case.*

*Proof.* Choose an actual generating tuple of $G$ of length $d$, by lifting a basis of $G/\Phi(G)$. For every epimorphism to $Q$, at most $q$ of these same generators have images generating $Q$: choose a basis from their images in $Q/\Phi(Q)$ and lift back to the selected indices. Padding by identities gives at most $(d+1)^q$ ordered tuples. Fix one such tuple $x = (x_1, \dots, x_q)$ and count the maps $f$ for which $f(x)$ generates $Q$.

Set $D = [\langle x\rangle^G, G] \le G'$ and $H = D \cap \ker f$. These are normal subgroups of the original $G$, and $f(D) = Q'$, so $[D : H] = |Q'|$. There are at most $|Q'|^\rho$ possible $H$. Indeed, for every $M \lhd G$ inside $G'$, the number of $G$-normal subgroups of index at most $2^a$ in $M$ is at most $2^{a\rho}$. To see this, every proper normal interval in a finite 2-group admits a bottom step of index two: choose a central involution in its nontrivial normal quotient. Each normal $J \le M$ has at most $2^{d_G(J)} - 1 \le 2^\rho - 1$ normal index-two children, since each is the kernel of a nonzero $G$-invariant character of $J$. Keeping the previous subgroups as well as these children multiplies the cumulative count by at most $2^\rho$ at each step, starting with the single subgroup $M$.

Now fix $H$ and let $C$ be the inverse image in $G$ of the centralizer of the tuple $xH$ in $G/H$. The conjugates of that tuple lie coordinatewise in $x_i D/H$, so

$$[G : C] \le |D/H|^q = |Q'|^q.$$

Every counted $f$ sends $C$ into $Z(Q)$, because it kills $H$ and $f(x)$ generates $Q$. Thus $G \to Q/Z(Q)$ is constant on left $C$-cosets, giving at most $|Q|^{|Q'|q}$ quotient maps. Normality of $C$ is not required. In a fixed quotient-map fibre, choose one map $f_0$ when the fibre is nonempty. The pointwise differences $f(g)f_0(g)^{-1}$ are central and form a homomorphism $G \to Z(Q)$; they determine $f$. There are at most $|Z(Q)|^d$ such homomorphisms, by the original generating tuple. Empty fibres contribute zero. Multiplying the tuple, normal-subgroup, quotient-map and central-fibre bounds proves (23.5). Finally $(d+1)^q \le (d+2)^{C_Q}$ and $|Q|^{|Q'|q} \le C_Q$. □

## 23.2 The finite rows and the recurrence

Use the masters $X = 8T26$, $J = 8T27$, $P = 8T35$ and the five degree-sixteen carriers. The finite normal-kernel certificate supplies their actual rows and their multiplicities. It includes every literal

normal kernel, by the central-involution closure argument in Lemma 24.5 below. The numerical reduction stated and certified in Appendix B gives the matrix

$$B = \frac{1}{64}\begin{pmatrix} 5 & 6 & 34 & 28 & 32 & 30 \\ 6 & 0 & 44 & 36 & 40 & 36 \\ 34 & 44 & 0 & 16 & 20 & 40 \\ 28 & 36 & 16 & 12 & 16 & 32 \\ 32 & 40 & 20 & 16 & 16 & 32 \\ 30 & 36 & 40 & 32 & 32 & 24 \end{pmatrix}. \tag{23.6}$$

$$\alpha = (5/8, 3/4, 0, 1/4, 1/4, 1/2), \quad \beta = (3/8, 1/2, 0, 1/4, 1/4, 1/2).$$

Here rows have physical scale $s_r = 1$ or 2. Before domination their pair coefficient is

$$\frac{\max\{h_r(c_s, g_s), h_s(c_r, g_r)\}}{s_r s_s},$$

and their marked coefficients are $k_r/s_r, q_r/s_r$. Each dominated row is moved to a row dominating all these coefficients simultaneously. Finally scale its accumulated mass by four, obtaining $p_i \geq 0$ with $\sum_i p_i = 4T$; this accounts for the factor $1/64$ in (23.6).

The crossing branch of $16T1332$ can be assigned directly to the sixth colour, without distinguishing its six possible pairs $(g, w)$. Indeed Lemma 23.9 and saturation give the coordinatewise upper row

$$(4-g,\ 6-g+w,\ \max(2, 4-g),\ \max(2, 4-g),\ 4-w,\ g), \qquad g, w \geq 1, \quad g+w \leq 4.$$

All these rows lie below $r_* = (3, 8, 3, 3, 3, 3)$ at scale two. For any displayed row $s = (k, n, m, a_2, c, g)$, with nonnegative slopes,

$$\max\{h_s(3,3), h_{r_*}(c, g)\} \leq \max\{3\min(n, k+m),\ 3(c+g)\}.$$

Checking this last expression on the forty-one displayed rows of Appendix B, with their stated colours $j$ and scales $s_s$, gives the bound $8(4s_s)B_{6j}$ in every column. The self support is at most $18 \leq 64B_{66} = 24$, and both marks are $3 \leq 8\alpha_6 = 8\beta_6 = 4$. This handles all crossing normals of that master at once, while retaining their separate original weights. The remaining branches are covered by Lemma 23.10. Its effective-polygon comparison sends every pair-family row, every above-derived row, and the small-radical rows into the twelve displayed H16 rows. The first two star intermediate rows go to H16 rows 2 and 4. The last star intermediate row is $r_\dagger = (3, 5, 3, 1, 4, 1)$, at scale two, and can also be assigned directly to colour six. Indeed

$$\max\{h_s(4,1), h_{r_\dagger}(c, g)\} \leq \max\{4k+m,\ 3(c+g)\}.$$

The forty-one displayed comparisons satisfy the same bound $8(4s_s)B_{6j}$. Its self support is at most $15 \leq 24$ and its pair support with $r_*$ is at most $18 \leq 24$; both marks are at most 4. Thus these two extra envelopes use the existing sixth colour and do not change the cone matrix or its physical normalization. No step replaces the original weighted normal family by a sum over distinct numerical labels.

**Lemma 23.12** (The complete marked recurrence)**.** *For every such master word, the logarithm of its full-subdirect count with $D$ is at most the maximum, over $0 \leq j \leq R$, $0 \leq \ell \leq R/2$ and its row masses, of*

$$\mathcal{E} = \ell(R/2 - \ell) + (j-\ell)(R-j) + \tfrac{1}{2}p^{\mathsf{T}}Bp + \sum_i p_i\{\ell(\alpha_i + \beta_i) + \alpha_i \max(j - 2\ell, 0)\}, \tag{23.7}$$

*plus $O((R+T)\log(R+T+2))$.*

*Proof.* Project first onto the complete carrier subgroup $H$. For this same $H$, Proposition 3.4 gives the sum of the marks $2^{(j-\ell)d(H)+\ell\tau(H)}$, with coefficient

$$\phi^{-2}\begin{bmatrix} c \\ \ell \end{bmatrix}_2 72^{\ell} 2^{(j-\ell)(R-j)}, \qquad c \le R/2.$$

For one carrier peel the lower projection $L$, literal axis $N$ and literal epimorphism $L \to G/N$ recover the subgroup uniquely. Thus (23.4) is used once, with no automorphism divisor. Lemma 23.1 shows that an arbitrary current mark $2^{xd+y\rho+z\tau}$, for $y, z \ge 0$, incurs at most

$$z(k+q_r) + h_r(x-z, y).$$

This remains true for negative $x$. For the initial mark $(x, y, z) = (j-\ell, 0, \ell)$ it is at most $\ell(k+q_r) + k\max(j-2\ell, 0)$. Every later epimorphism contributes the nonnegative tilt $(c_s, g_s)$. Subadditivity of the support function bounds the additional cost of an unordered pair of peels by the maximum in the definition of the pair coefficient. Summing pairs gives at most the quadratic term $\frac{1}{2}p^{\mathsf{T}}Bp$; adding diagonal self-pairs only enlarges it. The simultaneous row domination gives the last term of (23.7). Finally use $\begin{bmatrix} c \\ \ell \end{bmatrix}_2 \le \phi^{-1}2^{\ell(c-\ell)}$.

There are a fixed number of normal choices per coordinate, hence at most $2^{O(T)}$ histories. The polynomial factors in (23.4) cost $O(T\log(T+2))$; $72^{\ell}$, the Gaussian constants and the outer $(j, \ell)$ sum cost $O(R + \log(R+2))$. This proves the stated uniform error. The initial projection $H$ is recovered intrinsically, so the argument attaches its critical factors exactly once. □

**Theorem 23.13** (Critical–carrier reserve)**.** *Uniformly over all original carrier colour words of weight $T \ge 1$,*

$$\log \mathrm{SD}(D, \boldsymbol{G}) \le \frac{(R+4T)^2}{4} - \frac{25}{82}RT - \frac{29}{164}T^2 + O((R+T)\log(R+T+2)). \tag{23.8}$$

*The sum of their original action weights is*

$$A_T = [z^T]\exp\left(\frac{3}{64}z + \frac{65}{688128}z^2\right). \tag{23.9}$$

*Proof.* Appendix B proves, for every point of the two exhaustive cones $j \ge 2\ell$ and $j \le 2\ell$,

$$\mathcal{E} \le R^2/4 + \frac{139}{328}R\sum_i p_i + \frac{627}{2624}\left(\sum_i p_i\right)^2.$$

With $\sum_i p_i = 4T$, this is (23.8) for masters. There are fixed epimorphisms

$$X \to 8T26, \qquad J \to 8T27, 8T28, \qquad P \to 8T18, 8T29, 8T31, 8T35.$$

The inverse image under their product, acting identically on $D$ and all degree-sixteen factors, injects each original full-subdirect family into its master family. Its image recovers the original subgroup, and fullness of each replaced coordinate follows from surjectivity. This proves the bound for each original word.

The degree-eight normalizers have orders 192 for $8T18$, 384 for $8T31$, and 128 for the other five; their reciprocals sum to $3/64$. The degree-sixteen orders, in the displayed order, are $344064, 49152, 49152, 24576, 98304$, whose reciprocals sum to $65/688128$. Summing $\prod_U 1/(a(U)^{m_U}m_U!)$ proves (23.9). Each injection is used profile by profile with its original weight. Overlap between the images of different profiles causes no identification of the original colours. □

### 23.3 Adding every cyclic-four orbit

Let $a$ be the number of regular $C_4$ orbits, let $T$ be the carrier weight and put $N = R + 2a + 4T$. The exact original profile weight is

$$\frac{c_R}{8^a a!} A_T, \qquad c_R = [y^R] \exp(y/2 + y^2/6 + y^4/384). \tag{23.10}$$

**Lemma 23.14** (Cyclic-four comparison and Hall bound)**.** *For any finite 2-group $B$ of order $2^u$,*

$$\mathrm{SD}(D, C_4^a, B) \le \sum_{k=0}^{2a} \binom{2a}{k} \mathrm{SD}(D, C_2^k, B), \tag{23.11}$$

$$\log |\operatorname{Sub}(C_2^R \times C_4^a \times B)| \le \frac{u^2 + (R+a+u)^2 + a^2}{4} + O((a+u)\log(a+u+2) + \log(R+2)). \tag{23.12}$$

*In the first inequality $B$ may be a displayed word, with fullness required on each of its factors.*

*Proof.* For a central extension by $C_2$, the subgroups above a fixed subgroup of the quotient consist of its unique full preimage and its complements to $C_2$. The latter are absent or form a torsor of size $|\operatorname{Hom}(H, C_2)|$. The split extension realizes the upper bound exactly. Apply this successively to the central involutions in the $C_4$ coordinates, retaining the projection to $D \times B$ throughout. After dropping fullness only on these coordinates, the comparison group is $C_2^{2a}$. Its uniquely determined set of nonzero coordinate projections has size $k$, giving (23.11). For the terminal upper bound, this support decomposition is unnecessary: the terminal family already permits every image in its binary abelian coordinates. Regroup the new $2a$ binary coordinates with the original binary abelian part, retaining fullness on each nonabelian critical coordinate and each carrier factor. The resulting terminal family has critical rank $R + 2a$, with no binomial prefactor. This regrouping concerns the comparison family; the original orbit profile and its weight remain unchanged.

The same central comparison gives, for every finite group $E$,

$$|\operatorname{Sub}(E \times B)| \le |\operatorname{Sub}(E \times C_2^u)|.$$

Indeed, choose a central subgroup $C \le B$ of order two and compare $E \times B \to E \times (B/C)$ with its split extension. Apply induction to $B/C$, keeping $C_2 \times E$ as the unchanged exterior, and regroup the binary factors. This induction preserves every condition on the exact $E$-image; it does not require $E$ to be abelian.

Set $E = C_4^a \times C_2^R$ and $r = R + u$. Reduce the $C_4$ coordinates of $C_4^a \times C_2^r$ modulo two. Its kernel has binary dimension $r + a$. Above an actual image $U$ of dimension $j$, the squares span exactly the copy of $U$ in the doubled $C_4$ coordinates. The square obstruction therefore has dimension $j$. The exact central-lift formula, with $k$ the retained annihilator dimension, gives

$$|\operatorname{Sub}(C_4^a \times C_2^r)| = \sum_{j=0}^{a} \begin{bmatrix} a \\ j \end{bmatrix}_2 \sum_{k=0}^{r+a-j} \begin{bmatrix} r+a-j \\ k \end{bmatrix}_2 2^{jk}.$$

Use $\begin{bmatrix} n \\ k \end{bmatrix}_2 \le \phi^{-1} 2^{k(n-k)}$, where $\phi = \prod_{i \ge 1}(1 - 2^{-i})$. The exponent of each summand is at most

$$j(a-j) + k(r+a-j-k) + jk = j(a-j) + k(r+a-k) \le \frac{a^2 + (r+a)^2}{4}.$$

There are at most $(a+1)(r+a+1)$ summands. In particular we have the stronger explicit bound

$$|\operatorname{Sub}(C_2^R \times C_4^a \times B)| \le \frac{(a+1)(R+a+u+1)}{\phi^2} 2^{((R+a+u)^2+a^2)/4}. \tag{23.13}$$

The logarithm of the prefactor is $O((a+u)\log(a+u+2)+\log(R+2))$, uniformly also when $a=u=0$. Adding the nonnegative $u^2/4$ recovers (23.12) with the stated constants. □

**Theorem 23.15** (Complete mixture)**.** *The normalized mass, relative to $L_{2N}$, of all fixed-point-free binary subgroups whose orbit actions belong to the critical actions, regular $C_4$, and the twelve carriers, with some noncritical orbit, is $2^{-\Omega(N)}$. At a fixed profile the following stronger bounds hold uniformly:*

$$\begin{aligned} T \le a/140,\ a \ge 1: &\quad 2^{-aN/25+O(a\log(N+2))}, \\ T > a/140: &\quad 2^{-(29/656)TN+O(N\log(N+2))}. \end{aligned} \tag{23.14}$$

*Proof.* The carrier product $B$ has $u = \log|B| \le 7T$. Use the exterior-preserving central comparison in Lemma 23.14, and regroup its unrestricted binary abelian coordinates as above. The terminal reserve from Theorem 23.13, applied directly at critical rank $R+2a$, gives

$$\begin{aligned} \log \mathrm{SD}(D, C_4^a, B) &\le N^2/4 - \frac{25}{82}(R+2a)T - \frac{29}{164}T^2 + O(N\log(N+2)) \\ &\le N^2/4 - \frac{29}{656}TN + O(N\log(N+2)). \end{aligned}$$

The original noncritical profile weights have a uniform finite-sum bound without evaluating their normalizers. If $w_i$ is the original normalizer order of colour $i$, then $w_i \ge 1$, and any finite set $\mathcal{M}$ of multiplicity profiles satisfies

$$\sum_{\mathbf{m}\in\mathcal{M}} \prod_{i=1}^{13} \frac{1}{w_i^{m_i} m_i!} \le \prod_{i=1}^{13} \exp(1/w_i) \le e^{13}.$$

All thirteen colours remain separate, including those with the same master. Together with $c_R/c_N \le (2N)^{2a+4T}$, division by $c_N G_N$ and insertion of (23.10) proves the second envelope.

For the other envelope first project onto an arbitrary actual $H \le C_4^a \times B$. Its sectional binary rank is at most $a+u$: intersect an elementary section with $H \cap C_4^a$ and project the quotient to $B$. Hence $\tau(H) \le a+u$. Proposition 3.4, with the nonnegative $\delta$ and $d/2$ dropped, gives

$$N_D(H) \le X_R(d(H)) 2^{h_R(a+u+\log 72)+O(\log(R+2))}, \qquad h_R(x) = \max_{0\le\ell\le R/2}(x\ell - 3\ell^2/4) \le x^2/3.$$

The graph identity $\sum_{H\le C_4^a\times B} X_R(d(H)) = |\operatorname{Sub}(C_2^R \times C_4^a \times B)|$ then allows a single use of (23.12). Put $N_0 = R+2a$. Subtracting the resulting quadratic exponent from $N_0^2/4$ leaves exactly

$$aR/2 + a^2/6 - uR/2 - 7au/6 - 5u^2/6.$$

If $T \le a/140$, then $u \le a/20$, so this is at least $(19/40)aR + (17/160)a^2 \ge aN_0/20 \ge aN/25$; here $N \le 71N_0/70$. The fixed $\log 72$ correction costs $O(a+1)$, and all profile and coefficient costs are $O(a\log(N+2))$. This proves the first envelope, including $T=0$.

Let $C = 2a+4T$ be the noncritical half-support. For this finite alphabet, the entire range $0 < C \le \sqrt{N}/4$ has a direct estimate. Each original carrier action has binary character rank at

most three times its physical scale: apply the certified head bound to the top normal subgroup and inflate characters along its fixed master epimorphism. This is strictly less than half its original degree. The regular $C_4$ has rank one. For an actual full noncritical tail $J$, restrict to one of its original blocks. The restriction kernel acts faithfully on the original complement, so the half-degree bound for that kernel preserves the local gap and gives $d_2(J) \le C - 1$, even when $J$ is a proper subdirect product. Applied to the actual evaluation kernel, the same bound gives $\tau(J) \le C$. These marks concern $J$ itself, before any master pullback.

To count such tails, the master pullback is injective, and its source has order $2^q$ with $q = 2a + u \le 7C/4$. The central comparison above bounds its subgroup count by $G_q$. The terminal Hall estimate therefore bounds the model count by

$$2\phi^{-3}(R+1)(c+1)\, X_R(C-1)\, G_q\, 2^{(C+7)^2/3},$$

where $c$ is the number of nonabelian critical factors. The elementary Gaussian estimates give $X_R(C-1)/G_N \le \phi^{-2}(R+1)2^{-N/2+1/2}$ and $G_q \le \phi^{-1}(q+1)2^{q^2/4}$. Sum the critical profiles at their exact rank $R = N - C$, retaining $c_R$, and the noncritical profiles with their original weights as above. The logarithm of the normalized mass at this support is at most

$$-\frac{N}{2} + \frac{1}{2} + \frac{211}{192}C^2 + \frac{14}{3}C + \frac{49}{3} + C\log(2N) + O(\log(N+2)).$$

For $C \le \sqrt{N}/4$ the quadratic terms leave at most $-1325N/3072$, and the remaining error is uniformly $o(N)$. Thus this range, including its support-size sum, is exponentially small. This finite-alphabet argument uses neither the general equality classification nor the coarse ordinary subgroup-count estimate from Theorem 4.2.

If $C > \sqrt{N}/4$ and $T > a/140$, then $C < 284T$, so $T > \sqrt{N}/1136$. The negative term of the second envelope therefore has order at least $N^{3/2}$ and absorbs its error. The first envelope is exponentially small uniformly for $a \ge 1$. After summing the original multiplicity weights, at most $(N+1)^2$ parameter pairs $(a, T)$ remain. Their sum proves the theorem. □

The same weighted estimates give a direct extension to one natural $S_3$ orbit, with critical orbits and the original thirteen noncritical colours on the remaining points. Write $\mathbf{p}$ for the critical profile of rank $R$, keep the noncritical profile $\mathbf{q}$ fixed, and put $C = 2a + 4T > 0$ and $N = R + C + 1$. The physical degree is $2N + 1$. The exterior group is a 2-group. If $(g, d)$ lifts a three-cycle and $d^{2^k} = 1$, its $4^k$th power is $(g, 1)$. Thus every subgroup full on the $S_3$ coordinate contains $A_3$ on that coordinate alone. The sign quotient, followed by its inverse full preimage, identifies its full models bijectively with those of $(\mathbf{p} + \mathbf{e}_{C_2}, \mathbf{q})$. Every original critical and noncritical coordinate remains full; no full projection onto either entire class product is required.

The physical weight is still the original one. If $\omega(\mathbf{p}, \mathbf{q})$ denotes the reciprocal product of the original normalizers and occurrence factorials, then

$$\frac{\omega(\mathbf{p}, \mathbf{q})}{6} = \frac{p_{C_2} + 1}{3}\, \omega(\mathbf{p} + \mathbf{e}_{C_2}, \mathbf{q}).$$

The factor includes both the marker normalizer six and the new C2 occurrence factorial. The map $\mathbf{p} \mapsto \mathbf{p} + \mathbf{e}_{C_2}$ is injective into the critical profiles of rank $R + 1$, so positivity bounds the original weighted sum by $(R+1)/3$ times the complete shifted sum, with $\mathbf{q}$ unchanged. Its half-degree is exactly $(R+1)+C = N$. Moreover, $L_{2N+1} \ge (2N+1)!\, c_N G_N$, retaining the nonnegative additional term of the odd coefficient. The preceding weighted estimates therefore give a polynomial times $2^{-\Omega(N)}$ for each such bin. There are at most $(N+1)^2$ pairs $(a, T)$ with $C + 1 \le N$, and the weight

correction is at most $(N+1)/3$. Their cubic polynomial cost is absorbed by the exponential decay. This bounds the actual union of physical subgroups, allowing overlapping profile presentations. It is a specialization to this finite alphabet and leaves the general complete-family marker transfer of Section 28 unchanged.

Combining (23.14) with Propositions 22.1–22.4 proves the same negligible-mass conclusion for every fixed finite menu of simultaneous equal-degree quotient replacements retaining a positive fraction of the original noncritical support. The original small-support range is removed before paying its decoration fibre.

# 24 Complete binary counting

Let $F_N$ be the number of all fixed-point-free 2-subgroups of $S_{2N}$ divided by $L_{2N} = (2N)!c_N G_N$, and let $E_N$ be the same normalized count restricted to groups with a noncritical orbit. Write also $A_m = s_m/L_m$ for the complete normalized ordinary count. Thus $F_N = C_{2N}^{\mathrm{crit}} + E_N$ exactly. We prove the strictly forward ordinary-target recurrence needed in Section 32.

## 24.1 The original degree-thirty-two boundary

**Theorem 24.1** (Adaptive cover at degree sixteen)**.** *Let $T \le S_{16}$ be transitive and binary, and let $A$ be a section of its original permutation module whose action factors through a specified quotient $B$ of $T$. There are $C \le A^B$ and a specified permutation cover of $B$ of degree $v_0 \in \{8, 16\}$ such that*

$$v_0 + 2\dim C + 4\dim(A/C)^B \le 30. \tag{24.1}$$

*Proof.* Use $S, t, H, \mathcal{L}, \ell, H_0$ from Lemma 21.3. Put $d_0 = \dim H_0 \le 4$. For $a = 4$ the joint budgets $2^j B_{4-j}$, for $0 \le j \le 4$, are $(6, 6, 8, 8, 16)$. In particular $t, \ell \le 6$. If $d_0 \le 3$, taking $W = H$ gives $t + \ell \le 8$; taking a character line gives $j_0 \le 6 - t$. The separator in Lemma 21.2 supplies $q$ evaluations with kernel $C \le S$, and (5.4) gives $\dim C = t - q$, $\dim(A/C)^B = q$. If $t \le 5$, their cost is $2t + 2q \le t + \ell + 6 \le 14$. If $t = 6$, then $\ell \le 2$ and $j_0 = 0$, so $q \le 1$ and the same bound holds. The original degree-sixteen cover $T \to B$ therefore proves (24.1).

If $d_0 = 4$, the common kernel of $H_0$ has sixteen orbits on sixteen points. Faithfulness makes it trivial. Thus $T = C_2^4$ is regular; as every character in $H_0$ comes from $B$, also $B = T$. Four independent characters give a faithful action of this same $B$ on four disjoint pairs. Use it as the degree-eight cover. The original module still has $t, \ell \le 6$, $j_0 \le 6 - t$. For $t \le 5$ the separator cost is at most $t + \ell + 6 \le 17$; for $t = 6$ it has $q \le \lceil \ell/2 \rceil \le 3$ and cost at most 18. Adding the new cover degree gives at most 26. The module $A$ was never replaced by a section of the eight-point permutation module. □

For every transitive binary action $U$ of degree 32 and every $N \lhd U$, choose an invariant pair system with kernel $K$ and top $T \le S_{16}$. The groups $Q = U/N$, $A = KN/N$ and $B = U/(KN)$ meet the theorem's hypotheses. Since every simple binary module of a 2-group is trivial, $r(A/C) = \dim(A/C)^B$. Proposition 6.3 gives the same-source envelope with $v = v_0 + 2\dim C$, $\lambda = r(A/C)/2$ and $g = 32 - v - 8\lambda \ge 2$. The finite-menu proof of Theorem 6.5, restricted on its physical side to fixed-point-free binary sources, consequently gives

$$T_{32}(N) \le \eta_{32}(N) + K_{32}(N, N-16)F_{N-16}, \qquad \eta_{32}(N) = 2^{-\Omega(N^2)}, \quad K_{32}(N, N-16) = 2^{-\Omega(N)}. \tag{24.2}$$

Here $T_{32}$ denotes the family with a degree-32 orbit and none larger. The auxiliary graph may have fixed points, so its moment is bounded by the complete count $s_{b+qv}$ as in Proposition 6.3.

Only the physical cold complement is fixed-point-free. At this fixed width all action, normal, frame and cut choices form a finite menu, and may all be retained at a fixed constant cost. The coefficient is the sum of $\Delta(2N, 2N-32)D_C 2^{(4-g/16)(2N-32)}/a(U)$; its logarithm is at most $-g(2N-32)/16 + O(\log(N+2))$.

## 24.2 Uniformity over unbounded binary widths

Put $A_k = \sum_{j<k} B_j$, with $A_0 = 0$. These cumulative Boolean widths give the binary normal-count input directly.

**Lemma 24.2** (Original normal heads)**.** *For a transitive 2-group $U \le S_{2^k}$ and every original normal subgroup $N \lhd U$,*

$$\dim \mathrm{Hom}(N, C_2)^U \le A_k.$$

*Consequently $d(U) \le A_k$, and the number of original normal subgroups of $U$ is at most $2^{(2^k-1)A_k}$. Moreover $A_k = O(2^k/\sqrt{k+1})$, in particular $A_k = o(2^k)$.*

*Proof.* At degree one the group is trivial. Otherwise choose an original invariant pair system, with kernel $K$ and transitive top $T$ of degree $2^{k-1}$. The exact subgroup $N \cap K$ is a submodule of the pair permutation module, so its invariant character dimension is at most $B_{k-1}$ by the binary module bound. Restriction of invariant characters from $N$ has kernel the characters of the actual image $NK/K$, invariant under $T$. Thus the required head is at most $B_{k-1} + A_{k-1} = A_k$. Surjectivity of restriction is neither needed nor asserted.

Apply this to the original top subgroup to bound the ordinary character rank of $U$; Frattini lifting gives $A_k$ generators, padding by identities if necessary. To count normals, let $c_a$ count the original $U$-normal subgroups of index at most $2^a$. Every new subgroup at the next cutoff is an index-two child of one at the previous cutoff: for an original normal $H$, use an order-two subgroup of $Z(U/H)$. Each parent has at most $2^{A_k} - 1$ such children, from its nonzero invariant characters. Hence $c_0 = 1$ and $c_{a+1} \le 2^{A_k} c_a$. The bound follows from $|U| \le 2^{2^k-1}$. Finally the elementary central-binomial recurrence gives $(\lfloor j/2 \rfloor + 1)B_j^2 \le 4^j$. Splitting the cumulative sum at $k/2$ proves the stated estimate for $A_k$. □

**Lemma 24.3** (Zero cuts at large binary widths)**.** *Let $U$ be transitive and binary of degree $w = 2s$, where $s = 2^j$, and fix an original pair kernel $K$ and $N \lhd U$. For $A = KN/N$ and $B = U/(KN)$, the literal cut $C = 0$ has*

$$v = s, \qquad r(A) = \dim A^B \le B_j, \qquad g = w - v - 4r(A) \ge w/128 > 0 \quad (j \ge 9).$$

*For $j \ge 11$ the stronger bound $g \ge w/32$ holds. The corresponding coefficient remains $D = |A||H^1(B, A)|$, and the same-source moment weight is exactly $\Phi(J) = e_B(J)$.*

*Proof.* The invariant bound follows by taking the full preimage of $A^B$ in the original pair kernel and applying the binary module bound to that trivial quotient. Since $B$ is binary, its simple modules over $\mathbb{F}_2$ are trivial, so this invariant dimension is the Schur capacity. The original top $T \le S_s$ maps onto this same $B$; Proposition 6.3 applies to that faithful cover, without assuming a faithful degree-$s$ action of $B$ or a split extension. With $C = 0$ it gives the stated $D$ and $\Phi$.

Pascal's identity gives $B_{j+1} \le 2B_j$. The exact values $B_9 = 126$ and $B_{11} = 462$ therefore imply, respectively, $256B_j \le 63\,2^j$ for $j \ge 9$ and $64B_j \le 15\,2^j$ for $j \ge 11$. Substitution into $g = s - 4r(A)$ proves both bounds. □

Thus physical widths 1024 and 2048 already have zero-cut parameters $e = g/16 \geq 1/2$ and $e \geq 1$, respectively, and can be included in a fixed finite direct-fusion prefix. At every width at least 4096, the zero cut meets the uniform margin used below. Only widths $64, 128, 256, 512$ in the large-width route require the sharper cut construction; degree 32 remains the separate adaptive boundary above.

**Theorem 24.4** (Complete large-orbit deletion)**.** *Let $T_{64}(N)$ be the normalized family with some actual orbit of width at least 64. There are nonnegative $\eta_{64}$ and $K_{64}$, with $K_{64}(N, M) = 0$ unless $M \leq N - 32$, such that eventually*

$$T_{64}(N) \leq \eta_{64}(N) + \sum_{M<N} K_{64}(N, M) F_M, \qquad \eta_{64}(N) \leq 2^{-(2N)^2/2^{22}}, \quad \sum_M K_{64}(N, M) \leq 2^{-(2N)/512}. \tag{24.3}$$

*Proof.* Every nontrivial transitive 2-group has an invariant pair system: a central involution is fixed-point-free, since a central element fixing a point fixes its entire transitive orbit. An invariant pair system is determined by the partner of one point, so there are at most $w - 1$ systems. Retain all of them. Choose one auxiliary coordinate chart per system; in the sums below a frame means this chosen realization. Write $w = 2s \geq 64$. For every literal normal $N \lhd U$ and every pair kernel $K$, put

$$Q = U/N, \qquad A = KN/N, \qquad B = U/(KN), \qquad S = A^B.$$

For $s = 32$ use Theorem 21.5; the remaining finite widths $64 \leq s < 2048$ are covered by Theorem 21.4. For $s \geq 2048$ the zero cut in Lemma 24.3 already suffices. Retain all cuts $C \leq S$ satisfying $2c + 4r(A/C) \leq 15s/16$, where $c = \dim C$. This menu is nonempty and invariant under the original action normalizer. The original pair top is a specified degree-$s$ cover of $B$. Proposition 6.3 gives

$$e_Q(J) \leq D_C 2^{\lambda b} \Phi_C(J), \qquad \Phi_C(J) = e_B(J) 2^{cd_2(J)}, \quad D_C = |A/C||H^1(B, A/C)|, \quad \lambda = r(A/C)/2.$$

The module bound in Lemma 21.3 gives $t = \dim S \leq B_{\log s} \leq 5s/16$. Therefore

$$w/2 \leq v = s + 2c \leq 13w/16, \qquad g = w - v - 8\lambda \geq w/32, \qquad g \leq w - v \leq v. \tag{24.4}$$

We next bound the entire varying menu. Write $w = 2^k$. Lemma 24.2 bounds the number of literal normal subgroups of each action by $2^{(w-1)A_k}$ and gives an actual generating tuple of length $A_k$. Every such action is conjugate into a fixed Sylow subgroup of $S_w$, of order $2^{w-1}$. Choose one original tuple for each action class after this conjugation. Equal tuples generate the same subgroup, so the number of transitive action classes is also at most $2^{(w-1)A_k}$. This argument requires no separate soluble normal-generation estimate. Furthermore

$$t \leq B_{\log s} = O(s/\sqrt{\log s}), \qquad \log G_t = O(s^2/\log s),$$

$$\log D_C \leq \dim(A/C)(1 + d(B)) = O(w^2/\sqrt{\log w}).$$

The last inequality uses $d(B) \leq d(T) \leq A_{k-1}$ for the original transitive top $T$. More explicitly, every subspace of $\mathbb{F}_2^t$ has a spanning $t$-tuple, so there are at most $2^{t^2}$ cuts. Using the frame bound $w - 1 \leq 2^w$ and $1/a(U) \leq 1$, set

$$h(2^k) = 2(2^k - 1)A_k + B_{k-1}^2 + 2^{k-1}(1 + A_{k-1}) + 2^k = O(w^2/\sqrt{\log w}).$$

The cocycle bound here evaluates actual cocycles on generators of the same $B$ before passing to their cohomology quotient. These estimates give

$$\sum_{[U],\ \deg U=w} \frac{1}{a(U)} \sum_{\text{frames},N,C} D_C \le 2^{h(w)}. \tag{24.5}$$

Every normal and every cut is included before taking this bound.

The graph proof of Proposition 6.3 uses the same source $J$ for all maps. For binary $J$ and binary covers its graph is binary, but may have fixed points; in particular $\sum_{J\le S_b,\ J \text{ binary}} \Phi_C(J)^q \le s_{b+qv}$. Put $e = g/16$, $a_0 = v/8 + e$, and declare the entry hot if $\Phi_C(J) > 2^{a_0 b}$. Use $q = \max(1, \lceil 8eb/v^2 \rceil)$ in (6.4). For $b > 0$ its rounding square is at most $v^2/16$, while (24.4) gives

$$g/v \ge 1/26, \qquad b + qv \le 3b/2 + v \le 3n/2.$$

Consequently the main exponent of each hot term is at most

$$-2^{-18} b^2 - 87w^2/4096. \tag{24.6}$$

For $b = 0$ the strict hot event is empty. The normalized hot sum has the original factor

$$\frac{D_C}{b!\,a(U)\,c_{n/2}\,G_{n/2}} 2^{\lambda b - (q-1)a_0 b} s_{b+qv}.$$

Thus the coarse subgroup bound contributes $O(n^{3/2})$ and $c_{n/2}^{-1}$ contributes $O(n \log(n+2))$ to its logarithm; dropping $1/b! \le 1$ is harmless for this upper bound. For every $\delta > 0$ there is $H_\delta < \infty$ such that $h(w) \le \delta w^2 + H_\delta$ for all $w \ge 64$. Take $\delta = 1/128$ in (24.5). The retained negative terms in (24.6), together with $b^2 + w^2 \ge n^2/2$, absorb all errors and the at most $n$ widths, yielding the stated hot scalar.

On the complement of the union of all hot events, the complete cold coefficient for $b = n - w$ is

$$\Delta(n,b) \sum_{[U],\deg U=w} \frac{1}{a(U)} \sum_{\text{frames},N,C} D_C 2^{(w/8 - g/16)b}.$$

The source factorial cancels the $b!$ in the pointing factor once. The coefficient inequality $c_{b/2}/c_{n/2} \le (2N)^{w/2}$ and $\log(G_{b/2}/G_{n/2}) = -wb/8 - w^2/16 + O(1)$ give its logarithm at most

$$h(w) + (w/2)\log n + O(1) - wb/512 - w^2/16.$$

Use $h(w) \le w^2/128 + H$ and, uniformly for $64 \le w \le n$, absorb $(w/2)\log n + H + O(1)$ into $wn/8192$ for large $n$. The result is at most $-wn/1024$. Summing at most $n$ widths proves (24.3), conservatively with rate $n/512$.

The hot union is selected first over all actual entries; its complement is then bounded by cold pointing. Deleting an actual orbit leaves a complete fixed-point-free binary subgroup, so the cold target is precisely $F_{b/2}$. This establishes the inequality before any bound on $F$ has been assumed. □

## 24.3 Finite witness coverage and its universal interpretation

Let $\mathcal{A}$ be the seventeen-action alphabet in Theorem 23.15.

**Lemma 24.5** (Finite coverage principles)**.** *Fix a power-of-two degree $w$. A list of literal transitive 2-groups containing a Sylow subgroup of $S_w$, and closed up to supplied permutation conjugacy under all transitive index-two subgroups, contains every transitive binary action up to conjugacy. For any finite 2-group $U$, a list of literal normal subgroups containing 1 and closed under all inverse images of order-two subgroups of $Z(U/N)$ contains every normal subgroup of $U$.*

*Proof.* The iterated binary wreath group on $w$ points has order $2^{w-1}$, which is the full 2-part of $w!$. Hence every binary action is conjugate into this explicit Sylow group. In a finite 2-group any subgroup lies at the bottom of a chain of index-two subgroups: choose successive maximal proper overgroups. Every overgroup of a transitive subgroup is transitive. Induction down the chain proves the first claim. All index-two subgroups are exactly the kernels of the nonzero functionals on $U/\Phi(U)$, so this closure can be checked exhaustively.

If $N < L$ are normal in $U$, the nontrivial normal subgroup $L/N$ of $U/N$ meets its centre nontrivially, by the class equation. This intersection contains an involution. Its inverse image is a normal index-two extension of $N$ lying inside $L$. Repetition starting at 1 reaches every $L$, proving the second claim. □

**Lemma 24.6** (Uniform order stops at power-of-two degrees)**.** *Let $k \geq 4$, $w = 2^k$, and let $U \leq S_w$ be a faithful transitive binary action with $|U| \leq 2^{w/2}$. For every original normal subgroup $N \lhd U$, put $Q = U/N$, $a = \log_2 |Q|$, and $z = \dim \Omega_1 Z(Q)$. Then at least one of the sufficient fusion conditions*

$$2a < w, \qquad 38^{8z} < 2^w 25^{8z}$$

*holds. Consequently, for any fixed finite collection of such original actions, all their normal axes admit the original-weight fusion estimate of Theorem 6.5, with constants that may depend on that collection. In particular this applies at degree sixteen to every transitive binary action of order at most* 256*.*

*Proof.* If $N \neq 1$, then $|N| \geq 2$, so $|Q| = |U|/|N| \leq 2^{w/2-1}$ and $2a \leq w - 2 < w$. If $|U| \leq 2^{w/2-1}$, the same bound holds for every quotient, including $N = 1$.

It remains to consider $N = 1$ and $|U| > 2^{w/2-1}$. The elementary inequality $k + 1 \leq 2^{k-1}$ for $k \geq 4$ gives $2^{w/2-1} \geq 2^k = w$, so $|U| > w$. Every central subgroup acts freely on the original $w$ points: a central element fixing one point fixes its entire $U$-orbit, and faithfulness makes that element trivial. Hence $A = \Omega_1 Z(U)$ has order at most $w$. Equality would make $A$ transitive. Any element of $U$ fixing one point would then fix every point by commuting with $A$, so $U$ would act regularly and have order $w$, a contradiction. Thus $|A| < 2^k$ and $z < k$. For $k = 4$, the exact integer inequality $38^{24} < 2^{16} 25^{24}$ gives $38^{8z} < 2^{16} 25^{8z}$ for every $0 \leq z \leq 3$. For $k \geq 5$, induction from $k = 5$ gives $8(k-1) \leq 2^k = w$. If $z > 0$, use $38 < 50$ to obtain

$$38^{8z} < 2^{8z} 25^{8z} \leq 2^w 25^{8z},$$

since $8z \leq 8(k-1) \leq w$. The case $z = 0$ is immediate.

For the order condition, Proposition 6.4 supplies $D = 1$, $v = 2a$, $\lambda = 0$, and $\Phi(J) = 2^{ad_2(J)}$. These are graph-moment marker coordinates; they do not assert that $Q$ has a faithful action on $2a$ points. For the character condition, the actual-source estimate (24.7) below, with its stated nilpotent conjugacy-class bound, supplies $D = |\operatorname{Aut} Q|$, $v = 0$, $\Phi(J) = 1$, and $\lambda = z \log_2(38/25) < w/8$. This estimate also holds for arbitrary $J \leq S_b$: restriction to an original Sylow 2-subgroup preserves surjectivity to $Q$ and determines each map, since that Sylow subgroup maps onto the joint binary image of any two maps in $Q \times Q$. In both cases $w - v - 8\lambda > 0$, and all moments

concern the same original $J$. Sum over the complete fixed finite collection of these original pairs $(U, N)$ and apply Theorem 6.5. Its pointing weights retain $|N_{S_w}(U)|$, while the character branch retains $|\operatorname{Aut} Q|$. No quotient normalizer or smaller faithful action is substituted for the original action. □

**Lemma 24.7** (Finite binary entry theorem)**.** *For every transitive binary action $U$ of width $w \in \{2, 4, 8, 16\}$ outside $\mathcal{A}$, and every literal normal $N \lhd U$, at least one of the following witnesses exists for $Q = U/N$:*

1. *An invariant pair system with kernel $K$, an actual central cut $C \le (KN/N)^{U/(KN)}$, and an actual permutation cover of $B = U/(KN)$, such that $v_B + 2\dim C + 4\dim((KN/N)/C)^B < w$.*
2. *A faithful permutation representation of $Q$ of degree $v < w$.*
3. *With $z = \dim \Omega_1 Z(Q)$, the exact integer inequality $38^{8z} < 2^w 25^{8z}$.*
4. *An equal-degree full-subdirect replacement by a word in $\mathcal{A}$, with actual maps onto $Q$, retaining at least one quarter of its points in noncritical factors.*

*The last branch needs only the following four entries:*

| $U$ | $\lvert N\rvert$ | $Q$ | *replacement word* |
|---|---|---|---|
| $8T16$ | 2 | SmallGroup$(16, 3)$ | $8T27$ |
| $8T20$ | 2 | SmallGroup$(16, 3)$ | $8T27$ |
| $8T21$ | 2 | SmallGroup$(16, 3)$ | $8T27$ |
| $16T1086$ | 4 | SmallGroup$(256, 7667)$ | $C_4[4], D_8[4]^3$ |

*In the last row the cover is a proper full-subdirect subgroup of order 256 in the displayed product, not that whole product. Its four literal restriction images are, under explicit point relabellings, the original regular $C_4$ action and three copies of the critical $D_8$ action. Thus the carrier itself is the full physical profile with critical half-rank $R = 6$ and noncritical half-support $C = 2$. Reversible transport preserves fullness on all four blocks while retaining arbitrary correlations with the exterior coordinate.*

*Proof.* This is the finite computer-assisted lemma recorded by the literal binary entry certificate. We describe its mathematical coverage and the statements checked at each vertex; these are essential to its use. There are explicit iterated wreath roots in each of the four degrees. Each action record gives every transitive index-two child, an actual conjugator to its destination, and a checked basis of its Frattini quotient. Nonregular base actions retain their children. Regular transitive actions have no transitive proper subgroup. For an index-two child $K \le U$, a proposed conjugator to $T$ needs only the checks $g^{-1} t_j g \in K$ on generators $t_j$ of $T$, together with $2|T| = |U|$. They give $T \le gKg^{-1}$, and the actual index forces equality. No full element-by-element conjugacy table is needed. Each nonbase action record gives 1 and every central-involution extension of each normal vertex. The two closure checks prove exhaustive action and normal coverage by Lemma 24.5; catalogue identifiers play no role in these implications.

For the first branch the record supplies the pair partition, kernel $K$, intersection $M = K \cap N$ and lifted cut $M \le C_0 \le K$. The literal equalities $[C_0, U] \le M$ and $\ker(U \to \text{cover image}) = KN$ certify the same central section and quotient. Fixed sections are computed by commutators modulo $M$ and $C_0$, giving the integers $c, r$ and the strict gap. For a direct witness the supplied generator images define a map with kernel exactly $N$ and moved degree strictly below $w$. For a character

witness the centre and its involution subgroup are reconstructed in the literal quotient, and the displayed integer inequality is checked. For a transport witness both specified maps are onto the same literal quotient, the original map has kernel $N$, and each displayed projection of the actual cover group equals its assigned action. The degree and noncritical fraction are then checked on the actual orbit partition. The four exceptional charts satisfy these conditions, including the retained proper cover in degree sixteen. For $16T1086$, generator intertwiners identify the first restriction with the original regular $C_4$ and each of the other three restrictions with the canonical critical $D_8$. Surjectivity of the transported subgroup onto the actual carrier then gives fullness on every one of these four physical blocks. This argument never enlarges the proper subdirect carrier to the displayed product, and it leaves the exterior coordinate of every lift unchanged. Consequently this branch enters the same complete original-action profile family used in the carrier-mixture estimate, with the local increment $(R, C) = (6, 2)$, equivalently the parameter-bin coordinates $(R, a, T) = (6, 1, 0)$.

The other surviving split catalogue endpoint, $16T1332$, needs no replacement action. Its literal degree-sixteen action is already one of the original carrier colours, with the one-occurrence coordinates $(R, a, T) = (0, 0, 2)$. The explicit point chart puts that same subgroup in the complete carrier-mixture family, and conjugating the catalogue model changes only its physical labels. Simultaneously, every transported normal axis retains its effective bound by the fixed master-envelope menu. For global transport, use the quotient map by that transported normal on both sides of the same one-cell colour. This is a quotient-identity slot: its kernel is exactly the original axis, and its source equivalence is the same catalogue conjugacy. Thus this endpoint composes with the proper $16T1086$ slot and all degree-eight slots while preserving every exterior correlation. The physical-family witness and master-envelope row remain the upstream audit receipts; local membership in the mixture is not used as a substitute for the reversible slot. The committed finite records and checker in Appendix A supply these finite identities.

There is also a uniform proof for regular actions of width $w \geq 8$: choose a central involution and its coset pairs. Then $K = C_2$, $A = KN/N$ is central of dimension at most one, and one may take $C = A$. The quotient $B$ has its regular cover of degree at most $w/2$; $w/2 + 2 \dim A < w$. All binary actions of degrees two and four are already in $\mathcal{A}$. □

The first two witness types have the graph envelopes of Proposition 6.3, with $g > 0$. The third has a direct slope strictly below $w/8$. Indeed a 2-group $Q$ has a faithful tuple of $z$ irreducible complex characters: choose constituents whose central characters span the dual of $\Omega_1 Z(Q)$. A nontrivial normal subgroup meets $\Omega_1 Z(Q)$, so their common kernel is trivial. For a binary $J \leq S_b$, inflation along an epimorphism yields a tuple from $\mathrm{Irr}(J)^z$. Every tuple determines at most one kernel, and each kernel has exactly $|\operatorname{Aut} Q|$ epimorphisms to $Q$. The nilpotent conjugacy-class bound [Mar05, Theorem 1.5] therefore gives

$$|\operatorname{Epi}(J, Q)| \leq |\operatorname{Aut} Q|\, k(J)^z \leq |\operatorname{Aut} Q|\, 2^{z \log(38/25) b}. \tag{24.7}$$

The strict integer test is exactly $z \log(38/25) < w/8$. For $Q = 1$ use $e_Q(J) = 1$. These assertions concern the actual source $J$; no faithful action of an abstract quotient on its points is assumed.

## 24.4 Small widths and the complete recurrence

**Theorem 24.8** (Binary recurrence from the native S16 fibre)**.** *There are nonnegative $\eta_B(N)$ and $K_B(N, M)$, supported on $0 \leq M < 2N$, and constants $a_B, C_B > 0$, such that eventually*

$$E_N \leq \eta_B(N) + \sum_{M<2N} K_B(N, M) A_M, \qquad \eta_B(N) + \sum_{M<2N} K_B(N, M) \leq C_B 2^{-a_B N}. \tag{24.8}$$

*Proof.* Assign each noncritical binary subgroup to its first applicable owner: an actual orbit of width at least thirty-two, a direct degree-eight axis, a direct degree-sixteen axis, or the residual family in which every orbit has width at most sixteen and neither direct owner applies. Power-of-two orbit sizes prove that these cases cover the whole error family. In the residual case, zero old support is exactly the complete critical family, so the noncritical source has positive old support. This gives an injection into the sum of the four unmarked owner families. The first family is bounded by (24.3) and (24.2).

Make the degree-sixteen split on the original unmarked subgroup: the direct family consists of those subgroups for which some actual sixteen-point orbit and its literal Goursat axis has a direct witness. In the complementary family every such orbit-axis witness has a carrier witness; no distinguished orbit, point, chart, or normal axis is charged. This complement is automatically disjoint from the order-at-most-256 sector. Indeed the complete order/seven-character selection of Lemma 24.6 would make any such axis direct. Thus every surviving carrier witness has original restriction-image order greater than 256.

For the direct part of $S_{16}$, fix a literal transitive binary action $U \leq S_{16}$ and call a raw axis accepted when an original-weight direct entry exists on it. Entry existence transports through every equivalence of action groups, by the induced equivalence of the complete quotient. It is therefore invariant under the full normalizer of $U$. Consequently the predicate

$$K \longmapsto \{\text{the literal Goursat axis of } K \text{ is accepted}\}$$

is natural for physical pointing. Choose one entry on each accepted normal axis and assign the coefficient zero to every other axis. The latter coefficient is exact: its retained Goursat fibre is empty because decoding an epimorphism recovers that same rejected axis.

The original-normalizer physical count may now be applied before the axis sum. The first same-source moment of each selected entry gives its shifted complete count directly; no hot/cold split is needed. There are only finitely many literal subgroups of $S_{16}$ and finitely many normals in each, while every active entry has a strict positive gap. Thus their complete row is $2^{-\Omega(N)}$. An entry of even prefix degree $2r < 16$ points to ambient degree $2N - 16 + 2r$, so its half-degree lies between $N - 8$ and $N - 1$. The hypotheses only give $0 \leq r \leq 7$, so the aggregate support must allow the entire interval $N - 8, N - 7, \ldots, N - 1$; no argument reduces it to two special shifts. Only nonnegative targets are included. Zero-prefix character entries give the $N - 8$ endpoint and are covered by the same first-moment row. The actual orbit chart proves coverage of the original unmarked direct family, so no orbit, point, chart, action representative, or normalizer element is counted. Any preceding first-owner filter only decreases this family and uses the identical row.

In the remaining family every nonbase orbit has a transport witness for its exact axis. Apply all these maps simultaneously, leaving $\mathcal{A}$ factors as quotient-identity slots. The product of the embedded axes lies in $H$, so Proposition 22.1 recovers $H$ as the full preimage of the joint quotient image. Every replacement has the same physical degree, and retains at least one quarter of its noncritical support. The route is retained as Type-valued data until its local output chart has been formed. On a regular-$C_4$ route this chart removes the unique cell coordinate and applies the retained four-point chart. A degree-eight base route then applies its fixed base-action chart and the inverse catalogue conjugacy; each exceptional degree-eight route has one literal $8T27$ cell and applies only that inverse conjugacy. The $16T1332$ route is the analogous one-cell construction. For $16T1086$, the four displayed cells are first identified with the exact one-$C_4$/three-$D_8$ model, then carried through its checked sixteen-point physical chart and the inverse catalogue conjugacy. Thus each routed slot supplies an exact equivalence from its displayed output points to the points of the same original local action.

To identify the profile, enumerate the displayed cells of all replacement carriers by their original-action colours. More explicitly, if $X$ is the finite set of displayed cells and $\chi : X \to \mathcal{A}$ is the colour map, put $m(A) = |\chi^{-1}(A)|$. Enumerating each colour fibre gives the canonical equivalence

$$X \simeq \coprod_{A \in \mathcal{A}} \{1, \dots, m(A)\}.$$

Reindexing the dependent product along this equivalence turns cellwise fullness into full projection on every occurrence of the exact profile $m$. Pullback along the resulting type equalities restores the literal original colour on every cell, so no independently chosen occurrence matching remains in the data. The pullback from a witness action and axis to the canonical orbit axis leaves the underlying slot literally unchanged. Consequently its physical half-degree, retained support, old-support charge, and the two retention inequalities are reused verbatim; no numerical recalculation or profile comparison is hidden in this reindexing.

Finally assemble the local route-output charts along the canonical complete orbit chart of the original subgroup. This gives one fusion-natural equivalence from the completed-mixture point set directly to the original labels $\{1, \dots, 2N\}$. Simultaneous carrier transport may therefore be realized on those labels before insertion into the retained parameter bin. For a fixed supplied chart this original-label physical realization is injective, and it uses exactly the same critical-rank identity and the same retained bin as the canonical-label word target. The point chart is a deterministic composite of the retained routes and the original orbit chart, not a separately counted mark.

This gives a single whole-word closure statement. For arbitrary source groups $U_i$, arbitrary full proper subdirect carriers $C_i \leq \prod_j A_{\chi(i,j)}$, and surjections $U_i \twoheadrightarrow Q_i$ and $C_i \twoheadrightarrow Q_i$ onto the same quotients, the simultaneous full-preimage construction is injective on source subgroups with their exact kernel axes. If one displayed cell is noncritical, the fibre counts themselves make the completed-mixture support positive. Hence the whole decorated source family injects into the one physical parameter bin determined by $m$, and then into the completed mixture. This applies without commutativity to a proper nonabelian subdirect carrier such as the $16T1086$ cover. No carrier, orbit, chart, quotient, axis, or occurrence choice is counted as an additional marking. In the three degree-eight exceptional rows the checked full physical block is the original $8T27$ action, so it is a one-cell carrier; the $16T1086$ row remains its proper four-cell subdirect carrier. An original factor that is left in place is handled by the same one-cell construction, using its actual epimorphism to $Q$ on both sides. Thus its normal axis is retained rather than forced to be trivial. These identity slots and the four exceptional slots may be interleaved in one heterogeneous word, and one noncritical slot makes its completed-mixture support positive. The partition before this construction is intrinsic. A full source subgroup determines its complete tuple of embedded normal axes. Excluding the first direct owner therefore partitions the carrier residual by this canonical axis profile, without choosing or counting an axis. On each profile, the finite local theorem chooses a routed slot in every coordinate and the preceding whole-word injection applies. Retain the actual image of every fixed-profile transport until all route, block, and changed-axis data are forgotten together. Enlarging each fixed-profile image to the whole completed mixture before this step would charge arbitrary axes on untouched critical coordinates and is not used. The only routed word with no noncritical cell is the pure critical $E_8$ word, which belongs to the already counted critical family rather than to $S_{16}$.

It remains to identify the fibres after the auxiliary data are forgotten. Equality of the mixed table first aligns every changed route. The retained route-output charts identify the two native point sets with the same fixed original labels. Naturality of the displayed carrier action then transports equality of the fixed-label physical targets back to equality of the native targets. The

indexed full-preimage reflection theorem recovers the complete correlated word source, including every quotient-identity slot and its exact kernel axis. Finally the slot-preserving binary source reindex recovers the canonical flat source, and faithfulness of its literal orbit action recovers the original subgroup. Thus every fixed mixed-table/physical-target fibre has size one, also for the proper nonabelian $16T1086$ carrier and for unchanged critical coordinates.

For the simultaneous degree-eight/degree-sixteen word, each changed block is stored in a fixed nineteen-symbol record (degree tag, least point, route tag, and point representatives). At most one record is charged per unit of old half-support, so the common twenty-slot packing gives the explicit decoration factor $(2N+2)^{20C_{\mathrm{old}}}$, namely the $K=10$ instance of Proposition 22.2. The retained route-output charts show that no further point-chart field is needed in this record: the local charts assemble directly to the fusion-natural original-label target.

This construction is applied after disintegrating the unmarked positive family by its canonical old half-support. A positive regular-$C_4$, degree-eight, or degree-sixteen orbit contributes respectively $2, 4, 8$; twice this contribution is its literal orbit size, while a critical orbit contributes zero. Therefore every positive source has a unique integer $1 \le C_{\mathrm{old}} \le N$, and the positive family is exactly the disjoint sum of these fixed-support fibres. On each fibre the coordinate index, action types, and carrier products may depend on the already counted pair consisting of the mixed table and the physical target. The fibre-one result just proved gives the exact table-cardinality times target-cardinality bound. The outer aggregation is only the displayed sum over $C_{\mathrm{old}}$; it contains no factor counting orbit profiles or presentations.

The stronger profile envelopes (23.14) absorb the decoration before any of them is weakened. If the retained noncritical half-support is at most $\sqrt{N}/4$, the small-support estimate pays a subexponential decoration. Otherwise the Hall range pays it from the $aN$ reserve, and the complementary range pays it from the $TN$ carrier reserve. The remaining support and bin indices have cardinality at most $(N+1)^3$. Absorbing this polynomial leaves, in the explicit formal normalization, the rate $29/2980864$ in $N$. Hence the complete positive residual has normalized mass at most $2^{-(29/2980864)N}$ eventually. This includes unchanged cells by identity transport and uses the exact factor $(2N+2)^{20C_{\mathrm{old}}}$ above.

Consequently the residual $S_{16}$ family supplies the exponentially small scalar. The degree-eight and degree-sixteen direct owners supply their strictly forward original-weight rows, and the width-at-least-thirty-two owner supplies the remaining direct row. Every target is an ordinary subgroup count of degree $M < 2N$. Adding these four intrinsic owner families gives (24.8), with exponentially small scalar and aggregate row. No boundedness estimate for $A_M$ has been used to construct the recurrence; it is substituted directly into the ordinary master. □

## 25 Direct fusion for original binary orbits

The following cut-or-character alternative gives a shorter route to original local entries from width 32. It uses the deterministic separator and the actual action kernel, without the rank-moment calculation or the adaptive degree-32 cut. Its recurrence has complete ordinary subgroup counts at smaller degrees as targets. Thus it can be inserted directly in the ordinary recurrence. The complete binary-error recurrence of the preceding section also ends in complete ordinary targets; the theorem below is a parallel ambient estimate for sources having a large binary orbit. It may replace the corresponding large-orbit component when used, but it introduces neither a second target sequence nor an independent boundedness induction.

## 25.1 An original cut-or-character alternative

Write $B_j = \binom{j}{\lfloor j/2 \rfloor}$ and $\gamma = 5^{2/7}$. The permutation class estimate used in Proposition 8.3 implies the weaker bound $k(P)^7 \le 5^{2b}$ for every actual binary subgroup $P \le S_b$. Consequently, for a finite binary group $Q$, with $z = \dim \Omega_1 Z(Q)$, the original character envelope is

$$|\operatorname{Epi}(J,Q)| \le |\operatorname{Aut} Q|\, 2^{\alpha(z)b}, \qquad \alpha(z) = \frac{2z}{7}\log_2 5, \qquad J \le S_b. \tag{25.1}$$

Here $J$ is arbitrary: restriction to its original Sylow 2-subgroup is injective on epimorphisms to $Q$ and remains surjective onto $Q$. The same bound holds for any specified surviving subfamily by inclusion. This is the separate $\gamma$ envelope, not a replacement of every previously accepted 38/25 criterion.

**Theorem 25.1** (Original cut-or-character alternative)**.** *Let $U \le S_{2h}$ be transitive and binary, where $h = 2^k$ and $k \ge 4$. Fix an original invariant pair frame, with flip kernel $K$ and faithful transitive top $T \le S_h$. For each original $N \lhd U$, retain*

$$Q = U/N, \qquad A = K/(K \cap N), \qquad B = U/(KN),$$

*including the actual map $T \twoheadrightarrow B$ and the actual kernel chart $A \longrightarrow Q \twoheadrightarrow B$. At least one of the following alternatives holds.*

1. *There is a literal subspace $C \le A^B$ such that*

$$2\dim C + 4\dim(A/C)^B \le h - 2.$$

2. *The action of $T$ on $A$ is faithful, the whole center of $Q$ lies in $A$, and*

$$\Omega_1 Z(Q) = A^B, \qquad z = \dim \Omega_1 Z(Q) \le B_k, \qquad 5^{16z} < 2^{7(2h)}.$$

*In particular each original $N$ has a same-source fusion entry with an even prefix $v < 2h$ and a strictly positive gap. No splitting of $Q \to B$ is assumed.*

*Proof.* Put $S = A^B$, $t = \dim S$, $\ell = \dim(A/S)^B$, and $D_0 = 2B_{k-1}$. These are dimensions of the same actual section, so $t + \ell \le \dim A$. The original permutation-section bound and its one-character kernel restriction give $t \le B_k \le D_0$ and the complete displacement line-slice bound $D_0 - t$. The deterministic separator supplies $q$ independent functionals on $S$, with annihilator $C$, such that

$$\dim C = t - q, \qquad \dim(A/C)^B = q, \qquad 2q \le \ell + \max\{1, D_0 - t\}.$$

The fixed-quotient identity uses the full displacement slice; it does not merely test each scalar line separately.

If $2\dim A \le h$, the resulting even cost satisfies

$$2\dim C + 4\dim(A/C)^B = 2t + 2q \le t + \ell + D_0 + 1.$$

The inequalities $8D_0 \le 3h$ and $h \ge 16$ make the last expression at most $h - 1$. Evenness gives the first alternative. The inequality $t \le D_0$ is retained here; truncating a negative difference would not justify the displayed estimate.

Suppose instead that $2\dim A > h$. Let $H$ be the actual normal kernel of $T$ acting on $A$. If $H \ne 1$, its orbits on the original $h$ points have a common size $2^a \ge 2$: normality gives uniformity,

and faithfulness of $T$ excludes singleton orbits. The original permutation preimage of $A$, restricted to $H$, maps onto the trivial $H$-module $A$. Applying the orbit-section bound and $2B_a \le 2^a$ therefore gives $2\dim A \le h$, a contradiction. Thus $T$ acts faithfully on $A$. Since this action factors through the actual surjection $T \to B$, that map is injective as well.

For any central element of the original $Q$, its image in $B$ acts trivially on the literal kernel $A$. Faithfulness places the element in $A$. The actual kernel chart now identifies $Z(Q)$ with $A^B$; in particular this center is elementary. Finally $8B_k \le 3h$ gives $16z \le 3(2h)$, and $125 < 128$ gives the strict integer test. □

For the first alternative put $c = \dim C$, $r = \dim(A/C)^B$ and choose the entry

$$v = h + 2c, \quad \lambda = r/2, \quad D_C = |A/C|\,|H^1(B, A/C)|, \quad \Phi_C(J) = |\operatorname{Epi}(J, B)|\,2^{cd_2(J)}. \tag{25.2}$$

The original central extension and its actual quotient by $C$ give the envelope $D_C 2^{\lambda b}\Phi_C(J)$. Its full cocycle factor is retained. The faithful original cover $T \le S_h$ mapping onto the same $B$ gives all common-source moments

$$\sum_{J \le S_b} \Phi_C(J)^q \le s_{b+qv} \qquad (q \ge 1).$$

There is no assertion that $B$ itself has a faithful action of degree $h$. In the character alternative take

$$v = 0, \quad \lambda = \alpha(z), \quad D_C = |\operatorname{Aut} Q|, \quad \Phi_C(J) = 1.$$

Its moments equal $s_b$. In both alternatives define

$$e = \frac{2h - v - 8\lambda}{16} > 0.$$

The cut alternative has $2h - v - 8\lambda \ge 2$, but no such uniform lower bound is asserted for the character alternative. For each fixed finite collection of widths, positivity supplies a finite-menu minimum. All entries are chosen before the exterior degree and the exterior subgroup; the original normalizer is unchanged.

## 25.2 The ordinary forward recurrence for binary orbits

Let $s_n$ denote the number of all subgroups of $S_n$, let $L_n > 0$ be the exact parity-dependent benchmark, and put $\sigma_n = s_n/L_n$. Let $\mathcal{A}_n$ be the family of literal $H \le S_n$ having an actual orbit of size at least 32 whose restriction image is binary. The entire $H$, its complementary restriction, and their correlation are unrestricted.

**Theorem 25.2** (Original ordinary binary-orbit recurrence)**.** *There is a nonnegative row $K(n,m)$, with $K(n,m) = 0$ when $m \ge n$, such that for $n \ge 2048$,*

$$\frac{|\mathcal{A}_n|}{L_n} \le \sum_{m<n} K(n,m)\sigma_m. \tag{25.3}$$

*There are constants $C, \kappa > 0$ for which*

$$\sum_{m<n} K(n,m) \le C2^{-\kappa n}$$

*for all sufficiently large $n$. In particular the row sum is eventually at most $1/2$. Any subfamily defined by additional conditions on the original $H$ obeys the same recurrence by inclusion.*

*Proof.* An actual binary transitive orbit has power-of-two degree. For each width $w = 2h \in \{32, 64, 128, 256, 512\}$, choose one representative of each original action-conjugacy class and one invariant pairing for each representative. Apply Theorem 25.1 to every literal original normal subgroup. Relabelling the actual orbit and its actual complement gives physical coverage by these families. This is an upper union of original subgroups; no uniqueness of the chosen orbit or of its chart is required.

For a fixed representative $U$, normal axis $N$ and exterior degree $b = n - w$, use only the first same-source moment. Its exact contribution at the target degree $m = b + v$ is

$$\frac{n!}{b!\,|N_{S_w}(U)|}\, D_C 2^{\lambda b}\, \frac{L_{b+v}}{L_n}. \tag{25.4}$$

Every original normal is retained, including distinct summands with the same target degree. Since $v < w$, every such target is smaller than $n$. The exact benchmark ratio and $e > 0$ give exponential decay for this genuinely fixed finite menu, without an a priori estimate on the sequence $s_m$.

The widths 1024 and 2048 use their already proved positive zero-cut entries and the same finite first-moment argument. For widths at least 4096 use the original wide zero-cut installation and its proved normalizer-weighted menu aggregate. The actual zero entries are bounded by that full nonnegative row; numerical acceptance of extra entries is not used as a substitute for a local envelope. Its row sum has the proved uniform exponential bound.

Add these rows and cover the literal family $\mathcal{A}_n$ by the finite-width and wide sectors. Overlap only enlarges the upper bound. Adding their exponential estimates gives the asserted $C, \kappa$ and eventual contraction. All denominators in (25.4) remain the original symmetric normalizers. □

# 26 A joint cyclic-four character moment

For a finite group $J$, let $d(J), c(J)$ be the first two conjugate columns of the binary part of $J^{\mathrm{ab}}$, and put

$$q(J) = d(J) + c(J) = \log|\operatorname{Hom}(J, C_4)|, \qquad M_4(b, r) = \sum_{J \le S_b} 2^{rq(J)}, \qquad F(b, r) = \frac{b^2}{16} + \frac{br}{4} + \frac{r^2}{2}.$$

The two source columns must be retained jointly; the second column is not separately additive in an extension.

**Theorem 26.1.** *As $b \to \infty$,*

$$\log M_4(b, r) \le F(b, r) + o((b + r)^2),$$

*uniformly for $r/b$ in any fixed bounded interval. The range $r \ge b/2$ also satisfies the bound directly, without an upper restriction on $r/b$. Literal homomorphisms are counted, without a target automorphism divisor.*

## 26.1 The retained normal columns

We use the bounded-word counting construction of [RDT25, Sections 3–5 and Proposition 7.4], specifying the changes needed for the character weight. Fix a physical orbit bound $C$. In its nonexcess case, let $Q_i$ be quotients of actual transitive binary groups of degrees $n_i < C$, such that every $Q_i$-normal subgroup has normal-generator number at most $n_i/4$. Put $n = \sum_i n_i$. For an

efficiently ordered upper normal tableau write $N_i$ for its diagonal normal subgroup, $d_i = d_{Q_i}(N_i)$, and $c_i = \log|N_i|/n_i$. The efficient-generator division before the final maximization in that construction gives

$$E = \sum_{j<i}(c_i - c_j)n_j d_i, \quad 0 \le c_1 \le \cdots \le c_t \le 1, \quad d_i \le \min(n_i/4, c_i n_i) \tag{26.1}$$

as an upper bound on the logarithm of the number of groups with a fixed tableau. All tableau and ordering choices cost $O_C(n\log(n+2))$ by its normal-chain bound.

**Lemma 26.2.** *For a full subgroup $L$ with this tableau,*

$$d(L) \le D := \sum_i d_i, \qquad d(L) + c(L) \le S := \sum_i \min(2d_i, c_i n_i).$$

*Proof.* Filter $L$ by the kernels of successive tail projections. The $i$th quotient is $N_i$ with conjugation action supplied by the onto projection to $Q_i$. Restrict global characters $L \to C_4$ along this filtration. Two characters with the same previous restriction differ on the next term by an invariant character of $N_i$, hence by a character of $N_i/[N_i, Q_i]$. The latter abelian group has at most $d_i$ generators and order at most $2^{c_i n_i}$, so contributes at most $2^{\min(2d_i, c_i n_i)}$ maps. Multiplying the fibre bounds proves the second inequality. The same argument with $C_2$ proves the first. It bounds extendable global restrictions and does not presume that every invariant character extends. □

Put $T(t) = \sum_{c_i \ge t} n_i$, $0 < t < 1$. Layer-cake identities and (26.1) imply

$$E \le \frac{1}{4}\int_0^1 T(t)(n - T(t))\,dt, \qquad D \le \int_0^{1/4} T(t)\,dt, \qquad S \le \int_0^{1/2} T(t)\,dt. \tag{26.2}$$

For example the first integral before its factor $1/4$ is $\sum_{j<i}(c_i - c_j)n_i n_j$. It follows already that $E + rq(L) \le n^2/16 + nr/4 + r^2/2$, by maximizing separately on $0 < t < 1/2$ and $1/2 < t < 1$.

## 26.2 The abelian packet in the same tableau

Let $A$ be the product of the abelian physical coordinate quotients, on total support $a$, and put $B = A \times C_4^r$. The abelian quotient bound gives $\log|A| \le a/2$. For a fixed remaining projection $L$, the exact number of subgroups of $B \times L$ onto $L$ is

$$X_B(L) = \sum_{D \le B} |\operatorname{Hom}(L, B/D)|. \tag{26.3}$$

This follows by taking the actual intersection $D$ and the literal graph map. Write $x_j$ for the conjugate columns of $L_2^{\mathrm{ab}}$. The Hall subgroup formula, after dropping only the monotonicity restriction on quotient columns, bounds its logarithm by

$$\sum_j \Psi(B_j, x_j) + O_C(\log(a + r + 2)), \qquad \Psi(v, x) = \max_{0 \le y \le v} y(v - y + x) = \begin{cases} vx & v \le x, \\ (v+x)^2/4 & v \ge x. \end{cases}$$

There are boundedly many columns since the physical degrees are bounded. The Gaussian factors and the number of column profiles give the stated logarithmic error.

The derivative of $\Psi$ in its first argument increases in both arguments. Since $B_1 \ge B_j$ and $x_1 \ge x_j$, moving each unit of the original $A$-column mass into the first column can only increase the bound. Fill any remaining mass up to $a/2$. This gives

$$\log X_B(L) \le \Psi(a/2 + r, d(L)) + \Psi(r, c(L)) + O_C(\log(a + r + 2)). \tag{26.4}$$

Let the two maximizing Hall ranks be $\alpha \ge \beta$; this ordering is possible because both the ambient and source columns are ordered. Lemma 26.2 gives the joint inequality

$$\alpha d(L) + \beta c(L) \le (\alpha - \beta)D + \beta S \le \alpha \int_0^{1/4} T(t)\,dt + \beta \int_{1/4}^{1/2} T(t)\,dt.$$

Define

$$g_n(x) = \max_{0 \le z \le n} \{z(n-z)/4 + xz\} = \begin{cases} n^2/16 + nx/2 + x^2 & x \le n/4, \\ nx & x \ge n/4. \end{cases}$$

The tableau entropy and the full abelian fibre together are bounded by

$$\frac{1}{4} g_n(\alpha) + \frac{1}{4} g_n(\beta) + \frac{1}{2} g_n(0) + \alpha(a/2 + r - \alpha) + \beta(r - \beta).$$

Direct differentiation on the two branches proves

$$\max_{0 \le x \le v} \{x(v-x) + g_n(x)/4\} = (v + n/4)^2/4.$$

The maximizer is $x = v$ when $v \le n/4$, and $x = (v + n/4)/2$ otherwise. Relaxing the order between $\alpha, \beta$ now gives the upper bound

$$\frac{n^2}{32} + \frac{(a/2 + r + n/4)^2 + (r + n/4)^2}{4} = F(n + a, r) - \frac{na}{16} \le F(n + a, r). \tag{26.5}$$

This is the required abelian/nonabelian splice on one source.

## 26.3 Excessive quotients and bounded binary words

For a product $D$ of actual transitive binary actions of degree less than $C$, of total degree $b$, we claim

$$\log |\operatorname{Sub}(D \times C_4^r)| \le F(b, r) + \log \mathcal{P}(r) + O_C(b \log(b + r + 2)), \qquad \mathcal{P}(r) = 16(r+1)^2. \tag{26.6}$$

First condition on coordinate projections, refining any intransitive projection into its actual orbits. This costs $2^{O_C(b)}$ and does not increase physical support. Follow the literal normal-intersection construction in [RDT25, Proposition 7.4] on the physical coordinates, keeping the auxiliary factor $C_4^r$ throughout. Quotienting the embedded axes is injective because their product lies in the subgroup. In particular those axes are already killed by every auxiliary graph coordinate; no character is lost by quotienting first. The nonexcess branch is (26.5).

In the excessive branch choose the physical coordinate of degree $w$ in the priority of cases (IIa)–(IIc) of that proof. Its nonabelian quotient $Q$ has, by [RDT25, Theorem 5.7], a normal subgroup $F_0$ and parameters $e, z, g$ satisfying

$$z = \log |C_Q(F_0)|, \quad g = \log |[F_0, Q]|, \qquad ez + g \le w/2, \qquad z \le w/4.$$

The three values of $e$ are 3 for the degree-eight excess-two case, 1 for excess one, and $5/4$ for the remaining excess-two case; the excess-one branch retains the generator-equality hypothesis of that theorem.

In the remaining nonabelian projection select a maximal fully independent tail. Let $b_3$ be the degree of its dependent physical coordinates, and $b_4$ the degree of the independent tail together with every abelian physical coordinate. Thus $b_3 + b_4 = b - w$. The normal filtration in [RDT25,

Theorem 5.5] gives, on this same source, the bounds $eb_3/4 + b_4/2 + r$ for ordinary generators and $b_3/4$ for the derived-normal term in its epimorphism estimate. To check the auxiliary modification: the abelian-coordinate kernel contributes at most $b_4/2 + r$ generators, whereas commutators have trivial abelian coordinates, so the derived group projects isomorphically onto the derived nonabelian projection. Passing to quotients does not increase the relative normal-generator caps. No faithful action of $Q$ at degree $w$ is assumed.

The epimorphism estimate [RDT25, Theorem 4.6] consequently becomes

$$\log|\operatorname{Epi}(\text{source}, Q)| \le \frac{w(b_3+b_4)}{8} + \frac{wr}{4} + O_C(\log(b+r+2)).$$

This is paid by the exact reserve $F(b,r) - F(b-w,r) = w(b-w)/8 + wr/4 + w^2/16$. To perform the induction without repeating a terminal cost, retain one ordering and the whole monotone chain of original normal subgroups in each physical factor. If there are $t$ factors, each has order at most $2^C$, hence at most $2^{2^C}$ subgroups. A monotone chain is encoded by the entry time of each subgroup in its downward closure. The total history menu therefore has size at most

$$t!\,(t+1)^{2^{2^C}t}.$$

For one fixed history, the priority of an excessive quotient depends only on the current original normal subgroup and its original factor. Thus the same priority applies to every subgroup in that history, including its successive suffix projections. The literal decoder bounds each positive step by epimorphisms on that same suffix family; it does not introduce a fresh normal-subgroup menu.

Keep the initial support bound $b$ fixed in the induction. Each positive step pays at most $A_C(b+r+2)^{k_C}$, in addition to its exact difference of $F$ values, for constants depending only on $C$. At the stopping leaf, the nonexcess estimate contributes $T_C(b) = 2^{O_C(b\log(b+2))}$ once. The vertical group at zero remaining physical degree contributes at most $\mathcal{P}(r)2^{F(0,r)}$. Hence the fixed-history bound is

$$\mathcal{P}(r)\,T_C(b)\big(A_C(b+r+2)^{k_C}\big)^t 2^{F(b,r)}.$$

The quadratic differences telescope. Summing the single history menu, and using $t \le b$, gives (26.6). In particular neither the stopping cost nor $\mathcal{P}(r)$ is paid at every peel.

## 26.4 The weighted generation reduction

**Lemma 26.3** (Ordinary generation on the original orbits)**.** *Let $G$ be a finite soluble permutation group, and let $n_i$ be its nontrivial orbit lengths. Then*

$$d(G) \le 1280 \sum_i \frac{n_i}{\sqrt{\log n_i}}.$$

*Proof.* A soluble primitive group of degree $r$ has an actual regular elementary abelian normal subgroup. Its point stabilizer acts faithfully on that subgroup, giving $|P| \le r^{1+\lfloor \log r \rfloor}$. Consequently every composition series of $P$, and every subgroup's ordinary rank, has length at most $c(r) = (\log r)^2 + \log r$. This argument uses no sharp published primitive composition or normal-generation theorem.

Induct on the sum of the actual orbit lengths, retaining their joint faithful action. For a primitive head of degree $r$, the kernel of the remaining orbit actions embeds in that primitive image, so its ordinary rank is at most $c(r)$.

For an imprimitive head, take a minimal block of size $r$ and $s$ blocks. Keep the other orbit actions in the top group, even when that top acts nonfaithfully on these $s$ blocks. Along the actual chief series of the soluble primitive block component, each kernel is abelian. Its module generators supplement any subgroup already surjecting onto the quotient, so they lift ordinary generators. The proved induced-module bound and the preceding elementary composition bound give total cost at most

$$B(s)c(r), \qquad B(s) = \left\lceil \frac{4s}{\sqrt{\log s}} \right\rceil .$$

The two scalar inequalities, valid for $r, s \geq 2$, are

$$c(r) \leq \frac{1280r}{\sqrt{\log r}},$$
$$B(s)c(r) + \frac{1280s}{\sqrt{\log s}} \leq \frac{1280rs}{\sqrt{\log(rs)}}.$$

For the latter put $a = \log r$, $b = \log s$. Use $B(s) \leq 5s/\sqrt{b}$, $b \geq 1$, and

$$\sqrt{1+a}\left(1 + \frac{a^2+a}{256}\right) \leq 2^a.$$

Indeed $\sqrt{1+a} \leq e^{a/2}$ and $1+(a^2+a)/256 \leq (1+a/16)^2 \leq e^{a/8}$ for $a \geq 0$; now $\ln 2 \geq 5/8$ gives the assertion. Thus each compression decreases the total degree and preserves the claimed potential. Fixed points contribute zero and can be removed. □

**Lemma 26.4** (Bounded-section nilpotent core)**.** *Let $G$ be finite and soluble, of upper socle length at most $L$. Suppose every chief factor of every subgroup of $G$ has order at most $B$, and every abelian section has at most $M$ prime factors, counted with multiplicity. Then $G = \langle K, S\rangle$ for an actual nilpotent subgroup $K$ and a set $S$ with*

$$|S| \leq L(B!+1)\lfloor\sqrt{M}\rfloor.$$

*Proof.* Decompose the integral conjugation module of the actual socle $V$ into isotypic rows. Rows of multiplicity at most $\lfloor\sqrt{M}\rfloor$ share that many normal generators, by the projections onto distinct simple types. The sum of the multiplicities, weighted by the prime-factor lengths of their simple types, is at most $M$. Hence there are at most $\lfloor\sqrt{M}\rfloor$ remaining rows. Each simple type has at most $B$ elements. The action on these types has image of order at most $(B!)^{\lfloor\sqrt{M}\rfloor}$ and therefore needs at most $B!\lfloor\sqrt{M}\rfloor$ generators. Its kernel $C$ is the actual centralizer of the large rows and contains $V$. The normal image $C/V$ has upper socle length at most $L-1$; the section hypotheses pass to it. Apply induction there and lift its nilpotent core to a nilpotent supplement inside $C$. The large rows commute with that supplement, so adjoining them preserves nilpotence. Lift the quotient generators and adjoin the displayed action generators and the shared small-row generators. Their ordinary generated subgroup is $G$. This adds at most $(B!+1)\lfloor\sqrt{M}\rfloor$ generators at each of the $L$ steps. For $M = 0$ the group is trivial. □

*Proof of Theorem 26.1.* If $G = \langle K, x_1, \dots, x_l\rangle$, restriction and the displayed generator values determine a character. Thus

$$|\operatorname{Hom}(G, C_4)|^r \leq 4^{rl} |\operatorname{Hom}(K, C_4)|^r. \tag{26.7}$$

Choose one encoding for each $G$ and bound by the sum of all encodings. This avoids requiring any unproved closure property of a marked family.

For soluble groups in the original orbit-cutoff-$C$ containers, the actual orbit images embed in groups of order at most $C!$. Their subdirect socle length is at most $C!$, and every chief section of every subgroup has order at most $C!$. The proved abelian-quotient bound gives prime-factor length at most $2b$ for every abelian section. Lemma 26.4, with $L = B = C!$ and $M = 2b$, therefore supplies a nilpotent subgroup and $O_C(\sqrt{b})$ additional generators. The ambient soluble group has order $2^{O(b)}$. The tuple choices and (26.7) cost $2^{O_C(b^{3/2}+r\sqrt{b})}$. For a bounded-orbit nilpotent group, the Cartesian Sylow orbit grids of [RDT25, Lemma 2.7] compress the binary and odd actions to degrees $m, d$ with $m + d \le b$. The grid choices cost $2^{O_C(b\log(b+2))}$. The odd factor has no nontrivial $C_4$ character. Graph injection and (26.6) bound the binary moment by $2^{F(m,r)+o_C(b^2)}$ for $r = O(b)$; (1.1) bounds the odd factor by $2^{d^2/16+O(d^{3/2})}$. Since $F(m,r) + d^2/16 \le F(b,r)$, the bounded soluble case follows.

For the large-orbit step fix $\eta > 0$. Lemma 26.3 supplies a cutoff $C$ such that the large-orbit projection requires at most $\eta b/\delta$ generators, where $\delta = (\log 24)/3$ is the soluble order constant. Its kernel is an actual subgroup on the small support. Working in a product of maximal transitive soluble containers, the choices of lifts and their marked values cost at most $2^{\eta b^2+2\eta r b/\delta}$ by (26.7). Only a subquadratic container-menu exponent is needed. For every $\varepsilon > 0$, the labelled transitive-group count gives a constant $A_\varepsilon$ and at most $A_\varepsilon 2^{\varepsilon m^2}$ local containers on any $m$ labelled points. An orbit partition with fibre sizes $m_i$ therefore has at most $A_\varepsilon^b 2^{\varepsilon \sum_i m_i^2} \le A_\varepsilon^b 2^{\varepsilon b^2}$ choices. There are at most $b^b$ partition labels. Hence the complete original container menu has exponent $o(b^2)$; no sharper maximal-soluble class count is required. The bounded-source estimate is at most $F(b,r)$ since $F$ increases with physical degree. First take $b \to \infty$ with $\eta, C, \varepsilon$ fixed, and then $\eta, \varepsilon \to 0$. This proves the soluble result uniformly for $0 \le r \le K_0 b$. It does not assert a uniform $O(b^{3/2})$ error as $C$ varies.

Finally every finite group is generated by a soluble subgroup and one additional element by [RDT25, Theorem 8.4]. The additional costs $b!$ and $4^r$ are $2^{o(b^2)}$ in this range. For $r \ge b/2$, the direct bound $q(J) \le b/2$ gives $\log M_4(b,r) \le b^2/16 + br/2 + O(b^{3/2})$. Its leading expression is at most $F(b,r)$ because their difference is $r(2r-b)/4 \ge 0$. □

# 27 Markers, binary moments, and the one-triple audit

## 27.1 Exact natural-$S_3$ transport

**Proposition 27.1.** *Let $X \le T \times S_3^g$ be full on $T$ and every marker. Put $K_X = X \cap C_3^g$, let $\overline{X}$ be its image in $T \times C_2^g$, and let $L = \overline{X} \cap C_2^g$, of dimension $\ell$. Then*

$$\overline{X} \cong T \times L, \qquad X/O^2(X) \cong (T/O^2(T)) \times C_2^\ell, \qquad d_2(X) = d_2(T) + \ell, \qquad \tau_2(X) = \tau_2(T).$$

*For one marker the numbers of possible $X$ with $\ell = 1, 0$ are respectively $1$ and $2^{d_2(T)} - 1 + |\operatorname{Epi}(T, S_3)|$.*

*Proof.* The graph map $T \to C_2^g/L$ factors through $A_2(T)$ and has a linear lift, proving the first isomorphism. The kernel $K_X$ has odd order, so inflation is an isomorphism in binary cohomology. The maximal binary quotient is therefore that of $T \times L$. Künneth and the identical elementary factor show that the kernel defining $\tau_2$ is unchanged. For one marker Goursat's lemma gives: the full product; one common-$C_2$ graph for each nonzero binary character of $T$; and one literal graph for each epimorphism $T \to S_3$. These are disjoint. □

If $p_{T,g,\ell}$ counts the actual marker subgroups with this value of $\ell$, their exact attachment to a critical product $D$, with $f$ fixed points, has weight

$$\frac{1}{6^g g! f!} \sum_{\ell} p_{T,g,\ell} N_D((T/O^2(T)) \times C_2^{\ell}). \tag{27.1}$$

This identity does not license replacing an arbitrary complete exterior projection by its unmarked count.

## 27.2 A character moment on the complete binary error

Let $\mathcal{E}_N$ be all fixed-point-free 2-subgroups of $S_{2N}$ with at least one actual orbit outside Table 1, and put

$$E_N = \frac{|\mathcal{E}_N|}{L_{2N}}, \qquad pE_N = \frac{1}{L_{2N}} \sum_{Y \in \mathcal{E}_N} t(Y),$$

where $t(Y)$ is the number of its actual two-point orbits. Set $E_0 = pE_0 = 0$.

**Proposition 27.2** (Complete binary pair moment)**.** *For $N \geq 2$,*

$$pE_N \leq [7 + (6N)^{1/4}]E_N + r_N E_{N-1}, \qquad r_N = \frac{N-1}{4} \frac{c_{N-1} G_{N-1}}{c_N G_N}. \tag{27.2}$$

*The same estimate holds for the family of all fixed-point-free binary subgroups, with its corresponding counts.*

*Proof.* Each pair orbit supplies a nonzero character of $Y$. Let $q(Y)$ be the number of distinct such characters, $D(Y)$ the number of unordered pairs of equal characters, and $J_4(Y)$ the number of ordered selections of four pair orbits with independent characters. Then

$$t(Y) - q(Y) \leq D(Y), \qquad q(Y) \leq 7 + J_4(Y)^{1/4}.$$

For the second inequality, choose one original pair orbit for each distinct character. The span of at most three independent characters contains at most seven nonzero characters, so each of four successive choices has at least $q - 7$ possibilities. Thus $J_4(Y) \geq (q-7)^4$ when $q \geq 7$; the case $q < 7$ is immediate.

There is an injection from groups $Y$ with four ordered independent pair orbits to groups $H$ with one selected $E_8$ orbit and an ordering of its four center-pair orbits, at the same degree. For each ordered pair partition $\mathcal{P}$, fix a point chart once and use it to place an extraspecial group $E(\mathcal{P})$ on the eight selected points. Explicitly, inside $C_2^4 \rtimes S_4$, take the even flips semidirect the regular block $V_4$. Its centre has exactly these four pair orbits. Fix, once for each ordered partition, an isomorphism $E(\mathcal{P})/Z(E(\mathcal{P})) \cong C_2^4$. Replace $Y \leq C_2^4 \times J$ by its full preimage in $E(\mathcal{P}) \times J$. Independence gives the full $E_8$ projection. The centre of this local projection, its retained pair order, and the fixed isomorphism recover $Y$. The complete exterior projection is unchanged, so its noncritical orbit is retained. There are at most $\lfloor N/4 \rfloor$ possible selected $E_8$ orbits, each with 4! possible orders. Consequently

$$\sum_{Y \in \mathcal{E}_N} J_4(Y) \leq 6N|\mathcal{E}_N|.$$

Concavity of $x^{1/4}$ gives $\sum_Y q(Y) \leq [7 + (6N)^{1/4}]|\mathcal{E}_N|$.

Collapse two pair orbits carrying the same character to one pointed pair orbit. This operation and its inverse preserve a noncritical orbit. The exact labelled weight ratio is

$$\frac{1/(2^2\,2!)}{1/2} = \frac{1}{4}.$$

Thus

$$\frac{1}{(2N)!}\sum_{Y\in\mathcal{E}_N} D(Y) = \frac{1}{4(2N-2)!}\sum_{Y\in\mathcal{E}_{N-1}} t(Y) \le \frac{N-1}{4(2N-2)!}|\mathcal{E}_{N-1}|.$$

Divide by $c_N G_N$ and combine the two bounds. □

For $n = 2j+1$, put $p_{2j+1} = c_j + c_{j-1}/6$ and

$$\alpha_j = \frac{c_j}{p_{2j+1}}, \quad \beta_j = \frac{c_j}{3p_{2j+1}}, \quad u_j = \alpha_j + \beta_j[7 + (6j)^{1/4}], \quad v_j = \beta_j r_j. \tag{27.3}$$

A single fixed point contributes $\alpha_j E_j$. Replacing a pointed pair by a natural $S_3$ orbit has weight ratio $(1/6)/(1/2) = 1/3$ and contributes $\beta_j p E_j$. Thus their combined binary-error contribution is at most $u_j E_j + v_j E_{j-1}$. Section 33 gives $c_{j-1}/c_j \sim (96j)^{1/4}$, so $u_j \to 1$ and $v_j \to 0$. For the recurrence closure one can avoid this sharper asymptotic input: the elementary coefficient and Gaussian-ratio bounds give, uniformly,

$$0 \le u_j \le 4(j+1), \qquad 0 \le v_j \le 2\Phi^{-2}(j+1)^3.$$

Either fixed polynomial is absorbed by half of any positive exponential rate. Thus the odd transfer introduces no additional analytic frontier. In particular this is not a strict contraction on unrestricted complete counts. In the final assembly, the ordinary-target binary recurrence is applied at $j$ and $j-1$ before induction; the two target rows are padded to the ambient degree and these polynomial factors are absorbed into their exponential rates.

## 27.3 The split one-triple incidence

For a finite group $K$, a nonzero character $\chi : K \to C_3$ is split if it has a section. This is equivalent to being nonzero on an element of order three. The nonsplit characters, including zero, are therefore the annihilator of the span of the order-three elements in $A_3(K)$. They form a vector space. Write its dimension as $h_3(K)$ and set

$$\sigma_3(K) = 3^{d_3(K)} - 3^{h_3(K)}. \tag{27.4}$$

This counts oriented split nonzero characters exactly.

**Proposition 27.3** (Inverse-complement incidence)**.** *For any collection $\mathcal{P}$ of pairs $(K,\chi)$ with $K \le S_m$ and $\chi$ split and nonzero, define*

$$q(K,\chi) = |\{x \in K : |x| = 3,\ \chi(x) = 1\}|.$$

*Then*

$$|\mathcal{P}| = \sum_{|x|=3} \sum_{\substack{N^x = N,\ N\cap\langle x\rangle = 1 \\ (N\langle x\rangle, \chi_{N,x})\in\mathcal{P}}} \frac{1}{|\{y \in Nx : |y| = 3\}|}, \tag{27.5}$$

*where $\chi_{N,x}$ kills $N$ and sends $x$ to $1 \in C_3$. Furthermore, if $P \lhd K$ is a 2-group, inflation from $K/P$ preserves $d_3$, $h_3$, and $\sigma_3$.*

*Proof.* For each $(K,\chi)$ insert the identity $1=\sum_{x:|x|=3,\chi(x)=1} q(K,\chi)^{-1}$. With $N=\ker\chi$, the pair $(N,x)$ determines both $K$ and $\chi$, and the denominator is precisely the one displayed in (27.5). Every ternary character kills $P$. A split character upstairs has an order-three representative whose image is a representative downstairs. Conversely, choose an order-three element $y$ downstairs on which the character is nonzero and lift it to $x\in K$. Then $x^3\in P$, so $(x^3)^{2^a}=1$ for some $a$. The element $z=x^{2^a}$ satisfies $z^3=1$, and its character value is $2^a\chi(y)\neq 0$ in $\mathbb{F}_3$. Thus $z$ has order three and the inflated character splits. This proves splitness in both directions and hence all three equalities. □

These identities are required tests for applying a counting estimate to the one-external-triple case, customarily denoted $c=1$. One must bound the sum of (27.4), not an unmarked subgroup sum, and must retain the reciprocal in (27.5). For example, on the shared-top group $J=\mathbb{F}_4^r\rtimes C_3$ and target $\mathbb{F}_4^m\rtimes C_3$, above the fixed top map the exact cocycle and onto-lift counts are

$$4^{m(r+1)},\qquad 4^m\prod_{i=0}^{m-1}(4^r-4^i),$$

respectively. Maschke's theorem gives $H^1(C_3,\mathbb{F}_4^m)=0$; $\mathrm{Hom}_{C_3}(\mathbb{F}_4^r,\mathbb{F}_4^m)$ has $4^{mr}$ elements, and the $4^m$ coboundaries are literal distinct graphs. A second mark on the same $J$ produces a joint moment on that source. It cannot be replaced by two independently chosen source groups. These exact tests do not themselves prove that the high-incidence $c=1$ domain has been exhausted.

## 28 The complete marker sum

Here the binary exterior includes every even orbit, including the critical orbits and their already assembled attachments. Thus no terminal function is left to be multiplied into the counting comparisons below.

For $a\geq 1$ define the full-coordinate ternary fibre

$$L_3(a)=\sum_{\substack{K\leq\mathbb{F}_3^a\\ \mathrm{pr}_i K=\mathbb{F}_3\ (1\leq i\leq a)}} 3^{a-\dim K},\qquad L_3(0)=1.$$

For $g\geq q\geq 1$ put

$$H_{g,q}=\sum_{\substack{a_1+\cdots+a_q=g\\ a_i\geq 1}}\prod_{i=1}^{q}\frac{L_3(a_i)}{a_i!},\qquad H_{0,0}=1,$$

and put $H_{g,0}=0$ for $g>0$.

**Proposition 28.1** (Exact collapse of repeated signs)**.** *Let $\mathcal{Y}_m$ be a collection of fixed-point-free binary subgroups on $m$ points, closed under conjugation by $S_m$. Let $t_q(Y)$ count the $q$-element sets of actual pair orbits of $Y$ whose characters on $Y$ are distinct. For $n=m+3g+f-2q$, the number of subgroups obtained by replacing these selected pairs by $g$ natural $S_3$ orbits in $q$ sign classes, and adding $f$ fixed points, is exactly*

$$\frac{n!}{m!}\frac{2^q H_{g,q}}{6^g f!}\sum_{Y\in\mathcal{Y}_m} t_q(Y). \tag{28.1}$$

*Every group so counted recovers $Y$, the selected pairs, the sign classes, and all its ternary extension data.*

*Proof.* Write the original group as $X \le T \times S_3^g$, with $T$ its complete projection on the other moved orbits. The image $B$ of $X$ in $T \times C_2^g$ is a binary group. Its marker signs are nonzero characters. Group equal signs into classes of sizes $a_1, \dots, a_q$. Deleting repeated signs is injective on $B$; realize each retained sign on a pair. The resulting group is $Y \le S_m$ and the newly introduced pairs give the selected set.

Every $B$-invariant subspace of $C_3^g$ splits across these distinct sign isotypes by an explicit projection. Regard the signs as scalars in $\mathbb{F}_3$. For distinct signs $\chi, \psi$, choose $b \in B$ with $\chi(b) \neq \psi(b)$. The operator

$$(\chi(b) - \psi(b))^{-1}(b - \psi(b)I)$$

preserves the subspace, is the identity on the $\chi$-isotype and vanishes on the $\psi$-isotype. Multiplying one such diagonal operator for every other distinct sign gives the exact $\chi$-projection. These projections sum to the identity, with each distinct sign counted once, regardless of its coordinate multiplicity. In an isotype of dimension $a_i$, the intersection $K_i$ must have full coordinate projections. Indeed, lift an original marker 3-cycle and raise it to a sufficiently large power of 4: its binary image vanishes and the 3-cycle is unchanged. Conversely, a full ternary kernel coordinate together with a nontrivial sign generates the full original $S_3$ coordinate.

For a fixed invariant kernel $K = \bigoplus_i K_i$, let $P$ be the whole original preimage of $B$. Subgroups of $P$ with image $B$ and kernel $K$ correspond exactly to sections of $P/K \to B$. The fixed transpositions on the marker coordinates, together with the unchanged exterior coordinate, give one such section. Relative to this section the fibre is exactly $Z^1(B, \mathbb{F}_3^g/K)$, with no quotient-automorphism factor.

This cocycle count has an elementary inverse. On a nontrivial sign component $V_i = \mathbb{F}_3^{a_i}/K_i$, a cocycle $z$ vanishes on $\ker \chi_i$: there it is a homomorphism, and each value is annihilated by both a power of 2 and 3. Choose an actual $s_i \in B$ with $\chi_i(s_i) = -1$. The cocycle identity then gives

$$z(b) = \chi_i(b)z(s_i) - z(s_i) \qquad (b \in B).$$

Thus evaluation at $s_i$ is inverse to the principal-cocycle map, since $-1 - 1 = 1$ in $\mathbb{F}_3$. The distinct-sign projections descend to the original quotient; summing their projected evaluations gives the inverse for the whole quotient. Consequently the fibre has exactly $3^{g-\dim K} = \prod_i 3^{a_i - \dim K_i}$ elements, and the literal lift count is $\prod_i L_3(a_i)$.

The original marker weight is $1/(6^g g! f!)$. Assigning the displayed marker coordinates to the $q$ selected pairs contributes $g!/\prod_i a_i!$. A selected pair already has action weight $1/2$, so replacing the $q$ pairs contributes $2^q$. These give the coefficient in (28.1); the change of labelled support gives $n!/m!$; here closure under relabelling is essential. An ordering of the selected pairs only indexes the summation variables $a_i$; it creates no $q!$ factor. Projection recovers all the displayed data, proving equality. □

**Proposition 28.2** (A presentation bound for collapsed pairs)**.** *Under the hypotheses of Proposition 28.1, write $m = 2M$ and suppose $0 \le q \le M$. The same number of groups is at most*

$$\frac{n!}{m!}\frac{2^q H_{g,q}}{6^g f!}(M)_q\,|\mathcal{Y}_m|, \qquad (M)_q = \frac{M!}{(M-q)!}.$$

*This bound also follows directly by covering the groups with displayed collapsed presentations, without identifying equivalent presentations.*

*Proof.* For each original sign image, enumerate its $q$ distinct signs and assign each of the $g$ marker coordinates to its sign. These presentations cover the original family; counting all enumerations

gives an upper bound. For a fixed multiplicity profile, the assignments contribute $g!/\prod_i a_i!$ and the literal extension fibres contribute $\prod_i L_3(a_i)$. Thus their complete contribution is $g!H_{g,q}$.

Fix an exterior orbit profile with $p$ existing pair orbits. Merge the $q$ displayed collapsed pairs with those $p$ pairs, and retain the condition that the resulting subgroup belongs to $\mathcal{Y}_m$. Dropping distinctness and any conditions referring to the distinguished $q$ positions only enlarges this model family; its nonnegative presentation weights are kept unchanged. Relabelling invariance allows the ordinary full-profile labelling formula to be applied to it. The original pair divisor is $2^p p!$, whereas the merged pair divisor is $2^{p+q}(p+q)!$. Their ratio is

$$2^q \frac{(p+q)!}{p!} = 2^q (p+q)_q \le 2^q (M)_q.$$

Every other exterior normalizer and occurrence factorial is unchanged. Together with the original marker divisor $6^g g!$, the fixed-point divisor $f!$, and the labelled-support ratio $n!/m!$, this gives the claimed bound. Summing over exterior profiles counts each eligible collapsed profile at most once for the fixed value of $q$. □

**Theorem 28.3** (Complete noncanonical marker row)**.** *Write $n = 2N + \varepsilon$. For even $m = 2M < n$, define*

$$K_{\mathrm{mark}}(n,m) = \frac{c_M G_M}{p_n G_N} \sum_{g,f,q} \frac{2^q H_{g,q}}{6^g f!} \binom{M}{q}, \tag{28.2}$$

*where the sum is over nonnegative integers satisfying*

$$n = m + 3g + f - 2q, \quad \begin{cases} q = 0, & g = 0, \\ 1 \le q \le \min(g, M), & g > 0, \end{cases} \qquad D = \frac{g + f - \varepsilon}{2} \ge 1.$$

*For all sufficiently large $N$, at both parities,*

$$\sum_{\substack{m<n \\ m \text{ even}}} K_{\mathrm{mark}}(n,m) \le 2^{-N/10}. \tag{28.3}$$

*For collections $\mathcal{Y}_m$ closed under conjugation by $S_m$, the complete normalized count of the corresponding marker groups whose collapsed group lies in $\mathcal{Y}_m$ is at most $\sum_m K_{\mathrm{mark}}(n,m)|\mathcal{Y}_m|/L_m$. The same conclusions hold for the larger kernel $K^{+}_{\mathrm{mark}}(n,m)$ obtained by replacing $\binom{M}{q}$ with $(M)_q$.*

*Proof.* The counting comparison for $K_{\mathrm{mark}}$ follows from Proposition 28.1 and $t_q(Y) \le \binom{M}{q}$. In particular it does not assume boundedness of any normalized subgroup count.

Put $\lambda = \log_2 3$ and $k = g - q$. Counting ordered bases bounds the number of $d$-dimensional subspaces of $\mathbb{F}_3^a$ by $3^{d(a-d+1)}$: there are at most $3^{ad}$ ordered independent $d$-tuples, and each such subspace has at least $3^{d(d-1)}$ ordered bases. Dropping the full-coordinate condition therefore gives

$$L_3(a) \le (a+1) 3^{a^2/4 + a}.$$

For $a \ge 2$, use $a + 1 \le 3^{a-1}$ to obtain

$$L_3(a) \le 3^{(a-1)^2/4 + 15(a-1)/4} \le 2^{\lambda(a-1)^2/4 + 6(a-1)},$$

where $\lambda < 8/5$. The same bound holds at $a = 1$ because $L_3(1) = 1$ exactly. Consequently

$$H_{g,q} \le 2^{\lambda k^2/4 + 6k + g}.$$

Here $\sum_i (a_i-1)^2 \le k^2$, and the number of compositions of $g$ is at most $2^g$; dropping $\prod_i a_i!$ is harmless. For every term of (28.2),

$$M = N - D - k, \qquad 0 \le k \le \min(2D, N-D), \qquad g \le 2D+1.$$

The uniform Gaussian estimates leave quadratic exponent

$$Q = -\frac{N(D+k)}{2} + \frac{(D+k)^2}{4} + \frac{\lambda k^2}{4}.$$

We claim $Q \le -13ND/60$. Since $\lambda < 8/5$,

$$60Q + 13ND \le E := 15D^2 + 30Dk + 39k^2 - 17ND - 30Nk.$$

If $3D \le N$, factor this polynomial as

$$E = -(17D+30k)(N-3D) - (2D-k)(39k+18D) \le 0.$$

If $3D \ge N$, use instead

$$E = -(3D-N)(9N-8D) - (N-D-k)\big(39k + 9(N-D)\big) \le 0.$$

All displayed factors are nonnegative under the stated constraints, including the endpoint $M = 0$.

The coefficient recurrence gives $c_M/p_n \le (2N)^{D+k}$. Also $\binom{M}{q} \le (M)_q \le M^q \le N^q$, $q \le 3D$, and $6k+g \le 15D$. The omitted factor $2^q/(6^g f!)$ is at most one. For fixed $D$ there are at most $(2D+2)^2$ choices of $g, f, q$. The complete fixed-$D$ logarithm is therefore at most $-13ND/60 + O(D\log(N+2))$, with one absolute implied constant. Eventually the logarithmic error is at most $ND/60$, uniformly in $D$, so the fixed-$D$ logarithm is at most $-ND/5$. There are at most $N$ positive defects, and therefore the complete sum is at most $N2^{-N/5} \le 2^{-N/10}$ for all sufficiently large $N$. The same proof therefore bounds $K^{+}_{\mathrm{mark}}$; its counting comparison follows directly from Proposition 28.2. In the $D = 0$ states used below, $q$ is zero or one, so the falling factorial and binomial factors coincide. □

**Corollary 28.4** (Pure markers)**.** *The normalized mass of all subgroups whose even orbits are critical and whose marker profile has $D \ge 1$ is $O(2^{-N/10})$. This sum includes all actual critical attachments and both parities.*

*Proof.* Collapse yields only critical binary orbits. Theorem 1.1 bounds their complete normalized count uniformly, including all finite initial degrees. Apply Theorem 28.3 to this conjugation-invariant family. This argument counts already assembled groups, so it also includes non-Frattini central attachments. □

**Corollary 28.5** (One typed binary-error transfer)**.** *Suppose a family consists of fixed points, natural $S_3$ orbits, and binary orbits, at least one of which is noncritical. Its entire marker contribution is bounded by*

$$\sum_{\substack{m<n \\ m \text{ even}}} H_E(n,m) E_{m/2}, \qquad H_E(n,m) = K_{\mathrm{mark}}(n,m) + \mathbf{1}_{n=2N+1}\big(u_N \mathbf{1}_{m=n-1} + v_N \mathbf{1}_{m=n-3}\big),$$

*with $u_N, v_N$ from* (27.3)*. There is no scalar remainder.*

*Proof.* The noncritical binary orbit is unchanged by marker collapse, so the noncanonical part targets the complete conjugation-invariant family defining $E$. Any additional restrictions only decrease the original count. The only $D = 0$ marker states are no marker in even degree, one fixed point, and one $S_3$ in odd degree. The even state belongs to the complete binary source sector of the disjoint partition. The exact odd contributions are $\alpha_N E_N$ and $\beta_N p E_N$ by (28.1), and Proposition 27.2 gives the asserted two targets. Neither four-pair fusion nor duplicate collapse is required to preserve any earlier exclusion; they preserve a noncritical actual orbit, which is the defining condition of the complete target $E$. □

This corollary is an intermediate typed transfer into the complete binary error, not a recurrence used for induction. The final assembly immediately composes its zero-defect $E$-terms with Theorem 24.8, whose targets are ordinary counts.

For the final boundedness-first assembly we use a coarser but more compositional form of the same comparison. When $D \geq 1$, take $\mathcal{Y}_m$ to be the collection of all subgroups, so the marker kernel points directly to $A_m$. Only the zero-defect states retain the current and predecessor binary errors. Substituting the ordinary-target binary recurrence at those two indices then produces one strictly forward ordinary row.

## 29 The first split one-triple application

We consider subgroups with exactly one actual regular $C_3$ orbit and no actual natural $A_4$ orbit. On its complement of size $m = n - 3$, an external graph is determined by a literal nonzero character $\chi : K \to C_3$. Fix a generator of the displayed $C_3$ to specify its orientation. For split characters, let $\Theta(K)$ be the set whose complete graphs survive any preceding disjoint exclusions. The exact surviving character weight is

$$\sigma_3^{\mathrm{surv}}(K) = |\Theta(K)| = \sum_{\substack{\chi \neq 0 \\ \chi \text{ split}}} \mathbf{1}_{\{\mathrm{graph}(\chi) \text{ survives}\}}, \qquad 0 \leq \sigma_3^{\mathrm{surv}}(K) \leq 3^{d_3(K)} - 3^{h_3(K)}. \tag{29.1}$$

In particular the inequality need not be equality after exclusions. On a fixed labelled complement the surviving mass is precisely

$$\frac{n!}{6m!} \sum_{K \leq S_m} \sigma_3^{\mathrm{surv}}(K). \tag{29.2}$$

The sum restricts to residuals without a regular $C_3$ or natural $A_4$ orbit. The factor six is the original action normalizer, and there is one selected triple because the actual orbit is unique.

**Proposition 29.1** (Low one-triple cone)**.** *The split graphs with $20d_3(K) \leq 3m$ have normalized mass at most*

$$K_{\mathrm{lo}}(n,m)\frac{s_m}{L_m}, \qquad K_{\mathrm{lo}}(n,m) = \frac{1}{6}\Delta(n,m)3^{3m/20} \leq 2^{-\eta m + O(\log(n+2))}, \quad \eta = \frac{1}{4} - \frac{3\log_2 3}{20} > 0.$$

*Proof.* Apply (29.1) to each residual before enlarging its number to $s_m$. The labelled factorials cancel exactly as in (29.2). Here is an explicit bound covering every fixed deletion width $w$:

$$\Delta(b+w,b) \leq \phi^{-2} 2^{(w+1)/4}(b+w+1)^{w+2} 2^{-\lfloor w/2 \rfloor b/4}, \qquad \phi = \prod_{j \geq 1}(1 - 2^{-j}). \tag{29.3}$$

Indeed, the two guarded parity coefficients at rank $k$ both lie between $A_k$ and $(k+1)A_k$, since $A_{k-1} \leq 2kA_k$. The rank shift $t = \lfloor (b+w)/2 \rfloor - \lfloor b/2 \rfloor$ lies between $\lfloor w/2 \rfloor$ and $w$. Apply the

coefficient shift bound $A_k \le (2(k+t))^t A_{k+t}$ and the Gaussian ratio bound; their polynomial factors are at most $(b+w+1)^{w+2}$. The Gaussian exponent is at most $(w+1)/4 - \lfloor w/2 \rfloor b/4$, proving the display also at rank zero. For $w = 3$ this gives the required loss $m/4$. Finally $3^5 < 2^8$ gives $3\log_2(3)/20 < 6/25 < 1/4$, so the low-cone kernel is also $O(2^{-m/200})$ after absorbing its fixed polynomial factor. □

We next prove the earlier orbit estimates actually needed for the high cone. Only finitely many physical degrees occur in this application; no growing action menu is needed. Put $\gamma = (\log_2 3)/3$ and

$$Z_J(U) = \sum_{N \lhd U} |\operatorname{Epi}(J, U/N)|, \qquad a(U) = |N_{S_w}(U)|, \quad J \le S_b.$$

All target normal subgroups in this sum are literal. Fixing each target quotient map is the asymmetric Goursat convention, so there is no additional automorphism divisor.

We use one fixed-source reduction throughout these rows. Suppose an actual target extension has elementary $p$-kernel module $M$ and top $B$, and that $M$ embeds equivariantly in one copy of $\mathbb{F}_p[B]$. For an onto top map $\beta : J \twoheadrightarrow B$, evaluation at the identity coefficient injects the equivariant homomorphisms from $\ker\beta$ to $M$ into $\operatorname{Hom}(\ker\beta, \mathbb{F}_p)$. Indeed, equivariance and the surjectivity of $\beta$ recover every regular coordinate from that one coefficient. Consequently every surviving lift fibre has size at most

$$|M|\,|H^1(B, M)|\,|\operatorname{Hom}(\ker\beta, \mathbb{F}_p)|.$$

This remains valid for empty and nonsplit fibres. We first sum these character masses over the actual top epimorphisms, and only then over every literal original normal axis. The same complete source $J$ is retained in all terms; the fixed module and cohomology factors are absorbed into $D_U$. Thus the row-specific numerical work below concerns only the top maps and the character ranks of their actual kernels.

**Lemma 29.2** (Finite earlier orbit fibres)**.** *For the following actual actions, $Z_J(U)/a(U)$ is at most $D_U(1+b)2^{\alpha_U b}$, with a fixed constant $D_U$:*

| *action* | $w$ | $\alpha_U$ |
|---|---|---|
| $6T1, 6T4, 6T5, 6T6$ | 6 | $\gamma$ |
| $12T20, 12T85, 12T164$ | 12 | $2\gamma$ |
| $12T194$ | 12 | $16/15$ |
| $12T228, 12T229, 12T265$ | 12 | $1/2 + 5\log_2(3)/9$ |
| *transitive 3-group* | 9 | $5\log_2(3)/9$ |
| *transitive 3-group* | 27 | $5\log_2(3)/3$ |

*The degree-nine row and the three final degree-twelve actions require $J$ to have exactly one regular $C_3$ orbit and no natural $A_4$ orbit. Every other row permits arbitrary $J$. More precisely, the degree-nine numerical estimate consumes only the conjugacy-invariant consequence*

$$9d_3(J) \le 2b + 3;$$

*the displayed orbit pattern is the structural hypothesis which produces that budget on the retained source.*

*Proof.* We give the fibre arguments, including the shared source budgets. We use the internally proved bounds

$$d_2(F) \le b/2, \qquad d_3(F) \le b/3$$

for every literal permutation subgroup $F \le S_b$. They follow by restriction to an actual Sylow subgroup and degree induction on faithful $p$-group actions, and hence apply unchanged to literal kernels of the top maps below. For the few whole abelian target terms we separately use the abelianization-order bound $|J/J'| \le 3^{b/3}$ from [KP89]. Constants depending on a fixed target, including its finite normal menu, are absorbed only after its full quotient sum has been written.

For $6T1 = C_6$, the full fibre is $|\operatorname{Hom}(J, C_6)|$. For $6T5 = C_3 \wr C_2 \cong C_3 \times S_3$, its normal quotients are $1, C_2, C_3, C_6, S_3, C_3 \times S_3$. Set $F = J'J^2$ and $E = J/F$. The semisimple $\mathbb{F}_3E$-module $H_1(F; \mathbb{F}_3)$ has trivial multiplicity $d = d_3(J)$ and sign multiplicities $a_\theta$, with $d + \sum a_\theta \le b/3$. Coprime inflation–restriction gives exactly

$$|\operatorname{Epi}(J, S_3)| = 3\sum_{\theta \ne 0}(3^{a_\theta} - 1), \quad |\operatorname{Epi}(J, C_3 \times S_3)| = (3^d - 1)3\sum_{\theta \ne 0}(3^{a_\theta} - 1).$$

The product is onto because $C_3$ and $S_3$ have no common nontrivial quotient. Since $\sum(3^{a_\theta} - 1) \le 3^{\sum a_\theta} - 1$, each expression is at most $3^{b/3+1}$. The central and twisted characters have used one module budget.

The actions $6T4, 6T6$ are respectively $A_4$ in degree six and $C_2 \times A_4$ in degree six. Their binary bases are cyclic $\mathbb{F}_2[C_3]$-modules. Likewise $12T20, 12T85, 12T164$ have cyclic binary bases and elementary tops $C_3^2$, with the top action on the base allowed to have a kernel. Here is the requisite general estimate for $M \rtimes C_3^r$, $r = 1, 2$, with cyclic binary $M$. Each quotient is $V \rtimes C_3^h$, $h \le r$, with $V$ a multiplicity-one semisimple module. Use the same source $F = J'J^3$, $E = J/F$, $D = H_1(F; \mathbb{F}_2)$ for every top map. Write the trivial multiplicity of $D$ as $r_0$ and its nontrivial simple multiplicities as $a_\alpha$. Each nontrivial type has dimension two, and

$$r_0 + 2\sum_\alpha a_\alpha \le b/2.$$

For a fixed onto top map $\theta : E \to C_3^h$, surjective restrictions to $F$ number $(2^{r_0} - 1)^e \prod_i (4^{a_{\alpha_i(\theta)}} - 1)$, where $e \in \{0, 1\}$. If a nontrivial target constituent is retained, fix one of its two concrete ternary characters. Pulling it back to a specified source type permits at most $2 \cdot 3^{(h-1)d_3(J)}$ top maps. Bound the remaining factors using the one source budget and sum the retained type. The result is at most

$$(b/2)\, 3^{(h-1)d_3(J)} 2^{b/2}.$$

For each restriction and top map, the lift torsor has at most $|M|$ elements, by coprime cohomology on $E$. Abelian quotients are bounded by $|J/J'|^r$ (or by one such factor when $h = 0$). Thus all quotients obey the claimed $(1 + b)2^{r\gamma b}$ bound. This argument treats nonfaithful top actions as well as faithful ones.

For $12T194 = C_3 \wr A_4$, use its full ternary block kernel. Every quotient is an extension of a quotient $S$ of $A_4$ by a module subquotient $X$ of the natural permutation module $\mathbb{F}_3^4$. Every such $X$ embeds in $\mathbb{F}_3[S]$. To see this, first restrict to the normal $V_4$. Over a splitting field its trivial character gives one trivial $C_3$-module; its three nontrivial characters form one orbit and give a simple matrix-algebra component. Subquotients of the first embed in the regular $C_3$ block, and the second component is semisimple. The whole permutation module embeds in $\mathbb{F}_3[A_4]$ by coset sums. If the action factors through $S$, the image lies in the fixed space of the kernel, which is the inflated regular $\mathbb{F}_3[S]$-module. For a concrete top map $J \to S$ with kernel $F$, the restriction choices are therefore at most $3^{d_3(F)} \le 3^{b/3}$. The compatible lift torsor has a fixed target-dependent size: after the restriction is fixed, differences descend to $S$.

The quotient menu itself has a shorter structural proof. If $q : A_4 \twoheadrightarrow S$ and $N = \ker q$, then $|N|$ divides 12. A normal subgroup of order two is central, whereas $Z(A_4) = 1$. For $|N| = 3, 4, 6$, the quotient has order $4, 3, 2$, respectively, and is abelian. Thus

$$V_4 = [A_4, A_4] \leq N.$$

The orders three and six are impossible, and order four forces $N = V_4$. Hence $S$ is $A_4$, $C_3$, or trivial. For the latter two targets the internal ternary character-rank bound gives the required top count. The full target has a shorter same-source argument. Put $D = [J, J]$, identify $[A_4, A_4]$ with $V_4$, and fix one nonzero binary character $\eta : V_4 \to C_2$. Label an epimorphism $f : J \twoheadrightarrow A_4$ by

$$\chi_f = \eta \circ f|_D \in \mathrm{Hom}(D, C_2).$$

This label determines the literal kernel of $f$. Indeed, if $\chi_f = \chi_g$ and $x \in \ker f$, then surjectivity of $g$ gives, for every $a \in A_4$,

$$\eta([g(x), a]) = \eta(g([x, y])) = \eta(f([x, y])) = 0 \qquad (a = g(y)).$$

The action of $A_4/V_4 \cong C_3$ permutes the three nonzero elements and the three nonzero characters of $V_4$ freely. Consequently $\eta([v, a]) = 0$ for every $a \in A_4$ forces $v = 1$. Thus $x \in \ker g$, and symmetry gives $\ker f = \ker g$.

Every fixed kernel supports exactly $|\operatorname{Aut}(A_4)|$ literal epimorphisms. Applying the internal binary half-rank theorem to the same derived subgroup $D \leq S_b$ therefore gives

$$|\operatorname{Epi}(J, A_4)| \leq |\operatorname{Aut}(A_4)|\,|\mathrm{Hom}(D, C_2)| \leq |\operatorname{Aut}(A_4)|\,2^{\lfloor b/2 \rfloor}.$$

Since $1/2 < 8/15$, this embeds directly in the required quotient-top exponent, without a polynomial loss. For a fixed top map, the literal kernel contributes at most $3^{d_3(F)} \leq 3^{b/3} \leq 2^{8b/15}$. Summing the three actual quotient targets therefore gives $D2^{16b/15}$, as recorded in the table.

For degree-nine 3-groups, the target chart has two forms: an elementary ternary group of rank at most two, or an extension of $C_3$ by a module embedding in $\mathbb{F}_3[C_3]$. The source pattern is aggregated before any target is counted. If $o_0$ is its unique regular $C_3$ orbit, then for every literal orbit image and every original normal axis the local bounds have the form

$$9\delta_o \leq 2|o| + 3\mathbf{1}_{\{o=o_0\}}.$$

The faithful orbit filtration sums these inequalities on the same complete source $J$. In the present one-regular-$C_3$, no-natural-$A_4$ case, (9.5) reads

$$9d_3(J) \leq 2b + 3.$$

The arbitrary-normal block-section theorem below applies to both target forms at once. Specialised to degree nine, and retaining the same complete source $J$ on every normal axis, it gives a constant $D_U$ depending only on the fixed retained action such that

$$|\operatorname{Epi}(J, Q)| \leq D_U 3^{d_3(J)+b/3} \tag{29.4}$$

for every actual quotient target $Q$. Combining this with (9.5) gives the ternary exponent $5\log_2(3)/9$; when the descended extension has a binary kernel, the internally proved literal-kernel bound $d_2(\ker\beta) \leq b/2$ gives the mixed exponent $1/2 + 5\log_2(3)/9$.

For the three remaining degree-twelve actions, the actual action has three blocks of four points, binary translation base $W \le V_4^3$, and a 3-group top $P$ acting transitively on the nine nonzero local translations. Let $c$ be the number of actual regular-$C_3$ orbits of the complete source $J \le S_b$. The relative orbit filtration and the binary rank bound after deleting those same $3c$ points give the compatible inequalities

$$4d_3(J) \le b + c, \qquad 2d_2(\ker\theta) + 3c \le b \tag{29.5}$$

for every retained top epimorphism $\theta : J \twoheadrightarrow S$. The second inequality applies to the literal kernel of each $\theta$; it does not replace those kernels by $O^3(J)$.

The full module $V_4^3$ is cyclic: a nonzero local translation generates its local plane under the block stabilizer, and its top translates generate all three planes. Maschke's theorem makes every invariant submodule and quotient cyclic. Consequently the binary kernel $M$ of every normal quotient embeds, as an $\mathbb{F}_2[S]$-module, in one regular module. For fixed $\theta$ the regular-module evaluation theorem and the coprime cocycle torsor therefore give

$$\#\{\text{lifts over } \theta\} \le 2^6 2^{d_2(\ker\theta)}.$$

Apply (29.4) to the same $J$ and the actual nine-point action before enlarging the source class. With $L = \log_2 3$, the combined exponent is

$$\frac{b-3c}{2} + L\left(\frac{b+c}{4} + \frac{b}{3}\right) = \left(\frac{1}{2} + \frac{7L}{12}\right) b + \left(-\frac{3}{2} + \frac{L}{4}\right) c.$$

The coefficient of $c$ is negative. Since $L < 8/5$, the complete all-normal row has the uniform binary slope $43/30 < 3/2$. Thus the binary lift and ternary top are combined on one source before their common support parameter is discarded.

Finally we prove the degree-27 row without classifying its 3-groups. Let $U$ be transitive of degree $w = 3s$, choose the literal three-point block cover, and write $K$ for its complete correlated block kernel. For every original normal subgroup $N \triangleleft U$, put $M_N = K/(K \cap N)$ and $T_N = U/(KN)$. The reversible block chart retains this exact section, including proper diagonal and subdirect intersections. Its invariant dimension is at most $s/3$. The Schur lift count and the one-third rank bound for the actual Sylow subgroup of each source kernel give

$$|\operatorname{Epi}(J, U/N)| \le |\operatorname{Epi}(J, T_N)|\,|M_N|\,|H^1(T_N, M_N)|\,3^{sb/9}. \tag{29.6}$$

The module and cohomology factors are retained on each normal axis and are absorbed only after the finite original-normal sum.

It is useful to keep $d = d_3(J)$ visible. If the quotient envelope in degree $w$ is $D_U 3^{d + a_w b}$, then (29.6) gives $a_{3s} = a_s + s/9$, with $a_3 = 0$. Thus the relevant coefficients are

| $w$ | source class | $d + a_w b$ after the source bound | binary exponent |
|---|---|---|---|
| 3 | unrestricted base | $b/3$ | $8b/15$ |
| 9 | marked $c = 1$ | $5b/9 + O(1)$ | $8b/9$ |
| 9 | unrestricted internal | $2b/3$ | $16b/15$ |
| 27 | unrestricted | $5b/3$ | $8b/3$. |

The unrestricted degree-nine line is the one used inside degree twenty seven; the marked improvement is not imported into that recursion. For a fixed action all normal-axis module and cohomology factors are finite, and the original-weight physical sum gives the degree-27 row with exponent $8b/3$. □

**Lemma 29.3** (Actual action membership)**.** *Every exceptional action in Theorem 9.5, other than natural $C_3, A_4$, is accepted by Lemma 29.2. The action descriptions used there are certificates on the original permutation sets.*

*Proof.* For the degree-six actions the witnesses are, in order, regular $C_6$, the degree-six action of $A_4$, $C_3 \wr C_2$, and $C_2 \times A_4$ with its cyclic three-dimensional binary base. These witnesses also show that the first and third have normal Sylow 3-subgroup of index two, the second is an irreducible coprime affine action outside degree four, and the fourth has cyclic binary module and elementary ternary top. Thus they belong to the earlier physical action domains.

The degree-six ownership statement also has a structural proof. Choose an actual minimal block system. Highness rules out primitivity, and the degree product leaves either fibres of size three over two blocks or fibres of size two over three blocks. In the first case the faithful top has order two. A nontrivial sign on a three-point fibre annihilates the complete relative ternary head by the signed-coordinate argument, contradicting highness. Hence all fibre signs are trivial, the literal block kernel is a normal 3-group, and its index is two. Thus the whole $3 \times 2$ branch enters the first earlier domain without finite action recognition.

In the $2 \times 3$ branch, highness forces the faithful three-block top to have order three. The literal block kernel is its actual invariant submodule of the binary permutation module $\mathbb{F}_2^3$. Every invariant submodule of this local $C_3$-module is generated by the orbit of one vector. Schur–Zassenhaus supplies an actual complement to the binary kernel, and restriction identifies that complement with the three-block top. The additive orbit words then translate through the faithful pair-coordinate chart into products of actual conjugates of the distinguished kernel element. This is precisely the cyclic binary-module earlier-owner witness. Consequently the entire degree-six high branch is accepted structurally. The $6T4$ and $6T6$ certificates remain independent audits of the two familiar actions, but the $c = 1$ ownership implication no longer uses the degree-six catalogue.

Degree twelve also admits a structural reduction before any finite recognition is used. For an internally selected minimal block system, write $r$ for the fibre size, $s$ for the number of blocks, $v$ for the ternary weight of a chosen primitive-fibre chief series, and $t$ for the relative ternary head of the original normal top image. The high pair has relative rank two and

$$2 \le vB(s) + t, \qquad v \le \lfloor r/3 \rfloor, \qquad t \le C(s), \qquad rs = 12.$$

The $2 \times 6$ orientation has $v = 0$ and $t \le 1$, while the $6 \times 2$ orientation has $v \le 1$ and $t = 0$; both are impossible. Thus every high degree-twelve pair has a $3 \times 4$ or $4 \times 3$ minimal block system. In either case $B(s) = 1$, and equality is forced throughout:

$$\delta_A(N) = 2, \qquad v = 1, \qquad t = 1.$$

The rank-one four-block top is the natural $A_4$ action, while the faithful transitive rank-one three-block top has order three. Hence these are literal prime-base and $V_4$-block top geometries, rather than names assigned from the finite catalogue. Accordingly the remaining bounded recognition problem concerns only the two saturated prime-base and $V_4$-block geometries. The complete degree-twelve census below audits that structural fork and identifies the owner realized in each branch.

The prime-base branch retains more than the numerical equality above. A nontrivial sign on any three-point fibre would bound the original relative head by the top relative head. Since their dimensions are two and one, respectively, every fibre sign is trivial and the literal block kernel is a 3-group. Applying the exact normal-chain restriction formula to

$$N \cap \ker(A \longrightarrow A_4) \le N$$

shows that the retained restriction image on this physical intersection has dimension exactly one. In particular it contains an actual nonzero whole-ambient invariant character; this is the marking datum needed below, not merely a positive chief weight.

For reference, the natural $A_4$-module $\mathbb{F}_3^4$ has exactly four invariant submodules:

$$0, \qquad \langle(1,1,1,1)\rangle, \qquad \{x : \textstyle\sum_i x_i = 0\}, \qquad \mathbb{F}_3^4.$$

The augmentation module in the third position has full projection on every coordinate, so coordinate-surjectivity alone does not remove it. It has no nonzero $A_4$-invariant linear functional, whereas the diagonal and full modules do. Thus, once the literal fibre-coordinate action is identified with this natural module, the retained character forces the block kernel to be diagonal or full. This identification is now intrinsic. Choose one actual triple-fibre chart and transport it to the other fibres by ambient group elements. The vanished fibre signs make every transition even, so the resulting coordinate map injects the original block kernel $A_4$-equivariantly into $\mathbb{F}_3^4$. Its image and the image of every physical intersection with an original normal subgroup are invariant. The retained character therefore applies to the actual coordinate image, rather than to a substituted product module.

The same four-submodule classification removes a second, potentially case-by-case step. If $B, C \leq \mathbb{F}_3^4$ are $A_4$-invariant, the section $B/(B \cap C)$ embeds $A_4$-equivariantly back into $\mathbb{F}_3^4$: the required maps are the identity, the diagonal and augmentation projections, or zero. Suppose its action factors through an actual quotient $A_4 \twoheadrightarrow Q$. Evaluating an embedded vector at one coordinate and translating that functional through $Q$ gives an injective matrix-coefficient map

$$B/(B \cap C) \hookrightarrow \mathbb{F}_3[Q].$$

Thus every such descended section lies in one copy of the regular module of its actual quotient; no classification of the quotients of $A_4$ is needed. The arbitrary-normal quotient construction retains the other presentation of the same physical section. From the checked original kernel chart $\pi : G \twoheadrightarrow A_4$ and any $N \triangleleft G$, it constructs the literal extension

$$G/N \longrightarrow G/(\ker \pi\, N)$$

with kernel $\ker \pi/(N \cap \ker \pi)$, its quotient module and the original conjugation action. The same construction identifies its module with the invariant coordinate section above, constructs the onto top map $A_4 \twoheadrightarrow G/(\ker \pi\, N)$, and proves that the coordinate action factors through this actual quotient. The matrix-coefficient map therefore gives an injective equivariant map from the physical kernel of $G/N$ into $\mathbb{F}_3[G/(\ker \pi\, N)]$. This uses neither a splitting nor the assumption $N \leq \ker \pi$. Thus the remaining obligation in this orientation is the fixed-source owner sum, rather than quotient-module identification.

There is a parallel intrinsic reduction in the four-point-fibre branch. The physical intersection $N \cap \ker(A \to C_3)$ has relative ternary head one. An odd local image would propagate to every fibre by original ambient conjugation. Every normal subgroup of such an odd transitive four-point image has zero relative ternary head; the subdirect relative-head inequality would therefore force the displayed intersection head to vanish. Hence all literal fibre images are even. Set

$$W = \{g \in \ker(A \to C_3) : g^2 = 1\}.$$

Then $W \triangleleft A$ is an elementary binary group which embeds in the three actual Klein-four coordinates, so $|W| \leq 64$. The quotient $A/W$ is a 3-group, and Schur–Zassenhaus gives an actual ternary

complement. Moreover each block-kernel image is all of $A_4$: the positive intersection head supplies one full local image, and ambient conjugation transports fullness to the other fibres. Cubing an arbitrary lift of a local Klein translation kills every ternary coordinate and leaves the prescribed translation, so each literal map $W \to V_4$ is onto. This statement deliberately allows the proper submodules occurring in the degree-twelve rows: it does not replace $W$ by all of $V_4^3$. Put

$$\mathcal{T} = X \times (V_4 \setminus \{1\}),$$

where $X$ is the set of three actual four-point fibres. Ambient conjugation, written in arbitrary local point charts, defines an action on the nine elements of $\mathcal{T}$. Its kernel is exactly $W$: fixing all local translations forces the local permutation into the Klein group, and then faithfulness on the original points identifies the intrinsic core. Full local $A_4$ images make this action transitive. Hence the whole 3-group quotient $A/W$, rather than only the three-block top, has a faithful transitive degree-nine action. The elementary group $W$ also supplies its intrinsic binary conjugation module. For every original $N \lhd A$, the arbitrary-normal construction retains the literal extension

$$A/N \longrightarrow A/(WN), \qquad \ker = W/(W \cap N),$$

with the original representatives and conjugation action; the descended top is still a 3-group. No containment $N \leq W$, complement, full-product, or cyclic-module assumption is used. The remaining step in this orientation is its fixed-source top-character estimate.

The bounded degree-twelve certificate constructs $W = O_2(U)$ and a Sylow 3-complement for $12T20, 85, 164, 228, 229, 265$. It verifies that $W$ has three actual orbits of size four and local image $V_4$. Restricting translations to the blocks constructs a literal ambient $V_4^3$ normalized by $U$. The complement acts faithfully and transitively on its nine nonidentity local translations, and transitively on the three blocks. Its order, and the base order, are

| $U$ | $12T20$ | $12T85$ | $12T164$ | $12T228$ | $12T229$ | $12T265$ |
|---|---|---|---|---|---|---|
| $\lvert W\rvert$ | 4 | 16 | 64 | 64 | 64 | 64 |
| $\lvert P\rvert$ | 9 | 9 | 9 | 27 | 27 | 81. |

In the first three rows the complement is $C_3^2$, and an exhibited base vector has orbit spanning $W$; these are exactly the cyclic module witnesses used in the proof. In the last three rows the actual nine-point complement is, respectively, $9T6, 9T7, 9T17$; the proper/full binary-base domain applies. In $12T20, 12T194$, the normal ternary subgroup instead has four actual orbits of size three, local image $C_3$, and full natural $A_4$ block action. The standard transversal embedding puts the original group inside $C_3 \wr A_4$; the base is diagonal in the first case and full in the second. Thus $12T20$ also belongs to the earlier prime-base domain, and $12T194$ belongs to it directly.

These finite statements are part of the bounded action certificate: they verify actual blocks, local images, full projection and cyclic generators, rather than deciding membership from order or a catalogue label. They are separate from the exhaustive normal-pair computation which produced the high list. The structural three-by-three block argument in Theorem 9.5 places every degree-nine rank-two pair in the transitive 3-group owner, and its iterated reversible block-sign dichotomy places the degree-27 row in that same owner. The arbitrary-normal one-layer theorem applies uniformly to every actual transitive 3-group in those degrees and every literal quotient target. At degree nine the 8/9 row is used only after the surviving complete source is proved to satisfy $9d_3(J) \leq 2b + 3$. The one-regular-$C_3$, no-natural-$A_4$ pattern proves this inequality uniformly; the numerical row itself depends only on the inequality. Degree twenty seven uses the unrestricted internal degree-nine envelope. No further finite action classification is introduced here. □

**Corollary 29.4** (Ternary owner or capacity)**.** *Let $(A, N)$ be a transitive normal pair of degree $w$. Then at least one of the following holds:*

1. *$A$ is the natural $C_3$ or natural $A_4$ action;*
2. *the original action of $A$ is accepted by an earlier physical family in Lemma 29.2;*
3. $$20\delta_A(N) \le 3w.$$

*The owner in the second alternative is attached to the original permutation action, so deleting that actual orbit retains the complete complementary subgroup.*

*Proof.* If the capacity inequality fails, Theorem 9.5 puts the actual action on its necessary high list. The two natural actions give the first alternative, and Lemma 29.3 gives the second for every other entry. The structural degree-six and degree-twelve arguments in that lemma identify the owner intrinsically; the finite certificates only audit the resulting action charts. □

**Theorem 29.5** (First split one-triple estimate)**.** *The family of split one-triple graphs with exactly one actual regular $C_3$ orbit and no natural $A_4$ orbit admits a positive forward bound against complete lower-degree subgroup counts whose row tends to zero exponentially. After its low cone and the actual orbit families in Lemma 29.2 have been removed, its high cone is empty.*

*Proof.* For the global assembly there is a shorter dependency route. Full projection onto the displayed regular $C_3$ action makes its three-point block an actual orbit after every relabelling of the complete physical subgroup. Its orbit image has order three, so it is neither a 2-group nor the full $S_3$ marker action. Thus the entire noncritical family in this theorem is a literal subfamily of the outside-Fits frontier of Theorem 7.4. The general non-2 forward estimate therefore implies the present estimate directly. We retain the following finite-owner proof as the first required audit of its local certificates and original weights.

There is no separate high-rank split estimate at width three. The physical owner witness retains full projection onto the displayed outer regular $C_3$, so its displayed triple is an actual orbit. On the actual selected nested carrier, before passage to the enlarged canonical cell, the retained one-regular-$C_3$ pattern therefore forbids a second width-three ternary orbit. We remove that ternary-p-group label from the aligned index at this stage; we do not assert that the enlarged canonical width-three cell is empty. A transitive 3-group has odd degree, and hence the surviving ternary-p-group labels in the six-width menu have degrees nine and twenty seven. Their checked rows, together with the natural-$A_4$, degree-six, prime-base and binary-nine rows, exhaust the refined aligned index. Thus the finite-owner continuation closes without a shared-$C_3$ moment estimate for the discarded split cell.

Apply Proposition 29.1. For every remaining graph with an eligible actual orbit, point one such orbit of degree $w$. Its complete complement $J \le S_b$, $b = n - w$, includes the external triple and every other orbit and relation. The exact pointed comparison is

$$\sum_H k(H) = \sum_U \frac{n!}{b!a(U)} \sum_J Z_J^{\mathrm{surv}}(U), \qquad 0 \le Z_J^{\mathrm{surv}}(U) \le Z_J(U).$$

Here the surviving fibre retains the preceding exclusions on the complete pointed graph. The unrestricted fibre is used only for the upper bound. For each deletion, the second projection of the actual deleted fusion model is exactly the literal permutation image on the complementary points.

The complement orbits correspond injectively to the undeleted original orbits, and their complete induced actions are conjugate through the literal point charts. Deleting any indicated nonmarker action therefore preserves the unique-$C_3$, no-natural-$A_4$ pattern on this exact source, which yields $9d_3(J) \le 2b + 3$. The degree-nine row then consumes that numerical budget without carrying the stronger orbit description further. Only after the fibre has been bounded do we enlarge the number of such $J$ to $s_b$. Normalize once, obtaining terms

$$D_U(1 + b)\Delta(n, b)2^{\alpha_U b}\frac{s_b}{L_b}.$$

For even $w$ the Gaussian coefficient of $b$ is $w/8$; for odd $w$ its safe value in both parities is $(w-1)/8$. These are strictly larger than every corresponding $\alpha_U$ in the table. For a check using only rational numbers, $3^5 < 2^8$ bounds its five exponents by $8/15, 16/15, 25/18, 8/9, 8/3$, respectively. Each resulting gap is at least $1/9$. Equation (29.3) bounds the coefficient cost for each fixed deletion by a polynomial in $n$. There are only finitely many actions in degrees $6, 9, 12, 27$, so the whole forward row is $O(2^{-\varepsilon n})$ for some $\varepsilon > 0$. To form a disjoint physical partition, assign a graph to the first eligible family in a fixed order. Summing its intersections before applying the full pointed upper bound charges each family once. Multiple eligible orbits give only the positive pointing multiplicity.

Suppose now that a high graph survives. Its residual $K$ has $20d_3(K) > 3m$. Because projection of the complete graph onto $K$ is an isomorphism, the actual nonexternal orbit actions of the graph are exactly those of $K$. Apply Corollary 29.4 to each original orbit pair. The retained pattern excludes its first alternative, and the preceding physical removals exclude its second. Thus every orbit pair satisfies the capacity inequality. Equation (9.2) now gives

$$20d_3(K) \le \sum_i 20\delta_{A_i}(N_i) \le \sum_i 3 \deg A_i = 3m,$$

a contradiction. Fixed points have head zero and keep their degree in this inequality. Thus every surviving character set $\Theta(K)$ in the high cone is empty. This is a statement about complete marked graphs; it introduces no additional numerical kernel or scalar term. □

**Theorem 29.6** (Complete degree-three residual packet)**.** *Let $\mathcal{P}_n$ be the exact no-pair, no-complete-$C_3$ residual subfamily for which every orbit violating the post-$E_7$ alphabet has degree three. Then $|\mathcal{P}_n|/L_n$ admits a positive forward estimate whose scalar and total row are $O(2^{-n/1000})$. Its complement in the exact residual has a violating orbit of degree at least four.*

*Proof.* Every violating three-point orbit is the literal regular $C_3$ action: the alternative $S_3$ action conjugates its ternary characters to their inverses. Extract all $c > 0$ such orbits simultaneously and let $m$ be the degree of the retained tail, so $3c + m = n$.

If $m > 0$, every normal pair in the tail has relative ternary head at most $3m/20$. Otherwise the checked high-action degree list would produce a tail orbit of degree $3, 4, 6, 9, 12$, or $27$. Degree three would be one of the already extracted regular $C_3$ orbits, while every larger alternative would be another post-$E_7$ violation, contrary to the definition of $\mathcal{P}_n$. The simultaneous product-graph count, with the original occurrence factorials and normalizers retained, therefore gives the forward row at the literal smaller degree $m$. After exact benchmark cancellation its aggregate exponent is $11 - c(m + 3c)/100$; the polynomial profile count is absorbed uniformly, giving $O(2^{-n/500})$ and hence the stated $O(2^{-n/1000})$ row.

If $m = 0$, the action has the honest one-colour regular-$C_3$ profile. Its exact product-conjugation count is bounded by $(3c)!\,(c + 1)3^{c^2/4+c}$. Normalization by $L_{3c}$ gives an $O(2^{-n/1000})$ scalar. The

two cases partition $\mathcal{P}_n$. Finally, negating the defining universal condition selects a violating orbit whose degree is not three; every violation has degree at least three, so the selected degree is at least four. □

**The two exact incidence tests.** The preceding proof respects Proposition 27.3 on every disjoint piece. In particular, applying that proposition with $\chi \in \Theta(K)$ gives

$$\sum_K \sigma_3^{\mathrm{surv}}(K) = \sum_{|x|=3} \sum_{\substack{N^x=N,\ N\cap\langle x\rangle=1 \\ \chi_{N,x}\in\Theta(N\langle x\rangle)}} \frac{1}{|\{y \in Nx : |y| = 3\}|}.$$

The reciprocal is preserved. It is not harmless to drop it: for $K_0 = C_2^{3h} \rtimes \langle x\rangle$, with $x$ cycling $h$ triples of pair coordinates, and $K = K_0 \times A_5^2$ on $m = 6h + 10$ points, its indicated quotient coset contains exactly $4^h \cdot 21^2$ elements of order three. This already loses $m/3 - O(1)$ bits if the reciprocal is discarded.

Likewise the affine test on the same source $J = \mathbb{F}_4^r \rtimes C_3$ has, above the fixed top map to $\mathbb{F}_4^t \rtimes C_3$, exactly $4^{t(r+1)}$ cocycles and $4^t \prod_{i=0}^{t-1}(4^r - 4^i)$ onto lifts. A second mark stays on this same $J$. In the proof above the external triple is part of $J$ in every earlier fibre, including the degree-six affine fibre; it is never appended as an independent character maximum. The final zero-set argument makes all these joint high surviving incidences empty and therefore requires no factorization of their moments.

## 30 The complete macroscopic alternating packet

Write $A_4 = (\mathbb{F}_4, +) \rtimes C_3$ in its natural action on four points. Let $Z_{b,c,t}$ count subgroups in the fixed container $S_b \times C_3^c \times A_4^t$ full on the displayed $C_3, A_4$ coordinates and with residual projection having no natural $C_3$ or $A_4$ orbit. Put $n = b + 3c + 4t$.

**Theorem 30.1** (All image and source parameters summed)**.** *Uniformly for $b, c \geq 0$ and $t \geq 1$,*

$$\log Z_{b,c,t} \leq \frac{n^2}{16} - \frac{43t^2}{450} + O(b^{3/2} + t + \log(c + 2)). \tag{30.1}$$

*After the single original action factor $n!/(b!6^c c!24^t t!)$, the entire sector $t \geq n/1000$ has normalized mass at most $2^{-43n^2/(900\cdot 10^6)}$ eventually.*

*Proof.* Fix the literal residual group $J \leq S_b$. Put $F = J'J^3$, $E = J/F = \mathbb{F}_3^r$, $r = d_3(J)$, and $T = \mathbb{F}_3^t$. Conjugation makes $M = H_1(F, \mathbb{F}_2)$ a semisimple $\mathbb{F}_2E$-module. Since $\mathbb{F}_4$ contains all third roots of unity, its nontrivial simple modules have dimension two, are indexed by $h \in (E^* \setminus \{0\})/\{\pm 1\}$, and have endomorphism field $\mathbb{F}_4$. Write their multiplicities as $a_h$, and $A = \sum_h a_h$. The permutation section bound gives $A \leq d_2(F)/2 \leq b/4$; equation (9.5) gives $r \leq 2b/9$.

If $K \lhd J$ and $J/K$ is elementary ternary, then $K/F = D$ for a unique subspace $D \leq E$. Odd-order homology gives naturally

$$H_1(K, \mathbb{F}_2) = M_D.$$

The $D$-coinvariants retain precisely those simple types whose characters vanish on $D$, with the same multiplicities. Distinct retained types do not merge: equality up to sign on $E/D$ is already equality up to sign on $E$. This controls all image kernels from one fixed source module $M$.

An image $L \le J \times T$ onto $J$ contains $F \times 0$. Its reduction $\overline{L} \le E \times T$ is therefore equivalent to its annihilator, the graph

$$\overline{L}^{\perp} = \{(\Gamma(w), w) : w \in C_T\}, \qquad C_T \le T^*, \quad \Gamma : C_T \to E^*.$$

This is a bijection: projection of $\overline{L}$ onto $E$ is onto exactly when the annihilator meets $E^* \times 0$ trivially. Let $S = \{i : e_i \in C_T\}$, $s = |S|$, $u = t - s$, and $a = \dim C_T - s$. Fullness on packet coordinate $i$ says $\Gamma(e_i) \neq 0$ whenever $i \in S$. The exact rank is

$$d_3(L) = r + u - a.$$

The coordinates in $S$ are exactly those eligible for a residual source map; coordinate $i$ has source type $[\Gamma(e_i)]$. No eligible and ineligible coordinate can have the same packet isotype: equality up to sign would put $e_i \pm e_j$ in $\ker \Gamma \le C_T$, and hence put both in $C_T$.

We spell out the binary fibre above this same image. If $W \le V^t$, $V = (\mathbb{F}_4, +)$, is its intersection with the packet kernel, write $Q = V^t/W$. Subgroups above $L$ with intersection $W$ are the graphs of literal cocycles in $Z^1(L, Q)$. For $K = L \cap J$, the odd quotient $L/K \le T$ gives

$$H^1(L, Q) \cong \mathrm{Hom}_{L/K}(H_1(K, \mathbb{F}_2), Q), \qquad |B^1(L, Q)| = |Q|.$$

The second equality holds because all packet simple constituents are nontrivial and thus $Q^L = 0$. Coordinate fullness can only restrict this set. An isotype of packet multiplicity $m$ and source multiplicity $\alpha$ consequently contributes at most

$$F_\alpha(m) = \sum_{x=0}^{m} \begin{bmatrix} m \\ x \end{bmatrix}_4 4^{(\alpha+1)x} \le 2^{2\alpha m + m^2/2 + 4m}.$$

Indeed $\begin{bmatrix} m \\ x \end{bmatrix}_4 < 2 \cdot 4^{x(m-x)}$ and there are $m + 1$ summands. The term $+1$ counts the literal coboundaries; it is not another source-module choice.

Fix $C_T$ and extend its $s$ coordinate axes to a basis. Values of $\Gamma$ on the other $a$ basis vectors contribute $3^{ra}$. If the projective multiplicities of the nonzero axis values are $m_h$, the eligible fibre is at most $2^{2\sum_h a_h m_h + s^2/2 + 4s}$. Summing all its oriented axis values gives

$$2^{s^2/2 + 4s} \Big(2 \sum_h 4^{a_h}\Big)^s.$$

Integer convexity and the two physical rank bounds show

$$2 \sum_h 4^{a_h} \le 2 \left( \frac{3^r - 1}{2} + 4^A - 1 \right) \le 2^{b/2+2}.$$

The ineligible part has source multiplicity zero and contributes at most $2^{u^2/2 + 4u}$. Thus the whole fixed-$C_T$ sum is at most

$$2^{bs/2 + (s^2+u^2)/2 + (\log_2 3)ra + O(t)}.$$

For each exact axis set $S$, the number of possible $C_T$ of dimension $s + a$ is at most $\begin{bmatrix} u \\ a \end{bmatrix}_3$: quotient by the contained coordinate space, and omit the requirement of no additional axes. There are $\binom{t}{s}$ choices of $S$. Defining

$$J_*(x, y) = \max_{0 \le z \le y} z(x + y - z) = \begin{cases} xy, & x \ge y, \\ (x + y)^2/4, & x \le y, \end{cases}$$

and $\lambda = \log_2 3$, the all-image fibre over this $J$ is at most

$$2^{\max_{s+u=t}\{bs/2+(s^2+u^2)/2+\lambda J_*(r,u)\}+O(t)}. \tag{30.2}$$

This sums the image, source-column assignment, kernel subspace and cocycle on the same group.

The deficit in (30.2) from $bt/2+t^2$ is at least $\delta_0 t^2$, with $\delta_0 = (2-\lambda)/4$. For completeness, set $x = b/t$, $v = u/t$. The deficit divided by $t^2$ is at least

$$d(x,v) = 1/2 + v - v^2 + xv/2 - \lambda J_*(2x/9, v).$$

When $2x/9 \geq v$ this increases in $x$, so its minimum is at $x = 9v/2$, where it is $1/2+v+(5/4-\lambda)v^2 \geq 1/2$. On the other region it is concave in $x$, so check $x = 0$ and $x = 9v/2$. At zero it is $1/2 + v - (1+\lambda/4)v^2$, a concave quadratic whose minimum on $[0,1]$ is $\delta_0$.

Adjoin the $c$ external regular ternary coordinates. The binary packet extension does not change $d_3(L)$, and $d_3(L) \leq P = 2b/9 + t$. Its one exact graph fibre is at most $2^{\lambda J_*(P,c)+O(\log(c+2))}$. Its physical increment in $n^2/16$ is $3bc/8 + 3tc/2 + 9c^2/16$. If $c \leq P$, the difference is

$$(3/8 - 2\lambda/9)bc + (3/2-\lambda)tc + 9c^2/16 \geq -\frac{4}{9}(\lambda - 3/2)^2 t^2 > -t^2/225.$$

If $c \geq P$, the difference increases in $c$ and at $c = P$ is $P[(1/2-2\lambda/9)b+(33/16-\lambda)t] > 0$. Summing $J$ uses the independent coarse bound $s_b \leq 2^{b^2/16+O(b^{3/2})}$ exactly once. Since $\delta_0 - 1/225 > 43/450$, this proves (30.1).

Every actual group recovers the sets of its natural $C_3$ and $A_4$ orbits. Choosing these supports and actions contributes exactly $n!/(b!6^c c!24^t t!)$. The residual $J$ excludes those orbit types, so they are not selected a second time. Normalization by $L_n$ costs only $O(n\log(n+2))$ beyond the $n^2/16$ term. For $t \geq n/1000$ the main loss dominates this and the $O(n^{3/2})$ error uniformly; the $O(n^2)$ possible $(c,t)$ values are absorbed as well. This proves the complete scalar claim. □

## 31 The small natural alternating packet

Let $\mathcal{U}$ be the class of allowed transitive actions consisting of natural $C_3, A_4$ and actions $U$ for which every literal normal subgroup $N$ satisfies $\delta_U(N) \leq 3\deg(U)/20$. This is an explicit intrinsic action predicate. The following theorem does not assert that an arbitrary residual already lies in $\mathcal{U}$.

**Theorem 31.1** (Complete low-packet transfer)**.** *For subgroups all of whose actions lie in $\mathcal{U}$, the sector with fewer than $n/1000$ natural $A_4$ orbits and at least one natural $C_3$ or $A_4$ orbit has a complete ordinary forward row $O(2^{-n/500})$ and scalar at most $2^{-n^2/80000}$ eventually. The complementary sector with at least $n/1000$ natural $A_4$ orbits has the scalar in Theorem 30.1.*

*Proof.* An actual-orbit filtration gives, for a group $K$ with no natural $C_3$ orbit and $t$ natural $A_4$ orbits, $d_3(K) \leq 3\deg(K)/20 + 2t/5$. First delete all $c \geq 1$ natural $C_3$ orbits. The remaining degree is $b = n - 3c$ and the original-degree cutoff gives

$$d_3(K) \leq (3/20 + 2/5000)b + (6/5000)c.$$

Its exact unrestricted external fibre is $\sum_j \begin{bmatrix} c \\ j \end{bmatrix}_3 3^{jd_3(K)}$, at most $\rho_3(c+1)3^{cd_3(K)+c^2/4}$, where $\rho_3 = \prod_{i\geq 1}(1-3^{-i})^{-1}$. Thus the complete cell is

$$K_3(n, n-3c) = \frac{\Delta(n,n-3c)}{6^c c!}\rho_3(c+1)3^{(3/20+2/5000)cb+(1/4+6/5000)c^2}.$$

At both parities its Gaussian loss is at least $(3c-1)b/8 + 9c^2/16 - O(c+1)$ and its coefficient cost is $O(c\log(n+2))$. Using $\log_2 3 < 8/5$, the remaining linear and quadratic coefficients exceed $9/1000$ and $27/1000$, respectively. Consequently $K_3(n, n-3c) \le 2^{-cn/500}$ uniformly, including $b=0$; summing $c$ gives a geometric row.

In the remaining no-$C_3$ sector, point one actual $A_4$ orbit and let $J \le S_b$, $b = n-4$, be its complete complement. The exact fibre is $1 + |\operatorname{Epi}(J, C_3)| + |\operatorname{Epi}(J, A_4)|$. Call $J$ hot if some index-three normal subgroup $F$ has $d_2(F) > (1/4 + 1/200)b$. For fixed $e > 0$, the graph injection $(F, \phi : F \to C_2^r) \mapsto \operatorname{graph}\phi \le S_{b+2r}$ and the coarse subgroup bound, with $r$ nearest $2eb$, prove

$$|\{F \le S_b : d_2(F) > (1/4+e)b\}| \le 2^{b^2/16 - e^2b^2 + O_e(b^{3/2} + b\log(b+2))}.$$

Each $F$ has at most $b!$ index-three overgroups, by choosing a coset generator in $S_b$. The complete hot pointed count, using the elementary fibre bound $5 \cdot 2^{(1/2+(\log_2 3)/3)b}$ and its single factor $n!/(24b!)$, is at most $2^{-n^2/80000}$ after normalization.

For cold $J$ and one surjection $\theta : J \to C_3$, let $a_\theta$ be the matching $\mathbb{F}_4$ multiplicity in $H_1(\ker\theta, \mathbb{F}_2)$. The literal cocycle count gives $4(4^{a_\theta} - 1)$ surjections to $A_4$ above $\theta$, and $2a_\theta \le (1/4 + 1/200)b$. The residual still satisfies the original cutoff, so $d_3(J) \le (3/20 + 2/5000)b + 8/5000$. Its complete cold fibre is at most

$$z(b) = 5 \cdot 2^{(1/4+1/200)b + (\log_2 3)((3/20+2/5000)b + 8/5000)}.$$

The pointed coefficient is exactly $\Delta(n, n-4)z(n-4)/24$. Its source exponent is at most $0.49564b$, whereas deleting four points supplies Gaussian loss $b/2$ and polynomial coefficient cost. It is therefore at most $2^{-n/500}$ eventually. Choosing a deterministic actual $A_4$ orbit makes hot and cold physical inputs disjoint; enlarging each to all pointings pays each complete positive bound once. This proves the theorem. □

## 32 Exhaustive counting and boundedness-first assembly

Let $F_N$ be the normalized count of all fixed-point-free binary subgroups of $S_{2N}$, and retain the complete binary error $E_N$ from Section 27. Actual orbit projections give the exact disjoint partition

$$F_N = C_{2N}^{\mathrm{crit}} + E_N. \tag{32.1}$$

**Proposition 32.1** (Ordinary-target binary frontier transport)**.** *Put*

$$B_{2N} = E_N, \qquad B_{2N+1} = u_N E_N + v_N E_{N-1}.$$

*There are nonnegative functions $\eta_B^{\mathrm{amb}}(n)$ and $K_B^{\mathrm{amb}}(n,m)$, supported on $m < n$, and constants $a'_B, C'_B > 0$ such that eventually*

$$B_n \le \eta_B^{\mathrm{amb}}(n) + \sum_{m<n} K_B^{\mathrm{amb}}(n,m)A_m, \qquad \eta_B^{\mathrm{amb}}(n) + \sum_{m<n} K_B^{\mathrm{amb}}(n,m) \le C'_B 2^{-a'_B n}. \tag{32.2}$$

*Proof.* Theorem 24.8 gives

$$E_N \le \eta_B(N) + \sum_{m<2N} K_B(N,m)A_m, \qquad \eta_B(N) + \sum_{m<2N} K_B(N,m) \le C_B 2^{-a_B N}. \tag{32.3}$$

In even ambient degree use (32.3) directly. In odd degree apply the same recurrence at $N$ and $N-1$, multiply the two inequalities by $u_N$ and $v_N$, and pad both target rows by zero to the full range $m < 2N+1$. The coefficients are nonnegative, and Section 27 gives

$$u_N \le 4(N+1), \qquad v_N \le 2\Phi^{-2}(N+1)^3.$$

These fixed polynomial factors are absorbed by any smaller positive exponential rate. Since $n = 2N$ or $2N+1$, this proves (32.2). Thus the binary recurrence is transported before any boundedness of $F$, $E$, or $A$ is used. □

**Theorem 32.2** (Exhaustive ordinary counting estimate)**.** *Write $R_n = A_n - C_n^{\mathrm{crit}}$ for the normalized complement of the complete critical family. There are nonnegative functions $\eta(n)$ and $\mathcal{K}(n,m)$, supported on $m < n$, and constants $a, C > 0$ such that eventually*

$$0 \le R_n \le \eta(n) + \sum_{m<n} \mathcal{K}(n,m) A_m, \qquad \eta(n) + \sum_{m<n} \mathcal{K}(n,m) \le C 2^{-an}. \tag{32.4}$$

*The kernel is the sum of the outside-action row, the exact positive marker row, and the transported binary row of Proposition 32.1. Every forward edge has one complete ordinary target.*

*Proof.* First take all subgroups with an actual transitive orbit outside the alphabet consisting of binary groups, natural $C_3, A_4, S_3$, and fixed points. Theorem 20.1 bounds this complete family by an exponentially small ordinary forward row and a quadratically small scalar. Every pointing deletes its whole original orbit, so its exact weight is $n!/(b!a(U))$. A cold estimate uses it in the form $\Delta(n,b)/a(U)$ on $A_b$; a hot estimate retains the factor $1/(b!a(U)p_nG_{\lfloor n/2 \rfloor})$ on its complete source sum. These are bounds for already assembled subgroups, including their other orbits and any external characters. In the pair-action rows the faithful top is used only as a cover of the actual quotient $B_N = U/(KN)$; the counted extension remains $U/N \twoheadrightarrow B_N$. Thus this certificate choice introduces no new orbit occurrence or normalizer factor.

In the complementary alphabet let $c, t$ be the numbers of natural $C_3, A_4$ orbits. If $t \ge n/1000$, apply the complete scalar of Theorem 30.1. If $t < n/1000$ and $c + t > 0$, apply Theorem 31.1. Its intrinsic action condition holds here: every relative ternary head of a two-group is zero; the normals $1, C_3, S_3$ of natural $S_3$ also have zero relative ternary head, since $[C_3, S_3] = C_3$ and $S_3$ has no ternary abelian quotient. Fixed points contribute zero. This proves the condition for every residual action in the alphabet, without an assumption about which normal kernel is selected later. Add the ordinary row and scalar of this theorem once. The macro and small packet sectors are disjoint by their original-degree cutoff.

It remains that every moved orbit is binary or natural $S_3$. Write the marker counts as $g, f$, and $n = 2N + \varepsilon$. Under the exact collapse of Proposition 28.1, let $q$ be the number of distinct marker signs and put $D = (g + f - \varepsilon)/2$. This is an integer, because the binary exterior has even degree, and is nonnegative. When $D \ge 1$, the complete row in Theorem 28.3 applies directly with the collection of all collapsed subgroups. It contributes $\sum_{m<n} K_{\mathrm{mark}}(n,m) A_m$, and its row is exponentially small without any boundedness assumption on $A_m$.

When $D = 0$, in even degree there are no markers or fixed points. The complete binary partition gives $C_{2N}^{\mathrm{crit}} + E_N$, each once. In odd degree there is exactly one fixed point or one natural $S_3$ orbit. With critical binary orbits these are exactly the two disjoint odd parts of $C_n^{\mathrm{crit}}$. Otherwise their exact contributions are $\alpha_N E_N$ and $\beta_N p E_N$; the complete pair moment of Proposition 27.2 bounds their sum by $u_N E_N + v_N E_{N-1}$. Thus the entire zero-defect remainder is $B_n$. Proposition 32.1 replaces this term by one exponentially small scalar and one strictly forward ordinary row. No estimate for a restricted binary residual is substituted for the complete moment.

These sectors exhaust all actual subgroups and are pairwise disjoint before their positive enlargements. Within the first sector fix an order of the proved action families and assign a subgroup to the first one it meets, before testing temperatures. For each assigned family its hot part means that some accepted pointing is hot, and its cold part that none is hot. Each theorem's complete positive envelope is then paid once, over all its assigned inputs. Multiple orbit occurrences are already accounted for by pointing the original weight $\prod_U (a(U)^{m_U} m_U!)^{-1}$; auxiliary representations and alternative certificates contribute no new physical factorial. All subsequent sectors are restrictions of their proved positive envelopes and are likewise paid once.

Define $\eta$ to be the sum of the nonbinary, macro, small-packet, and transported binary scalars. Define $\mathcal{K}$ to be the sum of the nonbinary and small-packet ordinary rows, the positive marker row, and the transported binary row. Each summand is nonnegative, strictly forward, and exponentially small in total mass. Taking the minimum of their positive rates gives (32.4). □

*Proof of Theorem 1.3.* The critical ratios $C_n^{\mathrm{crit}}$ are bounded. Choose an onset at which the row in (32.4) is at most $1/2$, and let $M$ exceed all finitely many initial $A_m$ and twice a uniform bound for $C_n^{\mathrm{crit}} + \eta(n)$. The exact partition $A_n = C_n^{\mathrm{crit}} + R_n$ and induction in $n$ give $A_n \le M$ for every $n$. Substituting this established bound into (32.4) gives $R_n = O(2^{-cn})$ for some $c > 0$. The critical-family estimate now yields $|A_n - 1| = O(2^{-c'n})$. Taking the smaller positive rate proves the result at both parities. No boundedness premise was used in deriving any row or scalar; boundedness is a consequence of their eventual contraction. □

# 33 Evaluation of the benchmark

The results in this section estimate the explicit sequence $L_n$ in (1.2). Theorem 1.3 transfers them to the subgroup count with an exponentially smaller relative error.

## 33.1 The binary Gaussian sum

Define

$$\kappa_0 = \phi^{-1} \sum_{j \in \mathbb{Z}} 2^{-j^2}, \qquad \kappa_1 = \phi^{-1} \sum_{j \in \mathbb{Z}} 2^{-j(j-1)}. \tag{33.1}$$

Both series converge absolutely.

**Lemma 33.1.** *As $R \to \infty$,*

$$G_R = \kappa_{R \bmod 2}\, 2^{\lfloor R^2/4 \rfloor} (1 + O(R 2^{-R/2})).$$

*Proof.* The product formula reads

$$\begin{bmatrix} R \\ k \end{bmatrix}_2 = 2^{k(R-k)} \frac{\prod_{i=R-k+1}^{R} (1 - 2^{-i})}{\prod_{i=1}^{k} (1 - 2^{-i})}.$$

For central $k$ the fraction is $\phi^{-1}(1 + O(2^{-k} + 2^{-(R-k)}))$. When $R = 2m$, put $k = m + j$, giving exponent $m^2 - j^2$. When $R = 2m + 1$, put $k = m + j$, giving exponent $m(m+1) - j(j-1)$. On $|j| \le R/4$ the summable Gaussian weights times $2^{|j|}$ give error $O(2^{-R/2})$ up to an absolute parity constant. The remaining tails are $2^{-\Omega(R^2)}$ by (2.1). The stated, slightly weaker, error follows. □

## 33.2 One positive saddle

Let $\rho = \rho_R > 0$ be the unique solution of

$$R = \rho P'(\rho) = \frac{\rho}{2} + \frac{\rho^2}{3} + \frac{\rho^4}{96}, \qquad b(\rho) = \frac{\rho}{2} + \frac{2\rho^2}{3} + \frac{\rho^4}{24}. \tag{33.2}$$

Existence and uniqueness follow from strict monotonicity on the positive axis. In particular $\rho \sim (96R)^{1/4}$ and $b(\rho) \sim 4R$.

**Theorem 33.2** (Saddle formula)**.** *For $n = 2R + \varepsilon$,*

$$L_n = n! G_R \, \frac{(1+\rho/6)^{\varepsilon} e^{P(\rho)}}{\rho^R \sqrt{2\pi b(\rho)}} \left(1 + O(R^{-1})\right). \tag{33.3}$$

*In particular $c_{R-1}/c_R = \rho(1 + O(R^{-1}))$.*

*Proof.* Put $\lambda_1 = \rho/2$, $\lambda_2 = \rho^2/6$ and $\lambda_4 = \rho^4/384$, so that $R = \sum_k k\lambda_k$ and $b = \sum_k k^2\lambda_k$ satisfy $R \le b \le 4R$. Here and below $k \in \{1, 2, 4\}$. Define

$$A(\theta) = \sum_k \lambda_k (1 - \cos k\theta), \qquad B(\theta) = \sum_k \lambda_k (\sin k\theta - k\theta).$$

Then $P(\rho e^{i\theta}) - P(\rho) - iR\theta = -A + iB$. For any $d \in [0, 1]$, set

$$F_d(\theta) = (1-d)\cos B(\theta) + d\cos(\theta + B(\theta)), \qquad H_d(\theta) = e^{-A(\theta)} F_d(\theta).$$

Taking real parts in Cauchy's formula on $y = \rho e^{i\theta}$ gives

$$\frac{\rho^R [y^R](1+y/6)^{\varepsilon} e^{P(y)}}{(1+\rho/6)^{\varepsilon} e^{P(\rho)}} = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\theta)\, d\theta, \qquad d = \begin{cases} 0, & \varepsilon = 0, \\ \rho/(6+\rho), & \varepsilon = 1. \end{cases}$$

We estimate this integral uniformly for all $d \in [0, 1]$.

Use the fixed central interval $|\theta| \le \alpha = \pi/4$ and write $Q = b\theta^2/2$. The inequalities $u^2/8 \le 1 - \cos u \le u^2/2$ for $|u| \le \pi$, applied to each $k\theta$, give

$$R\theta^2/8 \le A \le Q, \qquad 0 \le Q - A = O(R\theta^4), \qquad |B| = O(R|\theta|^3).$$

The last two bounds follow from the fourth-order cosine and third-order sine remainders; their constants are absolute. Since $F_d$ is a convex combination of cosines,

$$|F_d| \le 1, \qquad 0 \le 1 - F_d \le B^2/2 + |\theta B| + \theta^2/2.$$

Together with $0 \le e^{-A} - e^{-Q} \le e^{-A}(Q - A)$, this yields

$$|H_d(\theta) - e^{-b\theta^2/2}| \le C e^{-R\theta^2/8} \big(\theta^2 + R\theta^4 + R^2\theta^6\big).$$

Integrating this bound gives $O(R^{-3/2})$, by the change of variable $u = \sqrt{R}\,\theta$. This argument requires no smallness assumption on $B$.

On the entire complement $\alpha \le |\theta| \le \pi$, the linear term alone gives

$$A(\theta) \ge \frac{\rho}{2}(1 - \cos\alpha) \ge \frac{\rho}{16}, \qquad |H_d(\theta)| \le e^{-\rho/16}.$$

Because $\rho \asymp R^{1/4}$, these tails are smaller than every fixed inverse power of $R$. The Gaussian tails outside $[-\alpha, \alpha]$ are $O(e^{-cR})$ for an absolute $c > 0$. Thus, uniformly in $d$,

$$\int_{-\pi}^{\pi} H_d(\theta)\, d\theta = \sqrt{\frac{2\pi}{b}} + O(R^{-3/2}) = \sqrt{\frac{2\pi}{b}}(1 + O(R^{-1})),$$

which proves (33.3). Finally, Cauchy's formula with amplitude $y/\rho$ gives $H_1$, at the same saddle $\rho_R$. Comparing it with $H_0$ gives

$$\frac{c_{R-1}}{\rho c_R} = \frac{\int_{-\pi}^{\pi} H_1(\theta)\, d\theta}{\int_{-\pi}^{\pi} H_0(\theta)\, d\theta} = 1 + O(R^{-1}). \qquad \square$$

### 33.3 An elementary four-periodic expression

Set

$$\begin{aligned} C_0 &= e^{-4/3}\kappa_0/2^{1/2}, \quad C_2 = e^{-4/3}\kappa_1/2^{3/4}, \\ C_1 &= \frac{e^{-4/3}\kappa_0\, 48^{3/8}}{6 \cdot 2^{7/16}}, \quad C_3 = \frac{e^{-4/3}\kappa_1\, 48^{3/8}}{6 \cdot 2^{11/16}}. \end{aligned} \tag{33.4}$$

For $\varepsilon = n \bmod 2$ define

$$M_n = C_{n \bmod 4}\, n^{3\varepsilon/8}\, 2^{n^2/16} \left(\frac{n^7}{48 \cdot 2^{\varepsilon} e^7}\right)^{n/8} \exp\left(2\sqrt{n/3} + \frac{(48n)^{1/4}}{2}\right). \tag{33.5}$$

**Theorem 33.3** (Elementary expansion)**.** *The exact benchmark satisfies*

$$\frac{L_n}{M_n} = 1 + \frac{6\varepsilon - 4}{(48n)^{1/4}} + O(n^{-1/2}). \tag{33.6}$$

*By Theorem 1.3, the same formula holds with $s_n$ in place of $L_n$, and* (33.3) *holds with $s_n$ on the left.*

*Proof.* Put $x = (96R)^{1/4}$ and $v = x(x - \rho)$. Only the first displacement of the saddle is needed:

$$\rho = x - 8x^{-1} + O(x^{-2}), \qquad v = 8 + O(x^{-1}). \tag{33.7}$$

Indeed, the equation is $\rho^4 + 32\rho^2 + 48\rho = x^4$. For large $x$ its positive root lies in $[x/2, x]$; substituting $x - 8/x$ leaves residual $O(x)$, and the derivative throughout the intervening interval is of order $x^3$. The mean value theorem proves (33.7).

Expand $\log_e \rho = \log_e x + \log_e(1 - v/x^2)$ to quadratic order. Since $v$ is bounded, the cubic remainder, multiplied by $R = x^4/96$, is $O(x^{-2})$. Direct substitution gives

$$\begin{aligned} P(\rho) - R\log_e \rho &= \frac{R}{4} - R\log_e x + \frac{x^2}{6} + \frac{x}{2} + \frac{v^2}{48} - \frac{v}{3} - \frac{v}{2x} + O(x^{-2}) \\ &= \frac{R}{4} - R\log_e x + \frac{x^2}{6} + \frac{x}{2} - \frac{4}{3} - \frac{4}{x} + O(x^{-2}). \end{aligned}$$

The cancellation is explicit: $v^2/48 - v/3 = -4/3 + (v-8)^2/48$. Also $b(\rho)/(4R) = 1 + O(x^{-2})$.

Define

$$T_R = \frac{1}{\sqrt{8\pi R}} \exp\left(-\frac{R}{4}\log_e R + \frac{1 - \log_e 96}{4} R + \frac{2\sqrt{6}}{3}\sqrt{R} + \frac{x}{2} - \frac{4}{3}\right).$$

The preceding estimates and the saddle formula yield

$$\log_e \frac{c_R}{T_R} = -\frac{4}{x} + O(x^{-2}). \tag{33.8}$$

The relative saddle error is $O(R^{-1}) = O(x^{-4})$ and is absorbed in this remainder. For the odd amplitude, the same first displacement gives

$$\log_e \frac{1+\rho/6}{x/6} = \frac{6}{x} + O(x^{-2}). \tag{33.9}$$

To make the parity conversion explicit, set $y = (48n)^{1/4}$ and

$$\begin{aligned}\Lambda_n = {} & (n+\tfrac{1}{2})\log_e n - n + \tfrac{1}{2}\log_e(2\pi) + \lfloor R^2/4 \rfloor \log_e 2 + \log_e \kappa_{R \bmod 2} \\ & + \log_e T_R + \varepsilon \log_e(x/6) + (6\varepsilon - 4)/x.\end{aligned}$$

Stirling's formula, Lemma 33.1 and the two expansions above give $\log_e L_n = \Lambda_n + O(n^{-1/2})$. Using $R = (n-\varepsilon)/2$ and the exact constants in (33.4), elementary algebra gives

$$\begin{aligned}\Lambda_n - \log_e M_n - \frac{6\varepsilon - 4}{y} = {} & \frac{n - 3\varepsilon + 4}{8} \log_e \frac{n}{n-\varepsilon} - \frac{\varepsilon}{8} \\ & + \frac{x^2 - y^2}{6} + \frac{x - y}{2} + (6\varepsilon - 4)\left(\frac{1}{x} - \frac{1}{y}\right).\end{aligned}$$

In even degree this is identically zero. In odd degree the first line is $O(n^{-1})$, while $y^4 - x^4 = 48$ bounds the remaining three differences by $O(n^{-1/2})$, $O(n^{-3/4})$ and $O(n^{-5/4})$, respectively. Thus

$$\log_e(L_n/M_n) = (6\varepsilon - 4)(48n)^{-1/4} + O(n^{-1/2}).$$

Exponentiation proves (33.6). Finally the exponential relative error of Theorem 1.3 is smaller than either polynomial error asserted here. □

*Remark* 33.4. The relative errors refer to three different approximations. The complete counting theorem gives exponential error against the exact coefficient benchmark $L_n$. Replacing the coefficient by its single-saddle expression gives $O(n^{-1})$. Replacing the saddle by the elementary expression $M_n$ gives an explicit term of order $n^{-1/4}$ and a remainder $O(n^{-1/2})$.

# A Finite evidence and reproduction

The accompanying computations distinguish three assertions: a literal record satisfies its algebraic equations; a stated finite domain is covered by such records; and an infinite counting theorem follows from that finite domain together with proved structural and summation lemmas. These assertions require different arguments.

The publication package contains the following mathematical data.

| Dataset | Mathematical role |
|---|---|
| Small binary actions | Literal actions through degree 16, normal axes, quotient maps, and the four local witness types. |
| Carrier data | Twelve noncritical actions, normal profiles, transition data, and specified quotient-replacement maps. |
| Nonbinary and rank data | Finite action and normal-rank domains used in the selected primitive and pair-top reductions. |
| Pair-top data | Concrete modules and constructive witnesses in the count-twelve and count-sixteen domains. |
| Four- and six-block data | Constructive classifications and actual quotient interfaces. |
| Zero-ternary frame data | Thirty complete binary module lattices and thirty-eight literal odd-subgroup/normalizer witnesses. |
| Critical and audit data | Quadratic realization constants, critical normalizers, and independent local chart controls. |

The data contracts specify literal point actions, normal subgroups, source and target maps, and the original symmetric normalizers. A quotient order or isomorphism-type name alone does not specify a counting witness. For a proper subdirect replacement, the image group and both its projection maps are part of the record.

For the critical quadratic constants of Lemma 2.3, one can enumerate $\mathrm{GL}(2,2)$ and $\mathrm{GL}(4,2)$ and test the equation $q(gv) = q(v)$ on every vector. This establishes the orders 2 and 72 for the displayed forms. The torsor argument in the lemma is what extends these finite calculations to arbitrary source dimension. Likewise the exact rational inequality (21.5) is a finite arithmetic certificate whose implication for all degree-32 sections is proved by the preceding incidence argument.

The classification libraries are pinned to TransGrp 3.6.5 and PrimGrp 3.4.4 [Hul23, HRDR23]; their mathematical construction and scope are described in [Hul05, HR20, RD05]. The recorded interpreter is GAP 4.13.1 [The24]. Where a coverage proof uses an independent constructive generation argument, the catalogue is a source of literal representatives, not its sole completeness argument.

All reproduction commands are local. The file `computations/README.md` documents the producers and independent checkers, and `certificates/manifests/finite_witnesses.json` binds their finite input and output files. Hashes identify bytes; they do not prove a mathematical theorem. The checks do not initialize or invoke a proof assistant.

The same-source $C_3$ example, the inverse-complement reciprocal, the proper-subdirect transport examples, and the binary duplicate factor $1/4$ are regression conditions on the counting maps. Passing these conditions alone does not establish an asymptotic theorem. The binary coverage principles are proved in Lemma 24.5; the universal mixture and recurrence arguments are proved in Theorems 23.15 and 24.8. The complete nonbinary exhaustion is proved in Theorem 20.1; the ordinary counting inequality and its boundedness-first induction are proved in Section 32. The supplementary affine-nine and narrow socket data provide independent local checks; the proof above uses the general frame and mixture arguments for those domains.

## A.1 Formal verification

The accompanying Lean development fixes Lean 4.30.0 and a precise mathlib revision. Its final theorem proves Theorems 1.3, 33.2 and 33.3 relative to seven explicit published mathematical inputs. Their complete formulations, Lean types and publication locators appear in `formal/README.md`.

No finite primitive or transitive catalogue correspondence is a parameter of the final theorem. The project-specific source, capacity, marked-moment and summation statements are proved internally. Both final exports have only the standard logical axioms of Lean in addition to their stated parameters.

An optional development starts from a pinned formulation of CFSG and proves model-specific inputs on its original concrete groups. In particular, for $q = p^f$ it establishes

$$|\operatorname{Out}(G_2(q))| \le 2f, \quad |\operatorname{Out}(F_4(q))| \le 2f,$$
$$|\operatorname{Out}({}^3D_4(q))| \le 3f, \quad |\operatorname{Out}({}^2E_6(q))| \le 6f.$$

The proofs include every characteristic, retain the actual signed root charts and inner scalar tori, and do not assume that arbitrary abstract automorphisms lift to the matrix source. Together with proved index and Sylow bounds, they remove these families from the optional reduction. For the original simple $G_2(q)$ model with $q \ge 3$, an actual singular-ray action gives $\ell < 3q^5$, while two disjoint Sylow images give $q^{12} \le |G_2(q)|$. Thus its least proper subgroup index also satisfies $\ell^2 < |G_2(q)|$, including $q = 3$. The original torus gives $q \le \ell$; together with the proved outer bound this discharges its full outer-order budget. Both residual model hypotheses now exclude $G_2$. For the original Ree group ${}^2G_2(q)$, the literal fixed group has a proved two-cell Bruhat decomposition, order $q^3(q-1)(q^3+1)$, and a soluble positive Sylow normalizer. Its two opposite positive subgroups have nontrivial normal image in the actual simple derived central quotient, giving the required generating Sylow witnesses there. Thus ${}^2G_2$ is also removed from the good-Sylow hypothesis. The actual Borel point action gives $\ell \le q^3 + 1$. The full derived centre is trivial and the fixed-to-derived index is at most four, so $4|{}^2G_2(q)| \ge q^3(q-1)(q^3+1)$. These bounds also prove $\ell^2 < |{}^2G_2(q)|$ for $q \ge 27$. The actual central root line and square torus give $q \le \ell$. The opposed Sylow pair's stabilizer restricts faithfully to the positive group; its characteristic layers and cubing map give central-axis rigidity. Counting this central action and charging the same inner derived torus proves $|\operatorname{Out}({}^2G_2(3^{2m+1}))| \le 2(2m+1)$. This discharges the outer-order budget as well, so both residual model hypotheses exclude ${}^2G_2$. The other Ree family, the Tits group and sporadic groups retain their full payloads. That reduction still has explicit residual model hypotheses and five published counting or generation inputs; it is not a proof from CFSG alone. The seven-input theorem stated above remains the canonical formalization of the asymptotic results.

The auxiliary class-count reductions include a shared basis-stabilizer argument: a finite faithful linear group preserving a nondegenerate bilinear form over a field of order $q$ has order at most $q^{n(n+1)/2}$ in dimension $n$. The same counting argument applies to separating sesquilinear pairings. It sharpens the classical-model order bounds while retaining the original socle and degree witnesses. A separate trace and centralizer argument proves $k(\mathrm{SL}_2(q)) \le q + 4$ over every finite field. A central-involution orbit count gives $2k(\mathrm{PSL}_2(q)) \le q + 5$ in odd characteristic. Together with actual torus and root actions and the small alternating cases, this closes the original rank-one component class budget over every permitted finite field, for both original degree forms. At field order nine, an order-360 index-gap argument either identifies the model with $A_6$ from an index-six subgroup or supplies the stronger proper-index lower bound eight. No finite subgroup census is assumed in this argument. Literal three-dimensional normal forms give the all-field bound $k(\mathrm{SL}_3(q)) \le \gcd(q-1, 3)(q^2 + 2q)$. The corresponding scalar-centralizer class-fibre bound and the sharper outer-order budget close the original rank-two component capacity as well. Over the two-element field, the proved order 168 and a faithful coset action force every proper index to be at least seven. The class equation also gives $k(S) \le |S|/b + 1$ for a center-free group whose proper indices are at least $b$; exact matrix orders and central quotients close further type-$A$ parameters with this shared test. The same class-equation capacity test closes seven further

orthogonal, twisted-orthogonal and unitary parameter cases, retaining the proved outer-order factor $8f$ in the twisted-orthogonal application. For the original odd-characteristic $C_2$ model, a coupled Heisenberg–$\mathrm{SL}_2$ stabilizer argument proves proper index at least $q^2$. One numerical induction then pays its component order cube for every $q \geq 7$. Over a prime field the coupled argument improves the index bound to $q^3$, which pays the two small odd fields as well. The whole odd-characteristic $C_2$ branch is therefore closed. The two binary type-$A$ parameters left by these order tests are closed by one cyclic-polynomial template. Equal multisets of actual cyclic factors give intertwining linear equivalences; the checked interior-bit covers have 18 codes in dimension four and 104 in dimension six. Surjective matrix and central-quotient maps transfer these bounds to the original projective models, without exceptional isomorphisms or enumeration of their matrices. In characteristic two, the actual symmetric shear subgroup and its general-linear normalizer give the $q^2$ proper-index bound for $q \geq 8$ as well. This argument retains the genuine prime-field annihilator of the actual stabilizer kernel, without asserting extension-field linearity. The same component budget therefore closes these $C_2$ capacities; field order four remains open. A sharper projective-level argument proves $q(q-1)^2 \leq 2[S : H]$ in characteristic two, giving index at least 18 at field order four, but this does not yet pay that case's class-count capacity. Separately, the half-degree ordinary-generator bound is proved for every finite nilpotent $J \leq S_n$, by sharing generators across its actual Sylow factors. The whole primitive-affine branch of the sharp generator induction is also proved from its strictly smaller-degree cases, using the complement action on nonidentity translations and, in odd characteristic, on inverse pairs. The whole intransitive branch and the binary-, ternary- and four-point-block branches are proved from the same induction premise, with the degree-three symmetric-group exception retained. For four-point blocks, the proof exhausts the actual zero, single and multiple cubic-kernel classes. The single-class case retains the genuine scalar centralizer; at two blocks an actual cyclic seed and a proper/top argument avoid imposing scalar linearity on the larger group. The block proofs use a shared polynomial-orbit generator bound for invariant subspaces of permutation and monomial modules. The same argument now covers all odd prime-power affine block components, retaining their actual inverse-pair quotient, binary sign kernel and prime-field translation module. The odd-block proof also covers a proper three-point top. An actual local-order argument proves the binary-affine block tail in dimensions at least seven, for every top; its remaining binary-affine fibre sizes are 8, 16, 32 and 64. For product actions with at least two socle factors, the component order budget pays the generator bound with the full factorial retained. The general class-count and nonnilpotent generation inputs remain explicit hypotheses. In the optional reduction, the already retained Tracey permutation input pays every transitive degree at least 22. Together with the proved intransitive induction, this reduces the sharp ordinary-generator input to transitive degrees below 22; that bounded input remains open.

The auxiliary original-$F_4$ source verification now also proves that its integral Serre lattice is spanned over $\mathbb{Z}$ by the displayed 52 divided-power vectors. Sparse integer action tables are proved from the original Serre relations; even second coefficients and zero third coefficients prove stability under every original divided power. Lattice minimality then gives exact spanning. Integer root and Cartan coordinates give a left inverse for the actual matrix images. Its split injection survives arbitrary scalar extension, proving faithfulness of the original image representation in every characteristic. In characteristic two, the original model is now identified with the literal folded matrix group, with trivial center. Its exceptional automorphism has the exact signed root parameters and squares to field Frobenius, without an ambient ordinary-$F_4$ simplicity assumption. The remaining Ree/Tits fixed-group bounds still require their own proofs. The pinned Ree and Tits definitions are now identified with the actual derived subgroup of this fixed group modulo its

full center. Collected middle-root words also construct an actual subgroup of order $q^2$ inside the literal fixed group, with injective parameters and proved multiplication and inverse laws. This is a bounded subgroup; it does not yet identify the whole positive group or its derived-center quotient.

Actual relative Weyl words conjugate the literal fixed positive subgroup onto the fixed negative subgroup. A constructed fixed-positive element has all six ambient simple-root leading coordinates equal to one. The required good-Sylow and numerical bounds for the original derived Ree/Tits models remain open.

The coarse all-subgroup estimate is derived internally from the retained soluble-generation input alone. For soluble primitive actions, the composition and subgroup-rank bounds used in its large-orbit step follow unconditionally from the actual affine structure. For every fixed orbit cutoff $C$, the actual bounded nilpotent family has size at most

$$2^{b^2/16+A_C b\log_2(b+2)}.$$

Its proof retains the binary and odd nilpotent sectors and their joint subdirect words. An actual soluble container menu of size $2^{O(b\log^4(b+2))}$, together with original-support compression and the large-orbit generator bound, gives $s_n \le 2^{n^2/16+\beta n\sqrt{n}}$. This coarse estimate is therefore not an additional parameter of the final formal theorem.

The same constructed soluble cover pays the marked large-orbit encoding: one containing group is recorded together with the original relative tuple and its character restrictions. Its subquadratic menu cost is absorbed uniformly in the marking parameter. Consequently the global marked-moment branch needs no labelled-transitive-count premise; that published input is retained only for the separate nonsoluble pre-$\mathrm{E}_7$ action menu.

Several classical results are proved internally: Burnside's two-prime solvability theorem; semisimplicity and conjugacy of Clifford constituents over an arbitrary field; the binary normal-subgroup bound

$$\#\{N \trianglelefteq G\}\, 2^{-\ell^2/4} \le 8\, 2^{2\ell} \qquad (|G| = 2^\ell);$$

simplicity of $\mathrm{PSL}_3(F)$; and $\mathrm{PGL}_2(\mathbf{F}_5) \cong S_5$. The projective proofs use elementary matrices and faithful projective actions, without ATLAS code. The selected proof ports preserve their Apache-2.0 attribution and exact revisions in the formal sources. Full abelianization-order bounds for faithful permutation actions, including higher prime-power cyclic factors, supply both abelian-quotient fields of the character interface. The faithful-character tuple theorem, nilpotent class-number bound and arbitrary-normal-target bounded-factor epimorphism theorem are also proved internally.

The refined induced-module theorem keeps one uniform generator capacity satisfying the half-dimension, logarithmic, coprime, primary and prime-power bounds simultaneously. Its proof uses actual Sylow–Mackey pieces, finite chain widths, Nakayama's lemma and coprime semisimplicity. The two soluble-orbit consumers are proved on the actual permutation module and on the actual globally normal ternary core. These replacements do not assert the stronger unused arbitrary-local per-orbit statement. Raw O'Nan–Scott structure is proved using actual minimal normal subgroups, the full socle, point-stabilizer maximality and the exact inner kernel in the outer-wreath action. The quantitative simple-index inequalities remain an explicit published input. They also imply $|\operatorname{Aut}(S)| \le |S|^2$ for every finite nonabelian simple $S$, so simple two-generation is not a separate input. The primitive relative ternary-head tail follows directly from actual affine orders and full-socle quotient orders, as in Lemma 9.3; no ordinary primitive normal-generation theorem is required.

Exact binary excess classification is now proved internally, including all degree and proper-normal restrictions. The proof retains the original pair kernels, joint parity relations and square

carries, and checks the finite scalar certificates in Lean's kernel. It also proves the full original abelianization consequences. Together with the intrinsic proper-normal bound, it supplies the three original dependent-word prices without changing their factors or maximal fully projecting tail. Roney-Dougal–Tracey's full binary normal-$F$ quotient theorem is also proved internally. Its commutator is the ambient $[F, Q]$; the centralizer, original quotient, exact excess alternatives and excess-one generator equality are all retained, including the degree-eight exception.

The last degree-32 excess-one configurations have kernel head and top rank $(4, 5)$ or $(3, 6)$. In the first, actual joint layer bounds give $|G'| \leq 1024$ and top order at most 1024; every nonabelian original quotient has centre at most 256. The small-denominator case uses the same original second centre: its constant-displacement kernel axis has order at most 16, and its image in the actual top centre has order at most 16. This joint bound avoids assuming faithfulness at a half-dimension boundary.

The second configuration has four actual kernel models. Their original normal quotients use joint source bounds or a proper normal subgroup constructed from an invariant interval $A < S$ in the actual pair module. The full lower layer and the same normal denominator are retained. A selected central constant flip has zero class in the original affine cocycle, proved by its actual commutator and annihilator identities; neither extension is assumed split. The mixed commutator and centralizer are then bounded together through that interval.

The current primitive consumers use actual linear and full-socle structure rather than bounded catalogue correspondence. This proves the ternary chief-weight and $3/10$ budgets and the ordinary Dixon order bound. The strict-head bound is used away from degrees 10 and 12; the complete physical $C_3$ source pays those two actual semisimple sectors separately. This route does not claim the stronger unused global strict-head theorem. The actual degree-eighteen bound and sufficient semisimple compression margin $3q \leq 2r$ complete the primitive consumers.

Small affine sources keep their actual block systems, translation kernels, normal axes and quotient fibres. Degree eight has complete checked linear and transitive action coverage. Degree sixteen uses the proved $\mathrm{GL}_4(\mathbf{F}_2) \cong A_8$ geometry and the original translation kernel; degree twenty-seven uses proved linear structure. The degree-five affine classification and degree-25 projective facts are internal. Degree 81 uses its complete linear-order budget; degrees 32 and 64 use original Clifford constituents, scalar fields and simple normal subgroups. The general affine component bound uses actual chief charges and full affine-order divisors. No primitive composition-length theorem is a parameter of the final exports. Complete original twelve-pair and sixteen-pair sources therefore require no action catalogue premise. Stronger optional classifiers and their finite witness packages remain separate from the final proof route.

Finally, the marked moment uses the now-derived coarse subgroup bound and the constructed soluble-container menu. Its soluble nilpotent core has the proved ordinary supplement bound $L(B!+1)\lfloor\sqrt{M}\rfloor$ on the original bounded-orbit source, with $L = B = C!$ and $M = 2b$. This changes only a fixed constant and does not claim the sharper unrestricted $4L\sqrt{M}$ theorem. The original cutoff, action weights, character restrictions and terminal attachment are preserved. Recursive local source/object verification and exported-declaration axiom audits accompany the kernel checks; detailed build records are kept outside the publication repository.

# B The finite carrier reduction and its two rational cones

The finite local input consists of the eight master groups and all their literal normal kernels. The respective numbers of kernels are $27, 13, 28, 2911, 2911, 2911, 2292, 2915$. Their generators and all central-involution extension edges are supplied in the carrier normal certificate. Lemma 24.5

proves completeness from these edges. For each kernel the checked tuple is $(k, n, m, a_2, c, g)$ as in Lemma 23.1 and (23.4). In particular the maximum defining $m$ is computed over the same complete normal list, and the original quotient is used for $c, g$.

For the five degree-sixteen groups, the structural proof in Lemma 23.10 gives exhaustive original-normal coverage without constructing these normal lists. Effective polygon inclusion reduces its rows to the displayed H16 rows and the two additional sixth-colour envelopes described in the proof above. The latter obey all displayed cross-column comparisons and their mutual comparison, so they use the same rational cone certificate below.

For the degree-sixteen groups the identities $\Phi(G) = G'$ permit $a_2 \le m$: here $N^2[N, G] \le N \cap G'$. Their rows are dominated by the following twelve scale-two rows; the other three blocks below are the scale-one rows of $X, J, P$. Domination is simultaneous on the pair supports and both marks $k, q = \max(m, a_2)$, including every cross column between degrees eight and sixteen.

| $H16$ | $k$ | $n$ | $m$ | $a_2$ | $c$ | $g$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 6 |
| 1 | 1 | 1 | 1 | 1 | 2 | 5 |
| 2 | 1 | 2 | 1 | 1 | 4 | 4 |
| 3 | 2 | 3 | 2 | 2 | 3 | 3 |
| 4 | 2 | 4 | 2 | 2 | 4 | 2 |
| 5 | 3 | 6 | 3 | 3 | 6 | 0 |
| 6 | 4 | 6 | 4 | 4 | 5 | 0 |
| 7 | 4 | 7 | 4 | 4 | 4 | 0 |
| 8 | 4 | 8 | 4 | 4 | 3 | 0 |
| 9 | 5 | 11 | 3 | 3 | 1 | 0 |
| 10 | 5 | 11 | 4 | 4 | 0 | 0 |
| 11 | 6 | 12 | 3 | 3 | 0 | 0 |

| $X$ | $k$ | $n$ | $m$ | $a_2$ | $c$ | $g$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 3 |
| 1 | 1 | 1 | 1 | 0 | 2 | 2 |
| 2 | 1 | 2 | 1 | 1 | 2 | 1 |
| 3 | 1 | 3 | 1 | 1 | 1 | 1 |
| 4 | 1 | 4 | 2 | 2 | 2 | 0 |
| 5 | 2 | 3 | 2 | 1 | 3 | 0 |
| 6 | 2 | 4 | 2 | 1 | 2 | 0 |
| 7 | 2 | 5 | 2 | 2 | 1 | 0 |
| 8 | 3 | 6 | 2 | 2 | 0 | 0 |

| $J$ | $k$ | $n$ | $m$ | $a_2$ | $c$ | $g$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 3 |
| 1 | 1 | 1 | 1 | 0 | 1 | 2 |
| 2 | 1 | 2 | 1 | 1 | 2 | 1 |
| 3 | 1 | 3 | 1 | 1 | 1 | 1 |
| 4 | 1 | 3 | 1 | 1 | 3 | 0 |
| 5 | 1 | 4 | 1 | 1 | 2 | 0 |
| 6 | 1 | 5 | 1 | 2 | 1 | 0 |
| 7 | 2 | 4 | 1 | 1 | 2 | 0 |
| 8 | 2 | 5 | 1 | 1 | 1 | 0 |
| 9 | 2 | 6 | 1 | 2 | 0 | 0 |

| $P$ | $k$ | $n$ | $m$ | $a_2$ | $c$ | $g$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 4 |
| 1 | 1 | 1 | 1 | 0 | 1 | 3 |
| 2 | 1 | 2 | 1 | 1 | 2 | 2 |
| 3 | 1 | 3 | 1 | 1 | 2 | 1 |
| 4 | 1 | 4 | 1 | 1 | 1 | 1 |
| 5 | 1 | 5 | 2 | 2 | 2 | 0 |
| 6 | 2 | 4 | 2 | 1 | 3 | 0 |
| 7 | 2 | 5 | 2 | 1 | 2 | 0 |
| 8 | 2 | 6 | 2 | 2 | 1 | 0 |
| 9 | 3 | 7 | 2 | 2 | 0 | 0 |

The normalized representatives are

| | $k$ | $n$ | $m$ | $a_2$ | $c$ | $g$ | $s$ |
|---|---|---|---|---|---|---|---|
| $H_9$ | 5 | 11 | 3 | 3 | 1 | 0 | 2 |
| $X_{\mathrm{whole}}$ | 3 | 6 | 2 | 2 | 0 | 0 | 1 |
| $P_0$ | 0 | 0 | 0 | 0 | 1 | 4 | 1 |
| $P_1$ | 1 | 1 | 1 | 0 | 1 | 3 | 1 |
| $P_2$ | 1 | 2 | 1 | 1 | 2 | 2 | 1 |
| $P_6$ | 2 | 4 | 2 | 1 | 3 | 0 | 1 |
| $P_{\mathrm{whole}}$ | 3 | 7 | 2 | 2 | 0 | 0 | 1 |

For completeness, the following list gives a dominating representative for each row in each displayed block, in its displayed order. The last representative has exactly the same normalized pair row and marks as $X_{\mathrm{whole}}$, so they are merged. All comparisons here are finite rational inequalities of the explicit support functions. They can be checked without enumerating the feasible integer pairs: for $x, y \ge 0$, the support of the row $(k, n, m, a_2, c, g)$ is at most

$$\min\{xk + ym,\ (x-y)_+k + (y-x)_+m + \min(x,y)n\}.$$

The first expression uses the two separate capacity bounds. For the second, use the joint bound on $\delta + \epsilon$ and the separate bound for the variable with the larger coefficient. Substitution of the displayed rational entries proves all $41^2$ ordered pair comparisons, with their original scales; the two mark comparisons are coordinatewise. This specialization is used only for the nonnegative quotient slopes, not to discard a negative first mark in the recurrence.

$$H16: \quad (P_0, P_1, P_2, P_2, P_2, P_6, P_6, P_6, P_6, H_9, X_{\mathrm{whole}}, X_{\mathrm{whole}}).$$

$$X: \quad (P_0, P_2, P_2, P_2, P_6, P_6, P_6, P_6, X_{\mathrm{whole}}).$$

$$J: \quad (P_0, P_1, P_2, P_2, P_2, P_2, P_6, P_6, P_6, X_{\mathrm{whole}}).$$

$$P: \quad (P_0, P_1, P_2, P_2, P_2, P_6, P_6, P_6, P_6, X_{\mathrm{whole}}).$$

The reduction of the actual degree-sixteen tuples to the twelve rows also checks all degree-eight columns, and the two marks, in the committed transition certificate. Thus the preceding reduction does not presume that domination inside a smaller submatrix survives the addition of new colours. Scaling and merging give exactly (23.6).

It remains to prove the scalar inequality used in Theorem 23.13. Put $p = \sum_i p_i$ and

$$\mathcal{D} = R^2/4 + (139/328)Rp + (627/2624)p^2 - \mathcal{E}.$$

In the first cone let $(L, J, S) = (\ell, j - 2\ell, R - j)$, and in the second let $(A, C, S) = (j, 2\ell - j, R - 2\ell)$. These are nonnegative in their respective cones. Let $z$ consist of $p_1, \ldots, p_6$ followed by the three corresponding cone coordinates. Expansion gives $5248\mathcal{D} = z^{\mathsf{T}} M_\nu z$. The upper left $6 \times 6$ block, common to both $M_\nu$, is

$$M_0 = \begin{pmatrix} 1049 & 1008 & -140 & 106 & -58 & 24 \\ 1008 & 1254 & -550 & -222 & -386 & -222 \\ -140 & -550 & 1254 & 598 & 434 & -386 \\ 106 & -222 & 598 & 762 & 598 & -58 \\ -58 & -386 & 434 & 598 & 598 & -58 \\ 24 & -222 & -386 & -58 & -58 & 270 \end{pmatrix}.$$

Write $M_\nu = \begin{pmatrix} M_0 & D_\nu \\ D_\nu^{\mathsf{T}} & E_\nu \end{pmatrix}$. The remaining blocks are

$$D_A = \begin{pmatrix} -400 & -528 & 1112 \\ -1056 & -856 & 1112 \\ 2224 & 1112 & 1112 \\ 912 & 456 & 1112 \\ 912 & 456 & 1112 \\ -400 & -200 & 1112 \end{pmatrix}, \qquad E_A = \begin{pmatrix} 5248 & 1312 & -1312 \\ 1312 & 1312 & -1312 \\ -1312 & -1312 & 1312 \end{pmatrix}.$$

$$D_B = \begin{pmatrix} -200 & -200 & 1112 \\ -528 & -528 & 1112 \\ 1112 & 1112 & 1112 \\ 456 & 456 & 1112 \\ 456 & 456 & 1112 \\ -200 & -200 & 1112 \end{pmatrix}, \qquad E_B = \begin{pmatrix} 1312 & 0 & -656 \\ 0 & 3936 & 1968 \\ -656 & 1968 & 1312 \end{pmatrix}.$$

The following rational matrices give positive-semidefinite parts.

$$P_A = \begin{pmatrix} 817 & 804 & -548 & -328 & -393 & -\frac{540}{11} & -896 & -688 & 688 \\ 804 & 1254 & -550 & -288 & -386 & -222 & -1129 & -856 & 856 \\ -548 & -550 & 1254 & 423 & 434 & -386 & 1836 & 1112 & -1112 \\ -328 & -288 & 423 & 577 & 237 & -\frac{3987}{44} & 563 & 323 & -323 \\ -393 & -386 & 434 & 237 & 490 & -58 & 646 & 363 & -363 \\ -\frac{540}{11} & -222 & -386 & -\frac{3987}{44} & -58 & 270 & -\frac{19363}{44} & -200 & 200 \\ -896 & -1129 & 1836 & 563 & 646 & -\frac{19363}{44} & 4645 & 1312 & -1312 \\ -688 & -856 & 1112 & 323 & 363 & -200 & 1312 & 1312 & -1312 \\ 688 & 856 & -1112 & -323 & -363 & 200 & -1312 & -1312 & 1312 \end{pmatrix}.$$

$$P_B = \begin{pmatrix} 729 & 721 & -469 & -236 & -318 & -\frac{602}{11} & -416 & -480 & 86 \\ 721 & 1254 & -550 & -296 & -386 & -222 & -565 & -585 & -2 \\ -469 & -550 & 1254 & 433 & 434 & -386 & 917 & 950 & -407 \\ -236 & -296 & 433 & 542 & 241 & -\frac{4061}{44} & 275 & 197 & -56 \\ -318 & -386 & 434 & 241 & 469 & -58 & 303 & 231 & -65 \\ -\frac{602}{11} & -222 & -386 & -\frac{4061}{44} & -58 & 270 & -\frac{2413}{11} & -\frac{10005}{44} & \frac{8581}{44} \\ -416 & -565 & 917 & 275 & 303 & -\frac{2413}{11} & 1254 & -134 & -813 \\ -480 & -585 & 950 & 197 & 231 & -\frac{10005}{44} & -134 & 2690 & 591 \\ 86 & -2 & -407 & -56 & -65 & \frac{8581}{44} & -813 & 591 & 891 \end{pmatrix}.$$

Subtraction gives $M_A - P_A \geq 0$ entrywise, with entries at most 2224, and $M_B - P_B \geq 0$ entrywise, with entries at most 1519. These assertions are direct comparisons of the displayed rational entries. Both matrices annihilate $v = (0, 17, 21, 0, 0, 44, 0, 0, 0)$; $P_A$ also annihilates

$w = (0,0,0,0,0,0,0,0,1,1)$. Delete coordinates six and nine from $P_A$. The successive leading principal determinants of the retained matrix are

$$817,\quad 378102,\quad 335067392,\quad 132631407594,$$
$$35034163435774,\quad 63581756789450739,\quad 2100371145348629273.$$

They are positive. The omitted coordinates of $(v, w)$ form $\operatorname{diag}(44, 1)$, so subtracting their unique linear combination carries any vector into this retained subspace without changing its quadratic value. Thus $P_A$ is positive semidefinite. Delete coordinate six from $P_B$. Its retained leading principal determinants are

$$729,\quad 394325,\quad 370093856,\quad 140632346331,$$
$$36346448139435,\quad 19760569015182064,$$
$$7216647669027584752,\quad 1335792379687239644084.$$

Again they are positive, and the omitted coordinate of $v$ is 44, so the same kernel elimination proves that $P_B$ is positive semidefinite. Hence $z^{\mathsf{T}} M_\nu z \geq 0$ for every nonnegative $z$. The two cones exhaust all $0 \leq j \leq R$, $0 \leq \ell \leq R/2$, including $j < \ell$, proving the required scalar inequality.

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